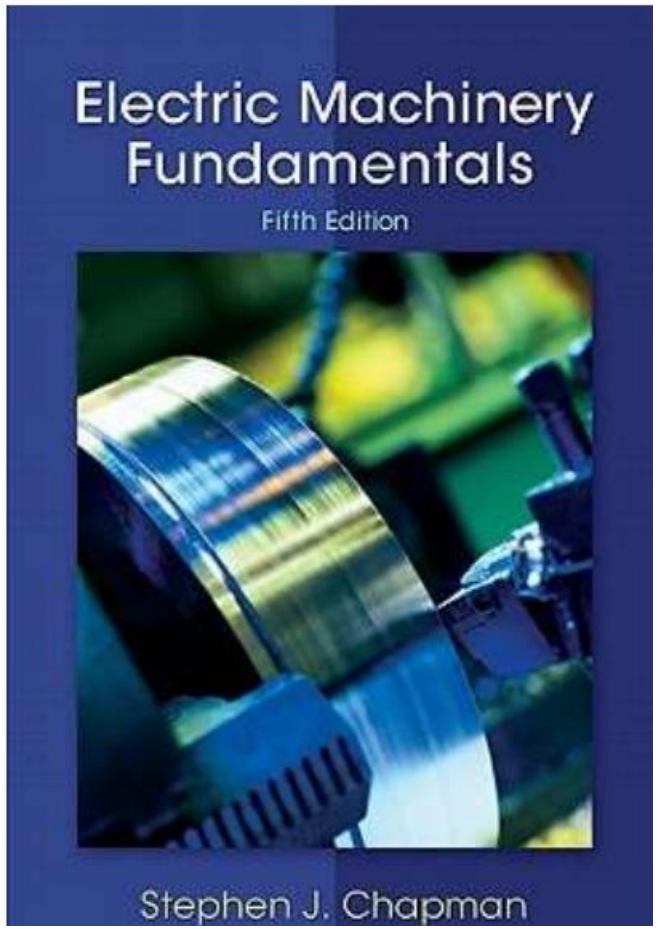
Text book and supplementary materials of this course

- Stephen J. Chapman



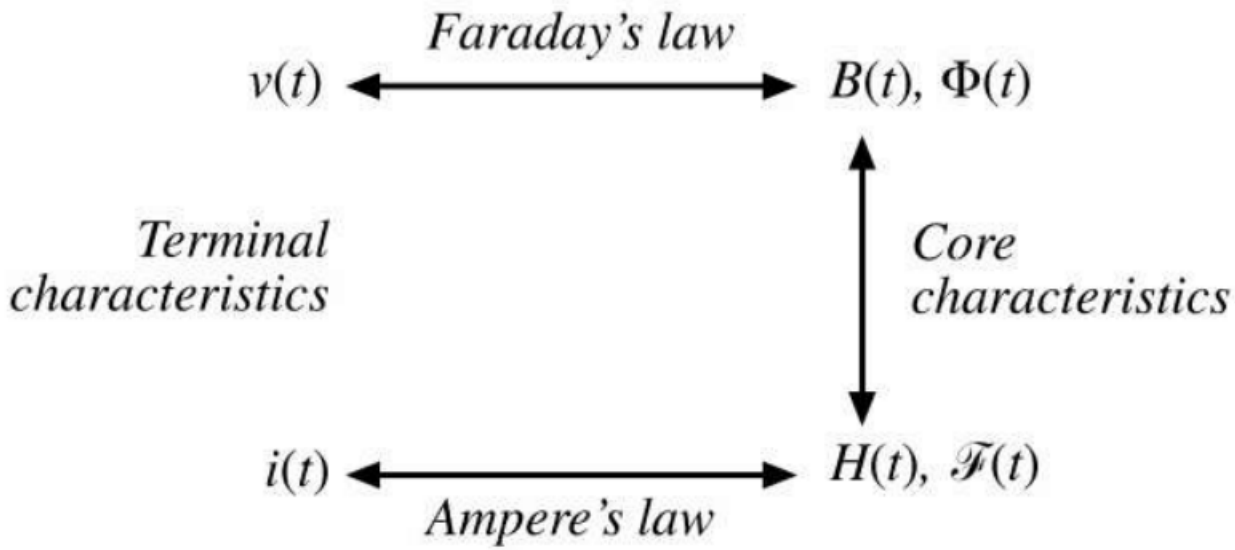
Electric Machinery

Course highlight

- ***Energy Conversion*** schemes are the key ideas introduced in this course
- Which types of energy conversion are concerned?
- Electric energy to electric energy
  - Transformer
- Electric energy to mechanical energy
  - Motor
- Mechanical energy to electric energy
  - Generator
- Magnetic energy is essential !

Course Outlines

- Magnetic field: Ampere's law
- Magnetic flux: magnetic material, hysteresis characteristics
- Transformer: Faraday's law, Len's law



Course Outlines

- Magnetic circuit
- Motor/generator: Induced voltage, induced force

- Ideal/non-ideal transformer
- Equivalent transformer circuit
- Voltage regulation, efficiency

Course Outlines

- AC machine : induction machine, synchronous machine
- DC machine : separated excited, shunt excited, series excited, compound excited
- How the motor works
  - Torque/speed
- How the generator works
  - Voltage/current

Course Outline

- Induction motor (IM) – the most widely used ac motor in the world
  - Structure and operation theories of IM
  - Equivalent circuit of IM
  - Torque/speed characteristics
  - Basic motor control
- Induction generator (seldom used)
  - Output voltage control
  - Voltage/current characteristics

Course Outline

- Synchronous generator (SG) – the most widely used generator in the world
  - Structure and operation theories of SG
  - Equivalent circuit of SG
  - Voltage/current characteristics
  - Parallel operation
- Synchronous motor
  - Operation principles
  - Starting of synchronous motor
  - Torque/speed characteristics

History of Electric Machinery

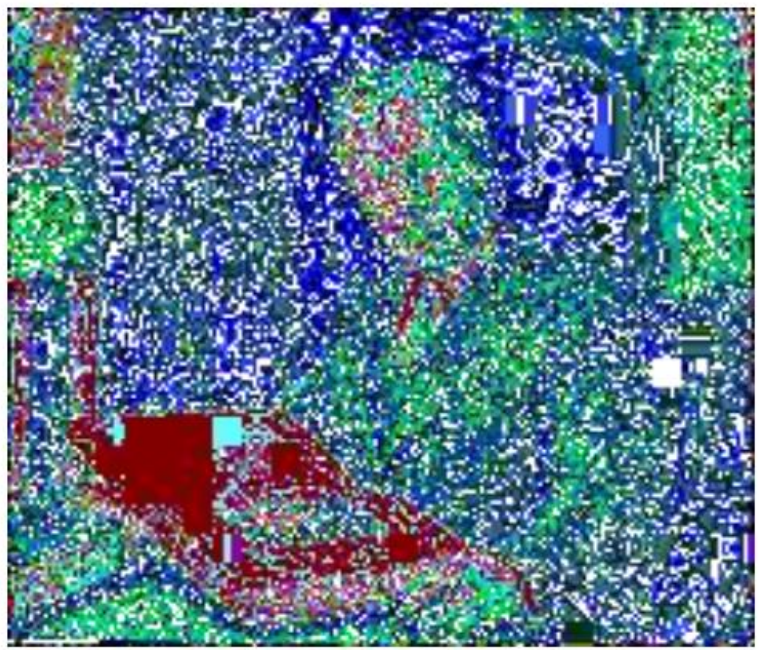
DC generator, driven by steam engines

1882 Thomas A. Edison opens Pearl St. Station, NYC

Waterwheel-driven DC generator installed in Appleton, Wisconsin

History of Electric Machinery

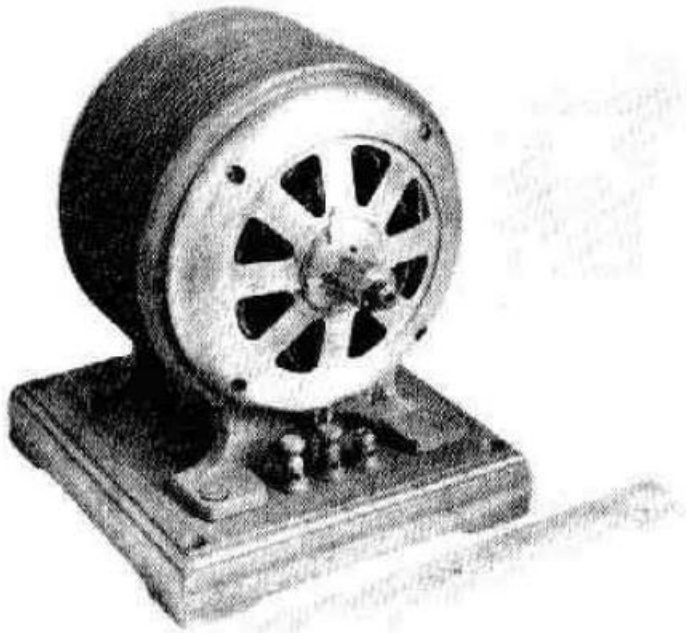
1884 Frank J. Sprague produces DC motor for Edison systems



History of Electric Machinery

1885 William Stanley develops commercially practical transformer

History of Electric Machinery



The TESLA AC Motor—1888.

1888

Nikola Tesla presents paper on two-phase ac induction and synchronous motors

Today's development

- DC Machine
  - Motor
  - Generator
- Transformer
  - Single phase
  - Three phases
- AC Machine
  - Synchronous machine – motor, generator
  - Asynchronous machine (induction machine) – motor, generator

Today's development and future trends

- Micro-step stepping motor
- Permanent magnet synchronous motor (PMSM)
  - Brushless dc motor (BLDCM)
- Linear motor
- Reluctance motor
  - Synchronous reluctance
  - Switched reluctance
- Ultrasonic motor
- Bionic robotics
- MEMS motor

Course relation

- It is the fundamental course of the electrical engineering
- Future courses
  - Power electronics
  - Motor control
  - Electric motor drive
  - Power systems
  - Renewable energy
  - Electrical vehicle

Chapter 1. Introduction to machinery principles

1. Rotation motion, Newton's law and power relationships
2. The magnetic field
3. Faraday's law
4. Produce an induced force on a wire
5. Produce an induced voltage on a conductor
6. Linear dc machine examples
7. Real, reactive and apparatus power in AC circuits

Chapter 1. Introduction to machinery principles

1. Rotation motion, Newton's law and power relationships
2. The magnetic field
3. Faraday's law
4. Produce an induced force on a wire
5. Produce an induced voltage on a conductor
6. Linear dc machine examples
7. Real, reactive and apparatus power in AC circuits

Rotation motion, Newton's law and power relationships

- Clockwise (CW) and Counterclockwise (CCW)
  - CCW is assumed as the positive direction, CW is assumed as the negative direction.
- Linear and rotation motion
  - Position and angular $\theta = \omega t$

    (meter) (degree or radian)
  - Speed and angular speed

$$v = \frac{dr}{dt} \qquad \omega = \frac{d\theta}{dt}$$

ω_m	angular velocity expressed in radians per second
f_m	angular velocity expressed in revolutions per second
n_m	angular velocity expressed in revolutions per minute

Rotation motion, Newton's law and power relationships

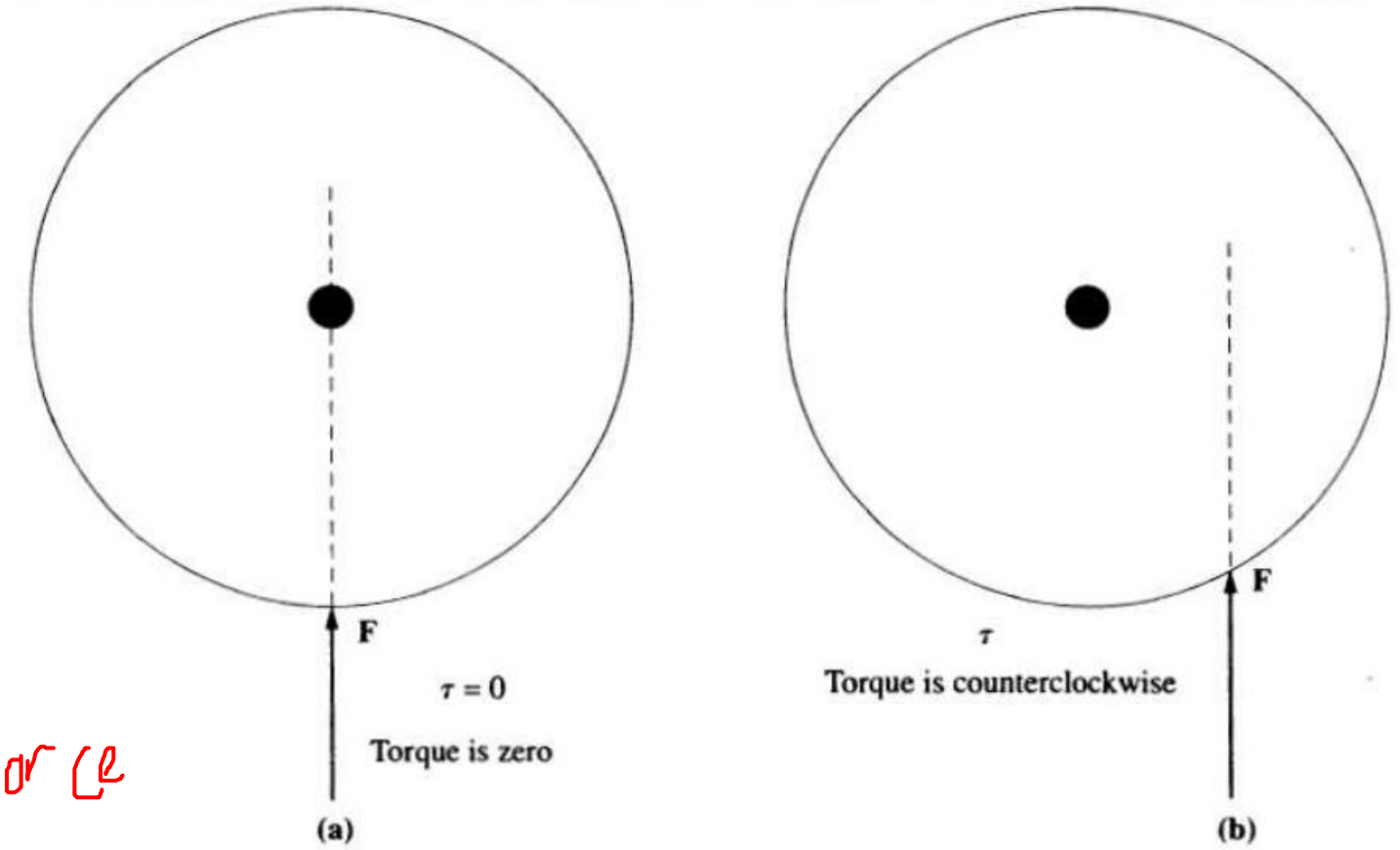
- relationships $n_m = 60 f_m$

$$f_m = \frac{\omega_m}{2\pi}$$

- Acceleration and angular acceleration

$$a = \frac{dv}{dt} \qquad \alpha = \frac{d\omega}{dt}$$

Torque



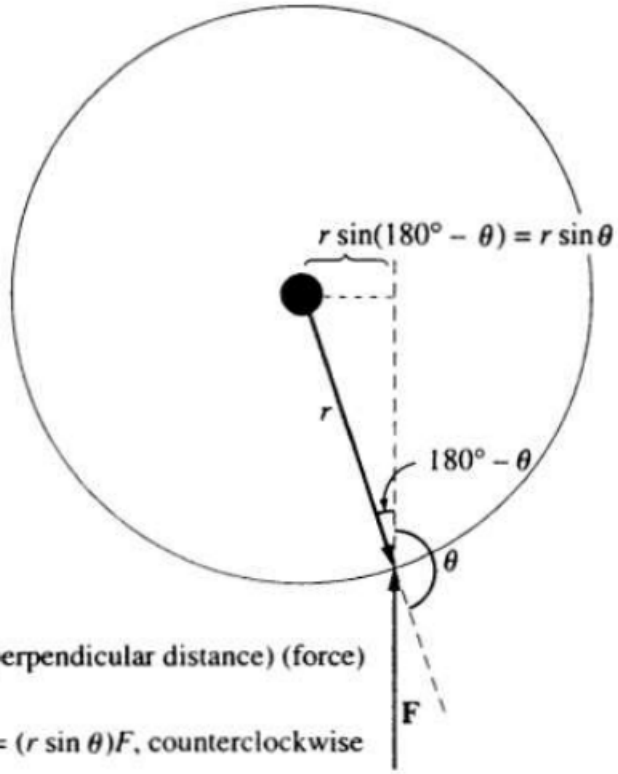
Twisting Force

Torque

$F = iLXB \Rightarrow iLB\sin\theta$

τ = (force applied)(perpendicular distance)
= $(F)\,(r \sin\theta)$
= $rF \sin\theta$

where θ is the angle between the vector **r** and the vector **F**.



τ = (perpendicular distance) (force)

$\tau = (r \sin\theta)F$, counterclockwise

$\tau = \alpha I \rightarrow$ moment of inertia (mass times)

$\rightarrow$ acceleration

Newton's law of rotation

1. Force $F = ma$

   F = net force applied to an object
   m = mass of the object
   a = resulting acceleration

2. Torque $\tau = J\alpha$

Torque and Work

For linear motion, work is defined as the application of a *force* through a *distance*. In equation form,

$$W = \int F\,dr \tag{1–9}$$

where it is assumed that the force is collinear with the direction of motion. For the special case of a constant force applied collinearly with the direction of motion, this equation becomes just

$$W = Fr \tag{1–10}$$

The units of work are joules in SI and foot-pounds in the English system.

For rotational motion, work is the application of a *torque* through an *angle*. Here the equation for work is

$$W = \int \tau\,d\theta \tag{1–11}$$

and if the torque is constant,

$$W = \tau\theta \tag{1–12}$$

Power (rate of doing work)

Power is the rate of doing work, or the increase in work per unit time. The equation for power is

$$P = \frac{dW}{dt} \tag{1–13}$$

It is usually measured in joules per second (watts), but also can be measured in foot-pounds per second or in horsepower.

By this definition, and assuming that force is constant and collinear with the direction of motion, power is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(Fr) = F\left(\frac{dr}{dt}\right) = Fv \tag{1–14}$$

Similarly, assuming constant torque, power in rotational motion is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau\left(\frac{d\theta}{dt}\right) = \tau\omega$$

$$P = \tau\omega \tag{1–15}$$

Equation (1–15) is very important in the study of electric machinery, because it can describe the mechanical power on the shaft of a motor or generator.

Conversion between watts and horsepower

1. Watts and horsepower

$$P\,(\text{watts}) = \frac{\tau\,(\text{lb-ft})\;n\,(\text{r/min})}{7.04}$$

$$P\,(\text{horsepower}) = \frac{\tau\,(\text{lb-ft})\;n\,(\text{r/min})}{5252}$$

1 HP ≈ 746 W

2. Conversion between two units

   5252 / 7.04 = 746.02

   1hp = 746W = 0.746kW

The magnetic field

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of *transformer action*.)
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of *motor action*.)
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of *generator action*.)

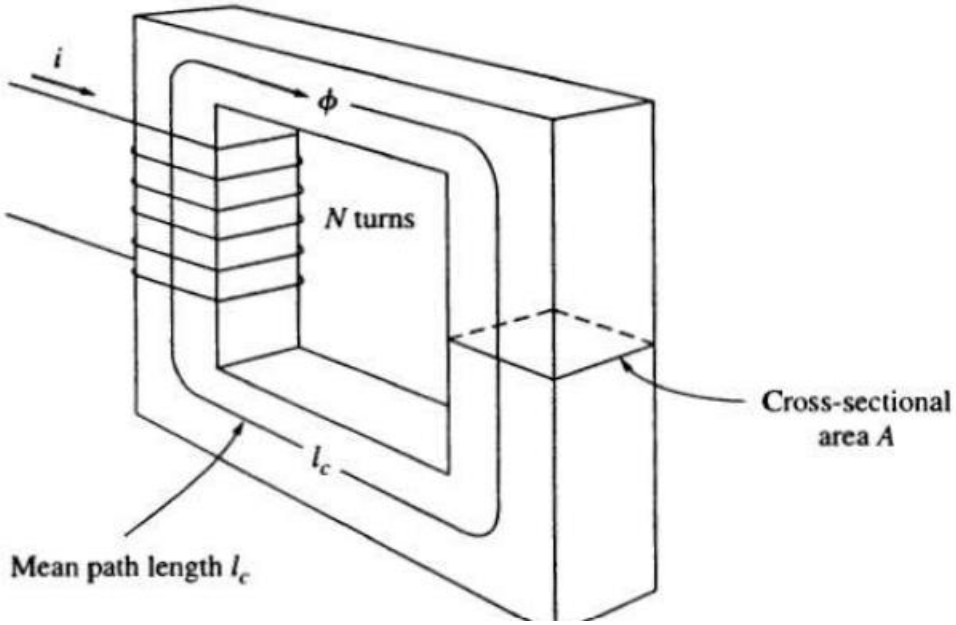
Produce a magnetic field – Ampere's law

1. The magnetic field is produced by ampere's law
2. The core is a ferromagnetic material

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{net}}$$

$$Hl_c = Ni$$

$$H = \frac{Ni}{l_c}$$



From the magnetic field to magnetic flux density

1. When the magnetic field is applied on a ferromagnetic material, the magnetic flux density **B** will be produced

$$\mathbf{B} = \mu\mathbf{H}$$

H = magnetic field intensity
μ = magnetic *permeability* of material
B = resulting magnetic flux density produced

Tesla

The permeability of free space is called μ_0, and its value is

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \qquad (1\text{–}22)$$

The permeability of any other material compared to the permeability of free space is called its *relative permeability:*

$$\mu_r = \frac{\mu}{\mu_0} \qquad (1\text{–}23)$$

Magnetic flux density and magnetic flux

1. Magnetic flux density

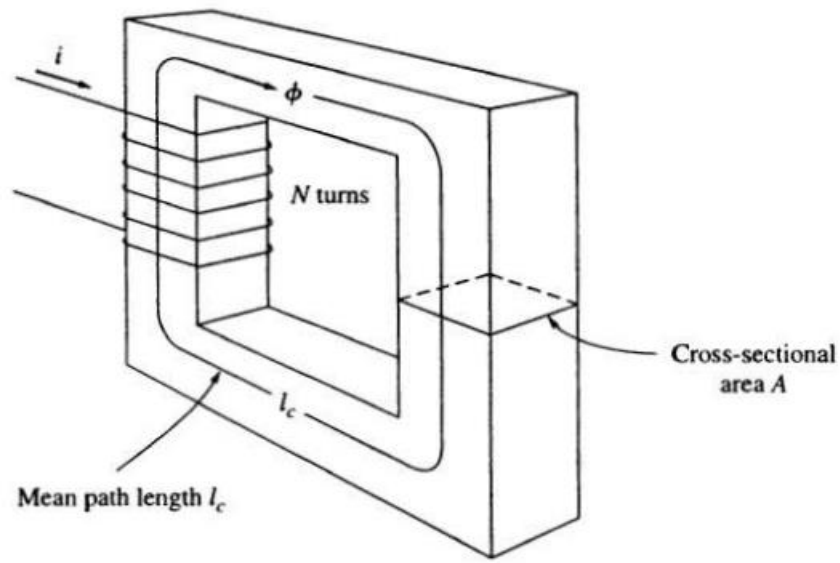
$$B = \mu H = \frac{\mu N i}{l_c}$$

2. Magnetic flux

$$\phi = \int_A \mathbf{B} \cdot d\mathbf{A}$$

$$\phi = BA$$

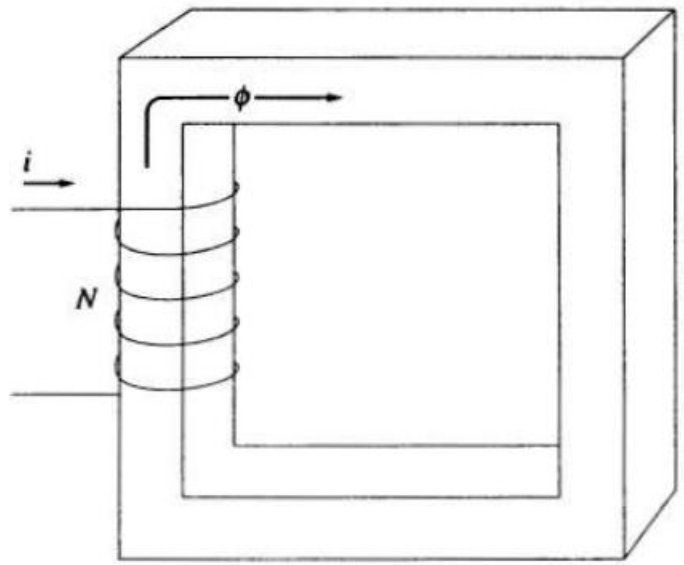
$$\phi = BA = \frac{\mu N i A}{l_c}$$



Magnetic Circuit – magnetomotive force

$\mathcal{F} = Ni$

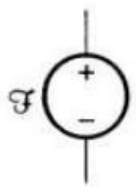
$\mathcal{F} = \phi \mathcal{R}$



$\mathcal{F}$ = magnetomotive force of circuit

ϕ = flux of circuit

$\mathcal{R}$ = *reluctance* of circuit



Magnetic circuit

1. Magnetic circuit

$$\phi = BA = \frac{\mu NiA}{l_c}$$

$$= Ni\left(\frac{\mu A}{l_c}\right)$$

$$\phi = \mathcal{F}\left(\frac{\mu A}{l_c}\right)$$

$$\mathcal{R} = \frac{l_c}{\mu A}$$

$$\mathcal{F} = \phi\,\mathcal{R}$$

Chapter 1. Introduction to machinery principles

1. Rotation motion, Newton's law and power relationships
2. The magnetic field
3. Faraday's law
4. Produce an induced force on a wire
5. Produce an induced voltage on a conductor
6. Linear dc machine examples
7. Real, reactive and apparatus power in AC circuits

Rotation motion, Newton's law and power relationships

Clockwise (CW) and Counterclockwise (CCW)

- CCW is assumed as the positive direction, CW is assumed as the negative direction.

Linear and rotation motion

- Position and angular

  (meter) (degree or radian)

- Speed and angular speed

$$v = \frac{dr}{dt} \qquad \omega = \frac{d\theta}{dt}$$

ω_m	angular velocity expressed in radians per second
f_m	angular velocity expressed in revolutions per second
n_m	angular velocity expressed in revolutions per minute

The magnetic field

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of *transformer action*.)
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of *motor action*.)
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of *generator action*.)

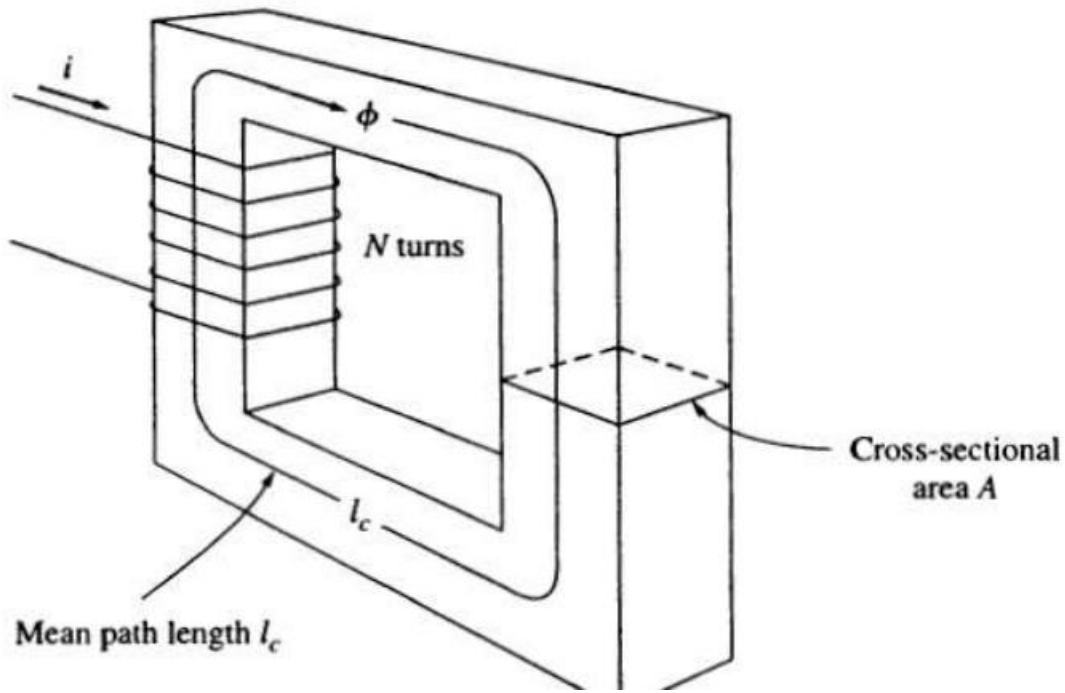
Produce a magnetic field – Ampere's law

1. The magnetic field is produced by ampere's law
2. The core is a ferromagnetic material

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{net}}$$

$$Hl_c = Ni$$

$$H = \frac{Ni}{l_c}$$



From the magnetic field to magnetic flux density

1. When the magnetic field is applied on a ferromagnetic material, the magnetic flux density **B** will be produced

$$\mathbf{B} = \mu\mathbf{H}$$

H = magnetic field intensity
μ = magnetic *permeability* of material
B = resulting magnetic flux density produced

The permeability of free space is called μ_0, and its value is

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \qquad (1\text{–}22)$$

The permeability of any other material compared to the permeability of free space is called its *relative permeability:*

$$\mu_r = \frac{\mu}{\mu_0} \qquad (1\text{–}23)$$

Magnetic flux density and magnetic flux

1. Magnetic flux density

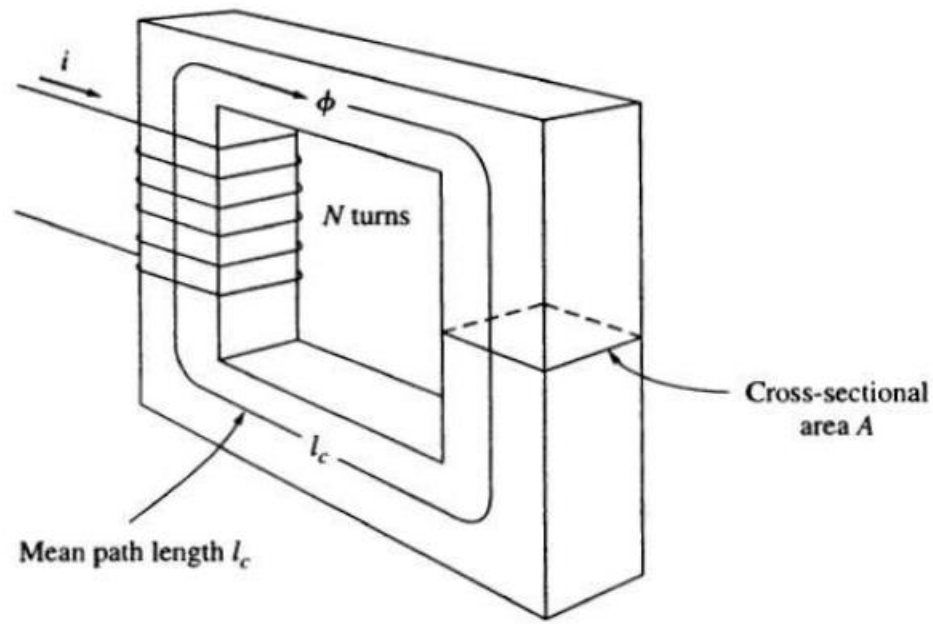
$$B = \mu H = \frac{\mu N i}{l_c}$$

2. Magnetic flux

$$\phi = \int_A \mathbf{B} \cdot d\mathbf{A}$$

$$\phi = BA$$

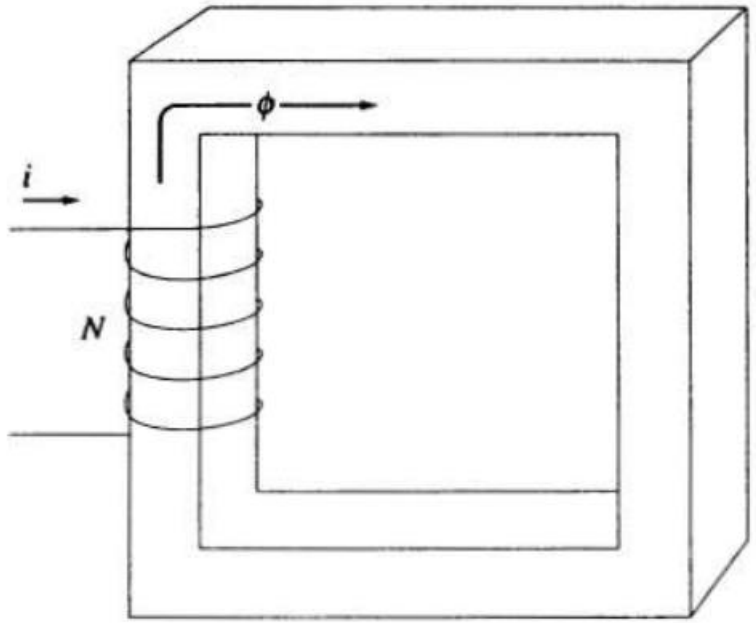
$$\phi = BA = \frac{\mu N i A}{l_c}$$



Magnetic Circuit – magnetomotive force

$\mathcal{F} = Ni$

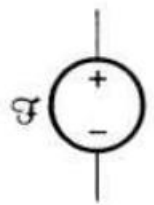
$\mathcal{F} = \phi \mathcal{R}$



$\mathcal{F}$ = magnetomotive force of circuit

ϕ = flux of circuit

$\mathcal{R}$ = *reluctance* of circuit



1. Magnetic circuit

$$\phi = BA = \frac{\mu NiA}{l_c}$$

$$= Ni\left(\frac{\mu A}{l_c}\right)$$

$$\phi = \mathcal{F}\left(\frac{\mu A}{l_c}\right)$$

$$\mathcal{R} = \frac{l_c}{\mu A}$$

$$\mathcal{F} = \phi\,\mathcal{R}$$

Electric circuit and magnetic circuit

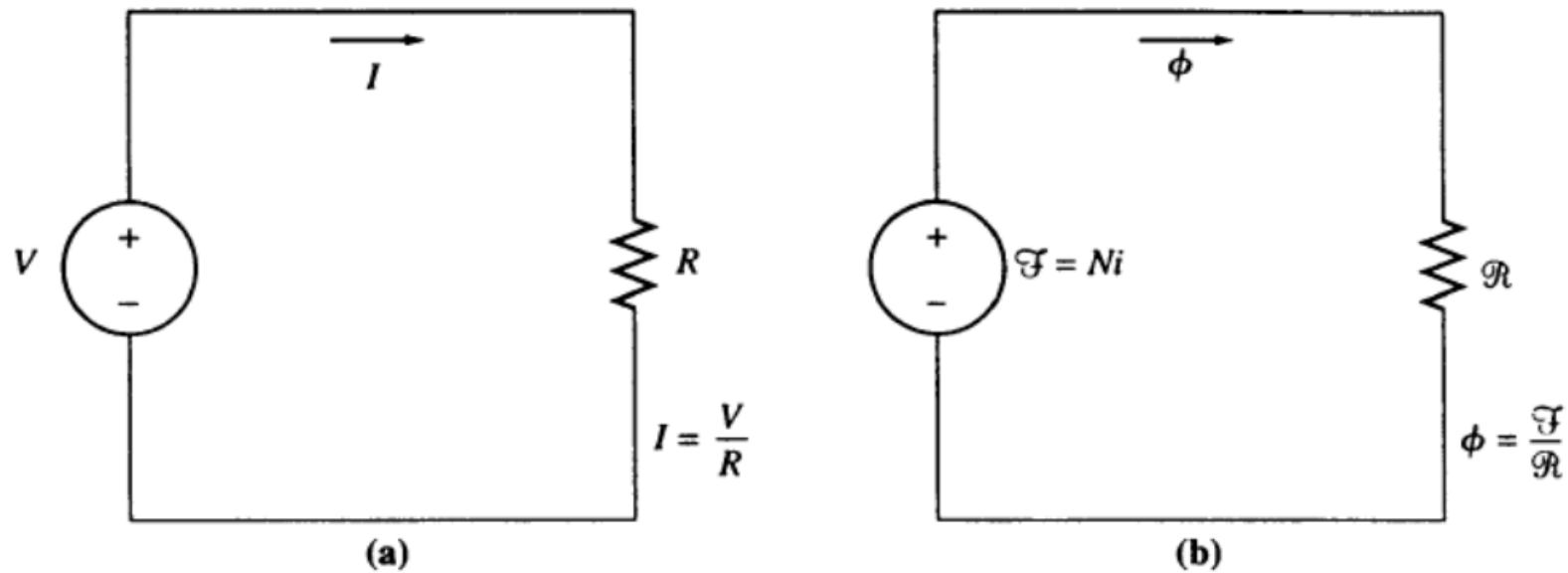


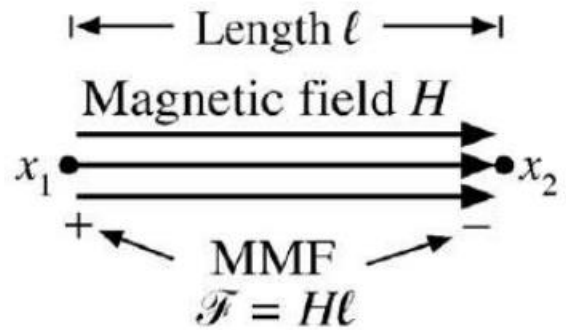
FIGURE 1–4
(a) A simple electric circuit. (b) The magnetic circuit analog to a transformer core.

Magnetic field H and magnetomotive force $\mathscr{F}$

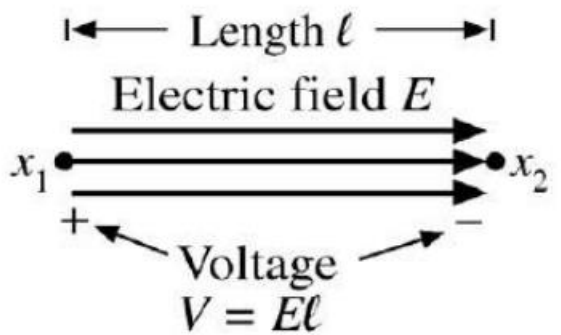
Magnetomotive force (MMF) $\mathscr{F}$ between points x_1 and x_2 is related to the magnetic field H according to

$$\mathscr{F} = \int_{x_1}^{x_2} \boldsymbol{H} \cdot d\boldsymbol{\ell}$$

Example: uniform magnetic field of magnitude H



Analogous to electric field of strength E, which induces voltage (EMF) V:

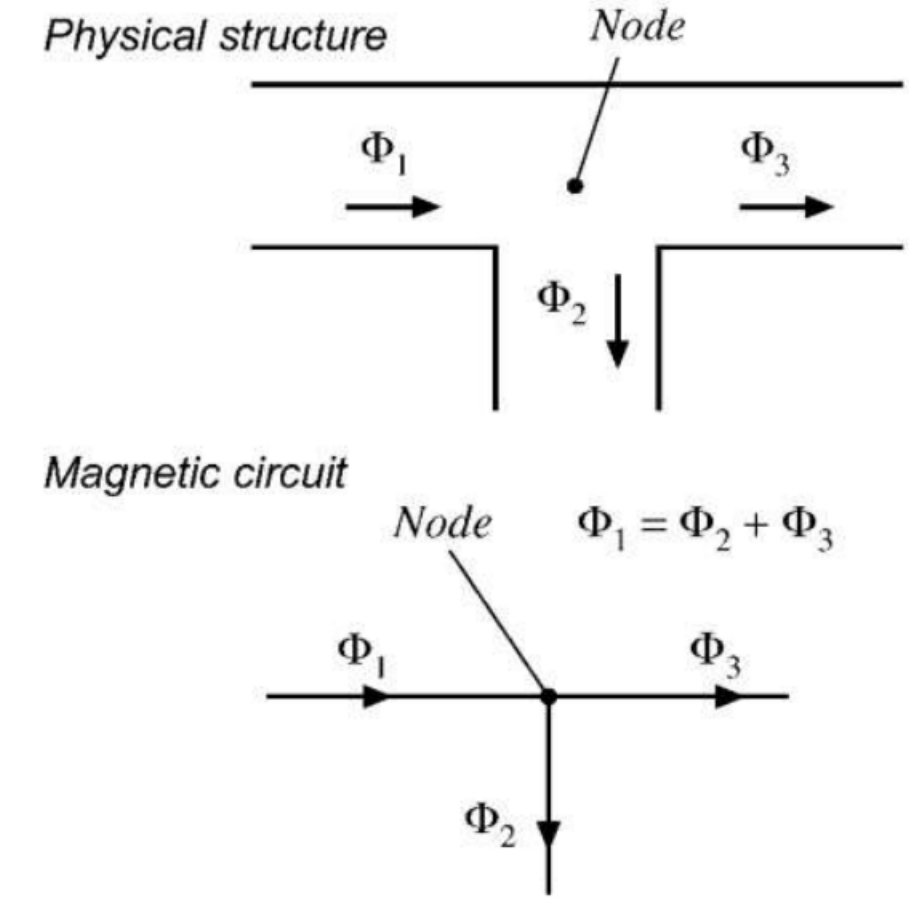


Magnetic analog of Kirchoff's current law

Divergence of $\boldsymbol{B} = 0$

Flux lines are continuous and cannot end

Total flux entering a node must be zero



Reluctance in magnetic circuit

1. Series connection

$$\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \cdots$$

2. Parallel connection

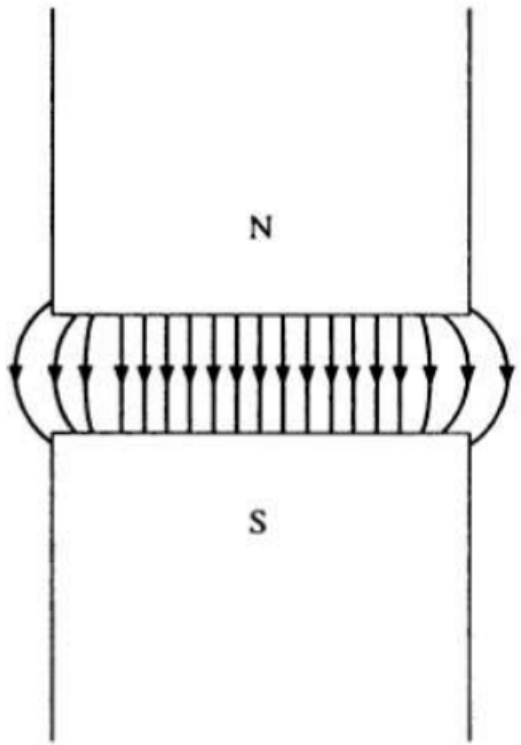
$$\frac{1}{\mathcal{R}_{eq}} = \frac{1}{\mathcal{R}_1} + \frac{1}{\mathcal{R}_2} + \frac{1}{\mathcal{R}_3} + \cdots$$

The errors in magnetic circuit computation

1. The magnetic circuit concept assumes that all flux is confined within a magnetic core. Unfortunately, this is not quite true. The permeability of a ferromagnetic core is 2000 to 6000 times that of air, but a small fraction of the flux escapes from the core into the surrounding low-permeability air. This flux outside the core is called *leakage flux,* and it plays a very important role in electric machine design.
2. The calculation of reluctance assumes a certain mean path length and cross-sectional area for the core. These assumptions are not really very good, especially at corners.
3. In ferromagnetic materials, the permeability varies with the amount of flux already in the material. This nonlinear effect is described in detail. It adds yet another source of error to magnetic circuit analysis, since the reluctances used in magnetic circuit calculations depend on the permeability of the material.

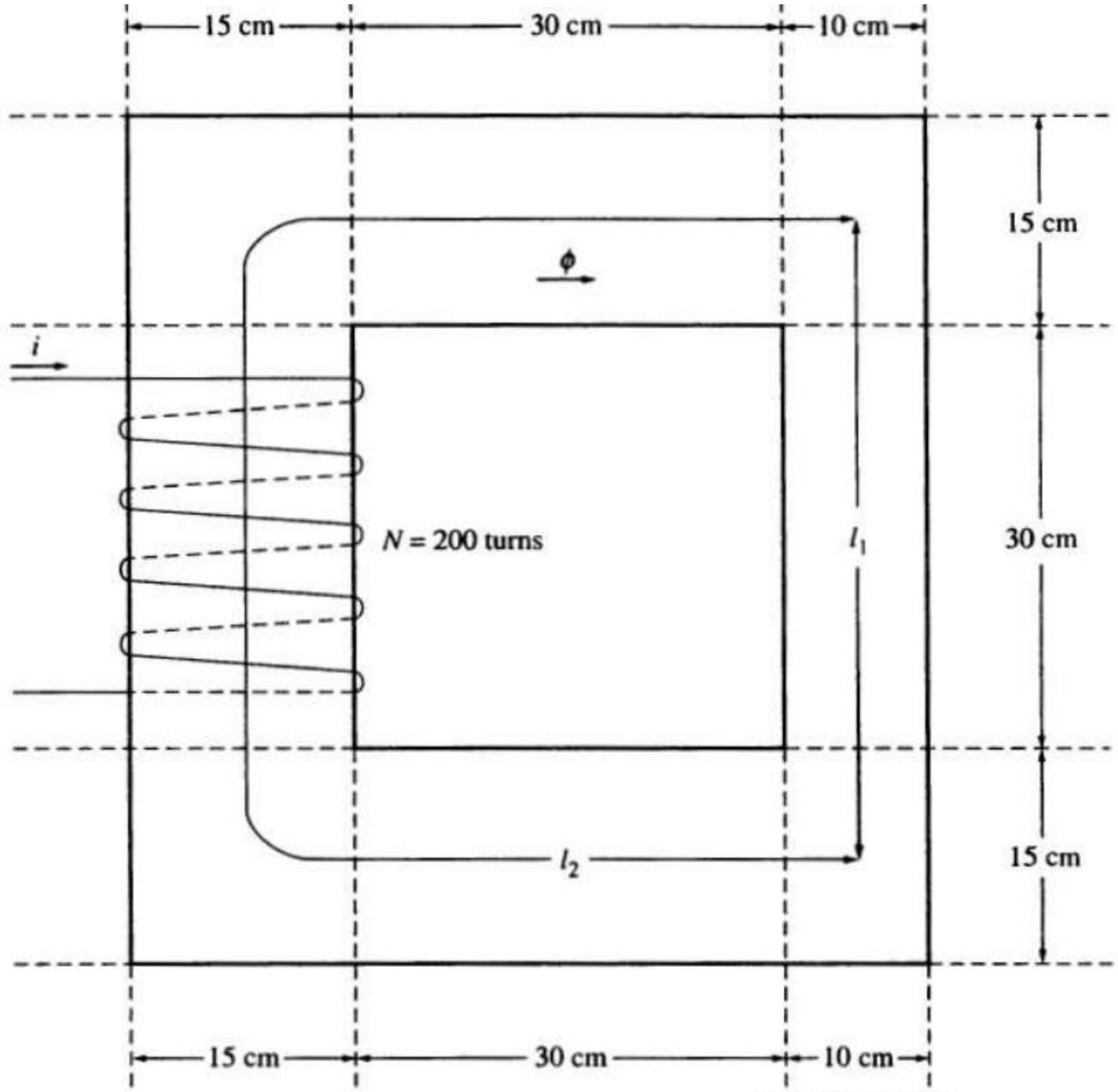
The errors in magnetic circuit computation

4. Air gap “fringing effect”



Example 1-1

Example 1–1. A ferromagnetic core is shown in Figure 1–7a. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200-turn coil wrapped around the left side of the core. Assuming relative permeability μ_r of 2500, how much flux will be produced by a 1-A input current?


15 cm
30 cm
10 cm
15 cm
30 cm
15 cm
ϕ
i
$N = 200$ turns
l_1
l_2
15 cm
30 cm
10 cm
Depth = 10 cm

(a)

The mean path length of region 1 is 45 cm, and the cross-sectional area is 10×10 cm $= 100$ cm^2. Therefore, the reluctance in the first region is

$$\mathcal{R}_1 = \frac{l_1}{\mu A_1} = \frac{l_1}{\mu_r \mu_0 A_1} \tag{1–32}$$

$$= \frac{0.45 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.01 \text{ m}^2)}$$

$$= 14{,}300 \text{ A} \cdot \text{turns/Wb}$$

The mean path length of region 2 is 130 cm, and the cross-sectional area is 15×10 cm $= 150$ cm^2. Therefore, the reluctance in the second region is

$$\mathcal{R}_2 = \frac{l_2}{\mu A_2} = \frac{l_2}{\mu_r \mu_0 A_2} \tag{1–32}$$

$$= \frac{1.3 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.015 \text{ m}^2)}$$

$$= 27{,}600 \text{ A} \cdot \text{turns/Wb}$$

Therefore, the total reluctance in the core is

$$\begin{aligned}\mathcal{R}_{eq} &= \mathcal{R}_1 + \mathcal{R}_2 \\ &= 14{,}300 \text{ A} \cdot \text{turns/Wb} + 27{,}600 \text{ A} \cdot \text{turns/Wb} \\ &= 41{,}900 \text{ A} \cdot \text{turns/Wb}\end{aligned}$$

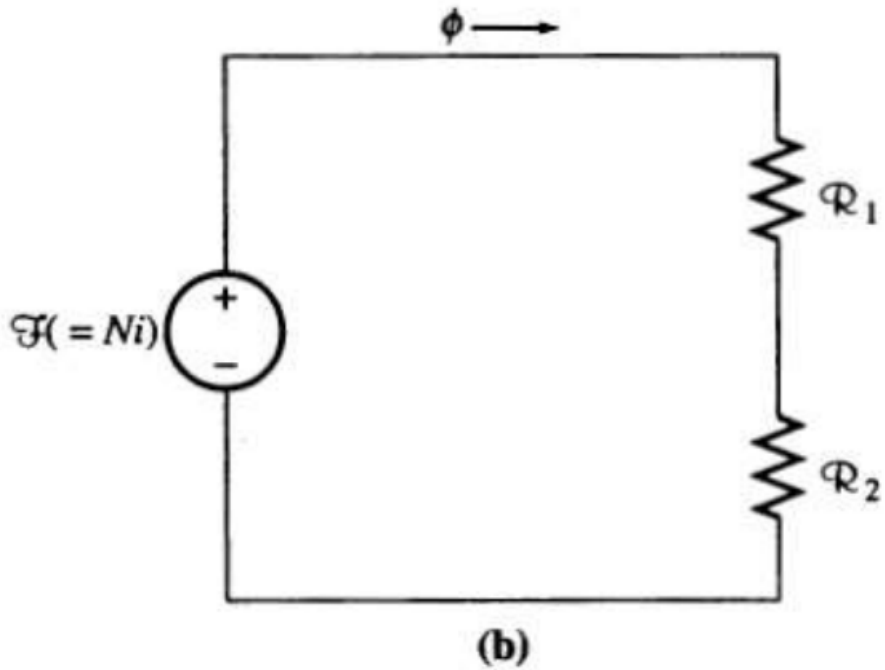
The total magnetomotive force is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

The total flux in the core is given by

$$\begin{aligned}\phi &= \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{41{,}900 \text{ A} \cdot \text{turns/ Wb}} \\ &= 0.0048 \text{ Wb}\end{aligned}$$

Magnetic circuit

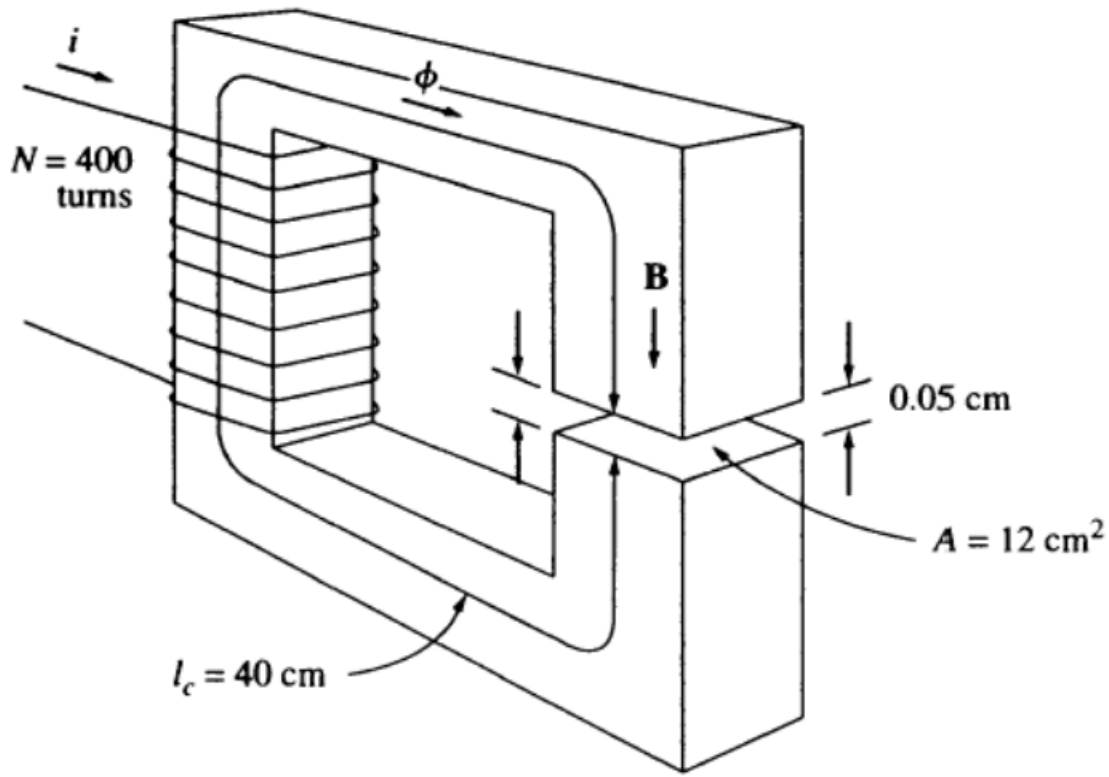


(b)

Example 1-2

Example 1–2. Figure 1–8a shows a ferromagnetic core whose mean path length is 40 cm. There is a small gap of 0.05 cm in the structure of the otherwise whole core. The cross-sectional area of the core is 12 cm^2, the relative permeability of the core is 4000, and the coil of wire on the core has 400 turns. Assume that fringing in the air gap increases the effective cross-sectional area of the air gap by 5 percent. Given this information, find (a) the total reluctance of the flux path (iron plus air gap) and (b) the current required to produce a flux density of 0.5 T in the air gap.

Example 1-2



Solution

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

(a) The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \tag{1–32}$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \tag{1–32}$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

Solution

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

(a) The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \tag{1–32}$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \tag{1–32}$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

(a) The ferromagnetic core of Example 1–2. (b) The magnetic circuit corresponding to (a).

Therefore, the total reluctance of the flux path is

$$\begin{aligned}\mathcal{R}_{eq} &= \mathcal{R}_c + \mathcal{R}_a \\ &= 66{,}300 \text{ A} \cdot \text{turns/Wb} + 316{,}000 \text{ A} \cdot \text{turns/Wb} \\ &= 382{,}300 \text{ A} \cdot \text{turns/Wb}\end{aligned}$$

Note that the air gap contributes most of the reluctance even though it is 800 times shorter than the core.

(b) Equation (1–28) states that

$$\mathcal{F} = \phi\mathcal{R} \tag{1–28}$$

Since the flux $\phi = BA$ and $\mathcal{F} = Ni$, this equation becomes

$$Ni = BA\mathcal{R}$$

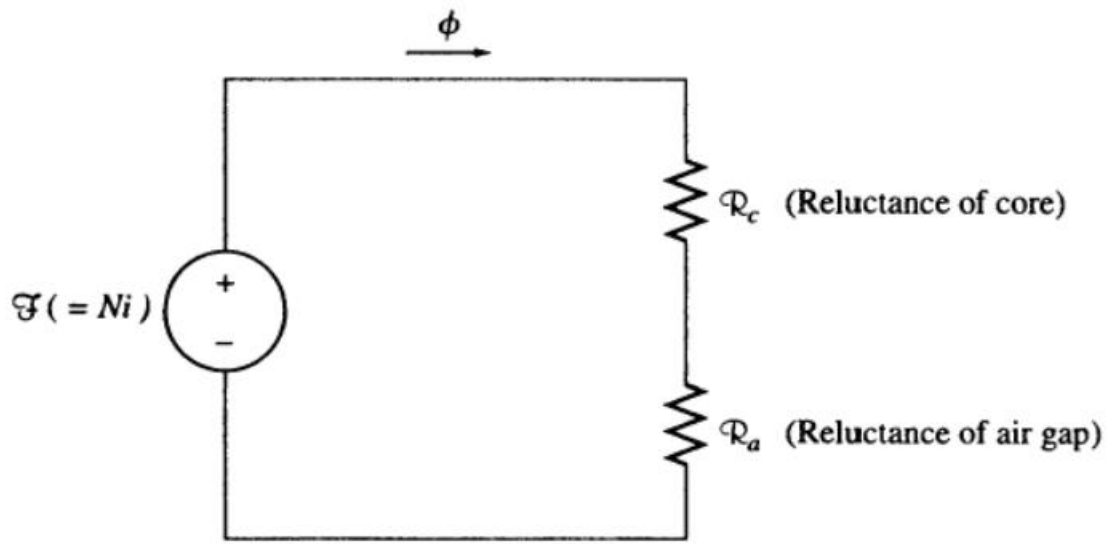
so

$$i = \frac{BA\mathcal{R}}{N}$$

$$= \frac{(0.5\ \text{T})(0.00126\ \text{m}^2)(383{,}200\ \text{A} \cdot \text{turns/ Wb})}{400\ \text{turns}}$$

$$= 0.602\ \text{A}$$

Notice that, since the *air-gap* flux was required, the effective air-gap area was used in the above equation.



(b)

Example 1-4

Example 1–4. Find the relative permeability of the typical ferromagnetic material whose magnetization curve is shown in Figure 1–10c at (a) $H = 50$, (b) $H = 100$, (c) $H = 500$, and (d) $H = 1000$ A • turns/m.

Solution

The permeability of a material is given by

$$\mu = \frac{B}{H}$$

and the relative permeability is given by

$$\mu_r = \frac{\mu}{\mu_0}$$

Thus, it is easy to determine the permeability at any given magnetizing intensity.

(a) At $H = 50$ A • turns/m, $B = 0.25$ T, so

$$\mu = \frac{B}{H} = \frac{0.25 \text{ T}}{50 \text{ A} \cdot \text{turns/m}} = 0.0050 \text{ H/m}$$

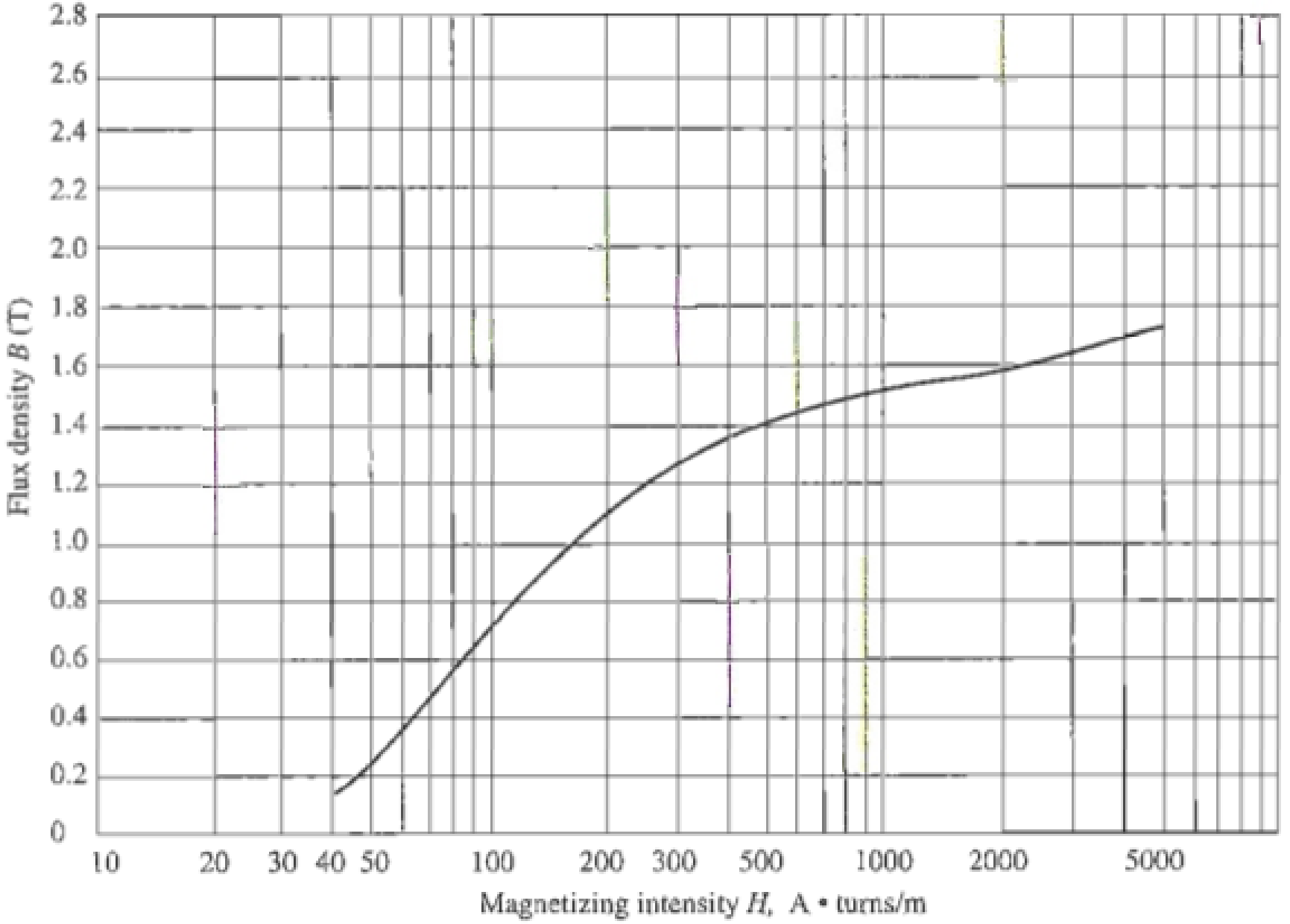
and

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.0050 \text{ H/m}}{4\pi \times 10^{-7} \text{ H/m}} = 3980$$

Example 1-5

Example 1–5. A square magnetic core has a mean path length of 55 cm and a cross-sectional area of 150 cm^2. A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve shown in Figure 1–10c.

(a) How much current is required to produce 0.012 Wb of flux in the core?
(b) What is the core's relative permeability at that current level?
(c) What is its reluctance?


Flux density B (T)
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
10
20
30
40
50
100
200
300
500
1000
2000
5000
Magnetizing intensity H, A • turns/m

(c)

Solution

(a) The required flux density in the core is

$$B = \frac{\phi}{A} = \frac{1.012 \text{ Wb}}{0.015 \text{ m}^2} = 0.8 \text{ T}$$

From Figure 1–10c, the required magnetizing intensity is

$$H = 115 \text{ A} \cdot \text{turns/m}$$

From Equation (1–20), the magnetomotive force needed to produce this magnetizing intensity is

$$\begin{aligned} \mathcal{F} &= Ni = Hl_c \\ &= (115 \text{ A} \cdot \text{turns/m})(0.55 \text{ m}) = 63.25 \text{ A} \cdot \text{turns} \end{aligned}$$

so the required current is

$$i = \frac{\mathcal{F}}{N} = \frac{63.25 \text{ A} \cdot \text{turns}}{200 \text{ turns}} = 0.316 \text{ A}$$

(b) The core's permeability at this current is

$$\mu = \frac{B}{H} = \frac{0.8\ \text{T}}{115\ \text{A} \cdot \text{turns/m}} = 0.00696\ \text{H/m}$$

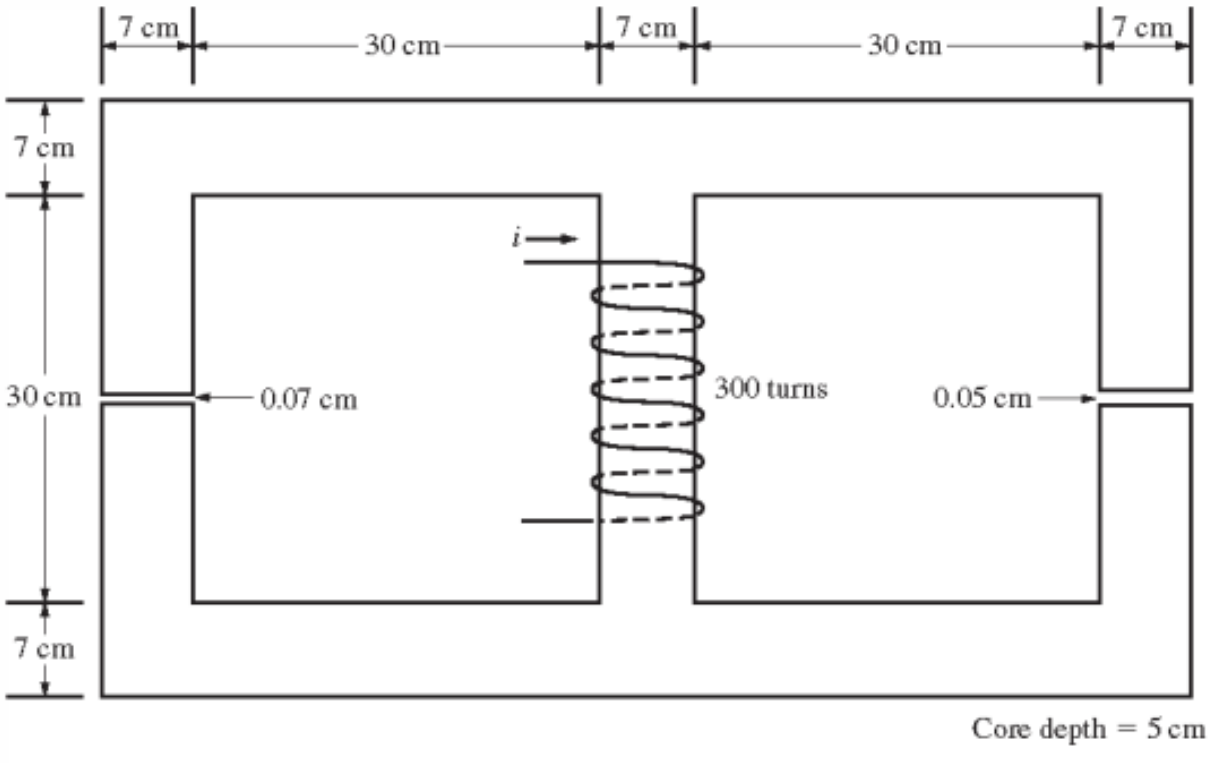
Therefore, the relative permeability is

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00696\ \text{H/m}}{4\pi \times 10^{-7}\ \text{H/m}} = 5540$$

(c) The reluctance of the core is

$$\mathcal{R} = \frac{\mathcal{F}}{\phi} = \frac{63.25\ \text{A} \cdot \text{turns}}{0.012\ \text{Wb}} = 5270\ \text{A} \cdot \text{turns/Wb}$$

A ferromagnetic core with a relative permeability of 1500 is shown in Figure P1-3. The dimensions are as shown in the diagram, and the depth of the core is 5 cm. The air gaps on the left and right sides of the core are 0.050 and 0.070 cm, respectively. Because of fringing effects, the effective area of the air gaps is 5 percent larger than their physical size. If there are 300 turns in the coil wrapped around the center leg of the core and if the current in the coil is 1.0 A, what is the flux in each of the left, center, and right legs of the core? What is the flux density in each air gap?



SOLUTION This core can be divided up into five regions. Let $\mathcal{R}_1$ be the reluctance of the left-hand portion of the core, $\mathcal{R}_2$ be the reluctance of the left-hand air gap, $\mathcal{R}_3$ be the reluctance of the right-hand portion of the core, $\mathcal{R}_4$ be the reluctance of the right-hand air gap, and $\mathcal{R}_5$ be the reluctance of the center leg of the core. Then the total reluctance of the core is

$$\mathcal{R}_{\text{TOT}} = \mathcal{R}_5 + \frac{(\mathcal{R}_1 + \mathcal{R}_2)(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4}$$

$$\mathcal{R}_1 = \frac{l_1}{\mu_r \mu_0 A_1} = \frac{1.11 \text{ m}}{(1500)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})} = 168 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_2 = \frac{l_2}{\mu_0 A_2} = \frac{0.0007 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})(1.05)} = 152 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_3 = \frac{l_3}{\mu_r \mu_0 A_3} = \frac{1.11 \text{ m}}{(1500)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})} = 168 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_4 = \frac{l_4}{\mu_0 A_4} = \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})(1.05)} = 108 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_5 = \frac{l_5}{\mu_r \mu_0 A_5} = \frac{0.37\ \text{m}}{(1500)(4\pi \times 10^{-7}\ \text{H/m})(0.07\ \text{m})(0.05\ \text{m})} = 56.1\ \text{kA} \cdot \text{t/Wb}$$

The total reluctance is

$$\mathcal{R}_{\text{TOT}} = \mathcal{R}_5 + \frac{(\mathcal{R}_1 + \mathcal{R}_2)(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} = 56.1 + \frac{(168+152)(168+108)}{168+152+168+108} = 204\ \text{kA} \cdot \text{t/Wb}$$

The total flux in the core is equal to the flux in the center leg:

$$\phi_{\text{center}} = \phi_{\text{TOT}} = \frac{\mathcal{F}}{\mathcal{R}_{\text{TOT}}} = \frac{(300\ \text{t})(1.0\ \text{A})}{204\ \text{kA} \cdot \text{t/Wb}} = 0.00147\ \text{Wb}$$

The fluxes in the left and right legs can be found by the "flux divider rule", which is analogous to the current divider rule.

$$\phi_{\text{left}} = \frac{(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} \phi_{\text{TOT}} = \frac{(168+108)}{168+152+168+108}(0.00147\ \text{Wb}) = 0.00068\ \text{Wb}$$

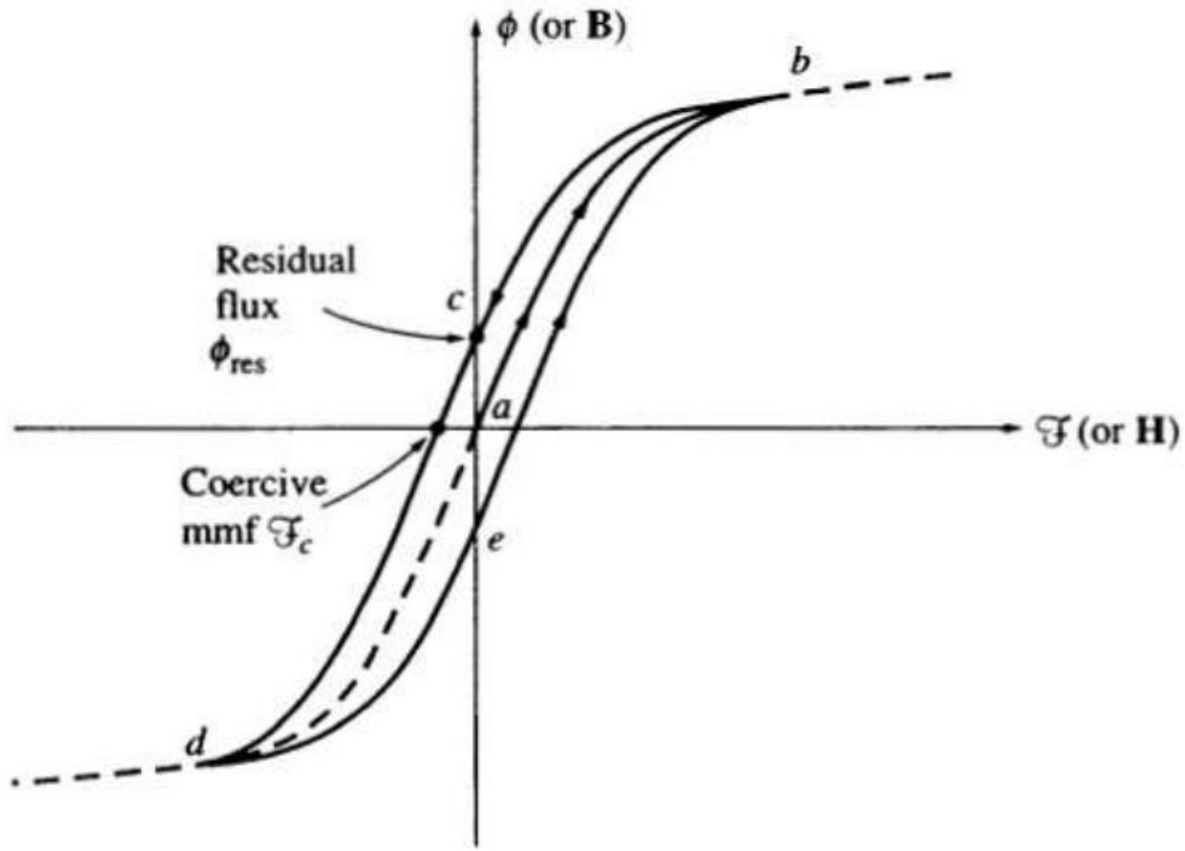
$$\phi_{\text{right}} = \frac{(\mathcal{R}_1 + \mathcal{R}_2)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} \phi_{\text{TOT}} = \frac{(168+152)}{168+152+168+108}(0.00147\ \text{Wb}) = 0.00079\ \text{Wb}$$

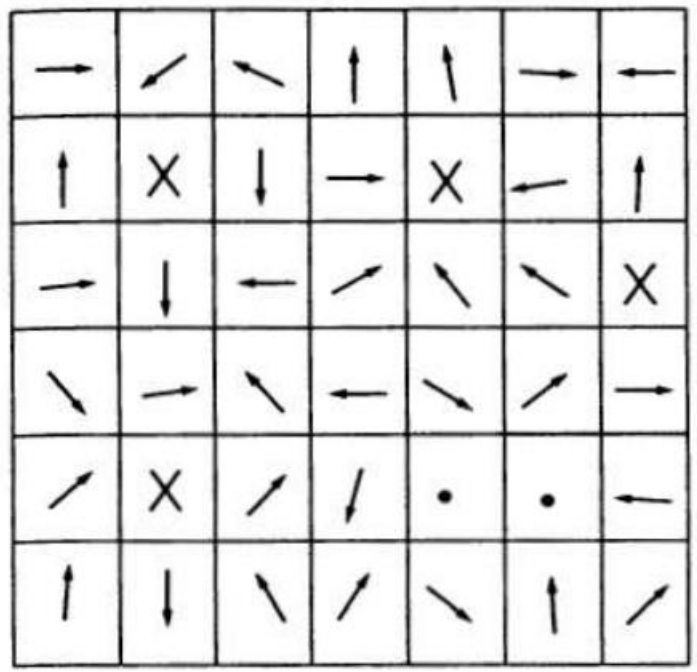
The flux density in the air gaps can be determined from the equation $\phi = BA$:

$$B_{\text{left}} = \frac{\phi_{\text{left}}}{A_{\text{eff}}} = \frac{0.00068\ \text{Wb}}{(0.07\ \text{cm})(0.05\ \text{cm})(1.05)} = 0.185\ \text{T}$$

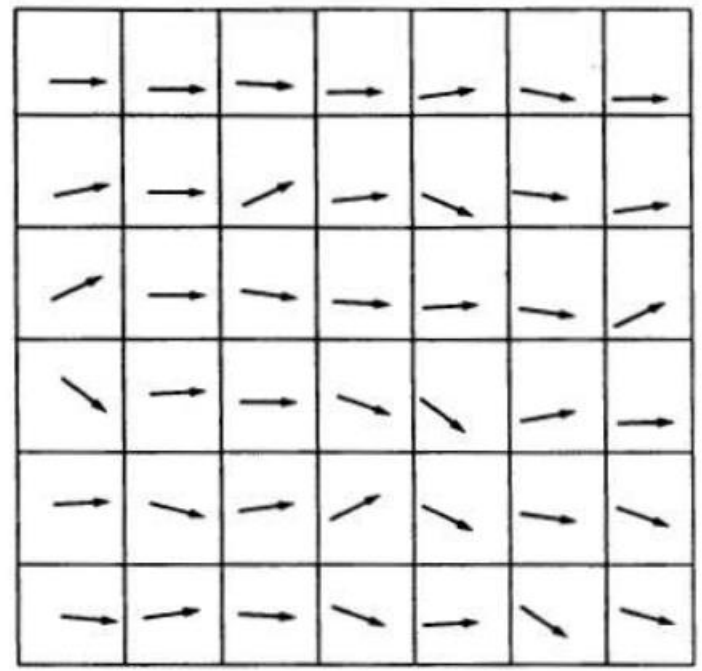
$$B_{\text{right}} = \frac{\phi_{\text{right}}}{A_{\text{eff}}} = \frac{0.00079\ \text{Wb}}{(0.07\ \text{cm})(0.05\ \text{cm})(1.05)} = 0.215\ \text{T}$$

Energy loss in ferromagnetic core – hysteresis loss



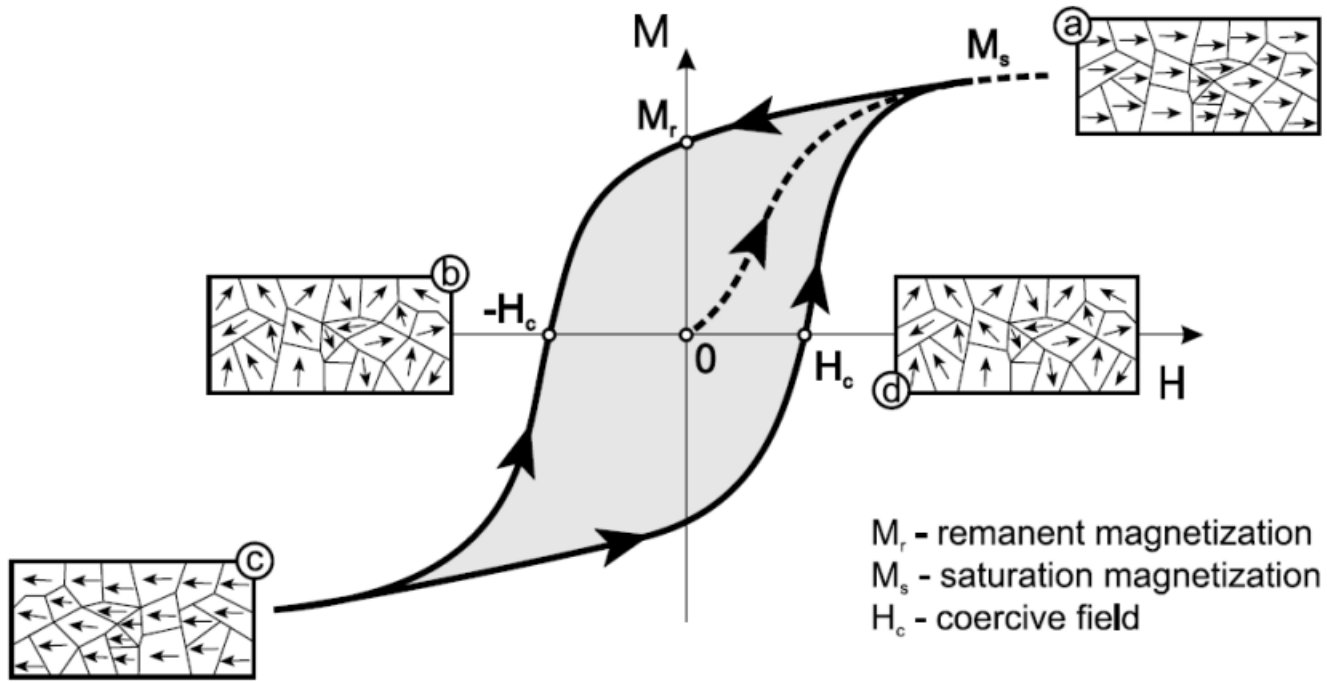


(a)



(b)

Magnetization curve



Faraday's law – induce voltage from a time-varying magnetic field

1. Induced voltage magnitude and polarity

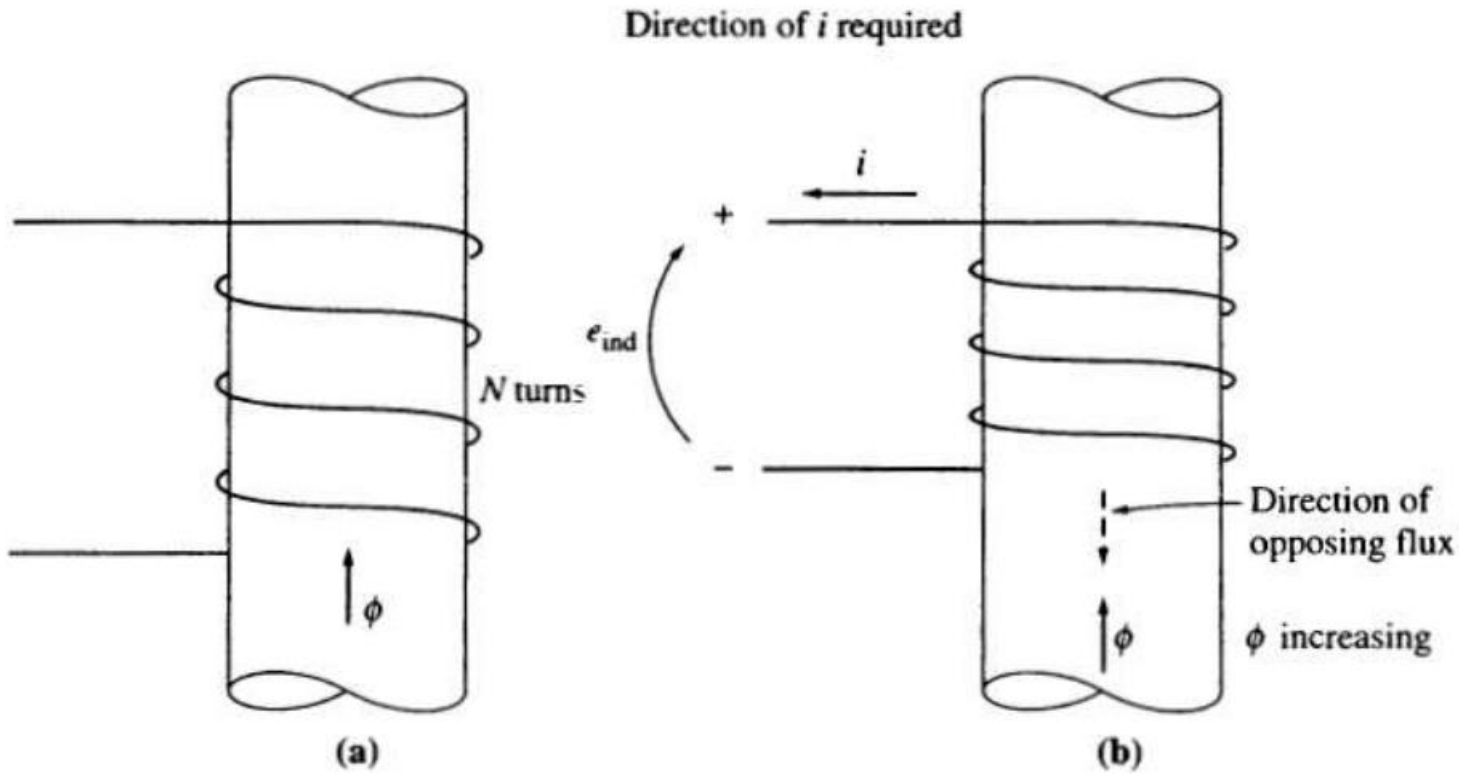
$$e_{\text{ind}} = -\frac{d\phi}{dt}$$

$$e_{\text{ind}} = \sum_{i=1}^{N} e_i$$

$$e_{\text{ind}} = -N\frac{d\phi}{dt}$$

e_{ind} = voltage induced in the coil
N = number of turns of wire in coil
ϕ = flux passing through coil

The induced voltage polarity – Lenz's law



Flux and flux linkage

The term in parentheses in Equation (1–40) is called *the flux linkage* λ of the coil, and Faraday's law can be rewritten in terms of flux linkage as

$$e_{\text{ind}} = \frac{d\lambda}{dt} \tag{1–41}$$

where

$$\lambda = \sum_{i=1}^{N} \phi_i \tag{1–42}$$

The units of flux linkage are weber-turns.

Example 1-6

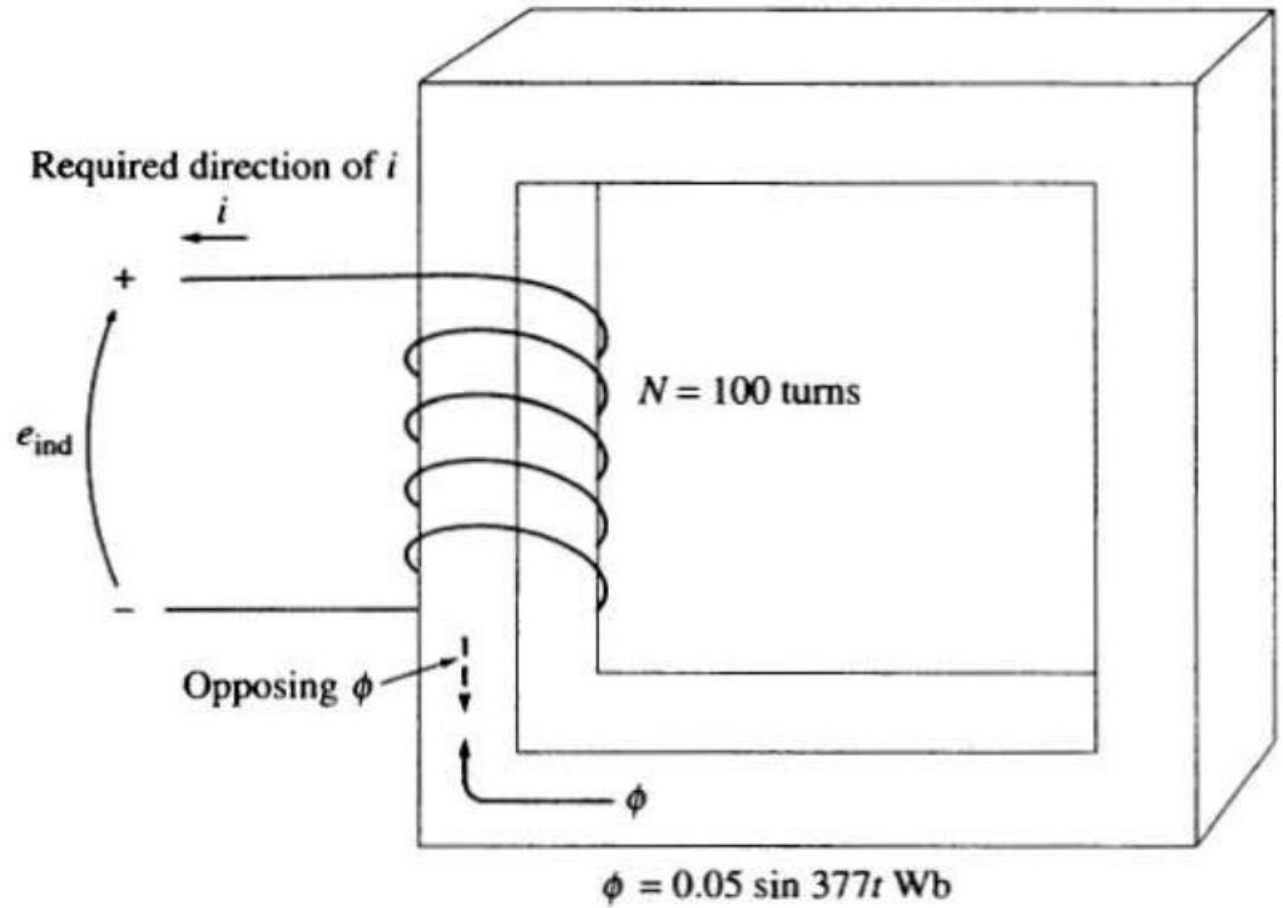
Example 1–6. Figure 1–15 shows a coil of wire wrapped around an iron core. If the flux in the core is given by the equation

$$\phi = 0.05 \sin 377t \quad \text{Wb}$$

If there are 100 turns on the core, what voltage is produced at the terminals of the coil? Of what polarity is the voltage during the time when flux is *increasing* in the reference

The core of Example 1–6. Determination of the voltage polarity at the terminals is shown.

direction shown in the figure? Assume that all the magnetic flux stays within the core (i.e., assume that the flux leakage is zero).


Required direction of i
i
+
e_{ind}
−
$N = 100$ turns
Opposing ϕ
ϕ
$\phi = 0.05 \sin 377t$ Wb

Solution

By the same reasoning as in the discussion on pages 29–30, the direction of the voltage while the flux is increasing in the reference direction must be positive to negative, as shown in Figure 1–15. The *magnitude* of the voltage is given by

$$e_{\text{ind}} = N\frac{d\phi}{dt}$$

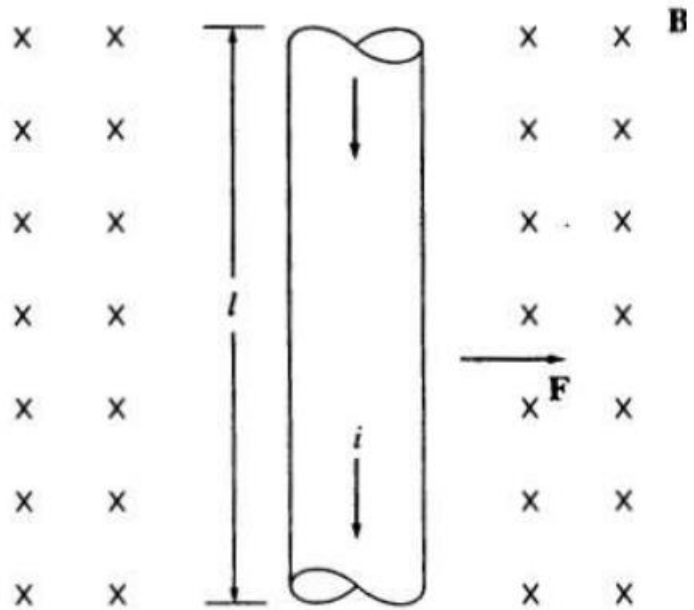
$$= (100 \text{ turns})\frac{d}{dt}(0.05 \sin 377t)$$

$$= 1885 \cos 377t$$

or alternatively,

$$e_{\text{ind}} = 1885 \sin(377t + 90°) \text{ V}$$

Produce an induced force on a wire



$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

$$F = ilB \sin \theta$$

i = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

Example 1-7

Example 1–7. Figure 1–16 shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?

Example 1-7

Solution

The direction of the force is given by the right-hand rule as being to the right. The magnitude is given by

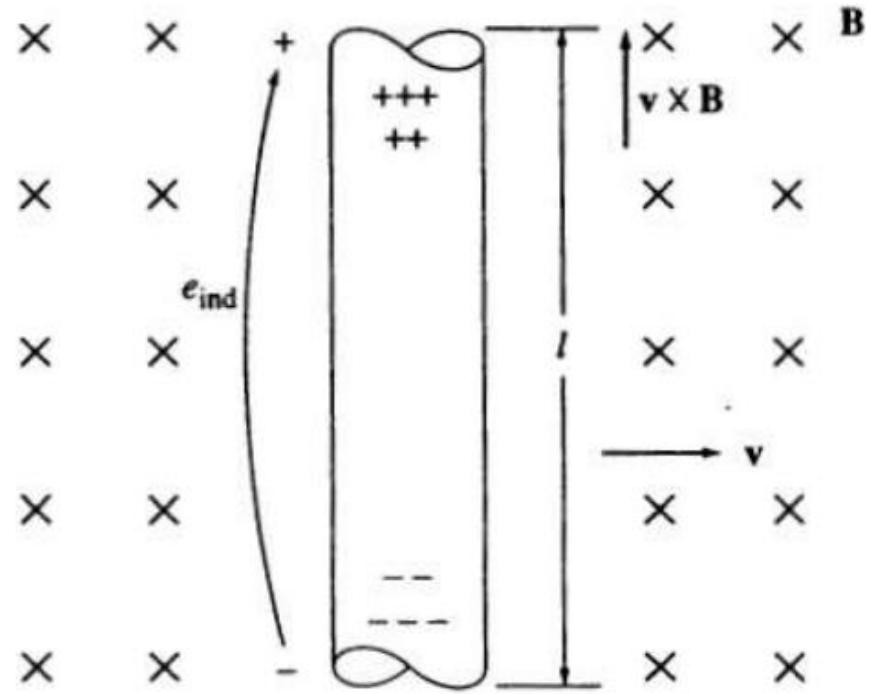
$$F = ilB \sin \theta \qquad (1\text{–}44)$$

$$= (0.5\ \text{A})(1.0\ \text{m})(0.25\ \text{T}) \sin 90° = 0.125\ \text{N}$$

Therefore,

$$\mathbf{F} = 0.125\ \text{N, directed to the right}$$

Induced voltage on a conductor



$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

v = velocity of the wire
B = magnetic flux density vector
l = length of conductor in the magnetic field

Example 1-8

Example 1–8. Figure 1–17 shows a conductor moving with a velocity of 5.0 m/s to the right in the presence of a magnetic field. The flux density is 0.5 T into the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?

Solution

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ in this example is up. Therefore, the voltage on the conductor will be built up positive at the top with respect to the bottom of the wire. The direction of vector **l** is up, so that it makes the smallest angle with respect to the vector $\mathbf{v} \times \mathbf{B}$.

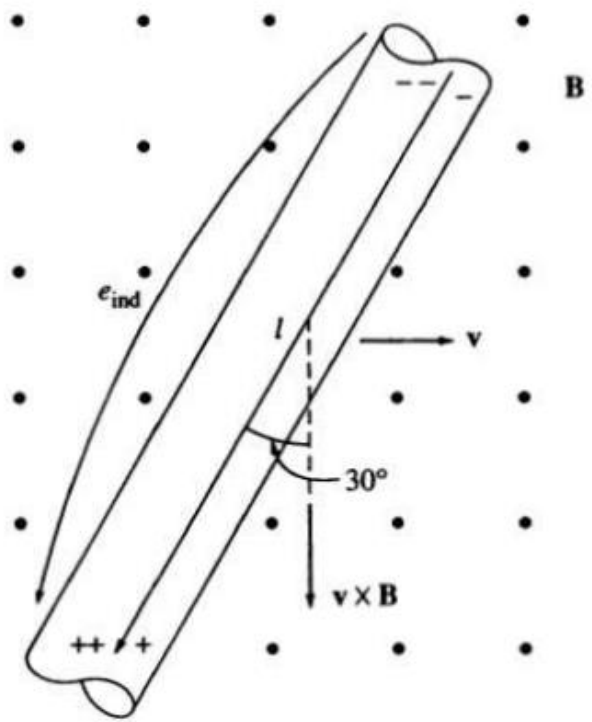
Since **v** is perpendicular to **B** and since $\mathbf{v} \times \mathbf{B}$ is parallel to **l,** the magnitude of the induced voltage reduces to

$$\begin{aligned} e_{\text{ind}} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 0° \\ &= vBl \\ &= (5.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \\ &= 2.5 \text{ V} \end{aligned}$$

Thus the induced voltage is 2.5 V, positive at the top of the wire.

Example 1-9

Example 1–9. Figure 1–18 shows a conductor moving with a velocity of 10 m/s to the right in a magnetic field. The flux density is 0.5 T, out of the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?



Solution

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ is down. The wire is not oriented on an up-down line, so choose the direction of l as shown to make the smallest possible angle with the direction of $\mathbf{v} \times \mathbf{B}$. The voltage is positive at the bottom of the wire with respect to the top of the wire. The magnitude of the voltage is

$$\begin{aligned} e_{\text{ind}} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 30° \\ &= (10.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \cos 30° \\ &= 4.33 \text{ V} \end{aligned}$$

The linear DC machine – a simple example

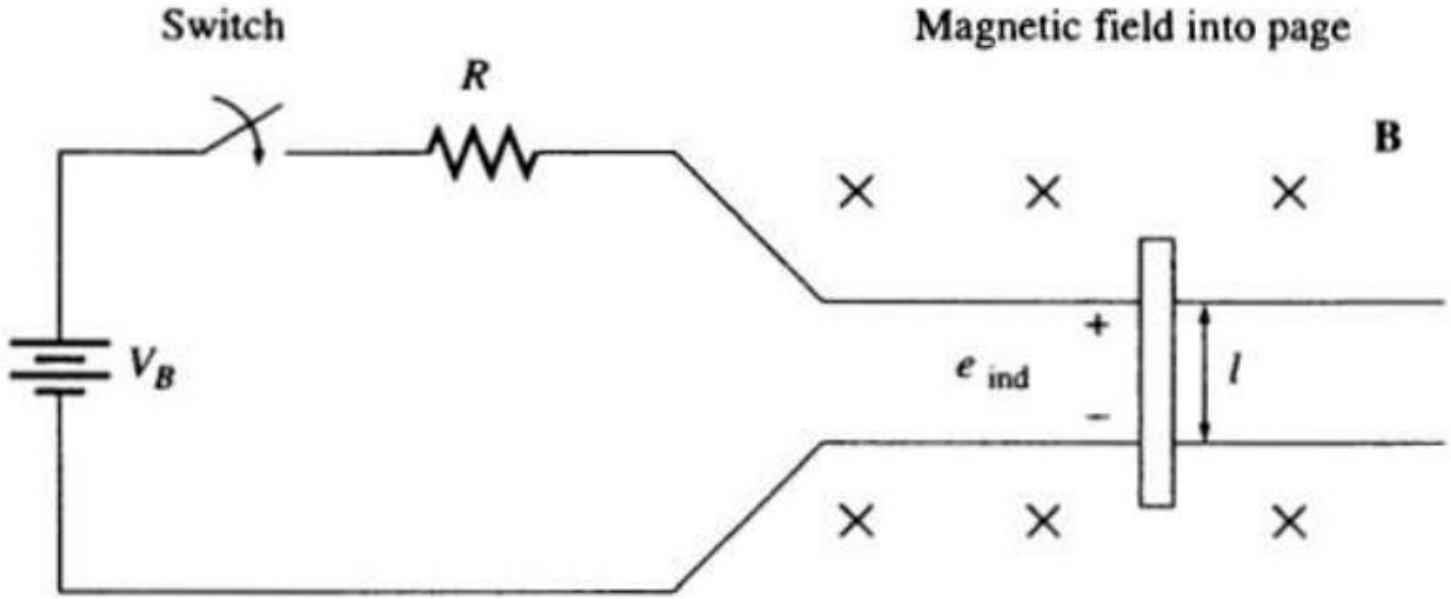


FIGURE 1–19
A linear dc machine. The magnetic field points into the page.

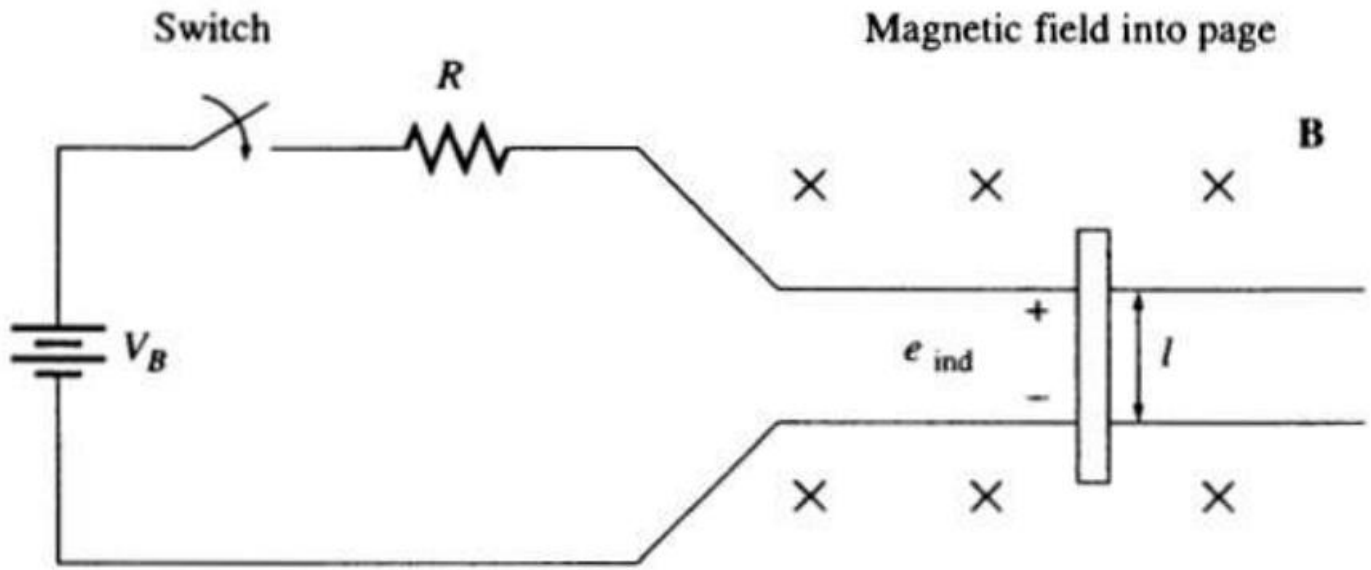


FIGURE 1–19
A linear dc machine. The magnetic field points into the page.

A *linear dc machine* is about the simplest and easiest-to-understand version of a dc machine, yet it operates according to the same principles and exhibits the same behavior as real generators and motors. It thus serves as a good starting point in the study of machines.

A linear dc machine is shown in Figure 1–19. It consists of a battery and a resistance connected through a switch to a pair of smooth, frictionless rails. Along the bed of this "railroad track" is a constant, uniform-density magnetic field directed into the page. A bar of conducting metal is lying across the tracks.

How does such a strange device behave? Its behavior can be determined from an application of four basic equations to the machine. These equations are

1. The equation for the force on a wire in the presence of a magnetic field:

$$\boxed{\mathbf{F} = i(\mathbf{l} \times \mathbf{B})} \qquad (1\text{–}43)$$

where $\mathbf{F}$ = force on wire

i = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

2. The equation for the voltage induced on a wire moving in a magnetic field:

$$\boxed{e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}} \qquad (1\text{–}45)$$

where e_{ind} = voltage induced in wire
$\mathbf{v}$ = velocity of the wire
$\mathbf{B}$ = magnetic flux density vector
$\mathbf{l}$ = length of conductor in the magnetic field

3. Kirchhoff's voltage law for this machine. From Figure 1–19 this law gives

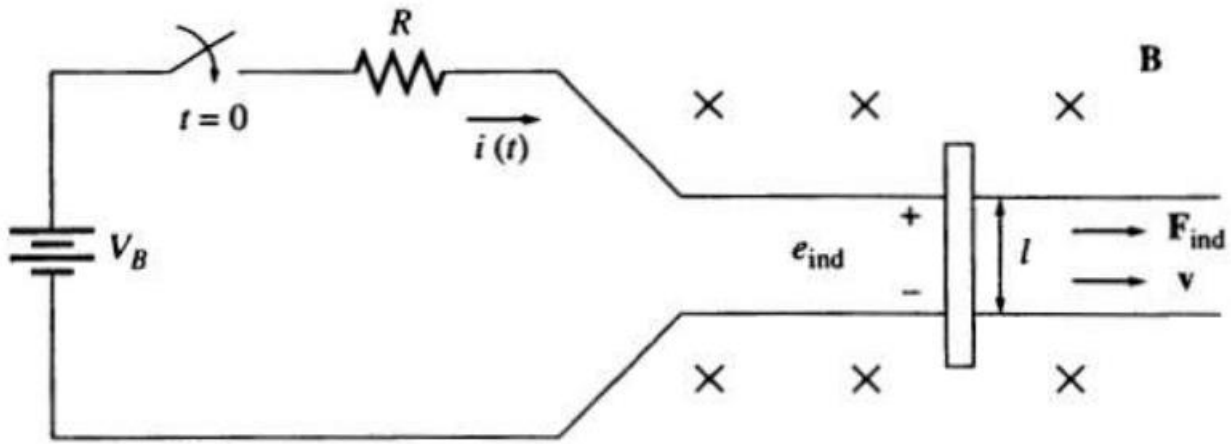
$$V_B - iR - e_{\text{ind}} = 0$$

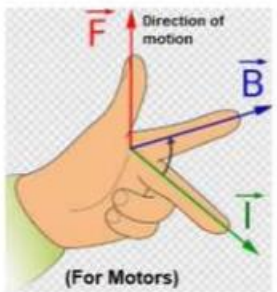
$$\boxed{V_B = e_{\text{ind}} + iR = 0} \qquad (1\text{–}46)$$

4. Newton's law for the bar across the tracks:

$$\boxed{F_{\text{net}} = ma} \qquad (1\text{–}7)$$

Starting a linear DC machine





Starting a linear DC machine

1. Current $i = \dfrac{V_B - e_{ind}}{R}$

2. Induced force $F_{ind} = ilB$ to the right

3. Induced voltage $e_{ind} = vBl$ positive upward

$$i\downarrow = \frac{V_B - e_{ind}\uparrow}{R}$$

Starting a linear DC machine

The result of this action is that eventually the bar will reach a constant steady-state speed where the net force on the bar is zero. This will occur when e_{ind} has risen all the way up to equal the voltage V_B. At that time, the bar will be moving at a speed given by

$$V_B = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl} \tag{1–50}$$

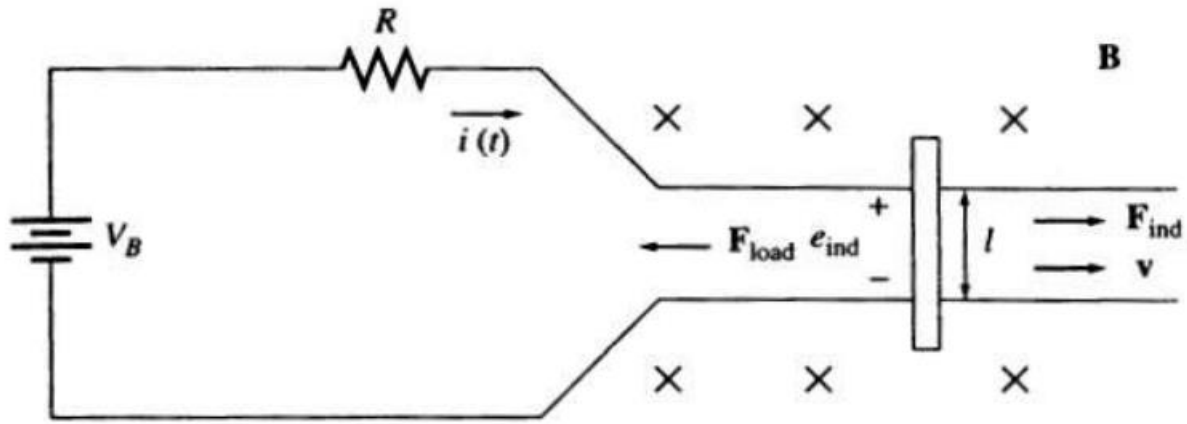
The bar will continue to coast along at this no-load speed forever unless some external force disturbs it. When the motor is started, the velocity v, induced voltage e_{ind}, current i, and induced force F_{ind} are as sketched in Figure 1–21.

Summarize of a dc machine starting

1. Closing the switch produces a current flow $i = V_B / R$.
2. The current flow produces a force on the bar given by $F = ilB$.
3. The bar accelerates to the right, producing an induced voltage e_{ind} as it speeds up.
4. This induced voltage reduces the current flow $i \searrow = (V_B - e_{ind} \uparrow)/R$.
5. The induced force is thus decreased ($F = i \downarrow lB$) until eventually $F = 0$. At that point, $e_{ind} = V_B$, $i = 0$, and the bar moves at a constant no-load speed $v_{ss} = V_B / Bl$.

Linear dc motor

- While the load is applied



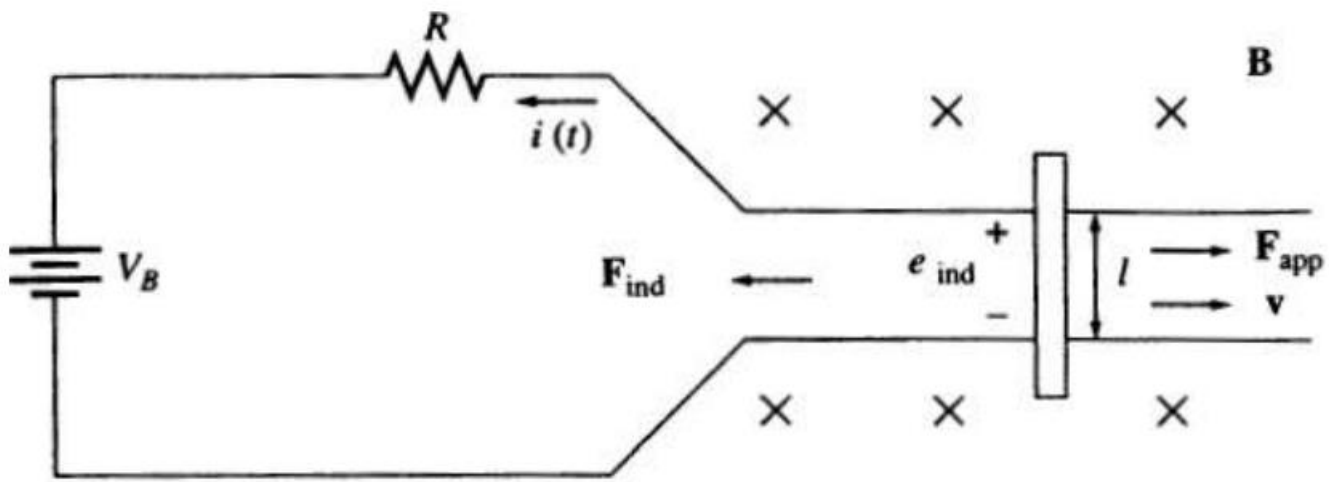
- The conversion power between mechanical and electrical $P_{conv} = e_{ind}i = F_{ind}v$

Summarize of a dc motor operation

1. A force $\mathbf{F}_{load}$ is applied opposite to the direction of motion, which causes a net force $\mathbf{F}_{net}$ opposite to the direction of motion.
2. The resulting acceleration $a = F_{net}/m$ is negative, so the bar slows down ($v\downarrow$).
3. The voltage $e_{ind} = v\downarrow Bl$ falls, and so $i = (V_B - e_{ind}\downarrow)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a lower speed v.
5. An amount of electric power equal to $e_{ind}i$ is now being converted to mechanical power equal to $F_{ind}v$, and the machine is acting as a motor.

Linear dc generator

- While the external force is applied on the moving direction

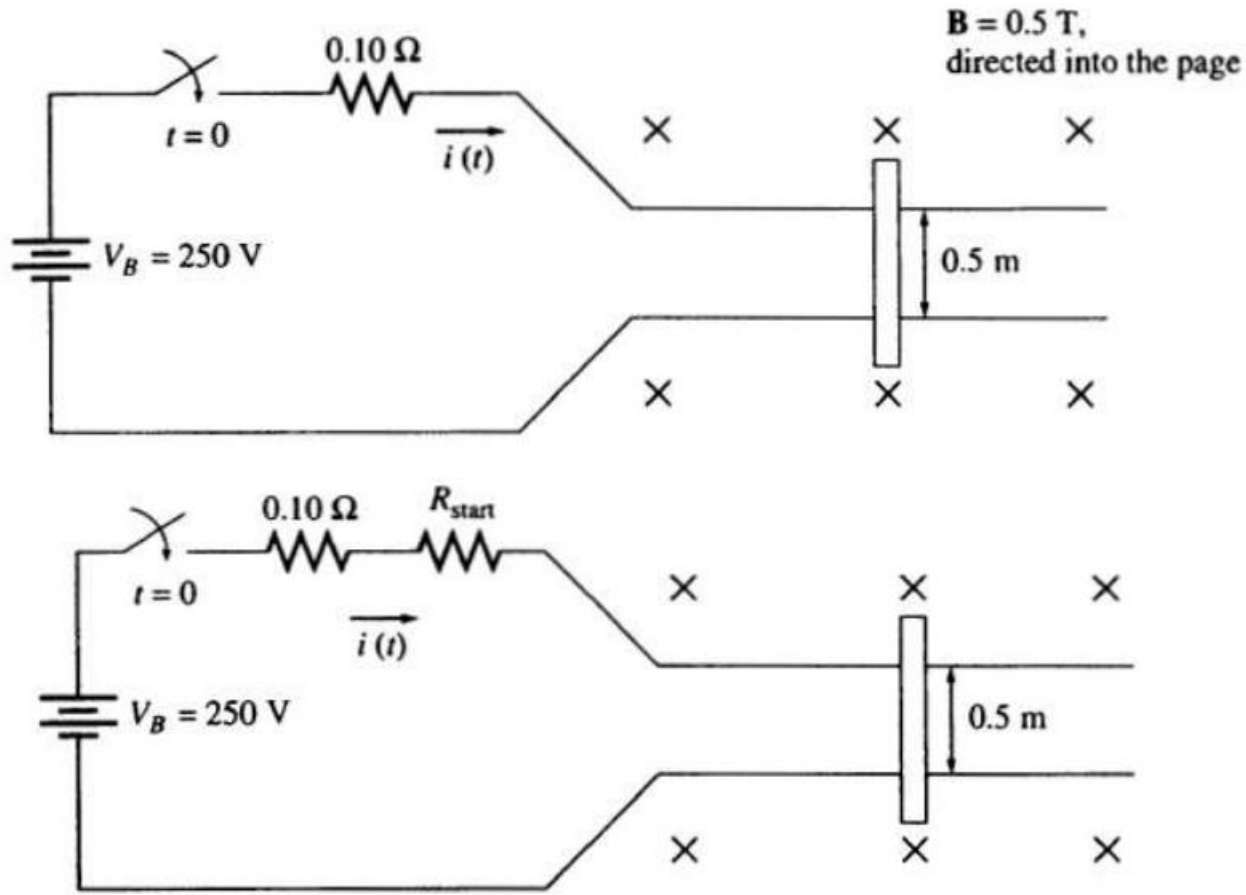


$$P_{conv} = \tau_{ind}\omega$$

Summarize of a dc generator operation

1. A force $\mathbf{F}_{app}$ is applied in the direction of motion; $\mathbf{F}_{net}$ is in the direction of motion.
2. Acceleration $a = F_{net}/m$ is positive, so the bar speeds up ($v\uparrow$).
3. The voltage $e_{ind} = v\uparrow Bl$ increases, and so $i = (e_{ind}\uparrow - V_B)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a higher speed v.
5. An amount of mechanical power equal to $F_{ind}v$ is now being converted to electric power $e_{ind}i$, and the machine is acting as a generator.

Starting problem of dc linear machine



Example 1-10

Example 1–10. The linear dc machine shown in Figure 1–27a has a battery voltage of 120 V, an internal resistance of 0.3 Ω, and a magnetic flux density of 0.1 T.

(a) What is this machine's maximum starting current? What is its steady-state velocity at no load?

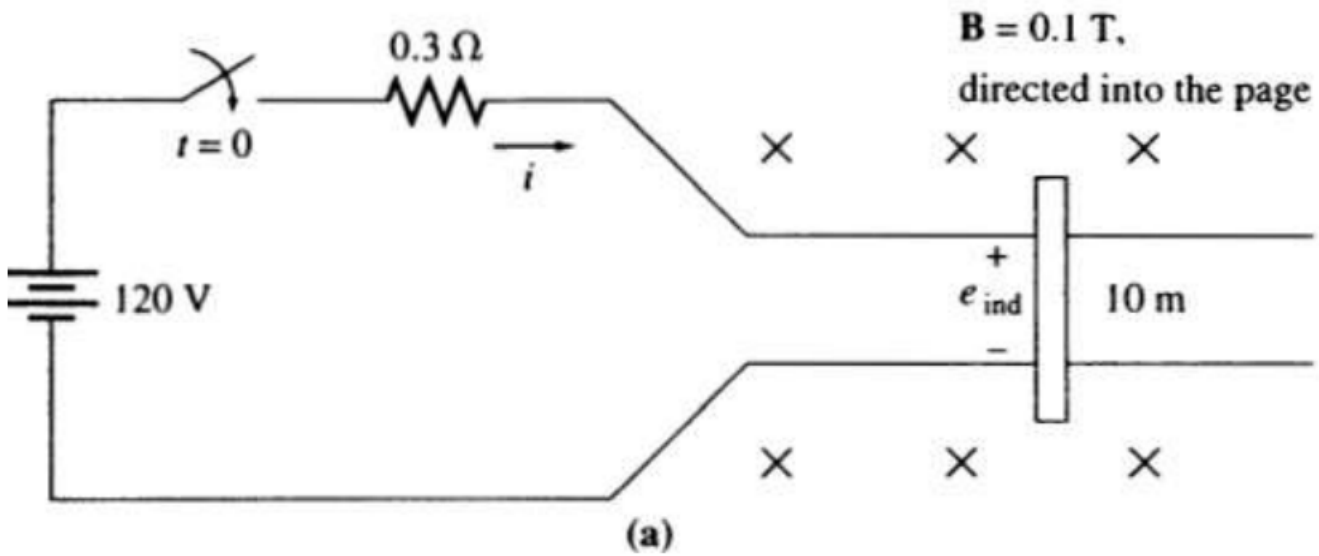
(b) Suppose that a 30-N force pointing to the right were applied to the bar. What would the steady-state speed be? How much power would the bar be producing or consuming? How much power would the battery be producing or consuming? Explain the difference between these two figures. Is this machine acting as a motor or as a generator?

(c) Now suppose a 30-N force pointing to the left were applied to the bar. What would the new steady-state speed be? Is this machine a motor or a generator now?

(d) Assume that a force pointing to the left is applied to the bar. Calculate speed of the bar as a function of the force for values from 0 N to 50 N in 10-N steps. Plot the velocity of the bar versus the applied force.

(e) Assume that the bar is unloaded and that it suddenly runs into a region where the magnetic field is weakened to 0.08 T. How fast will the bar go now?

Example 1-10



Solution

(a) At starting conditions, the velocity of the bar is 0, so $e_{ind} = 0$. Therefore,

$$i = \frac{V_B - e_{ind}}{R} = \frac{120\ \text{V} - 0\ \text{V}}{0.3\ \Omega} = 400\ \text{A}$$

When the machine reaches steady state, $\mathbf{F}_{ind} = 0$ and $i = 0$. Therefore,

$$VB = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl}$$

$$= \frac{120\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 120\ \text{m/s}$$

(*b*) Refer to Figure 1–27b. If a 30-N force to the right is applied to the bar, the final steady state will occur when the induced force $\mathbf{F}_{ind}$ is equal and opposite to the applied force $\mathbf{F}_{app}$, so that the net force on the bar is zero:

$$F_{app} = F_{ind} = ilB$$

Therefore,

$$i = \frac{F_{ind}}{lB} = \frac{30\text{ N}}{(10\text{m})(0.1\text{ T})}$$

$$= 30\text{ A} \quad \text{flowing up through the bar}$$

The induced voltage e_{ind} on the bar must be

$$e_{ind} = V_B + iR$$

$$= 120\text{ V} + (30\text{A})(0.3\ \Omega) = 129\text{ V}$$

and the final steady-state speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{129\text{ V}}{(0.1\text{ T})(10\text{ m})} = 129\text{ m/s}$$

The bar is *producing* $P = (129\text{ V})(30\text{ A}) = 3870\text{ W}$ of power, and the battery is *consuming* $P = (120\text{ V})(30\text{ A}) = 3600\text{ W}$. The difference between these two numbers is the 270 W of losses in the resistor. This machine is acting as a *generator.*

(c) Refer to Figure 1–25c. This time, the force is applied to the left, and the induced force is to the right. At steady state,

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{ind}}{lB} = \frac{30\text{ N}}{(10\text{ m})(0.1\text{ T})}$$

$$= 30\text{ A} \quad \text{flowing down through the bar}$$

The induced voltage e_{ind} on the bar must be

$$e_{ind} = V_B - iR$$

$$= 120\text{ V} - (30\text{ A})(0.3\ \Omega) = 111\text{ V}$$

and the final speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{111\text{ V}}{(0.1\text{ T})(10\text{ m})} = 111\text{ m/s}$$

This machine is now acting as a *motor*, converting electric energy from the battery into mechanical energy of motion on the bar.

(*e*) If the bar is initially unloaded, then $e_{ind} = V_B$. If the bar suddenly hits a region of weaker magnetic field, a transient will occur. Once the transient is over, though, e_{ind} will again equal V_B.

1.9 REAL, REACTIVE, AND APPARENT POWER IN SINGLE-PHASE AC CIRCUITS

$$S = VI \tag{1–62}$$

The units of apparent power are volt-amperes (VA), where 1 VA = 1 V × 1 A. As with reactive power, apparent power is given a distinctive set of units to avoid confusing it with real and reactive power.

$$P = VI \cos \theta \tag{1–60}$$

which is the *average* or *real* power (P) supplied to the load by term 1 of the Equation (1–59). The units of real power are watts (W), where 1 W = 1 V × 1 A.

The reactive power of a load is given by

$$Q = VI \sin \theta$$

$$P = I^2 Z \cos \theta$$

$$Q = I^2 Z \sin \theta$$

$$S = I^2 Z$$

where $|Z|$ is the magnitude of the load impedance Z.

Since the impedance of the load Z can be expressed as

$$Z = R + jX = |Z| \cos \theta + j|Z| \sin \theta$$

we see from this equation that $R = |Z| \cos \theta$ and $X = |Z| \sin \theta$, so the real and reactive powers of a load can also be expressed as

$$P = I^2R \tag{1–67}$$

$$Q = I^2X \tag{1–68}$$

where R is the resistance and X is the reactance of load Z.

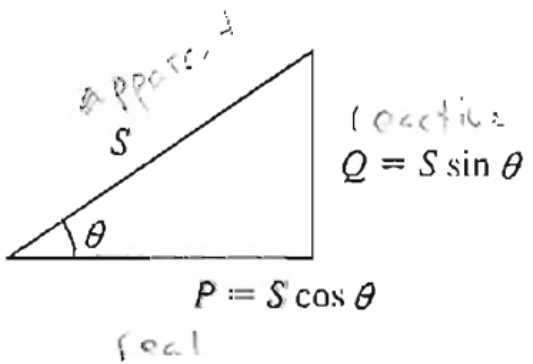
Complex Power

For simplicity in computer calculations, real and reactive power are sometimes represented together as a *complex power* **S**, where

$$\mathbf{S} = P + jQ \tag{1–69}$$

The complex power **S** supplied to a load can be calculated from the equation

$$\mathbf{S} = \mathbf{VI}^* \tag{1–70}$$
$$\mathbf{S} = VI\cos\theta + jVI\sin\theta$$



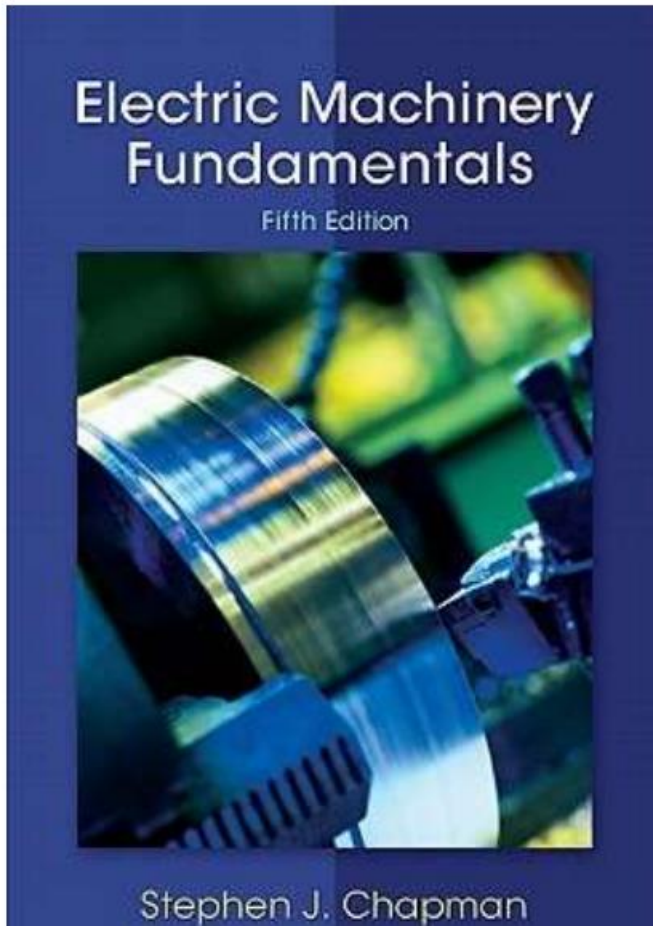
$$\cos\theta = \frac{P}{S}$$
$$\sin\theta = \frac{Q}{S}$$
$$\tan\theta = \frac{Q}{P}$$

FIGURE 1–33
The power triangle.

$$\text{PF} = \cos\theta$$

Text book and supplementary materials of this course

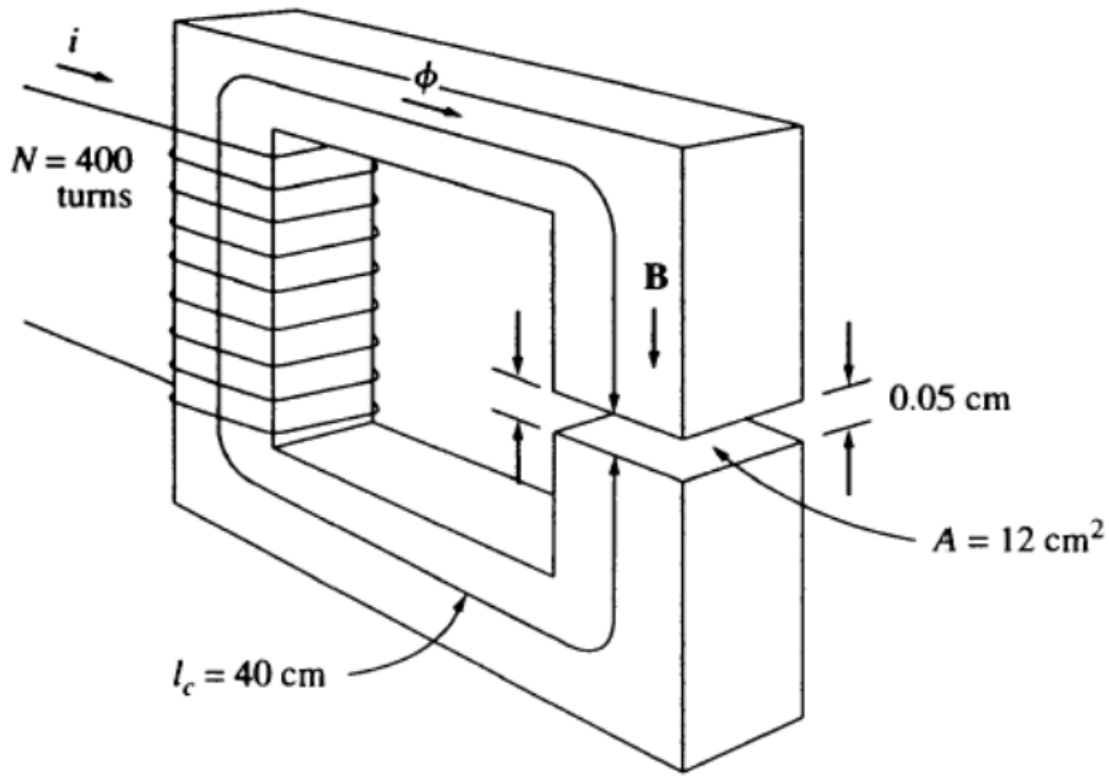
- Stephen J. Chapman



Example 1-2

Example 1–2. Figure 1–8a shows a ferromagnetic core whose mean path length is 40 cm. There is a small gap of 0.05 cm in the structure of the otherwise whole core. The cross-sectional area of the core is 12 cm^2, the relative permeability of the core is 4000, and the coil of wire on the core has 400 turns. Assume that fringing in the air gap increases the effective cross-sectional area of the air gap by 5 percent. Given this information, find (a) the total reluctance of the flux path (iron plus air gap) and (b) the current required to produce a flux density of 0.5 T in the air gap.

Example 1-2



Solution

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

(a) The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \tag{1–32}$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \tag{1–32}$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

Solution

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

(a) The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \tag{1–32}$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \tag{1–32}$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

(a) The ferromagnetic core of Example 1–2. (b) The magnetic circuit corresponding to (a).

Therefore, the total reluctance of the flux path is

$$\begin{aligned}\mathcal{R}_{eq} &= \mathcal{R}_c + \mathcal{R}_a \\ &= 66{,}300 \text{ A} \cdot \text{turns/Wb} + 316{,}000 \text{ A} \cdot \text{turns/Wb} \\ &= 382{,}300 \text{ A} \cdot \text{turns/Wb}\end{aligned}$$

Note that the air gap contributes most of the reluctance even though it is 800 times shorter than the core.

(b) Equation (1–28) states that

$$\mathcal{F} = \phi\mathcal{R} \tag{1–28}$$

Since the flux $\phi = BA$ and $\mathcal{F} = Ni$, this equation becomes

$$Ni = BA\mathcal{R}$$

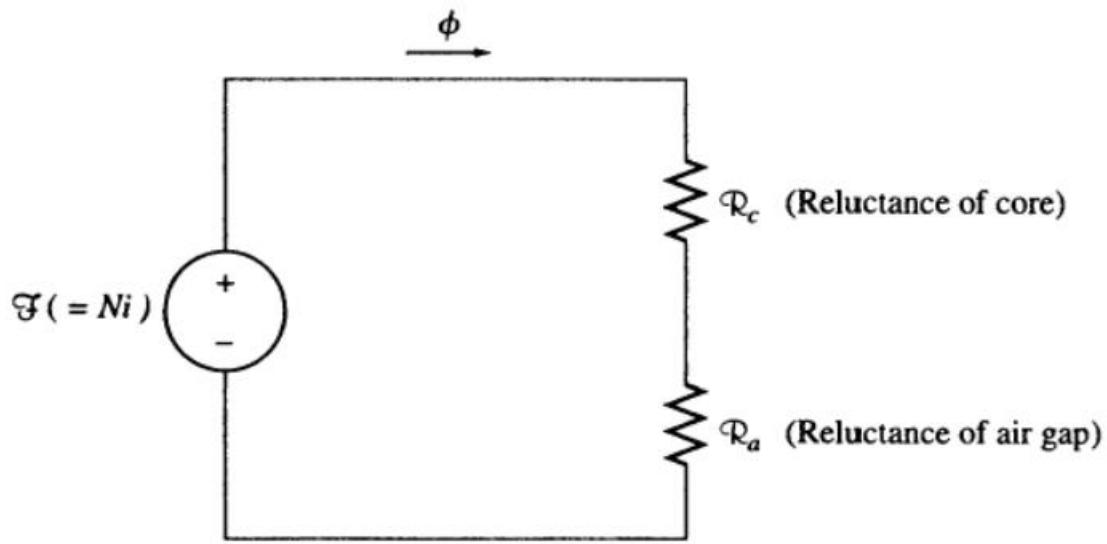
so

$$i = \frac{BA\mathcal{R}}{N}$$

$$= \frac{(0.5\ \text{T})(0.00126\ \text{m}^2)(383{,}200\ \text{A} \cdot \text{turns/ Wb})}{400\ \text{turns}}$$

$$= 0.602\ \text{A}$$

Notice that, since the *air-gap* flux was required, the effective air-gap area was used in the above equation.



(b)

Example 1-4

Example 1–4. Find the relative permeability of the typical ferromagnetic material whose magnetization curve is shown in Figure 1–10c at (a) $H = 50$, (b) $H = 100$, (c) $H = 500$, and (d) $H = 1000$ A • turns/m.

Solution

The permeability of a material is given by

$$\mu = \frac{B}{H}$$

and the relative permeability is given by

$$\mu_r = \frac{\mu}{\mu_0}$$

Thus, it is easy to determine the permeability at any given magnetizing intensity.

(a) At $H = 50$ A • turns/m, $B = 0.25$ T, so

$$\mu = \frac{B}{H} = \frac{0.25 \text{ T}}{50 \text{ A} \cdot \text{turns/m}} = 0.0050 \text{ H/m}$$

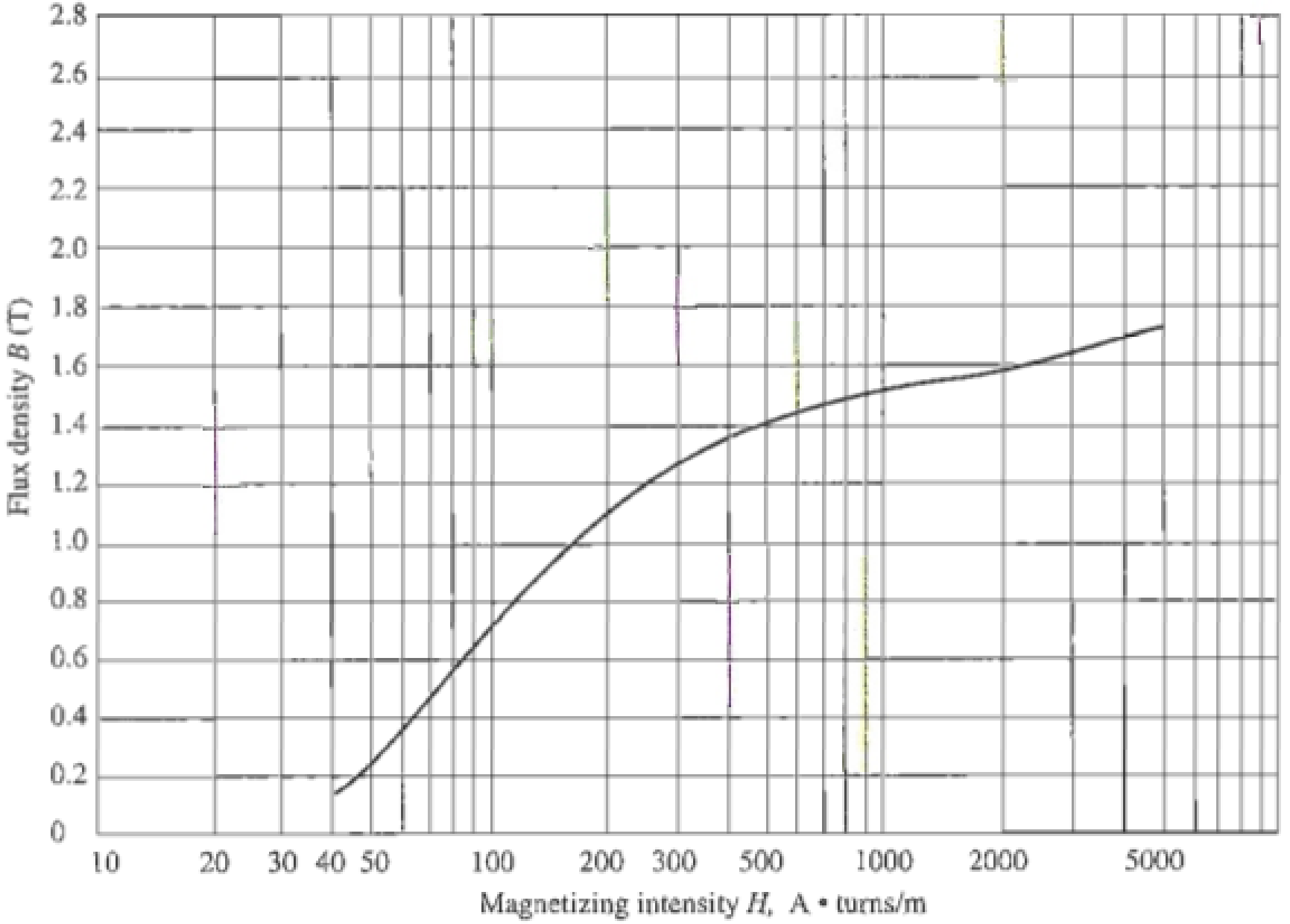
and

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.0050 \text{ H/m}}{4\pi \times 10^{-7} \text{ H/m}} = 3980$$

Example 1-5

Example 1–5. A square magnetic core has a mean path length of 55 cm and a cross-sectional area of 150 cm^2. A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve shown in Figure 1–10c.

(a) How much current is required to produce 0.012 Wb of flux in the core?
(b) What is the core's relative permeability at that current level?
(c) What is its reluctance?


Flux density B (T)
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
10
20
30
40
50
100
200
300
500
1000
2000
5000
Magnetizing intensity H, A • turns/m

(c)

Solution

(a) The required flux density in the core is

$$B = \frac{\phi}{A} = \frac{1.012 \text{ Wb}}{0.015 \text{ m}^2} = 0.8 \text{ T}$$

From Figure 1–10c, the required magnetizing intensity is

$$H = 115 \text{ A} \cdot \text{turns/m}$$

From Equation (1–20), the magnetomotive force needed to produce this magnetizing intensity is

$$\begin{aligned} \mathcal{F} &= Ni = Hl_c \\ &= (115 \text{ A} \cdot \text{turns/m})(0.55 \text{ m}) = 63.25 \text{ A} \cdot \text{turns} \end{aligned}$$

so the required current is

$$i = \frac{\mathcal{F}}{N} = \frac{63.25 \text{ A} \cdot \text{turns}}{200 \text{ turns}} = 0.316 \text{ A}$$

(b) The core's permeability at this current is

$$\mu = \frac{B}{H} = \frac{0.8\ \text{T}}{115\ \text{A} \cdot \text{turns/m}} = 0.00696\ \text{H/m}$$

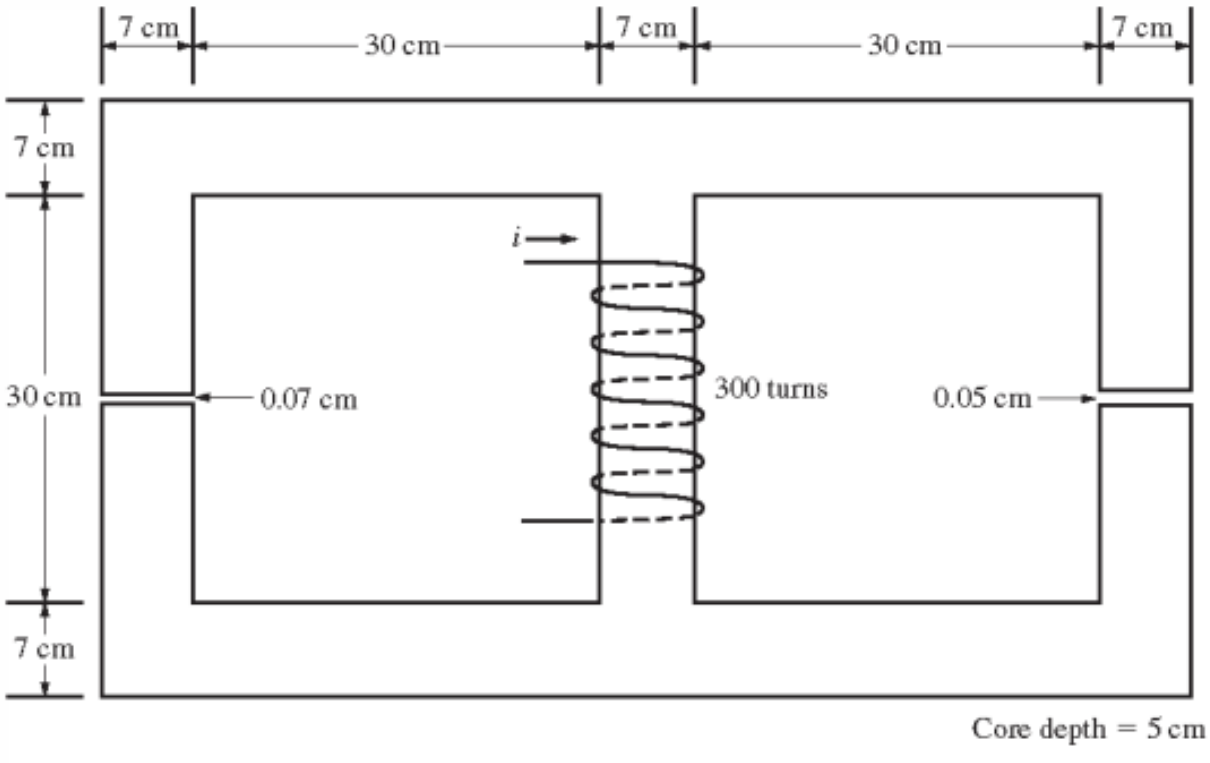
Therefore, the relative permeability is

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00696\ \text{H/m}}{4\pi \times 10^{-7}\ \text{H/m}} = 5540$$

(c) The reluctance of the core is

$$\mathcal{R} = \frac{\mathcal{F}}{\phi} = \frac{63.25\ \text{A} \cdot \text{turns}}{0.012\ \text{Wb}} = 5270\ \text{A} \cdot \text{turns/Wb}$$

A ferromagnetic core with a relative permeability of 1500 is shown in Figure P1-3. The dimensions are as shown in the diagram, and the depth of the core is 5 cm. The air gaps on the left and right sides of the core are 0.050 and 0.070 cm, respectively. Because of fringing effects, the effective area of the air gaps is 5 percent larger than their physical size. If there are 300 turns in the coil wrapped around the center leg of the core and if the current in the coil is 1.0 A, what is the flux in each of the left, center, and right legs of the core? What is the flux density in each air gap?



SOLUTION This core can be divided up into five regions. Let $\mathcal{R}_1$ be the reluctance of the left-hand portion of the core, $\mathcal{R}_2$ be the reluctance of the left-hand air gap, $\mathcal{R}_3$ be the reluctance of the right-hand portion of the core, $\mathcal{R}_4$ be the reluctance of the right-hand air gap, and $\mathcal{R}_5$ be the reluctance of the center leg of the core. Then the total reluctance of the core is

$$\mathcal{R}_{\text{TOT}} = \mathcal{R}_5 + \frac{(\mathcal{R}_1 + \mathcal{R}_2)(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4}$$

$$\mathcal{R}_1 = \frac{l_1}{\mu_r \mu_0 A_1} = \frac{1.11 \text{ m}}{(1500)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})} = 168 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_2 = \frac{l_2}{\mu_0 A_2} = \frac{0.0007 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})(1.05)} = 152 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_3 = \frac{l_3}{\mu_r \mu_0 A_3} = \frac{1.11 \text{ m}}{(1500)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})} = 168 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_4 = \frac{l_4}{\mu_0 A_4} = \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.05 \text{ m})(1.05)} = 108 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_5 = \frac{l_5}{\mu_r \mu_0 A_5} = \frac{0.37\ \text{m}}{(1500)(4\pi \times 10^{-7}\ \text{H/m})(0.07\ \text{m})(0.05\ \text{m})} = 56.1\ \text{kA} \cdot \text{t/Wb}$$

The total reluctance is

$$\mathcal{R}_{\text{TOT}} = \mathcal{R}_5 + \frac{(\mathcal{R}_1 + \mathcal{R}_2)(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} = 56.1 + \frac{(168+152)(168+108)}{168+152+168+108} = 204\ \text{kA} \cdot \text{t/Wb}$$

The total flux in the core is equal to the flux in the center leg:

$$\phi_{\text{center}} = \phi_{\text{TOT}} = \frac{\mathcal{F}}{\mathcal{R}_{\text{TOT}}} = \frac{(300\ \text{t})(1.0\ \text{A})}{204\ \text{kA} \cdot \text{t/Wb}} = 0.00147\ \text{Wb}$$

The fluxes in the left and right legs can be found by the "flux divider rule", which is analogous to the current divider rule.

$$\phi_{\text{left}} = \frac{(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} \phi_{\text{TOT}} = \frac{(168+108)}{168+152+168+108}(0.00147\ \text{Wb}) = 0.00068\ \text{Wb}$$

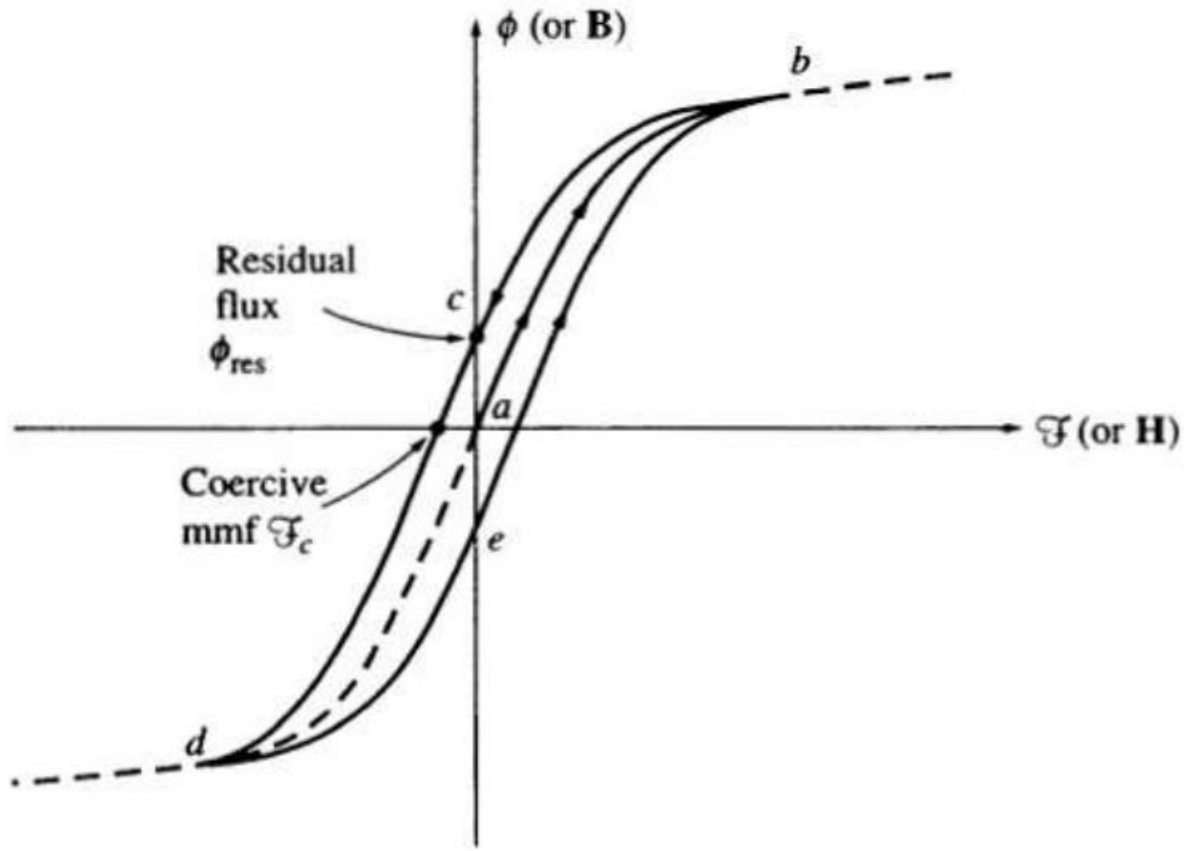
$$\phi_{\text{right}} = \frac{(\mathcal{R}_1 + \mathcal{R}_2)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} \phi_{\text{TOT}} = \frac{(168+152)}{168+152+168+108}(0.00147\ \text{Wb}) = 0.00079\ \text{Wb}$$

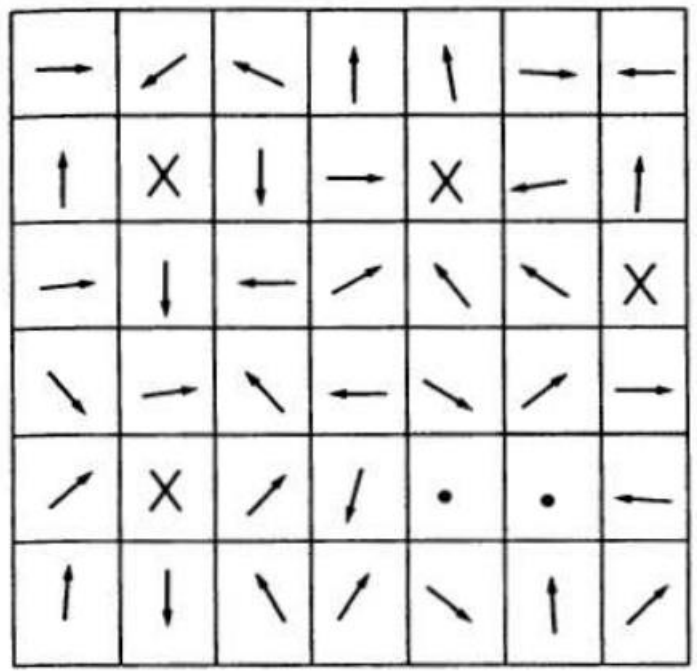
The flux density in the air gaps can be determined from the equation $\phi = BA$:

$$B_{\text{left}} = \frac{\phi_{\text{left}}}{A_{\text{eff}}} = \frac{0.00068\ \text{Wb}}{(0.07\ \text{cm})(0.05\ \text{cm})(1.05)} = 0.185\ \text{T}$$

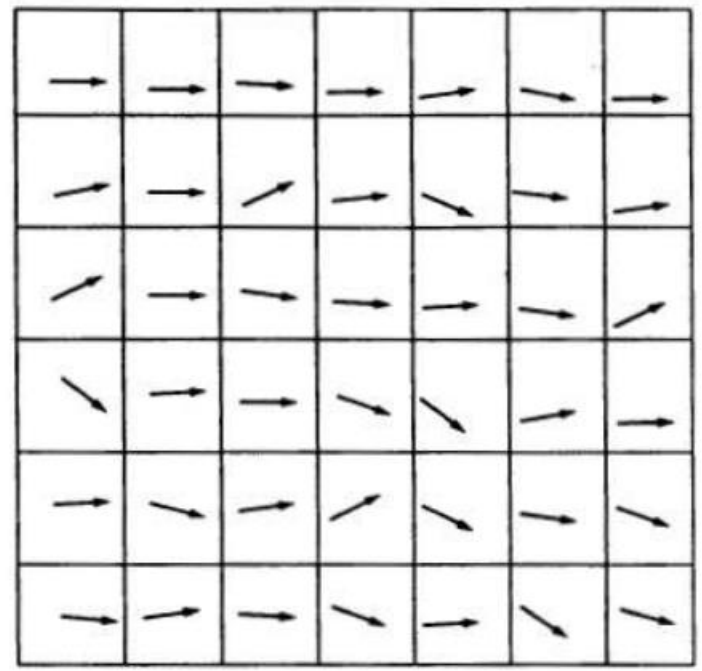
$$B_{\text{right}} = \frac{\phi_{\text{right}}}{A_{\text{eff}}} = \frac{0.00079\ \text{Wb}}{(0.07\ \text{cm})(0.05\ \text{cm})(1.05)} = 0.215\ \text{T}$$

Energy loss in ferromagnetic core – hysteresis loss



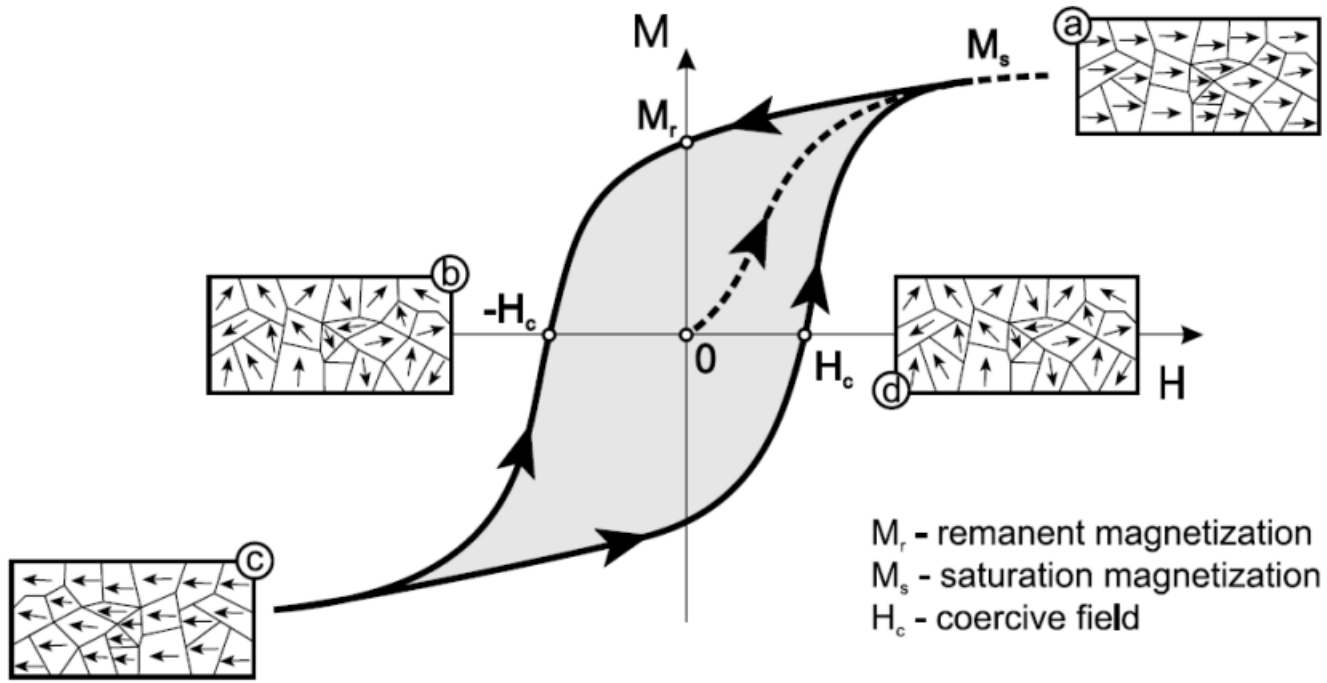


(a)



(b)

Magnetization curve



Faraday's law – induce voltage from a time-varying magnetic field

1. Induced voltage magnitude and polarity

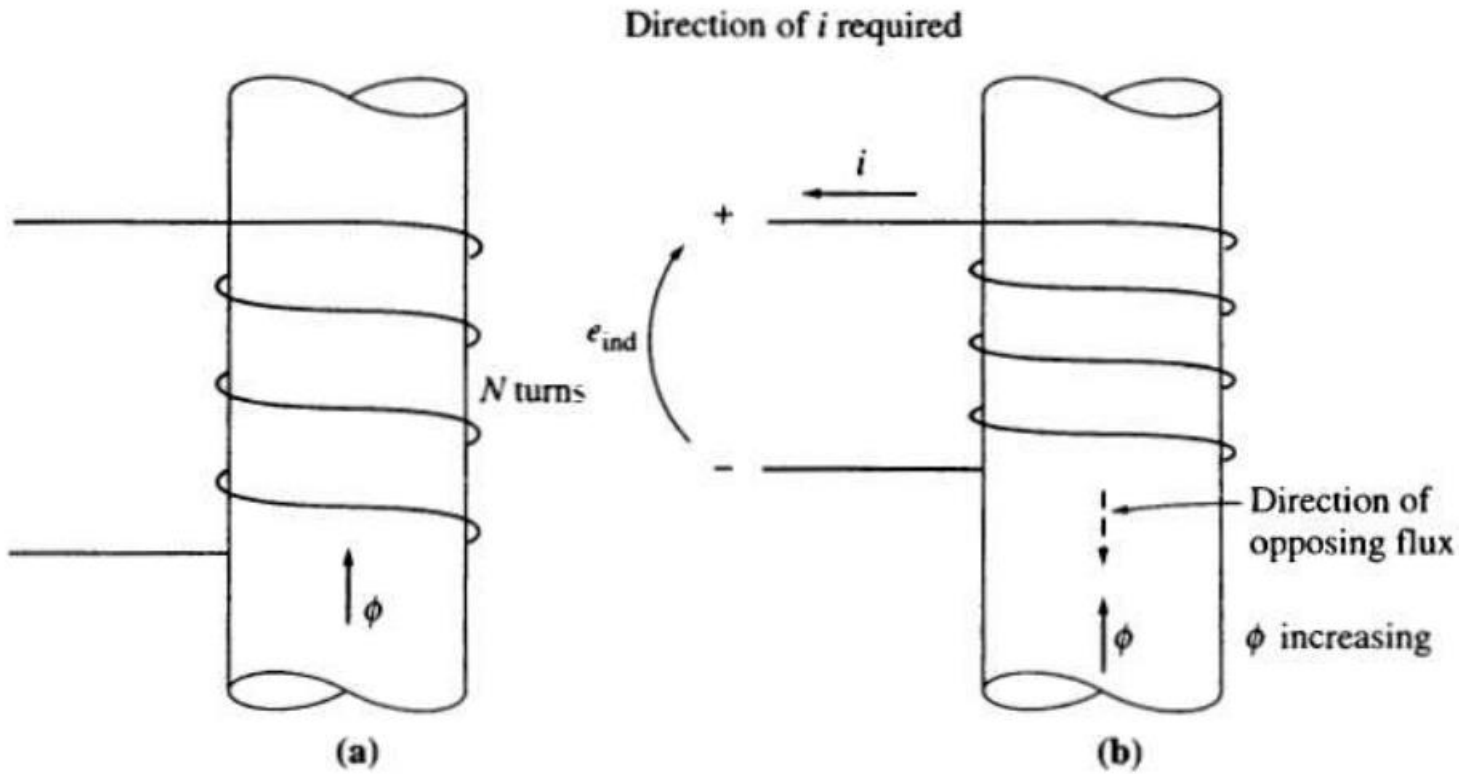
$$e_{\text{ind}} = -\frac{d\phi}{dt}$$

$$e_{\text{ind}} = \sum_{i=1}^{N} e_i$$

$$e_{\text{ind}} = -N\frac{d\phi}{dt}$$

e_{ind} = voltage induced in the coil
N = number of turns of wire in coil
ϕ = flux passing through coil

The induced voltage polarity – Lenz's law



Flux and flux linkage

The term in parentheses in Equation (1–40) is called *the flux linkage* λ of the coil, and Faraday's law can be rewritten in terms of flux linkage as

$$e_{\text{ind}} = \frac{d\lambda}{dt} \tag{1–41}$$

where

$$\lambda = \sum_{i=1}^{N} \phi_i \tag{1–42}$$

The units of flux linkage are weber-turns.

Example 1-6

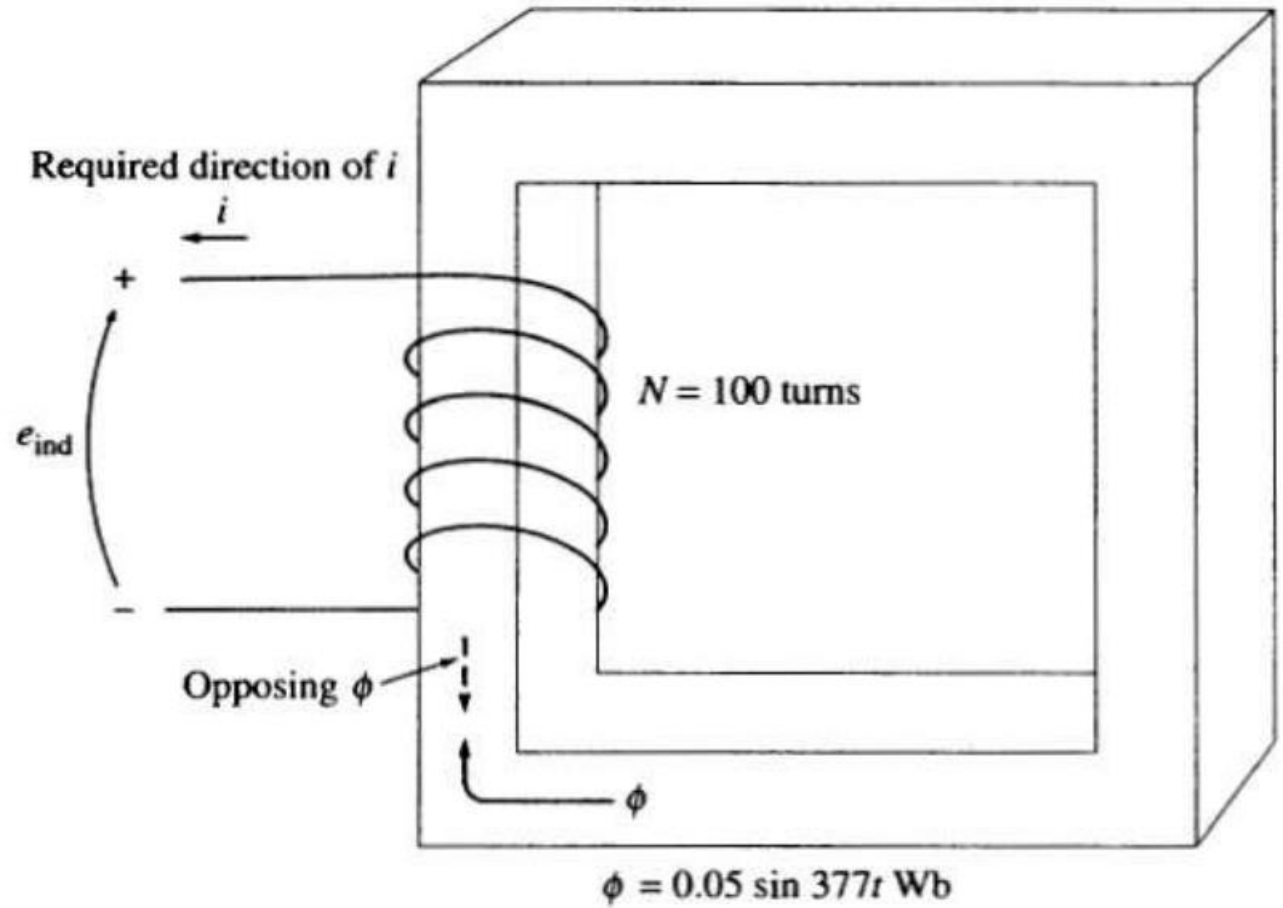
Example 1–6. Figure 1–15 shows a coil of wire wrapped around an iron core. If the flux in the core is given by the equation

$$\phi = 0.05 \sin 377t \quad \text{Wb}$$

If there are 100 turns on the core, what voltage is produced at the terminals of the coil? Of what polarity is the voltage during the time when flux is *increasing* in the reference

The core of Example 1–6. Determination of the voltage polarity at the terminals is shown.

direction shown in the figure? Assume that all the magnetic flux stays within the core (i.e., assume that the flux leakage is zero).


Required direction of i
i
+
e_{ind}
−
$N = 100$ turns
Opposing ϕ
ϕ
$\phi = 0.05 \sin 377t$ Wb

Solution

By the same reasoning as in the discussion on pages 29–30, the direction of the voltage while the flux is increasing in the reference direction must be positive to negative, as shown in Figure 1–15. The *magnitude* of the voltage is given by

$$e_{\text{ind}} = N\frac{d\phi}{dt}$$

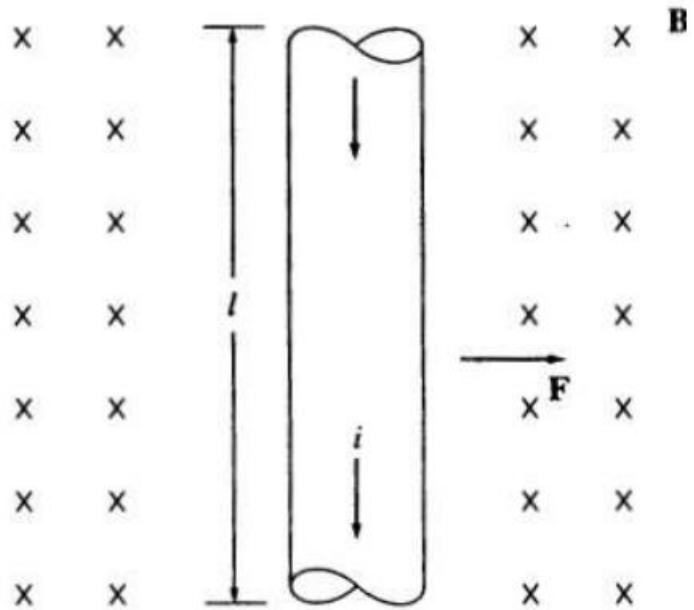
$$= (100 \text{ turns})\frac{d}{dt}(0.05 \sin 377t)$$

$$= 1885 \cos 377t$$

or alternatively,

$$e_{\text{ind}} = 1885 \sin(377t + 90°) \text{ V}$$

Produce an induced force on a wire



$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

$$F = ilB \sin \theta$$

i = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

Example 1-7

Example 1–7. Figure 1–16 shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?

Example 1-7

Solution

The direction of the force is given by the right-hand rule as being to the right. The magnitude is given by

$$F = ilB \sin \theta \qquad (1\text{–}44)$$

$$= (0.5\ \text{A})(1.0\ \text{m})(0.25\ \text{T}) \sin 90° = 0.125\ \text{N}$$

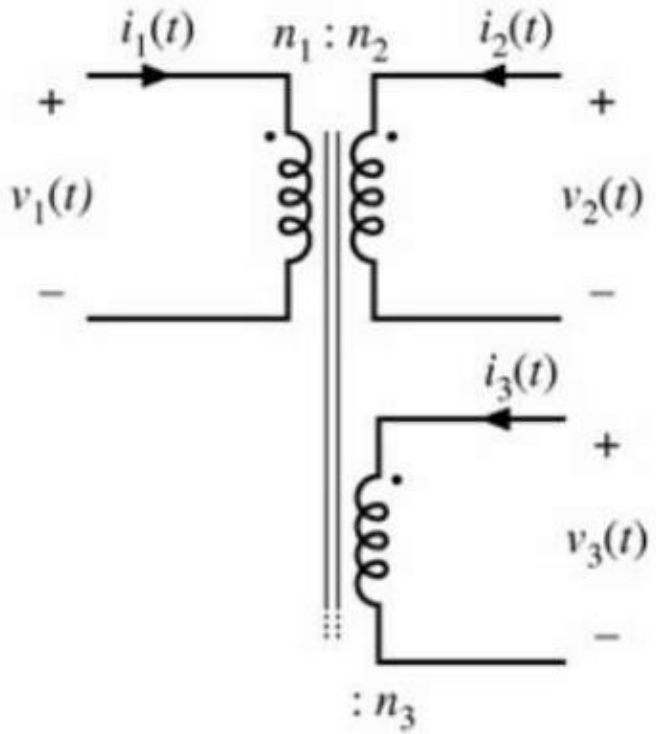
Therefore,

$$\mathbf{F} = 0.125\ \text{N, directed to the right}$$

Transformers

- The transformer winding connected to the power source is called the ***primary winding or input winding***
- The winding connected to the loads is called the ***secondary winding or output winding***
- ….***tertiary winding***

Multiple winding transformer



Why transformers are important to modern life

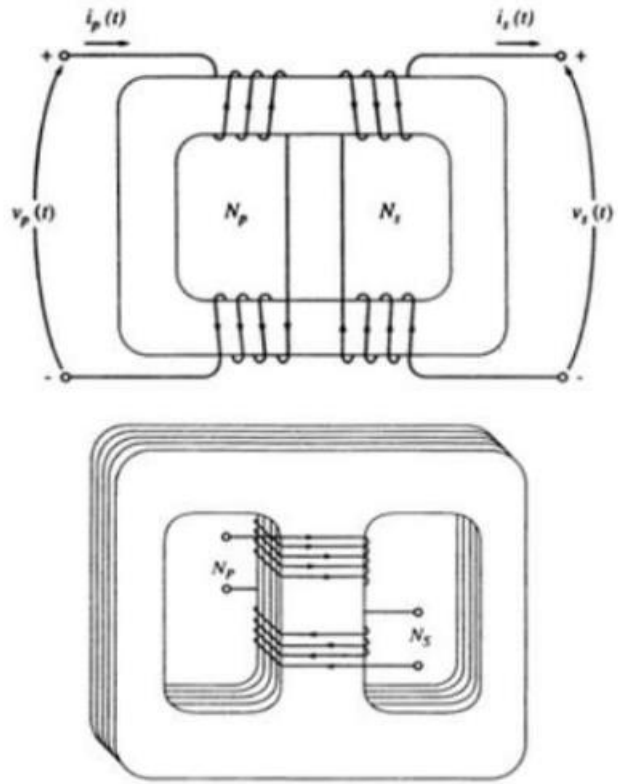
- The transformer ideally changes one ac voltage level to another voltage level without affecting the actual power supplied.
- The transformer can be used in distribution system for efficiency issues.
  - The step-up transformer decreases the line current and decreases the power loss on power line.
  - The transmission/distribution system with transformer can keep high efficiency

- Low voltage : 110V, 220V, 380V
- High voltage (HV): 11.4kV, 22.8kV (distribution system)
- Extra high voltage (EHV): 69kV, 161kV (transmission system)
- Ultra high voltage (UHV): 345kV (transmission system)

Types and construction of transformers

- **Core-form**: consists a simple rectangular laminated piece of steel with the transformer winding wrapped around two sides of the rectangle

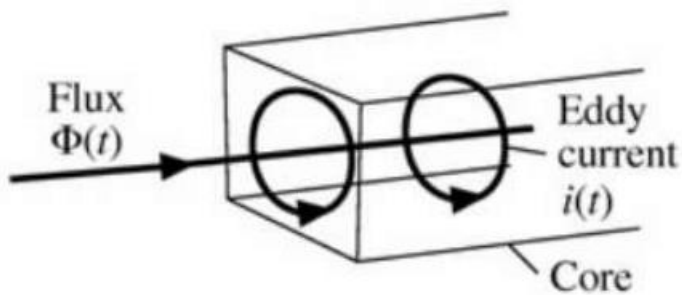
- **Shell-form** : consists three legs laminated core with winding wrapped around the center leg



Core material and eddy current

- The core is constructed of ***thin laminations*** electrically isolated from each other in order to minimize the ***eddy currents***

- Eddy current:

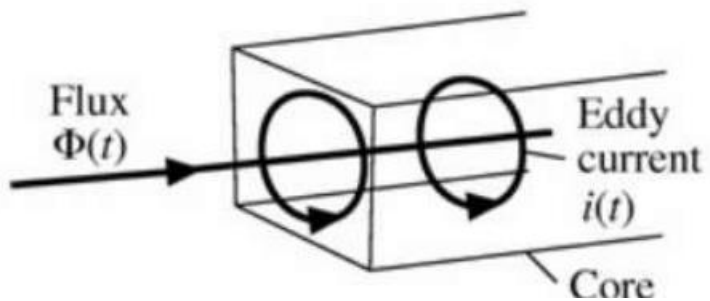


Eddy current loss $i^2(t)R$

Core material and eddy current

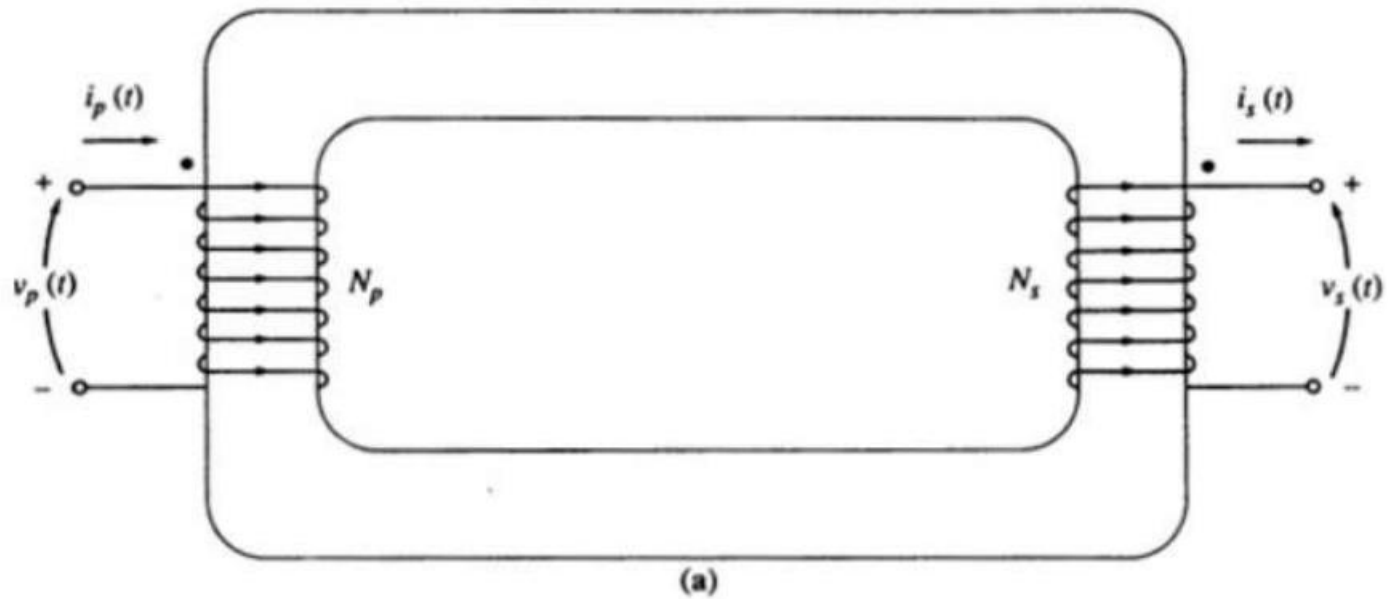
- The core is constructed of ***thin laminations*** electrically isolated from each other in order to minimize the ***eddy currents***

- Eddy current:



Eddy current loss $i^2(t)R$

The ideal transformer characteristics



(a)

The ideal transformer characteristics

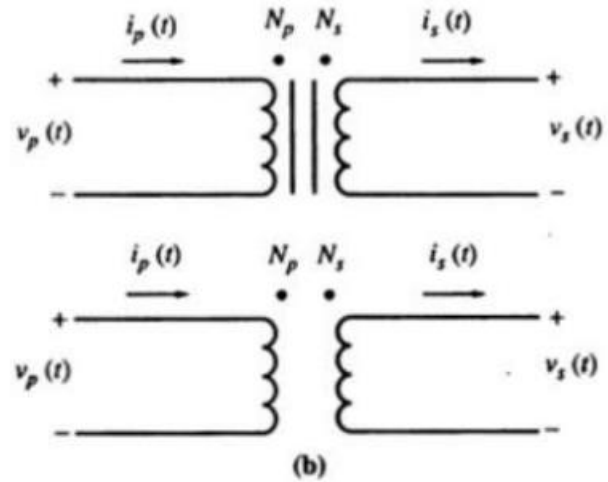
$$\frac{v_P(t)}{v_S(t)} = \frac{N_P}{N_S} = a$$

- Where $a = N_p/N_s$ is the turns ratio
- Energy balance relation

$$N_P i_P(t) = N_S i_S(t) \qquad \frac{i_P(t)}{i_S(t)} = \frac{1}{a}$$

- Phasor relation

$$\frac{\mathbf{V}_P}{\mathbf{V}_S} = a \qquad \frac{\mathbf{I}_P}{\mathbf{I}_S} = \frac{1}{a}$$



- The turns ratio ***a*** only effects the magnitude not the angle

Power in an ideal transformer

- The power supplied to the transformer by the primary circuit is given by the equation

$$P_{in} = V_P I_P \cos \theta_P \tag{2–6}$$

where θ_P is the angle between the primary voltage and the primary current. The power supplied by the transformer secondary circuit to its loads is given by the equation

$$P_{out} = V_S I_S \cos \theta_S \tag{2–7}$$

where θ_S is the angle between the secondary voltage and the secondary current. Since voltage and current angles are unaffected by an ideal transformer, $\theta_P - \theta_S = \theta$. The primary and secondary windings of an ideal transformer have the *same power factor.*

Power in an ideal transformer

$$P_{out} = V_S I_S \cos\theta \tag{2–8}$$

Applying the turns-ratio equations gives $V_S = V_P/a$ and $I_S = aI_P$, so

$$P_{out} = \frac{V_P}{a}(aI_P)\cos\theta$$

$$\boxed{P_{out} = V_P I_P \cos\theta = P_{in}} \tag{2–9}$$

Thus, *the output power of an ideal transformer is equal to its input power.*

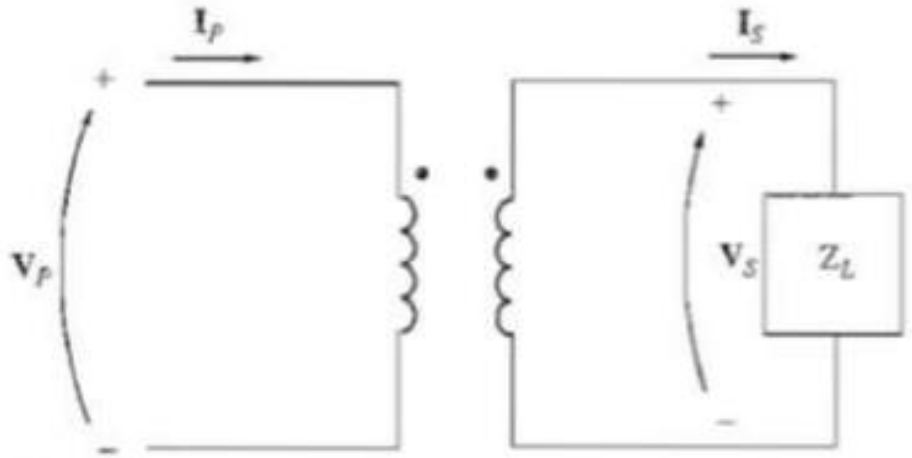
The same relationship applies to reactive power Q and apparent power S:

$$\boxed{Q_{in} = V_P I_P \sin\theta = V_S I_S \sin\theta = Q_{out}} \tag{2–10}$$

and

$$\boxed{S_{in} = V_P I_P = V_S I_S = S_{out}} \tag{2–11}$$

Impedance Transformation through a Transformer



The apparent impedance of the primary circuit

$$Z'_L = \frac{V_P}{I_P}$$

The impedance of the load (secondary circuit) $Z_L = \frac{V_S}{I_S}$

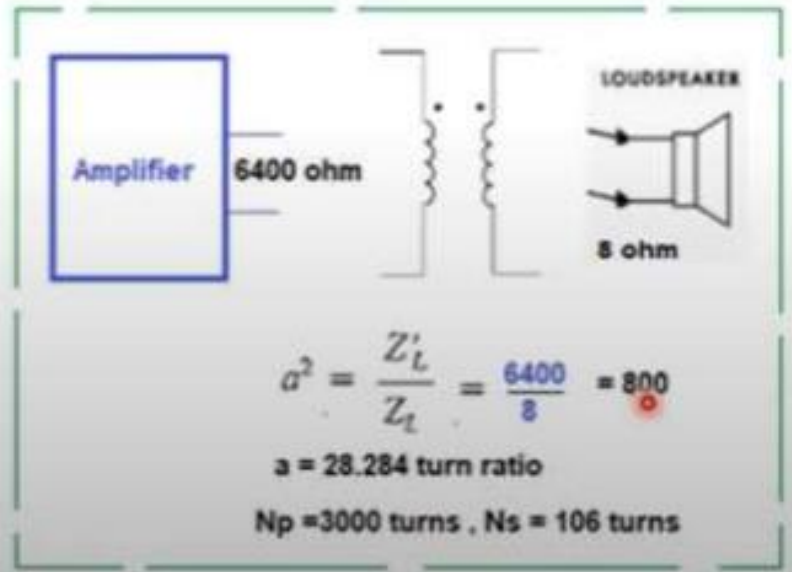
Since the primary voltage $V_P = aV_S$

and the primary current $I_P = \frac{I_S}{a}$

the apparent impedance of the primary is

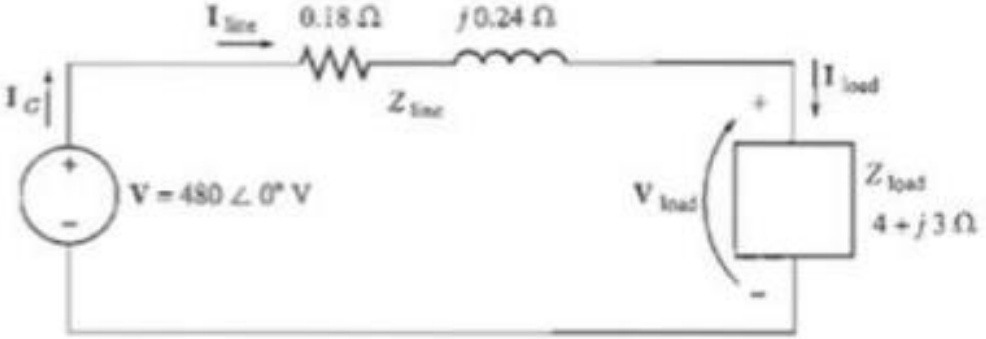
$$Z'_L = \frac{V_P}{I_P} = \frac{aV_S}{I_S/a} = a^2 \frac{V_S}{I_S} = a^2 Z_L$$

$$\boxed{Z'_L = a^2 Z_L}$$



Example 2–1.

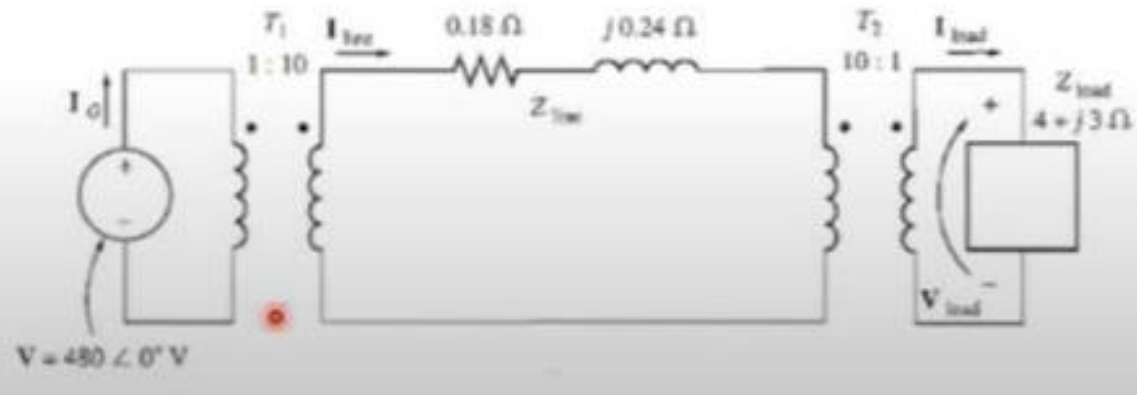
A single-phase power system consists of a 480-V 60-Hz generator supplying a load $Z_{load} = 4 + j3\ \Omega$ through a transmission line of impedance $Z_{line} = 0.18 + j0.24\ \Omega$.



(a) what will the voltage at the load be?

What will the transmission line losses be?

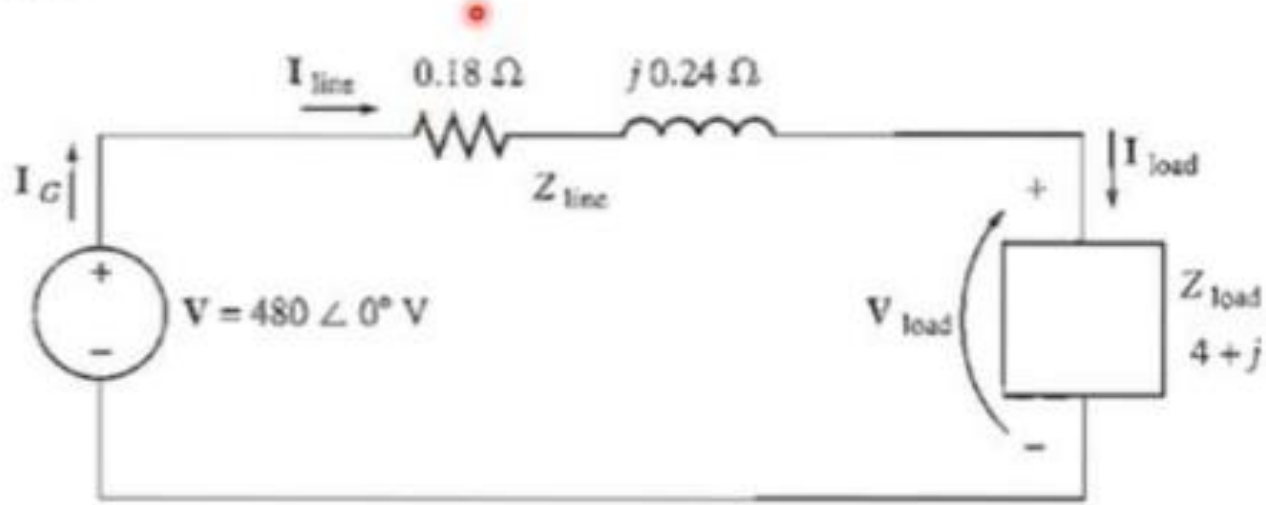
(b) Suppose a 1:10 step-up transformer is placed at the generator end of the transmission line and a 10:1 step-down transformer is placed at the load end of the line (as shown in Figure 2–6b).



(a) what will the voltage at the load be?

What will the transmission line losses be?

Solution:



Here $\mathbf{I}_G = \mathbf{I}_{line} = \mathbf{I}_{load}$

$$\mathbf{I}_{line} = \frac{\mathbf{V}}{Z_{line} + Z_{load}}$$

$$= \frac{480 \angle 0° \text{ V}}{(0.18\ \Omega + j0.24\ \Omega) + (4\ \Omega + j3\ \Omega)}$$

$$= \frac{480 \angle 0°}{4.18 + j3.24} = \frac{480 \angle 0°}{5.29 \angle 37.8°}$$

$$= 90.8 \angle -37.8° \text{ A}$$

Therefore the load voltage is

$$\mathbf{V}_{load} = \mathbf{I}_{line} Z_{load}$$

$$= (90.8 \angle -37.8° \text{ A})(4\ \Omega + j3\ \Omega)$$

$$= (90.8 \angle -37.8° \text{ A})(5 \angle 36.9°\ \Omega)$$

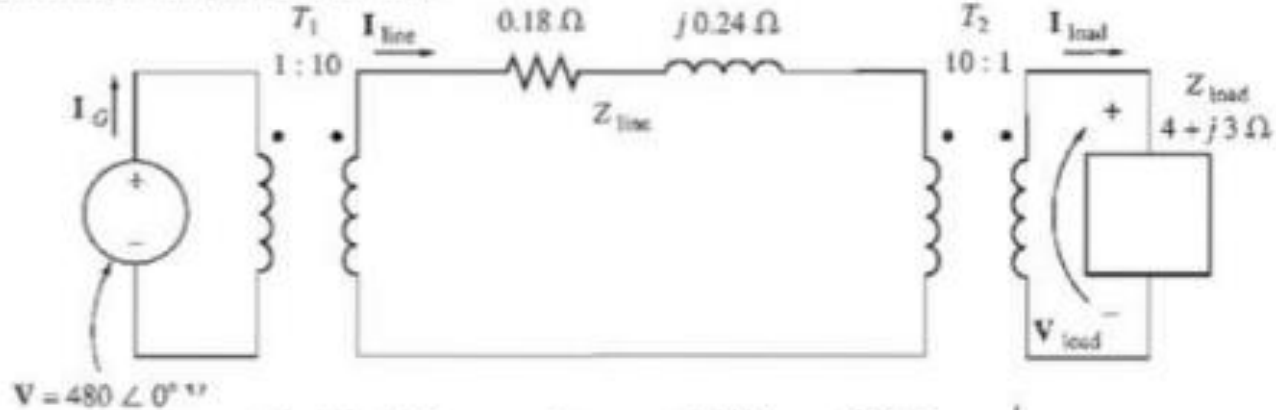
$$= 454 \angle -0.9° \text{ V}$$

and the line losses are (power is lost in resistive components only)

$$P_{loss} = (I_{line})^2 R_{line}$$

$$= (90.8 \text{ A})^2 (0.18\ \Omega) = 1484 \text{ W}$$

(b) Suppose a 1:10 step-up transformer is placed at the generator end of the transmission line and a 10:1 step-down transformer is placed at the load end of the line (as shown in Figure 2–6b).



What will the load voltage be now?

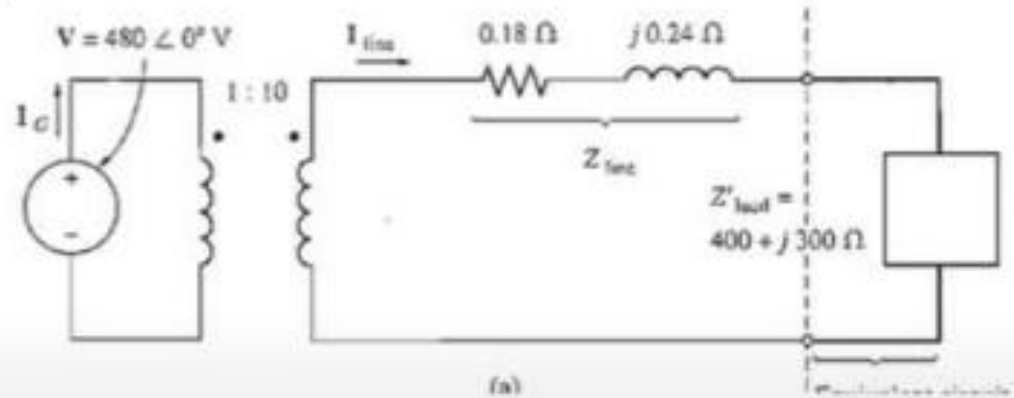
What will the transmission line losses be now?

Solution:

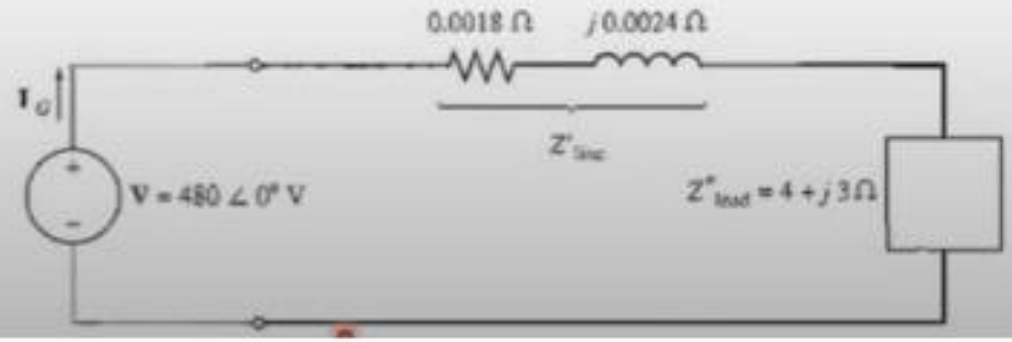
To analyze this system, it is necessary to convert it to a common voltage level.

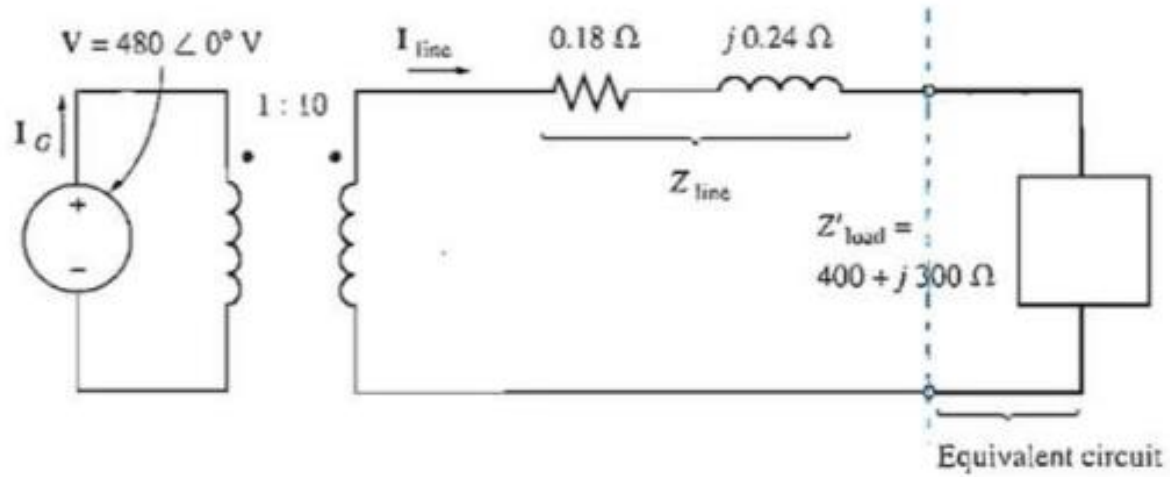
This is done in two steps:

1. Eliminate transformer T_2
   by referring the load over to the transmission line's voltage level.



2. Eliminate transformer T_1
   by referring the transmission line's elements and the equivalent load at the transmission line's voltage over to the source side.





$$Z'_{eq} = a^2 Z_{eq}$$

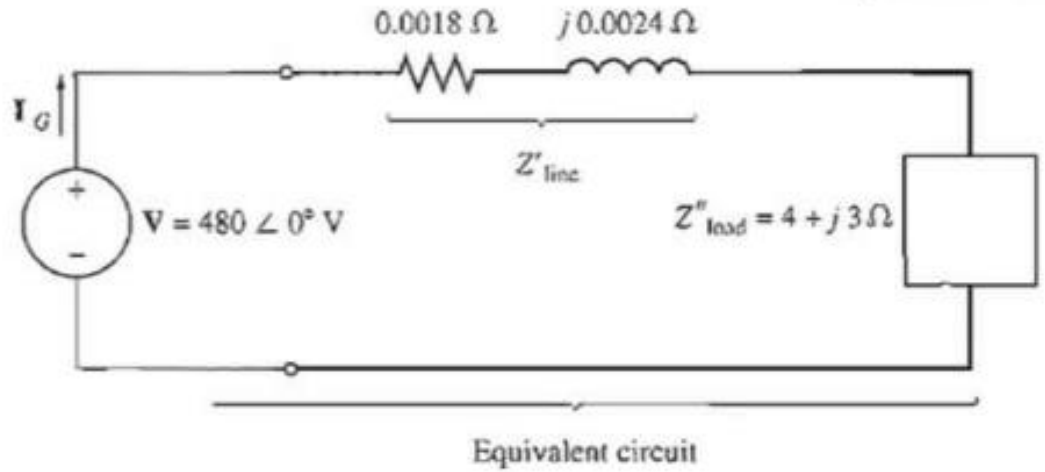
$$= a^2(Z_{line} + Z'_{load})$$

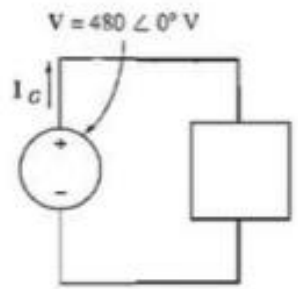
$$= \left(\frac{1}{10}\right)^2 (0.18\ \Omega + j0.24\ \Omega + 400\ \Omega + j300\ \Omega)$$

Note: Here 'a' is 1/10

$$= (0.0018\ \Omega + j0.0024\ \Omega + 4\ \Omega + j3\ \Omega)$$

$$= 5.003\ \angle 36.88°\ \Omega$$

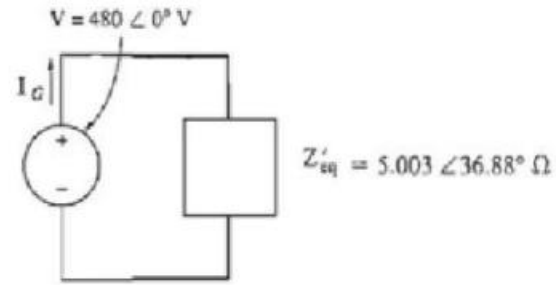




$Z'_{eq} = 5.003\ \angle 36.88°\ \Omega$

The generator's current is

$$I_G = \frac{480\angle 0^\circ \text{ V}}{5.003\angle 36.88^\circ\ \Omega} = 95.94\angle -36.88^\circ \text{ A}$$



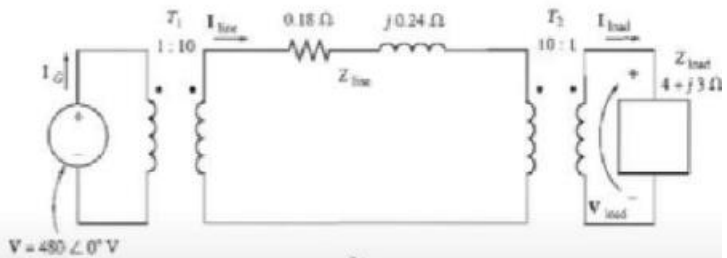
Knowing the current I_G, we can now work back and find I_{line} and I_{load}.

Working back through T_1, we get

$$N_{P1}I_G = N_{S1}I_{line}$$

$$I_{line} = \frac{N_{P1}}{N_{S1}}I_G$$

$$= \frac{1}{10}(95.94\angle -36.88^\circ \text{ A}) = 9.594\angle -36.88^\circ$$



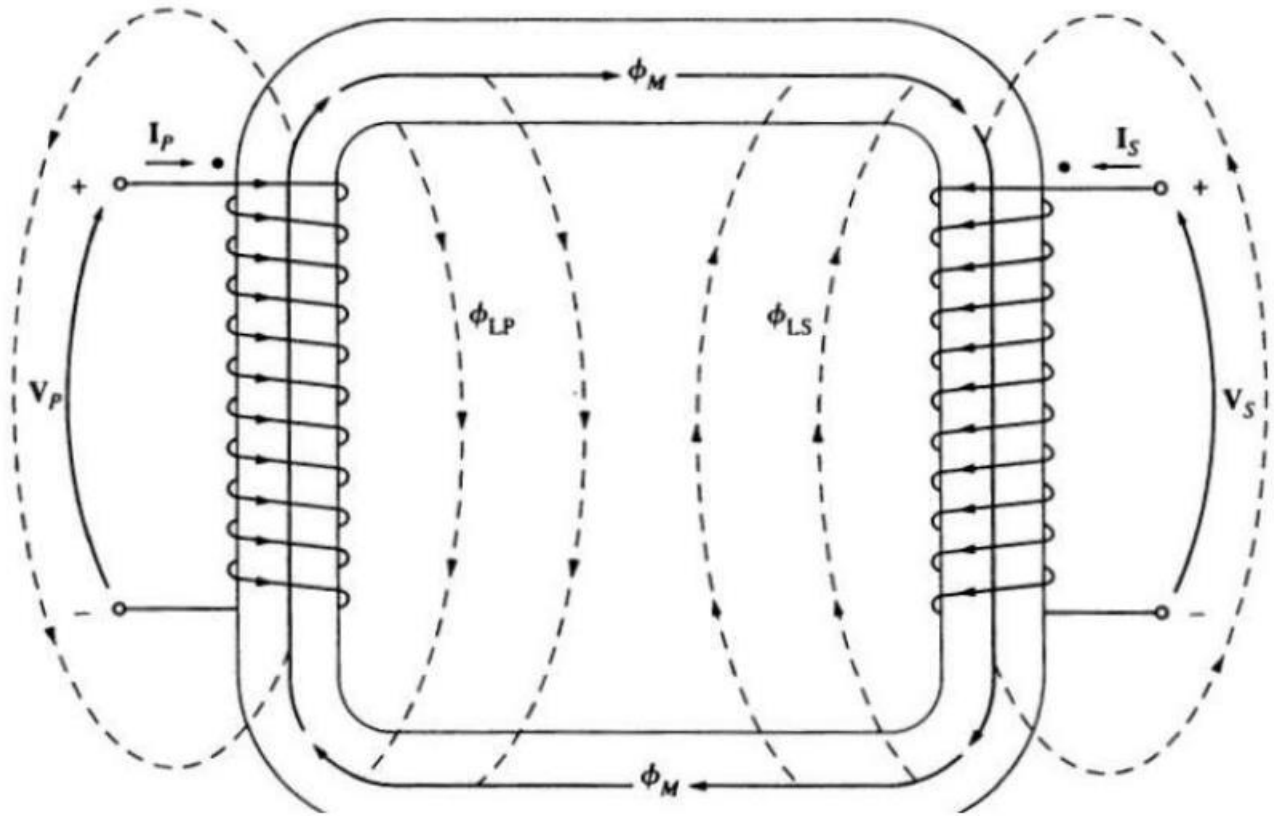
Working back through T_2 gives

$$N_{P2}I_{line} = N_{S2}I_{load}$$

$$I_{load} = \frac{N_{P2}}{N_{S2}}I_{line}$$

$$= \frac{10}{1}(9.594\angle -36.88^\circ \text{ A}) = 95.94\angle -36.88^\circ \text{ A}$$

Voltage relation between primary and secondary derived from Faraday's law



Voltage relation

$$e_{\text{ind}} = N\frac{d\bar{\phi}}{dt}$$

1. Induced voltage on each side $\bar{\phi} = \frac{1}{N_P}\int v_P(t)\,dt$

2. Primary side flux

$$\boxed{\bar{\phi}_P = \phi_M + \phi_{LP}}$$

$\bar{\phi}_P$ = total average primary flux
ϕ_M = flux component linking both primary and secondary coils
ϕ_{LP} = primary leakage flux

3. Secondary side flux

$$\boxed{\bar{\phi}_S = \phi_M + \phi_{LS}}$$

$\bar{\phi}_S$ = total average secondary flux
ϕ_M = flux component linking both primary and secondary coils
ϕ_{LS} = secondary leakage flux

Voltage relation

1. Induced voltage on primary side

$$v_P(t) = N_P \frac{d\bar{\phi}_P}{dt}$$

$$= N_P \frac{d\phi_M}{dt} + N_P \frac{d\phi_{LP}}{dt}$$

$$v_P(t) = e_P(t) + e_{LP}(t)$$

2. Induced voltage on secondary side

$$v_S(t) = N_S \frac{d\bar{\phi}_S}{dt}$$

$$= N_S \frac{d\phi_M}{dt} + N_S \frac{d\phi_{LS}}{dt}$$

$$= e_S(t) + e_{LS}(t)$$

Induced voltage relation - Induced by mutual flux

The primary voltage *due to the mutual flux* is given by

$$e_P(t) = N_P \frac{d\phi_M}{dt} \tag{2–25}$$

and the secondary voltage *due to the mutual flux* is given by

$$e_S(t) = N_S \frac{d\phi_M}{dt} \tag{2–26}$$

Notice from these two relationships that

$$\frac{e_P(t)}{N_P} = \frac{d\phi_M}{dt} = \frac{e_S(t)}{N_S}$$

Therefore,

$$\boxed{\frac{e_P(t)}{e_S(t)} = \frac{N_P}{N_S} = a} \tag{2–27}$$

Terminal voltage relation - Neglecting the leakage flux

This equation means that *the ratio of the primary voltage caused by the mutual flux to the secondary voltage caused by the mutual flux is equal to the turns ratio of the transformer*. Since in a well-designed transformer $\phi_M >> \phi_{LP}$ and $\phi_M >> \phi_{LS}$, the ratio of the total voltage on the primary of a transformer to the total voltage on the secondary of a transformer is approximately

$$\frac{v_P(t)}{v_S(t)} = \frac{N_P}{N_S} = a \tag{2–28}$$

The smaller the leakage fluxes of the transformer are, the closer the total transformer voltage ratio approximates that of the ideal transformer discussed in Section 2.3.

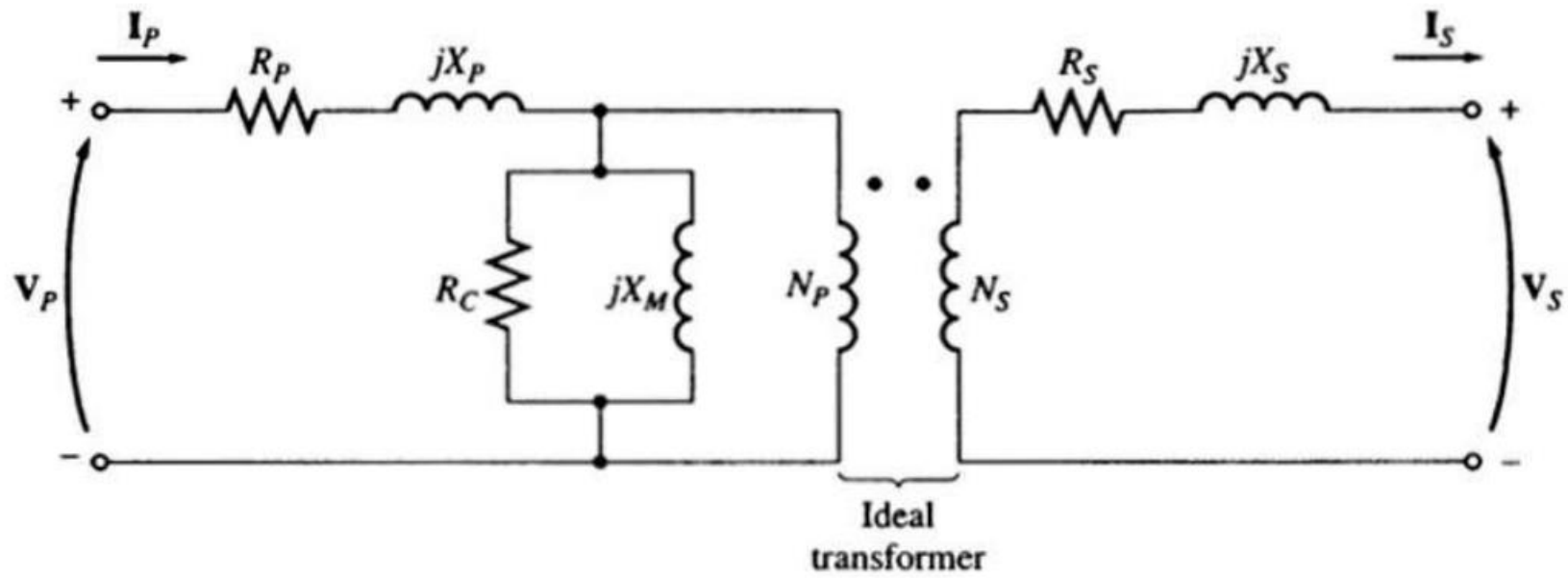
The equivalent circuit of a transformer – to model the non-ideal characteristics

1. *Copper (I^2R) losses.* Copper losses are the resistive heating losses *in the primary and secondary windings* of the transformer. They are proportional to the square of the current in the windings.
2. *Eddy current losses.* Eddy current losses are resistive heating losses *in the core* of the transformer. They are proportional to the square of the voltage applied to the transformer.
3. *Hysteresis losses.* Hysteresis losses are associated with the rearrangement of the magnetic domains in the core during each half-cycle, as explained in Chapter 1. They are a complex, nonlinear function of the voltage applied to the transformer.
4. *Leakage flux.* The fluxes ϕ_{LP} and ϕ_{LS} which escape the core and pass through only one of the transformer windings are leakage fluxes. These escaped fluxes produce a *self-inductance* in the primary and secondary coils, and the effects of this inductance must be accounted for.

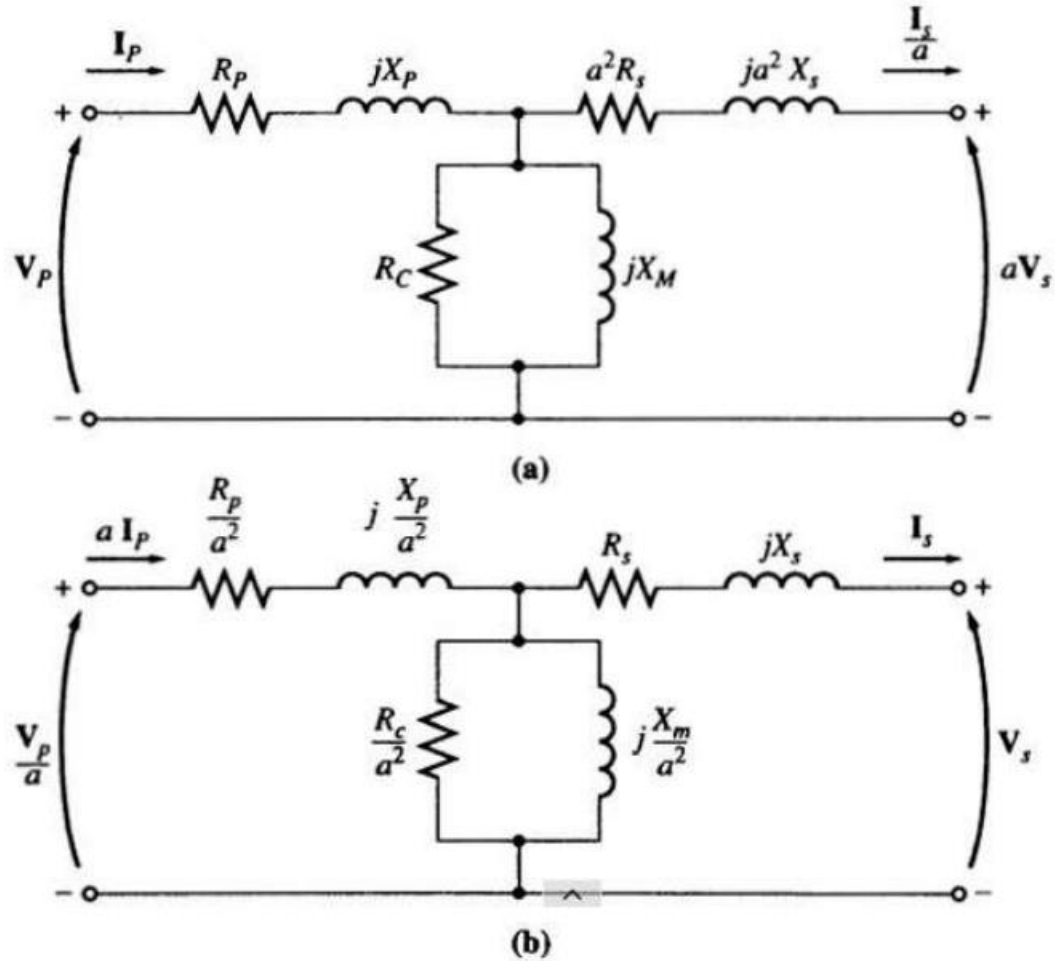
Modeling excitation current and copper loss

- The hysteresis and eddy currents is in-phase with input voltage (modeled as a shunt resistor R_c)
- The magnetization current is lagging input voltage by 90 degrees (modeled as a shunt inductor X_m)
- The copper loss can be modeled as the series resistors R_p and R_s

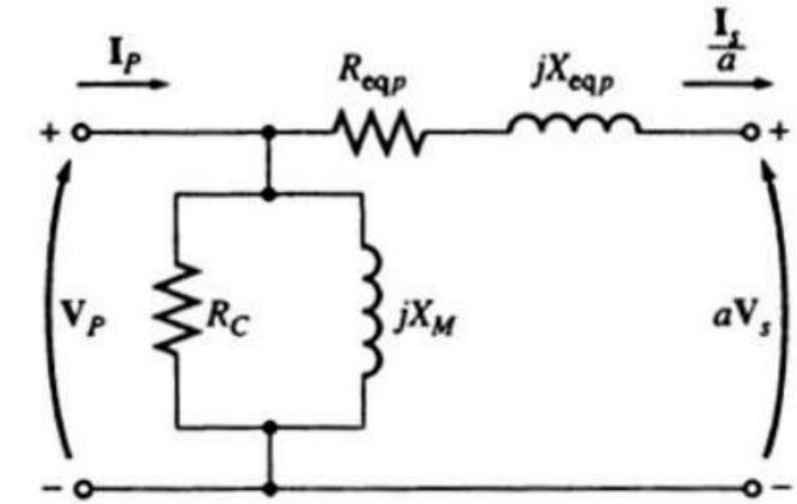
The resulting equivalent circuit



Impedance transform to primary or secondary side

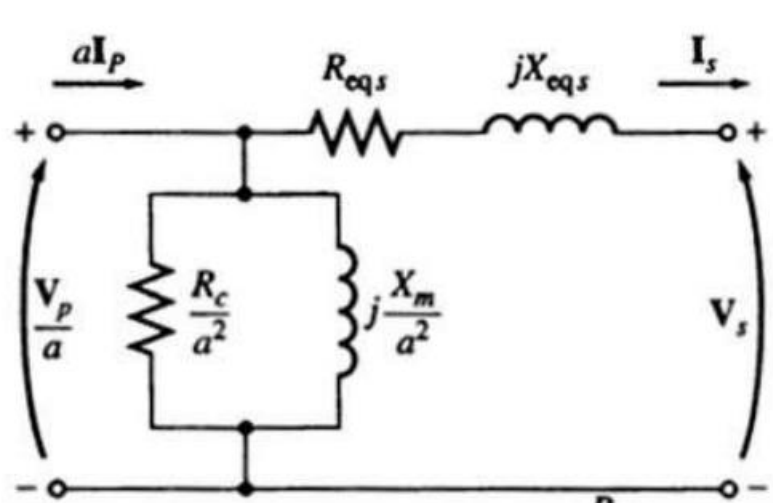


Approximate equivalent circuit



$$R_{eqp} = R_p + a^2 R_s$$

$$X_{eqp} = X_p + a^2 X_s$$



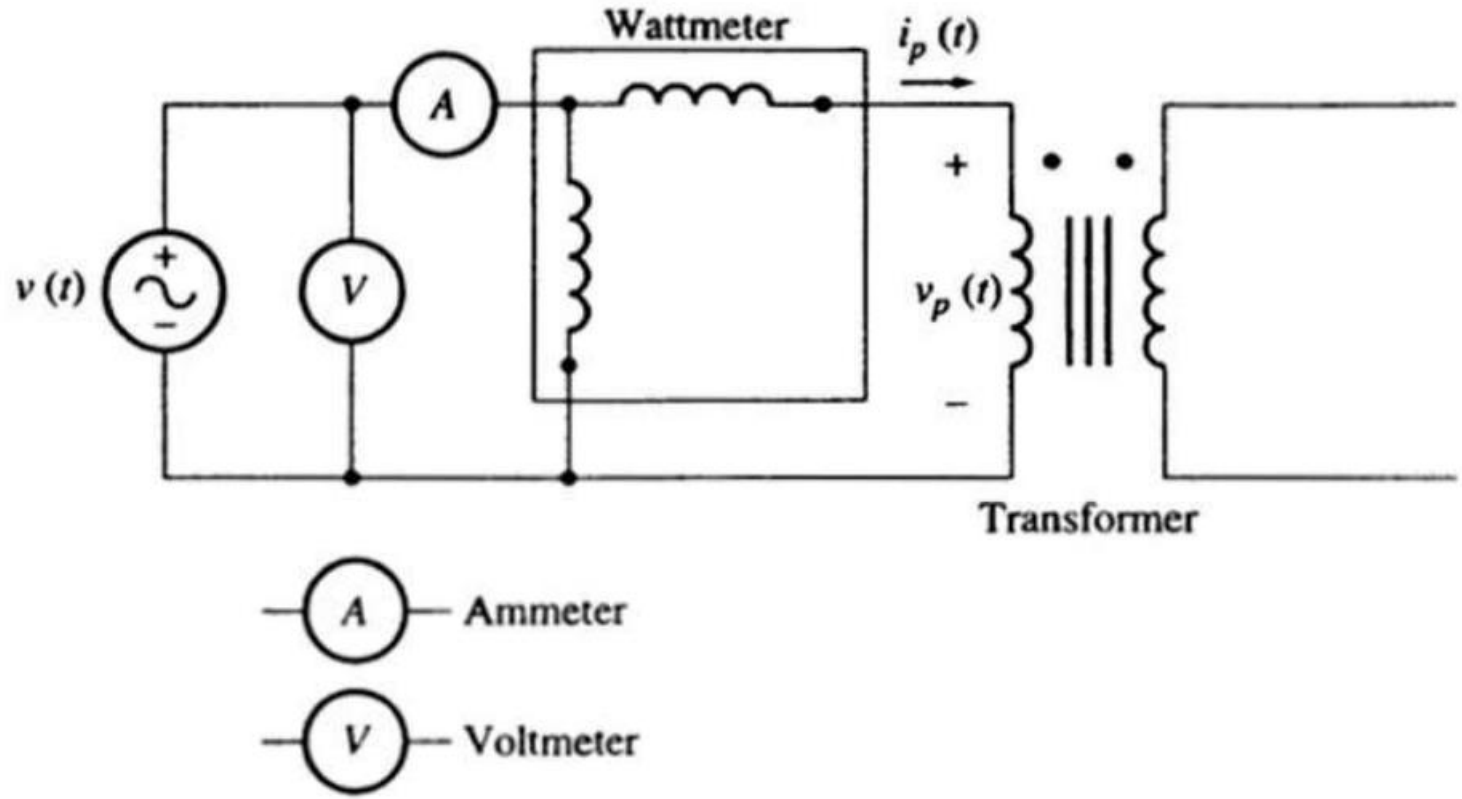
$$R_{eqs} = \frac{R_p}{a^2} + R_s$$

$$X_{eqs} = \frac{X_p}{a^2} + X_s$$

Measure the equivalent circuit parameters

- There are two types of measurements used for determination the equivalent circuit parameters
- ***Open circuit test*** – used to measure excitation branch
- ***Short circuit test*** – used to measure series branch

Open circuit test



Open circuit test

- Under the open circuit condition, all the input current flows through the excitation branch (Since the current is small, the test usually performs at high voltage side)

$$Y_E = G_C - jB_M$$
$$= \frac{1}{R_C} - j\frac{1}{X_M}$$

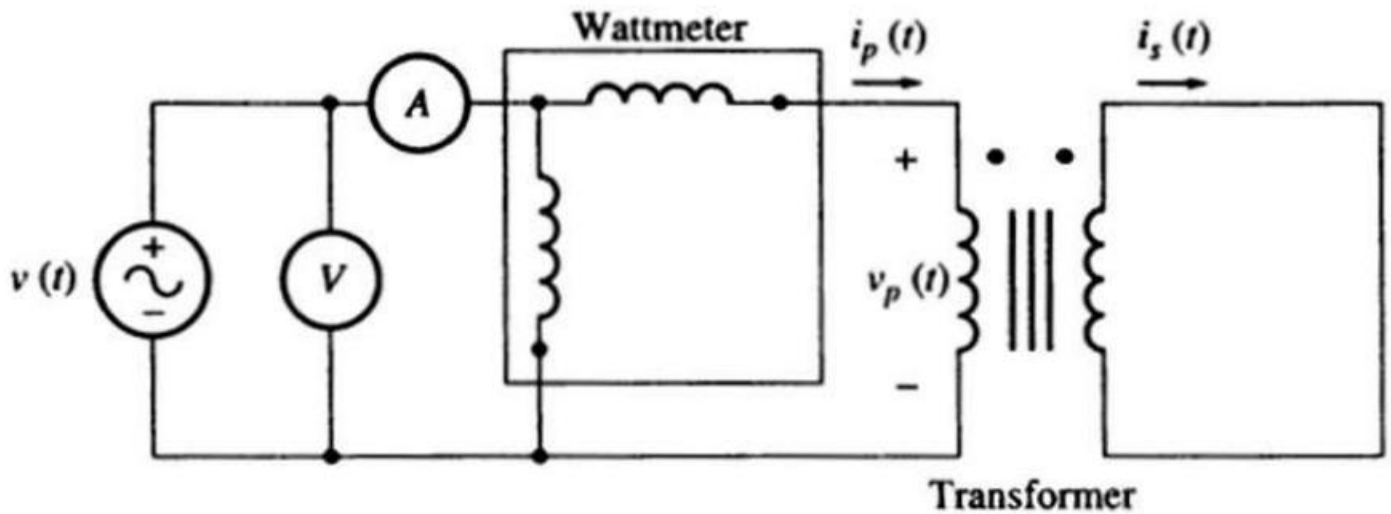
$$|Y_E| = \frac{I_{OC}}{V_{OC}}$$

$$\theta = \cos^{-1}\frac{P_{OC}}{V_{OC}I_{OC}}$$

$$\text{PF} = \cos\theta = \frac{P_{OC}}{V_{OC}I_{OC}}$$

$$Y_E = \frac{I_{OC}}{V_{OC}}\angle -\theta$$
$$= \frac{I_{OC}}{V_{OC}}\angle -\cos^{-1}\text{PF}$$

Short circuit test



Short circuit test

- At secondary side short circuit condition, the input voltage must be a very low value to prevent input large short circuit current (usually performs at low voltage side)

$$|Z_{SE}| = \frac{V_{SC}}{I_{SC}}$$

$$\text{PF} = \cos\theta = \frac{P_{SC}}{V_{SC} I_{SC}} \qquad \theta = \cos^{-1} \frac{P_{SC}}{V_{SC} I_{SC}}$$

$$Z_{SE} = \frac{V_{SC} \angle 0°}{I_{SC} \angle -\theta°} = \frac{V_{SC}}{I_{SC}} \angle \theta°$$

$$Z_{SE} = R_{eq} + jX_{eq}$$
$$= (R_P + a^2 R_S) + j(X_P + a^2 X_S)$$

Example 2–2.

The equivalent circuit impedances of a 20-kVA, 8000/240-V, 60-Hz transformer are to be determined.

The open-circuit test and the short-circuit test were performed on the primary side of the transformer,

and the following data were taken:

Open-circuit test (on primary)	Short-circuit test (on primary)
$V_{OC} = 8000$ V	$V_{SC} = 489$ V
$I_{OC} = 0.214$ A	$I_{SC} = 2.5$ A
$V_{OC} = 400$ W	$P_{SC} = 240$ W

Find the impedances of the approximate equivalent circuit referred to the primary side, and sketch that circuit.

Solution: $P = VI\cos\theta$

$$\text{PF} = \cos\theta = \frac{P_{OC}}{V_{OC} I_{OC}}$$

$$= \cos\theta = \frac{400\text{ W}}{(8000\text{ V})(0.214\text{ A})}$$

$$= 0.234 \text{ lagging}$$

R_C & X_M can be calculated from admittance $Y_E = \frac{1}{R_C} - j\frac{1}{X_M}$

$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -\theta$$

$$= \frac{I_{OC}}{V_{OC}} \angle -\cos^{-1}\text{PF}$$

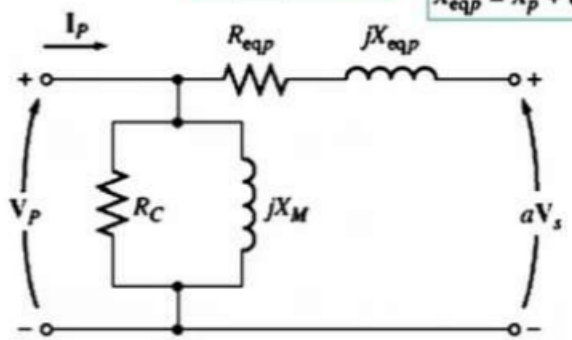
$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -\cos^{-1}\text{PF} = \frac{0.214\text{ A}}{8000\text{ V}} \angle -\cos^{-1} 0.234$$

$$= 0.0000268 \angle -76.5°\ \Omega$$

$$= 0.0000063 - j0.0000261 = \frac{1}{R_C} - j\frac{1}{X_M}$$

$R_{eqp} = R_p + a^2 R_s$

$X_{eqp} = X_p + a^2 X_s$



$$R_C = \frac{1}{0.0000063} = 159\text{ k}\Omega$$

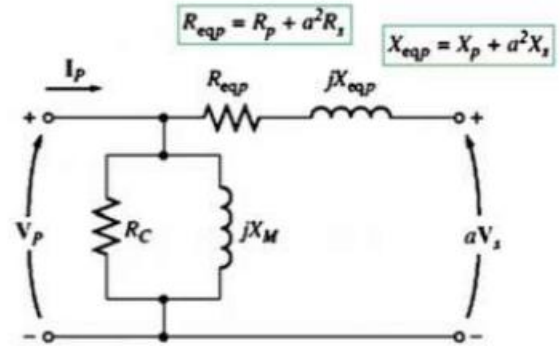
$$X_M = \frac{1}{0.0000261} = 38.4\text{ k}\Omega$$

R_{eq} & X_{eq} can be calculated from series impedance. $Z_{SE} = R_{eq} + jX_{eq}$

$$Z_{SE} = \frac{V_{SC}\angle 0^\circ}{I_{SC}\angle -\theta^\circ} = \frac{V_{SC}}{I_{SC}}\angle \theta^\circ$$

$$= \frac{V_{SC}}{I_{SC}}\angle \cos^{-1} \text{PF}$$

Open-circuit test (on primary)	Short-circuit test (on primary)
$V_{OC} = 8000$ V	$V_{SC} = 489$ V
$I_{OC} = 0.214$ A	$I_{SC} = 2.5$ A
$V_{OC} = 400$ W	$P_{SC} = 240$ W



The power factor during the *short-circuit* test is

$$\text{PF} = \cos\theta = \frac{P_{SC}}{V_{SC}I_{SC}}$$

$$= \cos\theta = \frac{240\text{ W}}{(489\text{ V})(2.5\text{ A})} = 0.196 \text{ lagging}$$

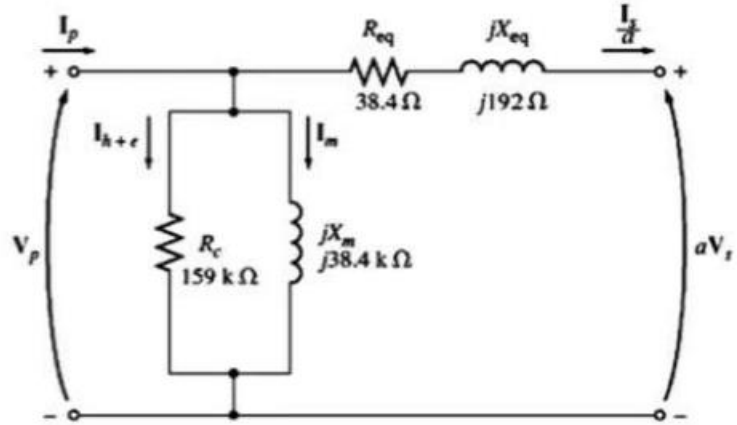
The series impedance is given by

$$Z_{SE} = \frac{V_{SC}}{I_{SC}}\angle \cos^{-1} \text{PF}$$

$$= \frac{489\text{ V}}{2.5\text{ A}}\angle 78.7^\circ$$

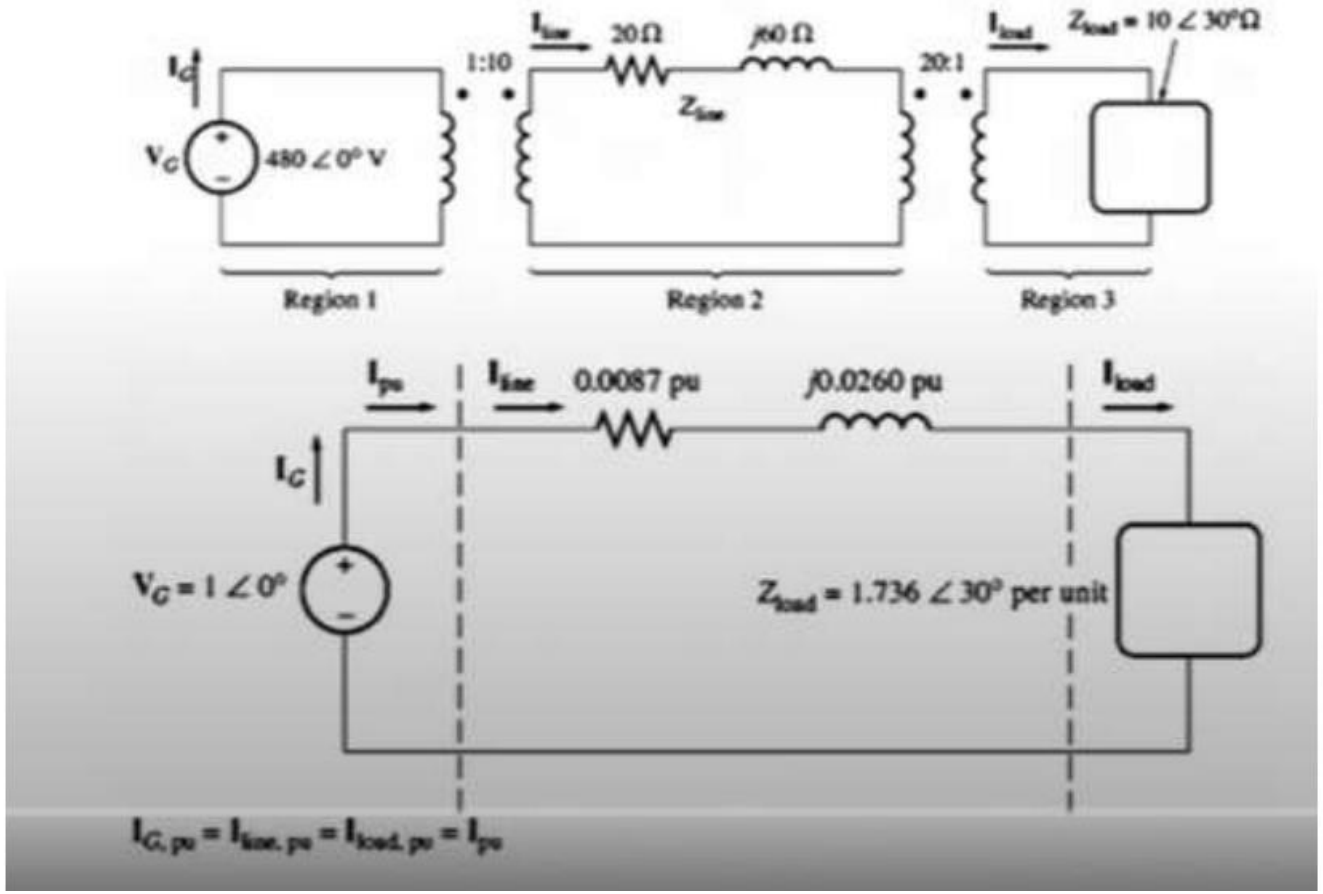
$$= 195.6\angle 78.7^\circ = 38.4 + j192\ \Omega$$

$$R_{eq} = 38.4\ \Omega \qquad X_{eq} = 192\ \Omega$$



Per-Unit

System of Measurement



Impedance transformation (from secondary to primary and vice versa) is tedious

Because such impedance transformations can be avoided, circuits containing many transformers can be solved easily with less chance of error. This method of calculation is known as the *per-unit* (*pu*) *system* of measurements.

In the per-unit system, the voltages, currents, powers, impedances, and other electrical quantities are not measured in their usual SI units (volts, amperes, watts, ohms, etc.).

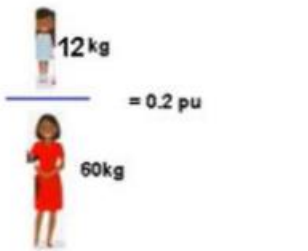
Instead, *each electrical quantity is measured as a decimal fraction* of some base level.

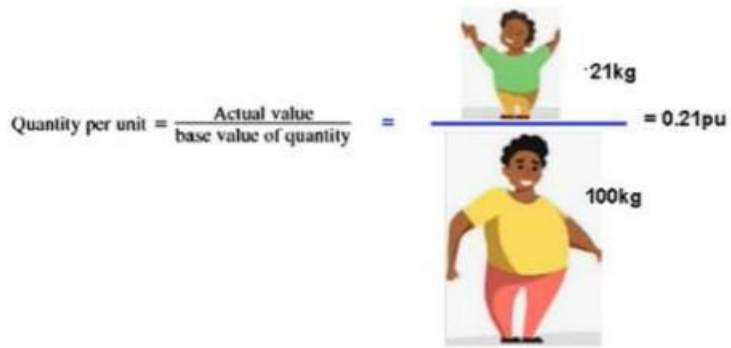
Any quantity can be expressed on a per-unit basis by the equation

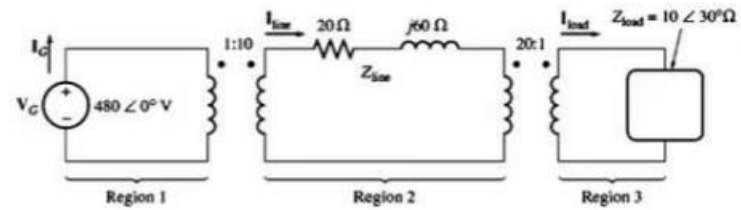
$$\text{Quantity per unit} = \frac{\text{Actual value}}{\text{base value of quantity}} \qquad (2\text{–}53)$$

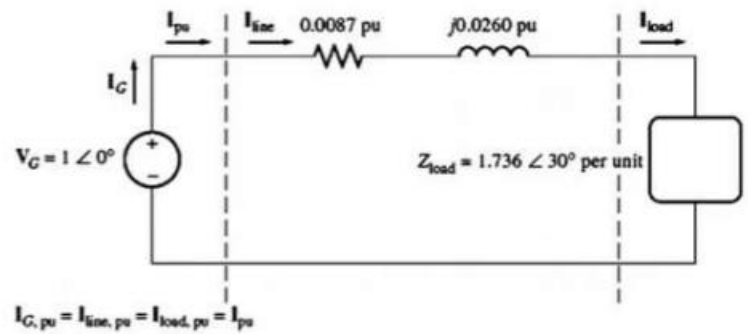
Any quantity can be expressed on a per-unit basis by the equation

$$\text{Quantity per unit} = \frac{\text{Actual value}}{\text{base value of quantity}} \qquad (2\text{–}53)$$









Example 2–3.

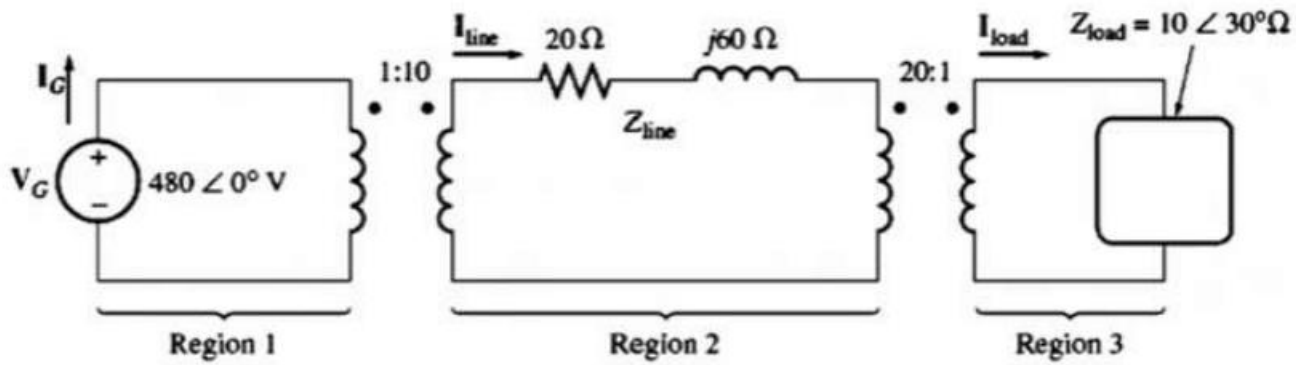
A simple power system is shown in Figure 2–22.

This system contains a 10 kVA , 480-V generator
connected to an ideal 1:10 step-up transformer,
a transmission line, an ideal 20:1 step-down transformer, and a load.

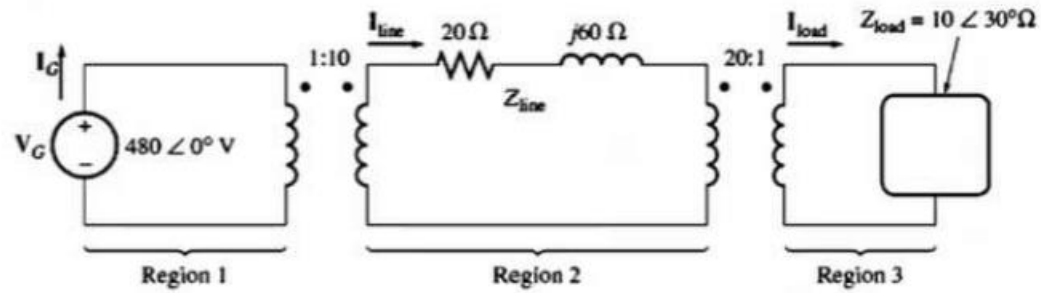
The impedance of the transmission line is $20 + j60\ \Omega$,
and the impedance of the load is $10\angle 30°\Omega$.

The base values for this system are chosen to be 480 V and 10 kVA at the generator.

(a) Find the base voltage, current, impedance, and apparent power at every point in the power system.
(b) Convert this system to its per-unit equivalent circuit.
(c) Find the power supplied to the load in this system.
(d) Find the power lost in the transmission line.



(a) Find the base voltage, current, impedance, and apparent power at every point in the power system.



Solution:

(a)

In the generator region, $V_{base} = 480$ V and $S_{base} = 10$ kVA,

$$\text{so} \qquad I_{base\,1} = \frac{S_{base}}{V_{base\,1}} = \frac{10{,}000 \text{ VA}}{480 \text{ V}} = 20.83 \text{ A}$$

$$Z_{base\,1} = \frac{V_{base\,1}}{I_{base\,1}} = \frac{480 \text{ V}}{20.83 \text{ A}} = 23.04\ \Omega$$

The turns ratio of transformer T_1 is $a = 1/10 = 0.1$.

in the transmission line region

the base voltage

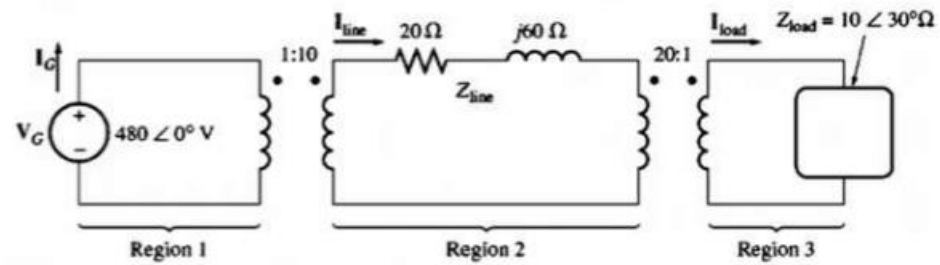
$$V_{base\,2} = \frac{V_{base\,1}}{a} = \frac{480 \text{ V}}{0.1} = 4800 \text{ V}$$

The other base quantities are

$S_{base\,2} = 10$ kVA (remains constant through out)

$$I_{base\,2} = \frac{10{,}000 \text{ VA}}{4800 \text{ V}} = 2.083 \text{ A}$$

$$Z_{base\,2} = \frac{4800 \text{ V}}{2.083 \text{ A}} = 2304\ \Omega$$



The turns ratio of transformer T_2 is $a = 20/1 = 20$,

in the load region is

the base voltage $V_{base\,3} = \frac{V_{base\,2}}{a} = \frac{4800\ \text{V}}{20} = 240\ \text{V}$

The other base quantities are

$S_{base\,3} = 10\ \text{kVA}$ (remains constant through out)

$I_{base\,3} = \frac{10{,}000\ \text{VA}}{240\ \text{V}} = 41.67\ \text{A}$

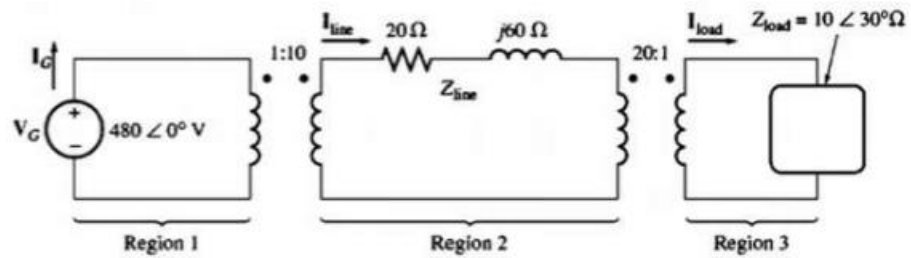
$Z_{base\,3} = \frac{240\ \text{V}}{41.67\ \text{A}} = 5.76\ \Omega$

(b) Convert this system to its per-unit equivalent circuit.

generator region, $V_{base} = 480\ \text{V}$

transmission line region $Z_{base\,2} = 2304\ \Omega$

load region $Z_{base\,3} = 5.76\ \Omega$



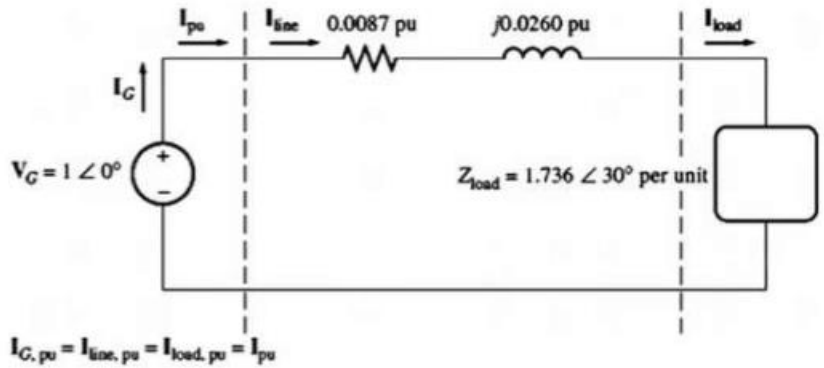
(b) To convert a power system to a per-unit system, each component must be divided by its base value in its region of the system.

The *generator's* per-unit volt- age is

$$V_{G,pu} = \frac{\text{actual value}}{\text{base value}} = \frac{480 \angle 0°\ \text{V}}{480\ \text{V}} = 1.0 \angle 0°\ \text{pu}$$

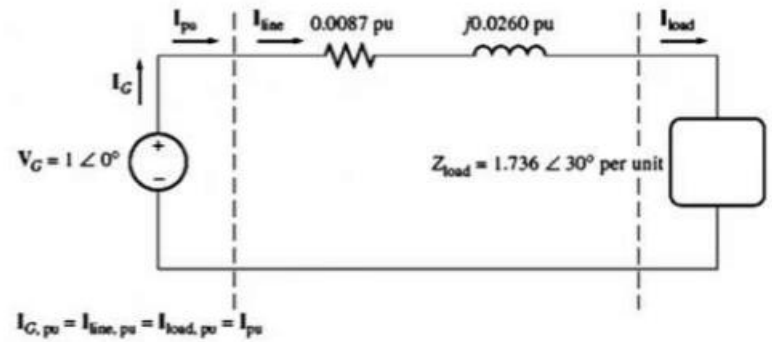
$$Z_{line,} = \frac{\text{actual value}}{\text{base value}} = \frac{20 + j60\ \Omega}{2304\ \Omega} = 0.0087 + j0.0260\ \text{pu}$$

$$Z_{load,pu} = \frac{\text{actual value}}{\text{base value}} = \frac{10 \angle 30°\ \Omega}{5.76\ \Omega} = 1.736 \angle 30°\ \text{pu}$$



(c) Find the power supplied to the load in this system.

The per-unit power of the load is
$P_{load,pu} = I_{pu}^2 R_{pu}$



The current flowing in this per-unit power system is

$$\mathbf{I}_{pu} = \frac{\mathbf{V}_{pu}}{Z_{tot,pu}}$$

$$= \frac{1 \angle 0°}{(0.0087 + j0.0260) + (1.736 \angle 30°)}$$

$$= 0.569 \angle -30.6° \text{ pu}$$

Therefore, the per-unit power of the load is

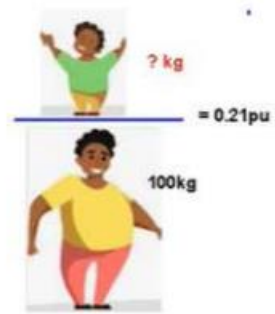
$$P_{load,pu} = I_{pu}^2 R_{pu} = (0.569)^2(1.503) = 0.487$$

$Z_{load,pu} = 1.736 \angle 30°$ pu $= 1.5 + j0.868$

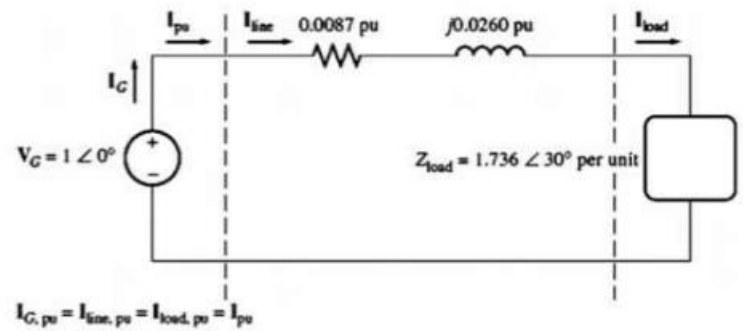
Therefore, the actual power supplied to the load is

$$P_{load} = P_{load,pu} S_{base} = (0.487)(10{,}000 \text{ VA})$$

$$= 4870 \text{ W}$$



(d) Find the power lost in the transmission line.



The per-unit power lost in the transmission line is

$$P_{line,pu} = I_{pu}^2 R_{line,pu} = (0.569)^2(0.0087) = 0.00282$$

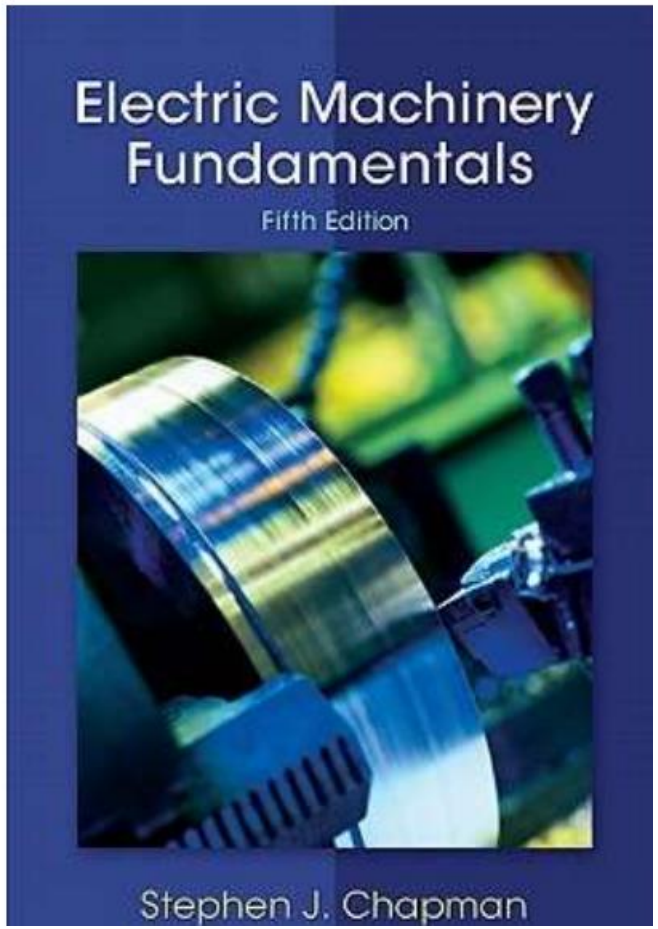
the actual power lost in the transmission line is

$$P_{line} = P_{line,pu} S_{base} = (0.00282)(10{,}000 \text{ VA})$$
$$= 28.2 \text{ W}$$

When only one device (transformer or motor) is being analyzed,
its own ratings are usually used as the base for the per-unit system.

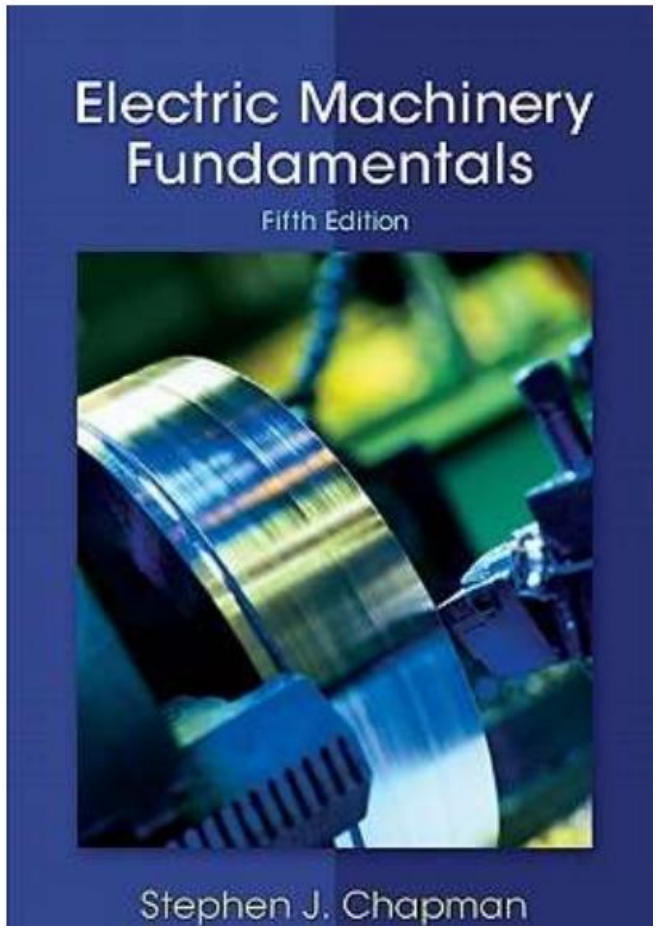
Text book and supplementary materials of this course

- Stephen J. Chapman



Text book and supplementary materials of this course

- Stephen J. Chapman



Transformer
Voltage Regulation

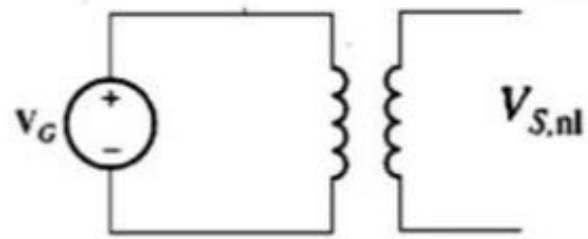
&

Phasor Diagram

2.7 TRANSFORMER VOLTAGE REGULATION AND EFFICIENCY

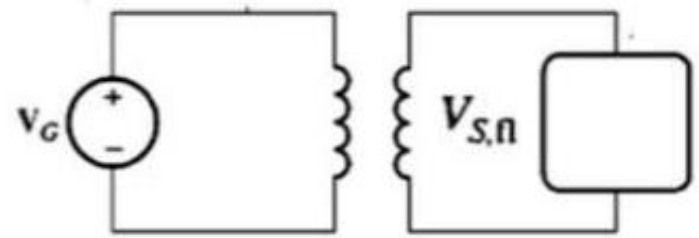
Voltage regulation (VR) is a quantity that compares the output voltage of the transformer at no load with the output voltage at full load.

$$\text{VR} = \frac{V_{S,\text{nl}} - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\%$$



Since at no load, $V_S = V_P/a$,
the voltage regulation can also be expressed as

$$\boxed{\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\%}$$



Usually it is a good practice to have as small a voltage regulation as possible.

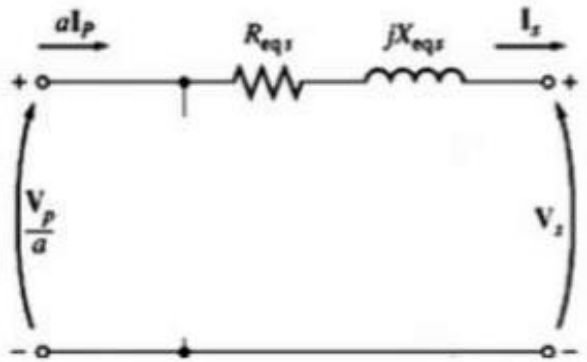
How can the voltage regulation of a transformer be determined? (Next Slide)

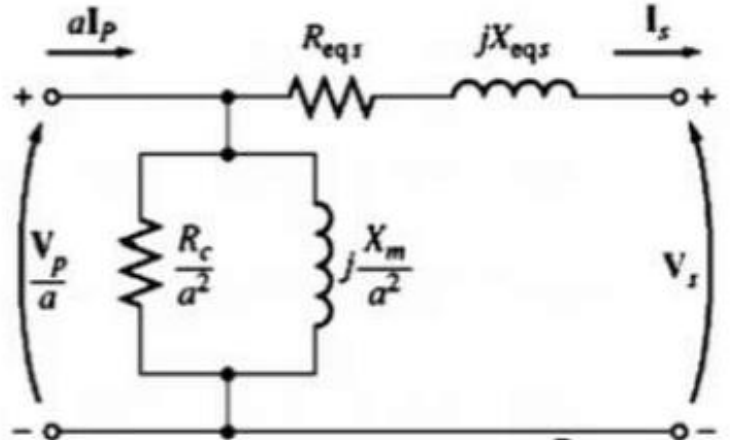
How can the voltage regulation of a transformer be determined?

To determine the voltage regulation of a transformer, it is necessary to understand the voltage drops within it.

The effects of the excitation branch on transformer voltage regulation can be ignored.

so only the series impedances need be considered.





(b) Approximate transformer models. referred to the secondary side

The easiest way to determine the effect of the impedances and the current phase angles on the transformer voltage regulation is to examine a *phasor diagram*.

In all the following phasor diagrams, the phasor voltage $\mathbf{V}_S$ is assumed to be at an angle of 0°, and all other voltages and currents are compared to that reference.

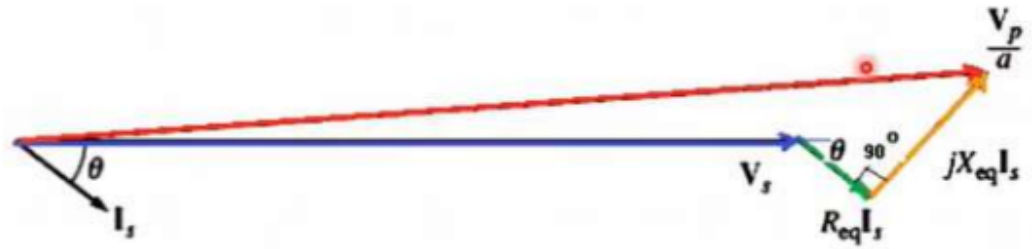


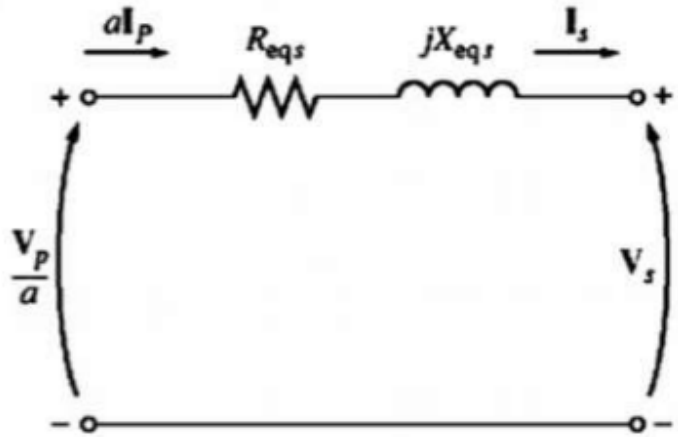
FIG 2–26 Phasor diagram of a transformer operating at a lagging power factor.

By applying Kirchhoff's voltage law to the equivalent circuit in Figure 2–18(d) the primary voltage can be found as

$$-\frac{\mathbf{V}_P}{a} + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S + \mathbf{V}_S = 0$$

$$\frac{\mathbf{V}_P}{a} = \mathbf{V}_S + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S$$

A transformer phasor diagram is just a visual representation of this equation.



2–18(d)

$$\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\%$$

It is easy to see that $V_P/a > V_S$ for lagging loads,

So, VR > 0.

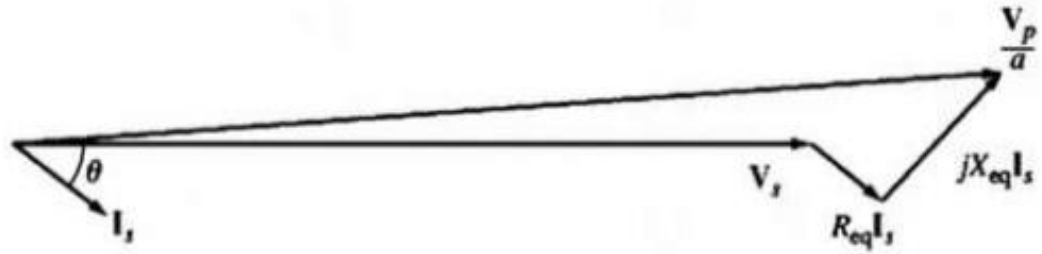
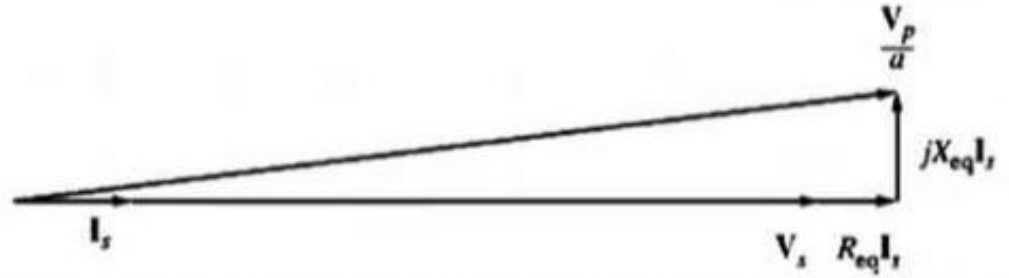


FIGURE 2–26 Phasor diagram of a transformer operating at a lagging power factor.

At unity power factor

$$V_P/a > V_S$$

so VR > 0.



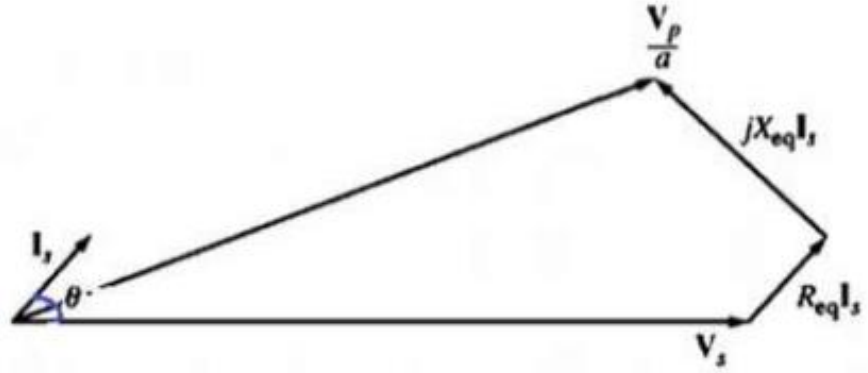
Phasor diagram of a transformer operating at unity power factor. (means $\theta = 0$)

However, this time the voltage regulation is a smaller number than it was with a lagging current.

If the secondary current is leading, the secondary voltage can actually be *higher* than the referred primary voltage.

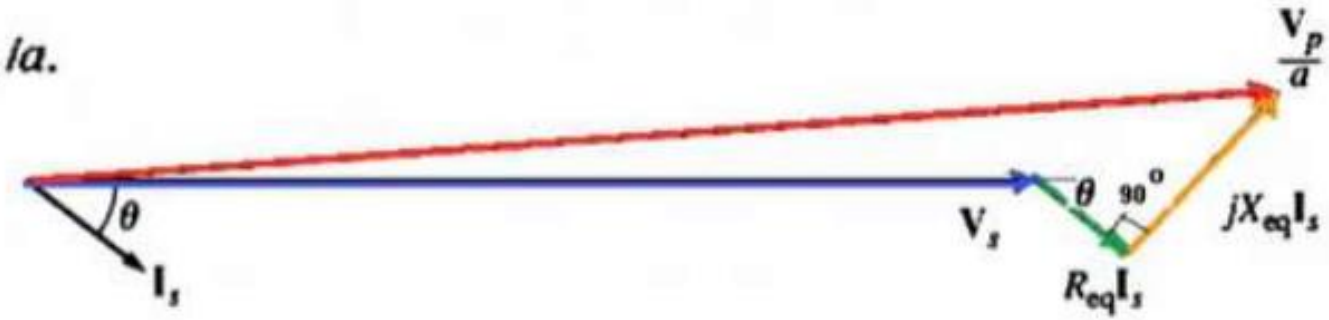
If this happens, the transformer actually has a *negative* voltage regulation.

Negative voltage regulation means the voltage increases with the load. undesirable condition

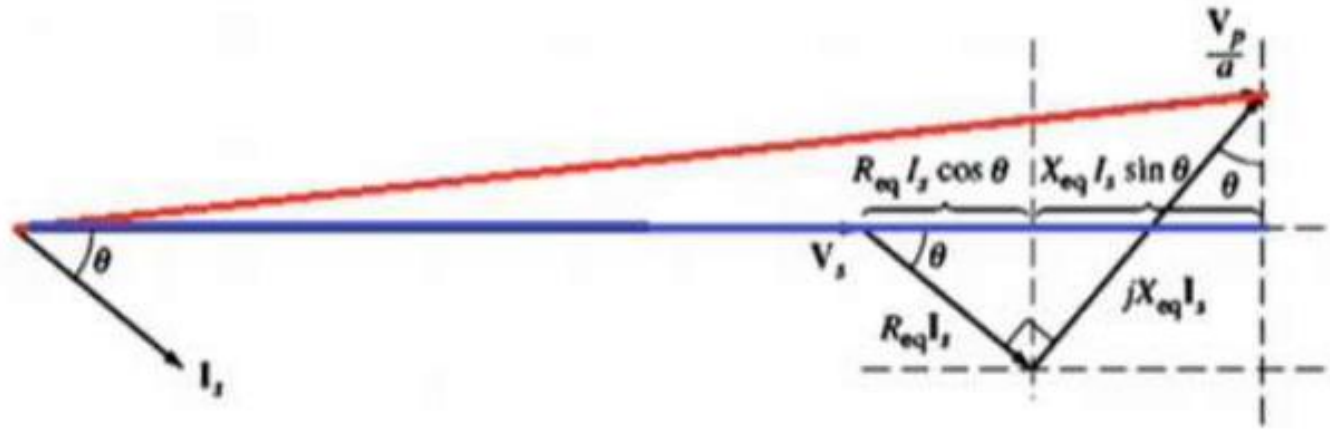


Phasor diagram of a transformer operating at leading power factor.

Derivation of the approximate equation for V_P/a.



$$\frac{\mathbf{V}_p}{a} = \mathbf{V}_S + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S$$



$$\frac{V_p}{a} \approx \mathbf{V}_s + R_{eq}\,I_s\cos\theta + X_{eq}\,I_s\sin\theta$$

Transformer Efficiency

Transformers are also compared and judged on their efficiencies.

The efficiency of a device is defined by

$$\underset{\text{(eta)}}{\eta} = \frac{P_{out}}{P_{in}} \times 100\%$$

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100\% \qquad \boxed{P_{in} = P_{out} + P_{loss}}$$

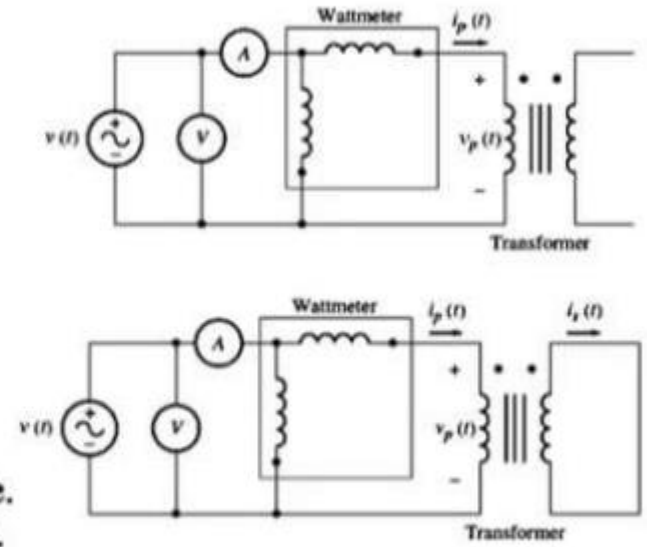
Example 2–5.

A 15-kVA, 2300/230-V transformer is to be tested to determine

(a) its excitation branch components,
(b) its series impedances,
(c) its voltage regulation.

The following test data have been taken from the primary side of the transformer:

Open-circuit test	Short-circuit test
$V_{OC} = 2300$ V	$V_{SC} = 47$ V
$I_{OC} = 0.21$ A	$I_{SC} = 6.0$ A
$P_{OC} = 50$ W	$P_{SC} = 160$ W



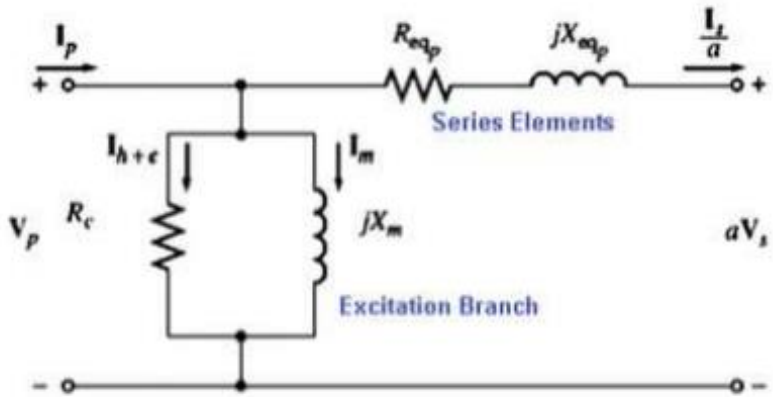
(a) Find the equivalent circuit of this transformer referred to the high-voltage side.
(b) Find the equivalent circuit of this transformer referred to the low-voltage side.
(c) Calculate the full-load voltage regulation at 0.8 lagging power factor, 1.0 power factor, and at 0.8 leading power factor.
(d) Plot the voltage regulation as load is increased from no load to full load at power factors of 0.8 lagging, 1.0, and 0.8 leading.
(e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

Solution:

(a) Find the equivalent circuit of this transformer referred to the high-voltage side.

2300/230-V transformer

(Primary side)



[this part is exactly same as example 2.2]

(a) The excitation branch values of the transformer equivalent circuit can be calculated from the *open-circuit test* data, and the series elements can be calculated from the *short-circuit test* data.

From the open-circuit test data, the open-circuit impedance angle is

$$\theta_{OC} = \cos^{-1}\frac{P_{OC}}{V_{OC}I_{OC}} \qquad [P = VI\cos\theta]$$

$$= \cos^{-1}\frac{50\text{W}}{(2300\text{ V})(0.21\text{ A})} = 84°$$

Open-circuit test
$V_{OC} = 2300$ V
$I_{OC} = 0.21$ A
$P_{OC} = 50$ W

The excitation admittance is thus

$$Y_E = \frac{I_{OC}}{V_{OC}}\angle -84° = \frac{0.21\text{ A}}{2300\text{ V}}\angle -84°$$

$$= 9.13 \times 10^{-5}\angle -84°\,\Omega = 0.0000095 - j0.0000908\,\Omega = \frac{1}{R_C} - j\frac{1}{X_M}$$

The elements of the excitation branch referred to the primary are

$$R_C = \frac{1}{0.0000095} = 105\text{ k}\Omega$$

$$X_M = \frac{1}{0.0000908} = 11\text{ k}\Omega$$

From the short-circuit test data, the short-circuit impedance angle is

Short-circuit test
$V_{SC} = 47$ V
$I_{SC} = 6.0$ A
$P_{SC} = 160$ W

$$\theta_{SC} = \cos^{-1}\frac{P_{SC}}{V_{SC}I_{SC}}$$

$$= \cos^{-1}\frac{160\text{ W}}{(47\text{ V})(6\text{ A})} = 55.4°$$

The equivalent series impedance is thus

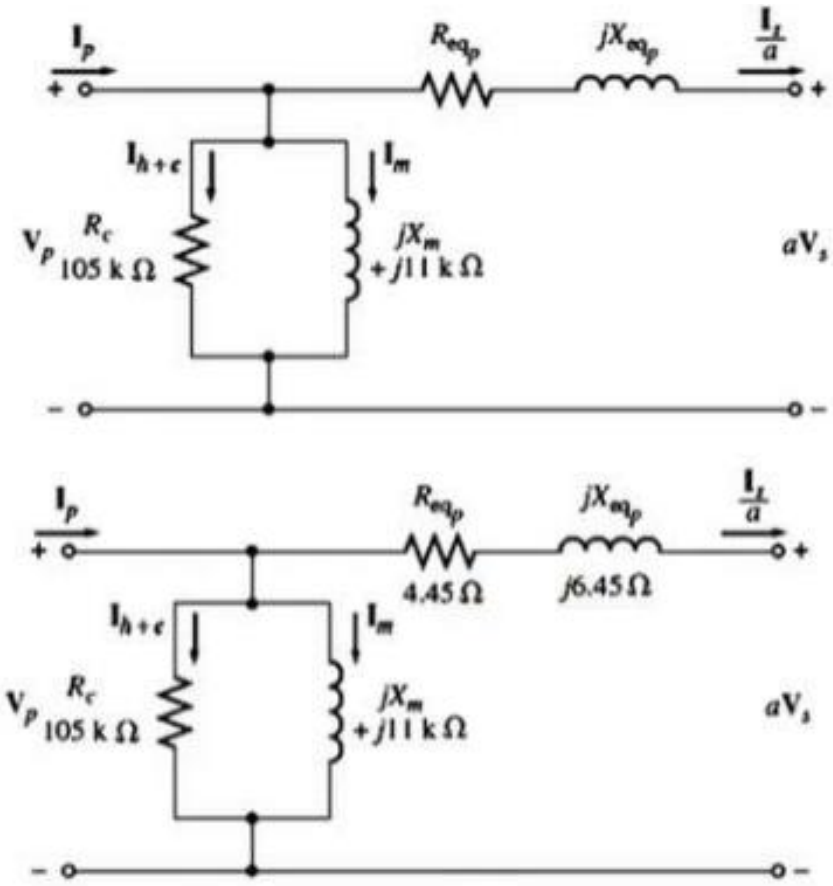
$$Z_{SE} = \frac{V_{SC}}{I_{SC}}\angle\theta_{SC}$$

$$= \frac{47\text{ V}}{6\text{ A}}\angle 55.4°\ \Omega$$

$$= 7.833\angle 55.4° = 4.45 + j6.45$$

The series elements referred to the primary are

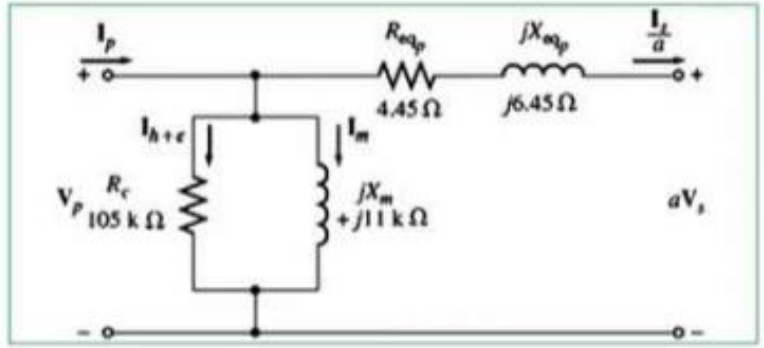
$$R_{eq} = 4.45\ \Omega \qquad X_{eq} = 6.45\ \Omega$$

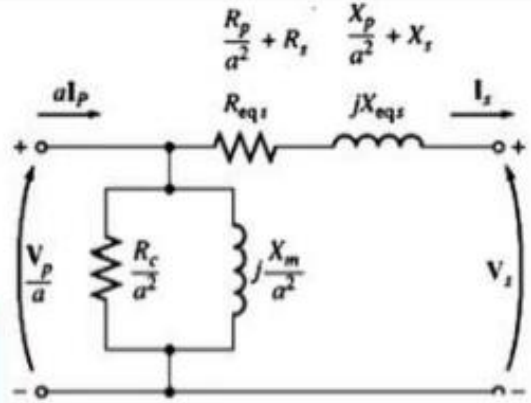


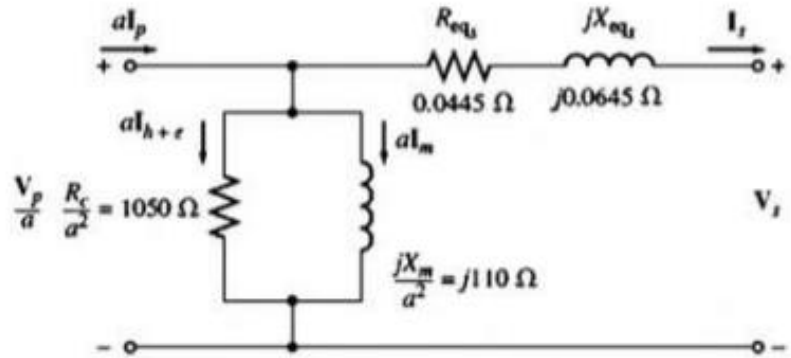
(b) Find the equivalent circuit of this transformer referred to the low-voltage side.

(b) To find the equivalent circuit referred to the low-voltage side, it is simply necessary to divide the impedance by a^2.

Since $a = N_P/N_S = 10$, the resulting values are





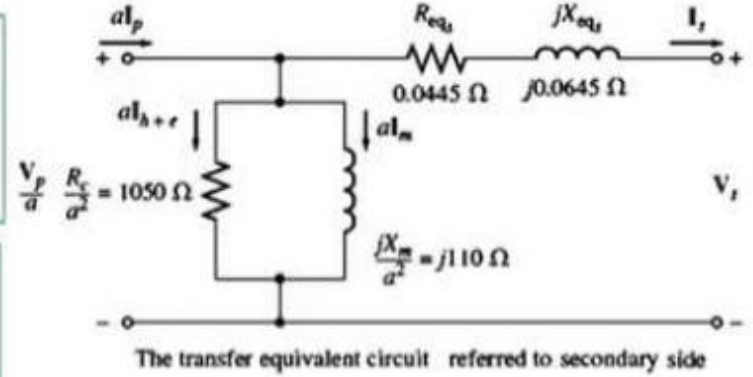


The transfer equivalent circuit referred to secondary side

(c) Calculate the full-load voltage regulation at
0.8 lagging power factor,
1.0 power factor, and at
0.8 leading power factor.

Since at no load, $V_S = V_P/a$,
the voltage regulation can also be expressed as

$$\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\%$$



The transfer equivalent circuit referred to secondary side

A 15-kVA, 2300/230-V transformer

To calculate V_P/a, use Equation (2–64):

$$\frac{\mathbf{V}_P}{a} = \mathbf{V}_S + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S$$

current $\mathbf{I}_S$ = ?

The full-load current on the secondary side of this transformer is

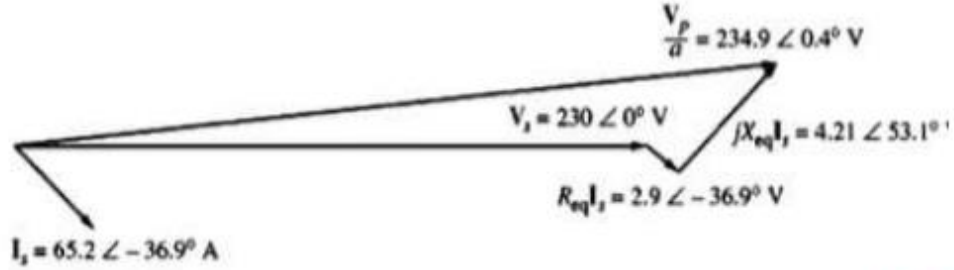
$$I_{S,\text{rated}} = \frac{S_{\text{rated}}}{V_{S,\text{rated}}} = \frac{15{,}000\ \text{VA}}{230\ \text{V}} = 65.2\ \text{A}$$

At PF = 0.8 lagging,

current $\mathbf{I}_S = 65.2 \angle -36.9°$ A. $\angle -\cos^{-1}$ PF

Therefore,

$$\frac{\mathbf{V}_P}{a} = 230\angle 0°\ \text{V} + (0.0445\ \Omega)(65.2\angle -36.9°\ \text{A}) + j(0.0645\ \Omega)(65.2\angle -36.9°\ \text{A})$$
$$= 230\angle 0°\ \text{V} + 2.90\angle -36.9°\ \text{V} + 4.21\angle 53.1°\ \text{V}$$
$$= 234.85\angle 0.40°\ \text{V}$$



The resulting voltage regulation is

$$\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\% = \frac{234.85\ \text{V} - 230\ \text{V}}{230\ \text{V}} \times 100\% = 2.1\%$$

At PF = 1.0, current $\mathbf{I}_S = 65.2 \angle 0° \text{ A}$. Therefore,

$$\frac{\mathbf{V}_P}{a} = 230 \angle 0° \text{ V} + (0.0445\ \Omega)(65.2 \angle 0° \text{ A}) + j(0.0645\ \Omega)(65.2 \angle 0° \text{ A})$$

$$= 230 \angle 0° \text{ V} + 2.90 \angle 0° \text{ V} + 4.21 \angle 90° \text{ V}$$

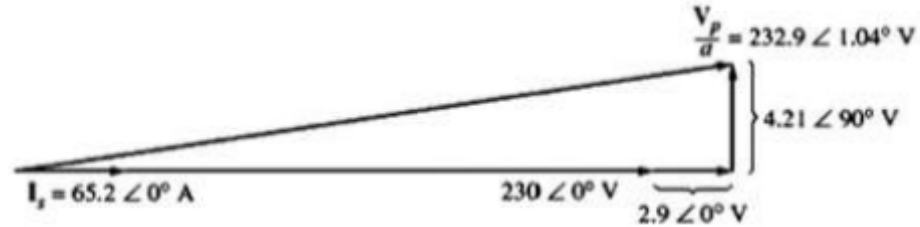
$$= 232.94 \angle 1.04° \text{ V}$$

The resulting voltage regulation is

$$\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\% = \frac{232.94 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = 1.28\%$$

To calculate V_P/a, use Equation (2–64):

$$\frac{\mathbf{V}_P}{a} = \mathbf{V}_S + R_{\text{eq}}\mathbf{I}_S + jX_{\text{eq}}\mathbf{I}_S$$



At PF = 0.8 leading, current $\mathbf{I}_S = 65.2 \angle 36.9° \text{ A}$. Therefore,

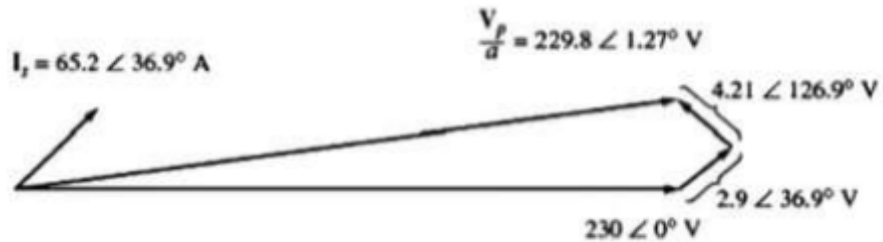
$$\frac{\mathbf{V}_P}{a} = 230 \angle 0° \text{ V} + (0.0445\ \Omega)(65.2 \angle 36.9° \text{ A}) + j(0.0645\ \Omega)(65.2 \angle 36.9° \text{ A})$$

$$= 230 \angle 0° \text{ V} + 2.90 \angle 36.9° \text{ V} + 4.21 \angle 126.9° \text{ V}$$

$$= 229.85 \angle 1.27° \text{ V}$$

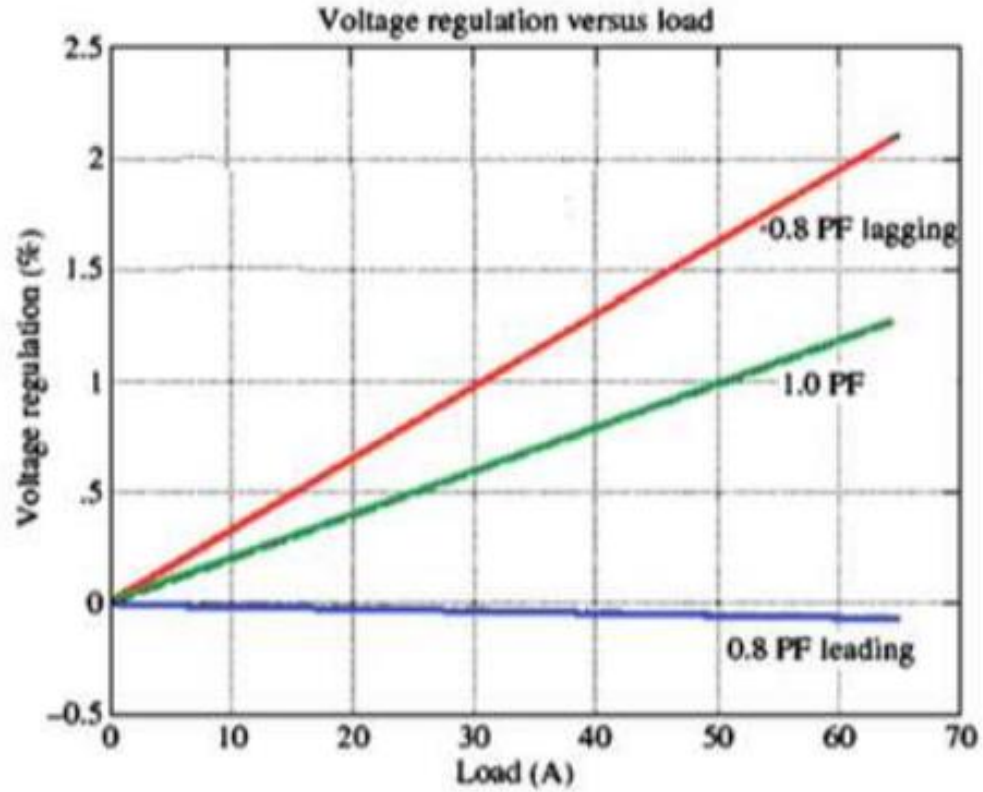
The resulting voltage regulation is

$$\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\% = \frac{229.85 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = -0.062\%$$



(d) Plot the voltage regulation as load is increased from no load to full load at power factors of 0.8 lagging, 1.0, and 0.8 leading.

'd) The best way to plot the voltage regulation as a function of load is to repeat the calculations in part *c* for many different loads using MATLAB.

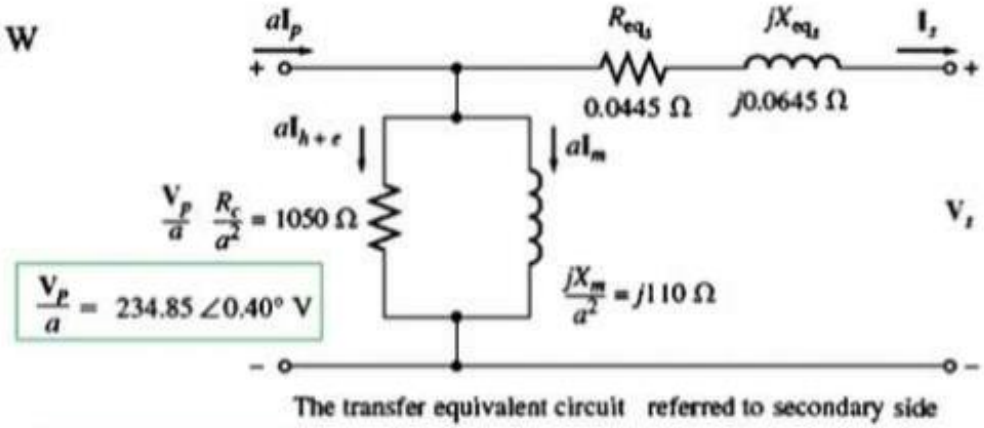


(e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

$$\eta = \frac{V_S I_S \cos\theta}{P_{Cu} + P_{core} + V_S I_S \cos\theta} \times 100\%$$

The copper losses are

$$P_{Cu} = (I_S)^2 R_{eq} = (65.2\ \text{A})^2(0.0445\ \Omega) = 189\ \text{W}$$



$\frac{V_p}{a} = 234.85 \angle 0.40°$ V

The transfer equivalent circuit referred to secondary side

The core losses are given by

$$P_{core} = \frac{(V_p/a)^2}{R_C} = \frac{(234.85\ \text{V})^2}{1050\ \Omega} = 52.5\ \text{W}$$

The output power of the transformer at this power factor is

$$P_{out} = V_S I_S \cos\theta$$
$$= (230\ \text{V})(65.2\ \text{A})\cos 36.9° = 12{,}000\ \text{W}$$

$V_s = 230 \angle 0°$ V
$I_s = 65.2 \angle -36.9°$ A.

Therefore, the efficiency of the transformer at this condition is

$$\eta = \frac{V_S I_S \cos\theta}{P_{Cu} + P_{core} + V_S I_S \cos\theta} \times 100\% = \frac{12{,}000\ \text{W}}{189\ \text{W} + 52.5\ \text{W} + 12{,}000\ \text{W}} \times 100\%$$
$$= 98.03\%$$

Transformer taps and voltage regulation

- The taps of transformer is used to change the effective turns ratio of transformer

Example 2–6. A 500-kVA, 13,200/480-V distribution transformer has four 2.5 percent taps on its primary winding. What are the voltage ratios of this transformer at each tap setting?

Solution

The five possible voltage ratings of this transformer are

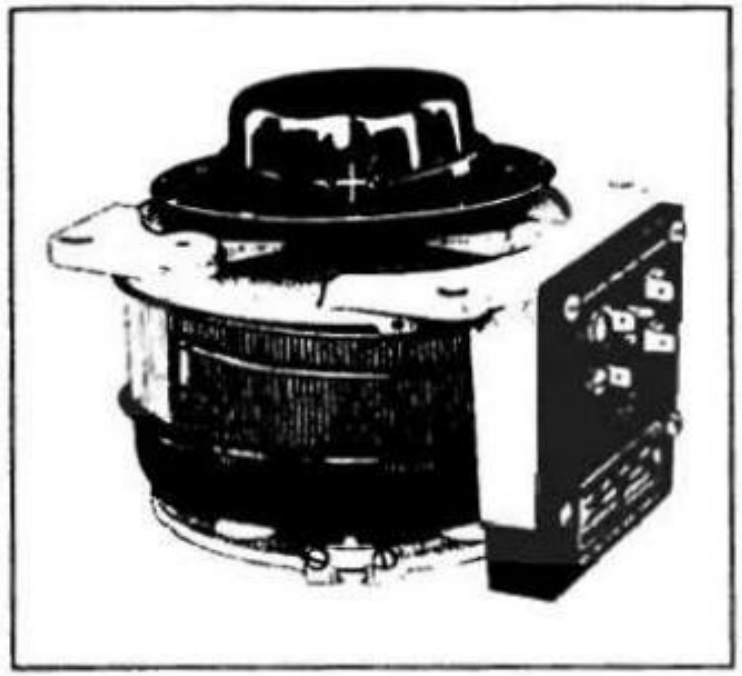
+5.0% tap	13,860/480 V
+2.5% tap	13,530/480 V
Nominal rating	13,200/480 V
−2.5% tap	12,870/480 V
−5.0% tap	12,540/480 V

TCUL and voltage regulator

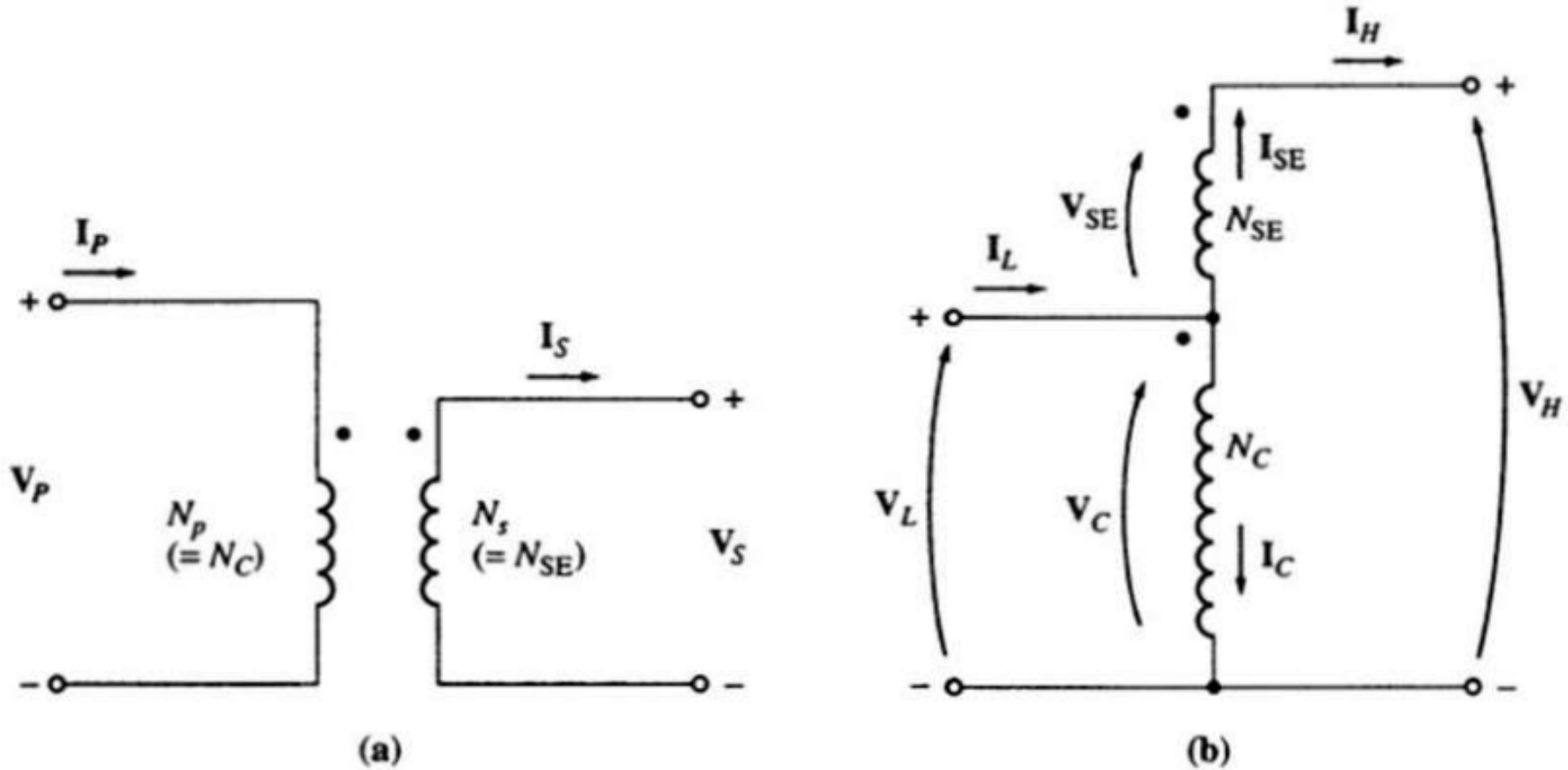
1. The ***tape changing under load*** (TCUL) is a transformer with the ability to change taps while power is connected to it
2. The ***voltage regulator*** is the TCUL with voltage sensing circuitry that automatically change taps to maintain the output voltage level

The autotransformer

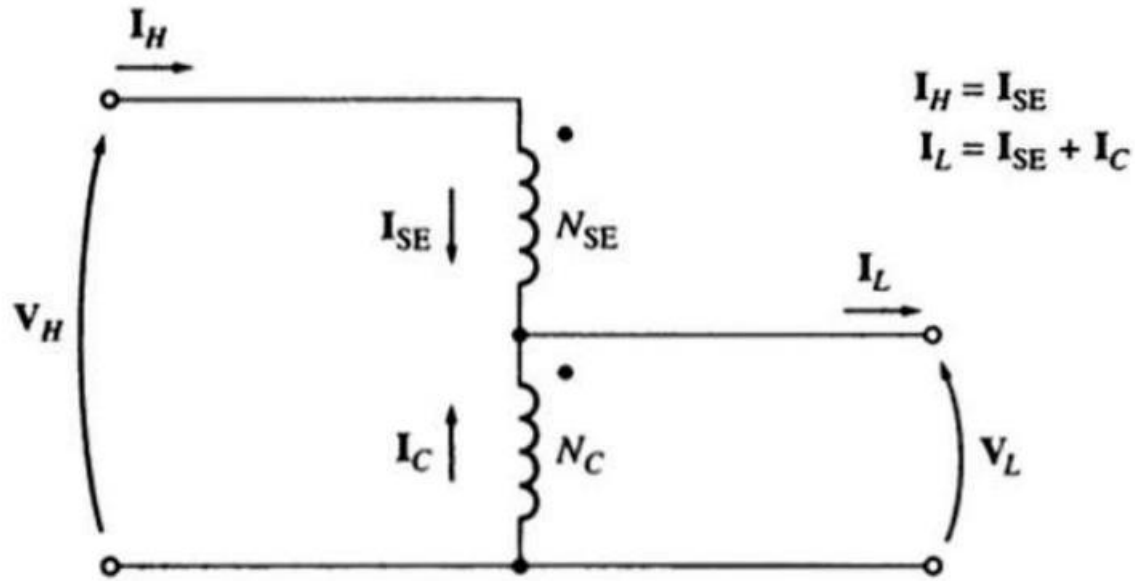
- Continuously tune the output voltage magnitude
- The size of auto transformer is smaller than the size of conventional transformer
- Output terminal is not electrical isolation



The step-up connection of autotransformer from conventional transformer



The step-down connection of autotransformer



Voltage and current in both coils

- ***Coil*** voltage and current in Nc and Nse still follow the voltage and current relation

$$\frac{\mathbf{V}_C}{\mathbf{V}_{SE}} = \frac{N_C}{N_{SE}}$$

$$N_C \mathbf{I}_C = N_{SE} \mathbf{I}_{SE}$$

- The autotransformer ***terminal*** voltage and current

$$\mathbf{V}_L = \mathbf{V}_C$$

$$\mathbf{V}_H = \mathbf{V}_C + \mathbf{V}_{SE}$$

$$\mathbf{I}_L = \mathbf{I}_C + \mathbf{I}_{SE}$$

$$\mathbf{I}_H = \mathbf{I}_{SE}$$

Terminal voltage and current relation of autotransformer

$$\mathbf{V}_H = \mathbf{V}_C + \mathbf{V}_{SE} \tag{2–71}$$

But $\mathbf{V}_C/\mathbf{V}_{SE} = N_C/N_{SE}$, so

$$\mathbf{V}_H = \mathbf{V}_C + \frac{N_{SE}}{N_C}\mathbf{V}_C \tag{2–74}$$

Finally, noting that $\mathbf{V}_L = V_C$, we get

$$\mathbf{V}_H = \mathbf{V}_L + \frac{N_{SE}}{N_C}\mathbf{V}_L$$

$$= \frac{N_{SE} + N_C}{N_C}\mathbf{V}_L \tag{2–75}$$

or

$$\boxed{\frac{\mathbf{V}_L}{\mathbf{V}_H} = \frac{N_C}{N_{SE} + N_C}} \tag{2–76}$$

Terminal voltage and current relation of autotransformer

$$\mathbf{I}_L = \mathbf{I}_C + \mathbf{I}_{SE} \tag{2–72}$$

From Equation (2–69), $\mathbf{I}_C = (N_{SE}/N_C)\mathbf{I}_{SE}$, so

$$\mathbf{I}_L = \frac{N_{SE}}{N_C}\mathbf{I}_{SE} + \mathbf{I}_{SE} \tag{2–77}$$

Finally, noting that $\mathbf{I}_H = \mathbf{I}_{SE}$, we find

$$\mathbf{I}_L = \frac{N_{SE}}{N_C}\mathbf{I}_H + \mathbf{I}_H$$

$$= \frac{N_{SE} + N_C}{N_C}\mathbf{I}_H \tag{2–78}$$

or

$$\boxed{\frac{\mathbf{I}_L}{\mathbf{I}_H} = \frac{N_{SE} + N_C}{N_C}} \tag{2–79}$$

Apparatus power rating advantage in autotransformer

- There are two types of rating
  - Power rating on terminals Sio
  - Power rating on windings Sw
- Terminals power rating Sio

$$S_{in} = V_L I_L \quad S_{out} = V_H I_H \qquad S_{in} = S_{out} = S_{IO}$$

- Winding power rating Sw

$$S_W = V_C I_C = V_{SE} I_{SE}$$

Apparatus power rating advantage in autotransformer

- Relation

$$\begin{aligned} S_W &= V_C I_C \\ &= V_L(I_L - I_H) \\ &= V_L I_L - V_L I_H \end{aligned}$$

$$\begin{aligned} S_W &= V_L I_L - V_L I_L \frac{N_C}{N_{SE} + N_C} \\ &= V_L I_L \frac{(N_{SE} + N_C) - N_C}{N_{SE} + N_C} \\ &= S_{IO} \frac{N_{SE}}{N_{SE} + N_C} \end{aligned}$$

$$\boxed{\frac{S_{IO}}{S_W} = \frac{N_{SE} + N_C}{N_{SE}}}$$

Example 2-7

Example 2–7. A 100-VA 120/12–V transformer is to be connected so as to form a step-up autotransformer (see Figure 2–34). A primary voltage of 120 V is applied to the transformer.

(a) What is the secondary voltage of the transformer?

(b) What is its maximum voltampere rating in this mode of operation?

(c) Calculate the rating advantage of this autotransformer connection over the transformer's rating in conventional 120/12–V operation.

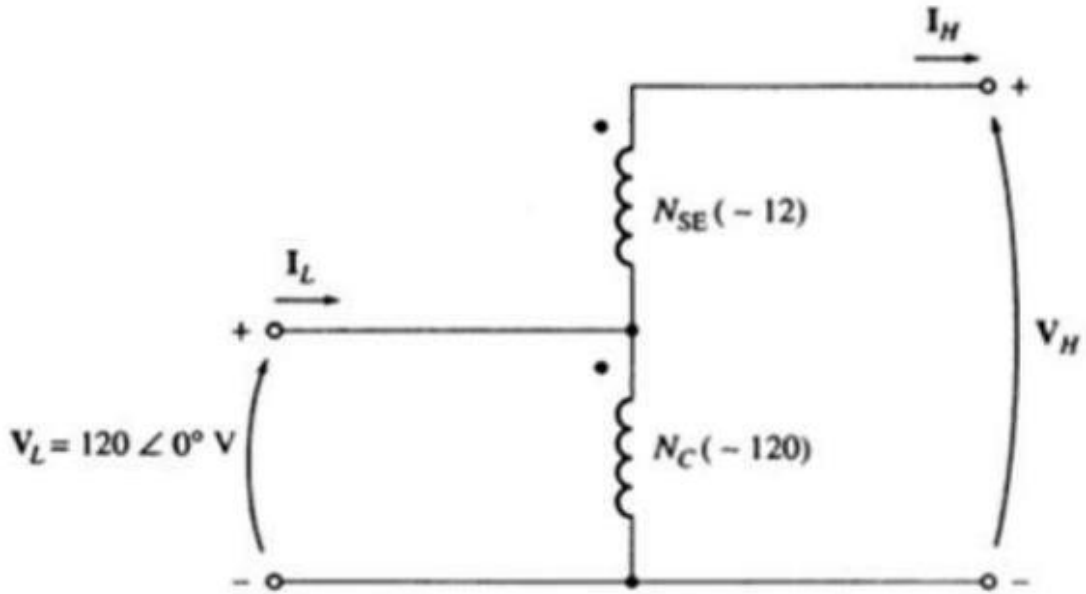
Solution

To accomplish a step-up transformation with a 120-V primary, the ratio of the turns on the common winding N_C to the turns on the series winding N_{SE} in this transformer must be 120:12 (or 10:1).

(a) This transformer is being used as a step-up transformer. The secondary voltage is V_H, and from Equation (2–75),

$$\mathbf{V}_H = \frac{N_{SE} + N_C}{N_C}\mathbf{V}_L \qquad (2\text{–}75)$$

$$= \frac{12 + 120}{120}\,120\text{ V} = 132\text{ V}$$



(b) The maximum voltampere rating in either winding of this transformer is 100 VA. How much input or output apparent power can this provide? To find out, examine the series winding. The voltage V_{SE} on the winding is 12 V, and the voltampere rating of the winding is 100 VA. Therefore, the *maximum* series winding current is

$$I_{SE,max} = \frac{S_{max}}{V_{SE}} = \frac{100 \text{ VA}}{12\text{V}} = 8.33 \text{ A}$$

Since I_{SE} is equal to the secondary current I_S (or I_H) and since the secondary voltage $V_S = V_H = 132$ V, the secondary apparent power is

$$S_{out} = V_S I_S = V_H I_H$$

$$= (132 \text{ V})(8.33 \text{ A}) = 1100 \text{ VA} = S_{in}$$

(c) The rating advantage can be calculated from part (*b*) or separately from Equation (2–86). From part *b*,

$$\frac{S_{IO}}{S_W} = \frac{1100 \text{ VA}}{100 \text{ VA}} = 11$$

From Equation (2–86),

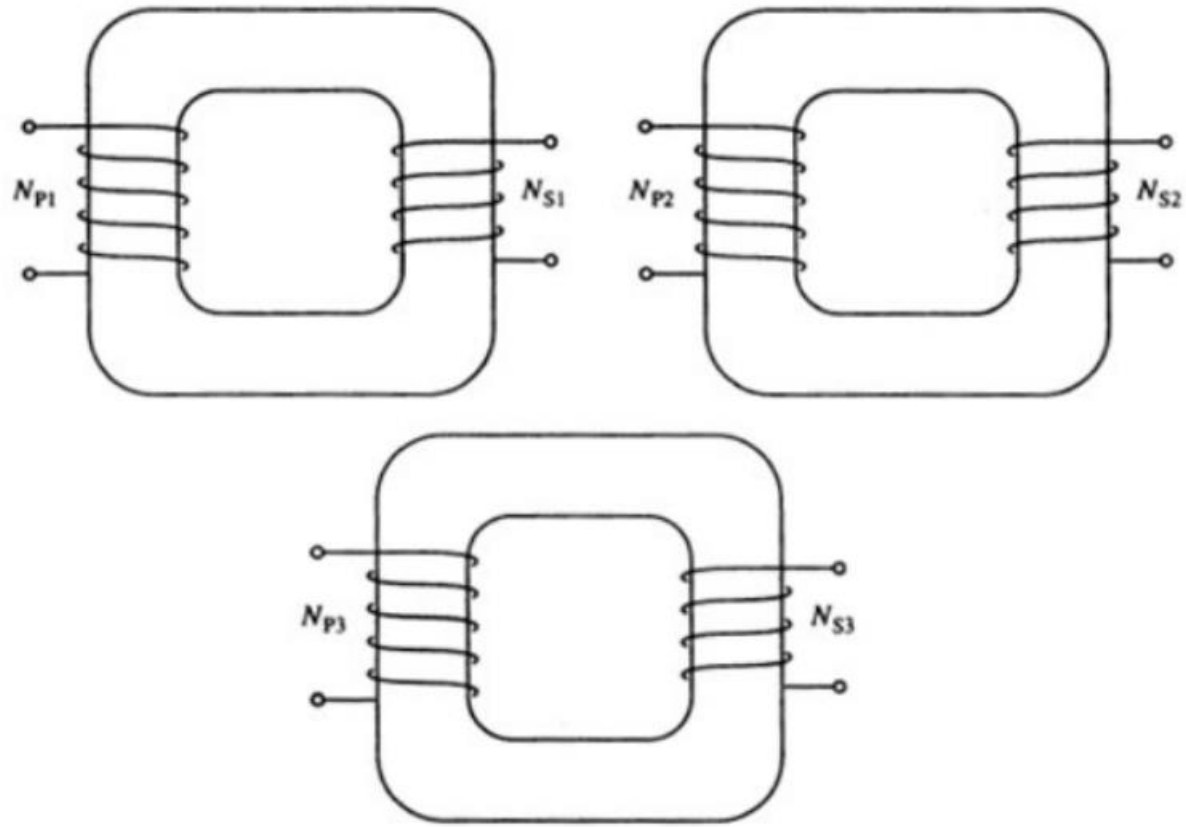
$$\frac{S_{IO}}{S_W} = \frac{N_{SE} + N_C}{N_{SE}} \qquad (2\text{–}86)$$

$$= \frac{12 + 120}{12} = \frac{132}{12} = 11$$

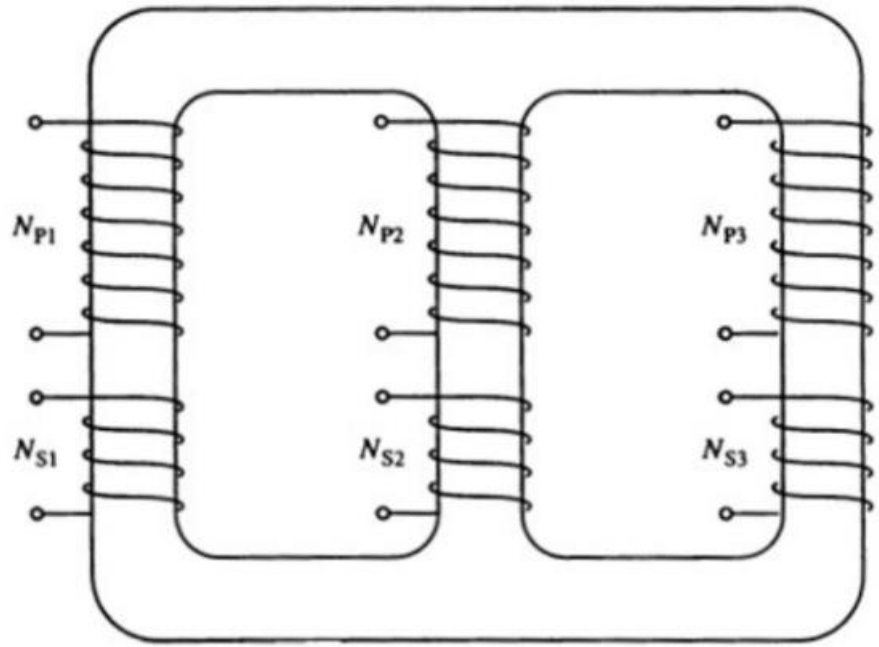
Autotransformer summarize

- When two voltages are fairly close to each other
- The power advantage is very large
- There is a direct physical connection between primary and secondary sides.
- The autotransformer is a convenient and inexpensive way to tie **nearly equal** two voltages together
- The electrical isolation of two sides is lost

Three-phase transformer – three single-phase transformer banks



Three-phase transformer – one three-lags transformer banks

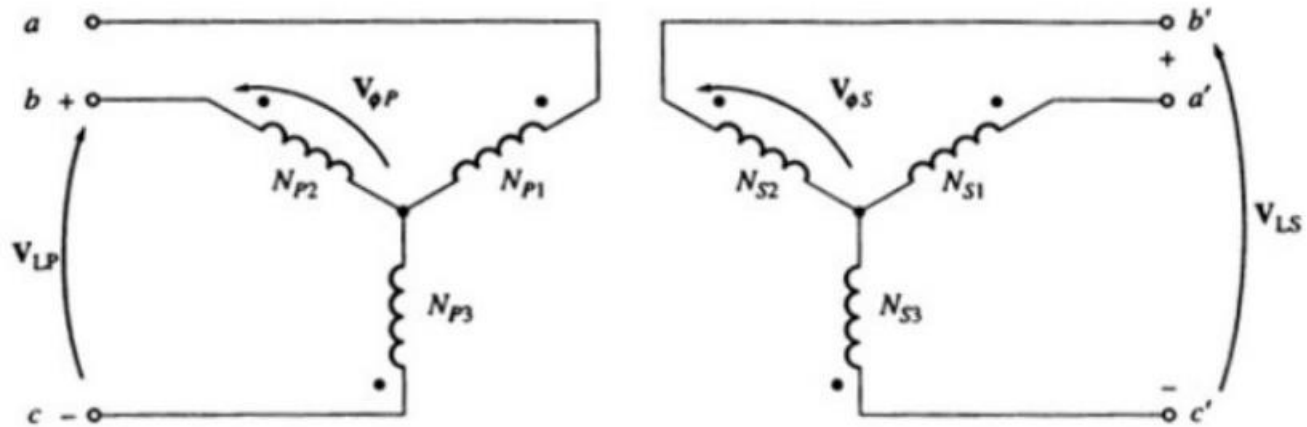


Three types of terminal connection

A three-phase transformer consists of three transformers, either separate or combined on one core. The primaries and secondaries of any three-phase transformer can be independently connected in either a wye (Y) or a delta (Δ). This gives a total of four possible connections for a three-phase transformer bank:

1. Wye–wye (Y–Y)
2. Wye–delta (Y–Δ)
3. Delta–wye (Δ–Y)
4. Delta–delta (Δ–Δ)

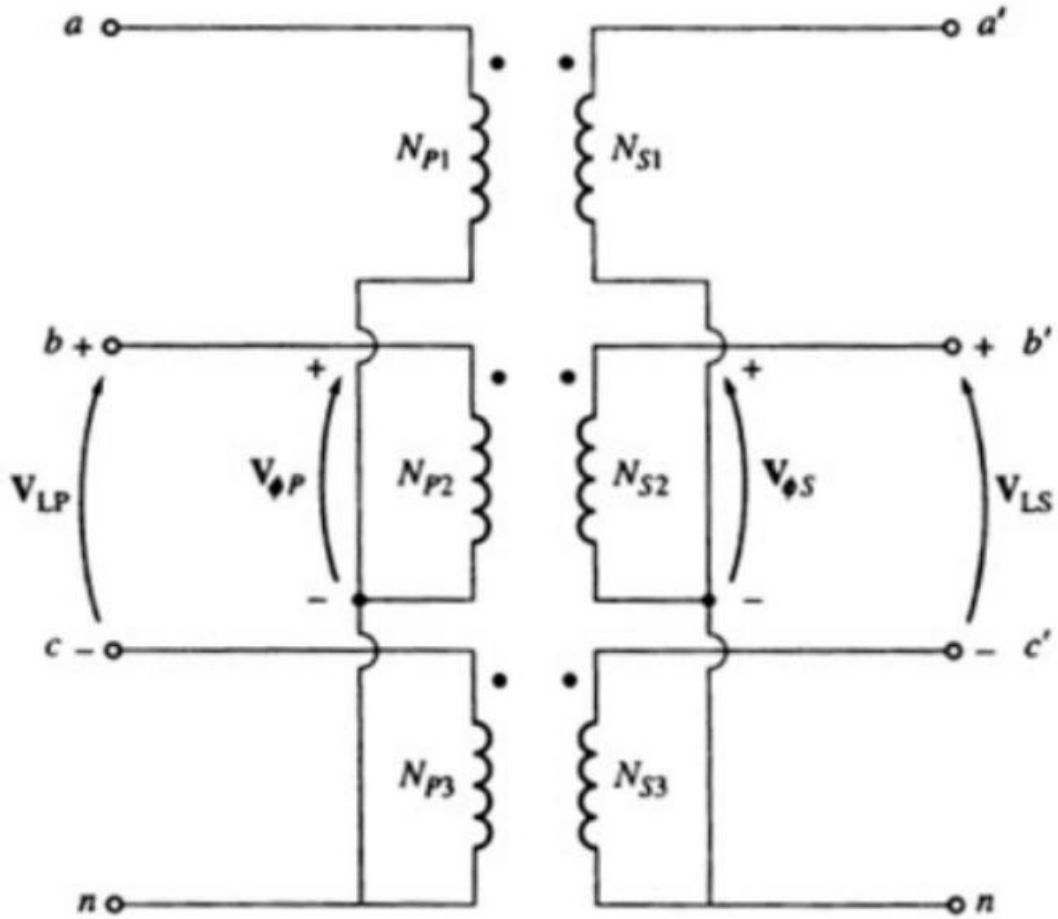
Wye-wye connection



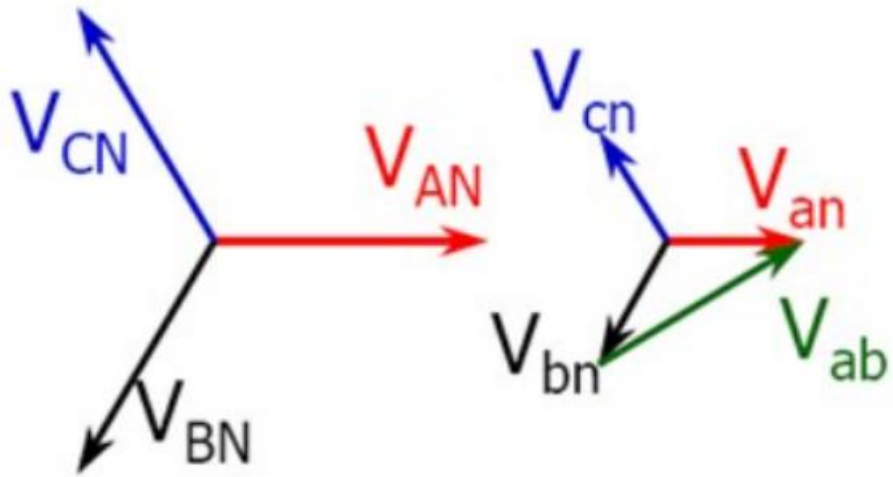
$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{\sqrt{3}V_{\phi S}} = a \qquad \text{Y–Y}$$

1. If loads on the transformer circuit are unbalanced, then the voltages on the phases of the transformer can become severely unbalanced.
2. Third-harmonic voltages can be large.

Wye-wye connection



Wye-wye connection



$$V_{ab} = V_{an} - V_{bn}$$

$$V_{bc} = V_{bn} - V_{cn}$$

$$V_{ca} = V_{cn} - V_{an}$$

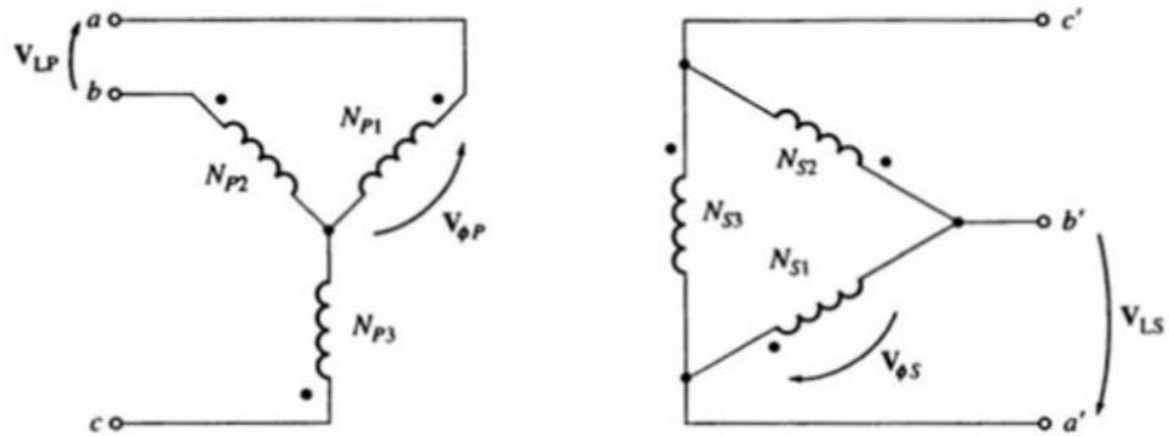
$$V_{AN} = aV_{an}$$

$$V_{AB} = aV_{ab}$$

Solving the unbalance and 3rd harmonic problems in wye-wye connection

1. ***Solid ground the neutral of transformers***: solve the unbalance problem and support a return path to the 3rd harmonic component.
2. ***Add a Δ-connected third winding***: Since 3rd harmonic components are in-phase in each branch of Δ-connection, 3rd harmonic components will be limited in Δ-connection as the circulating current.

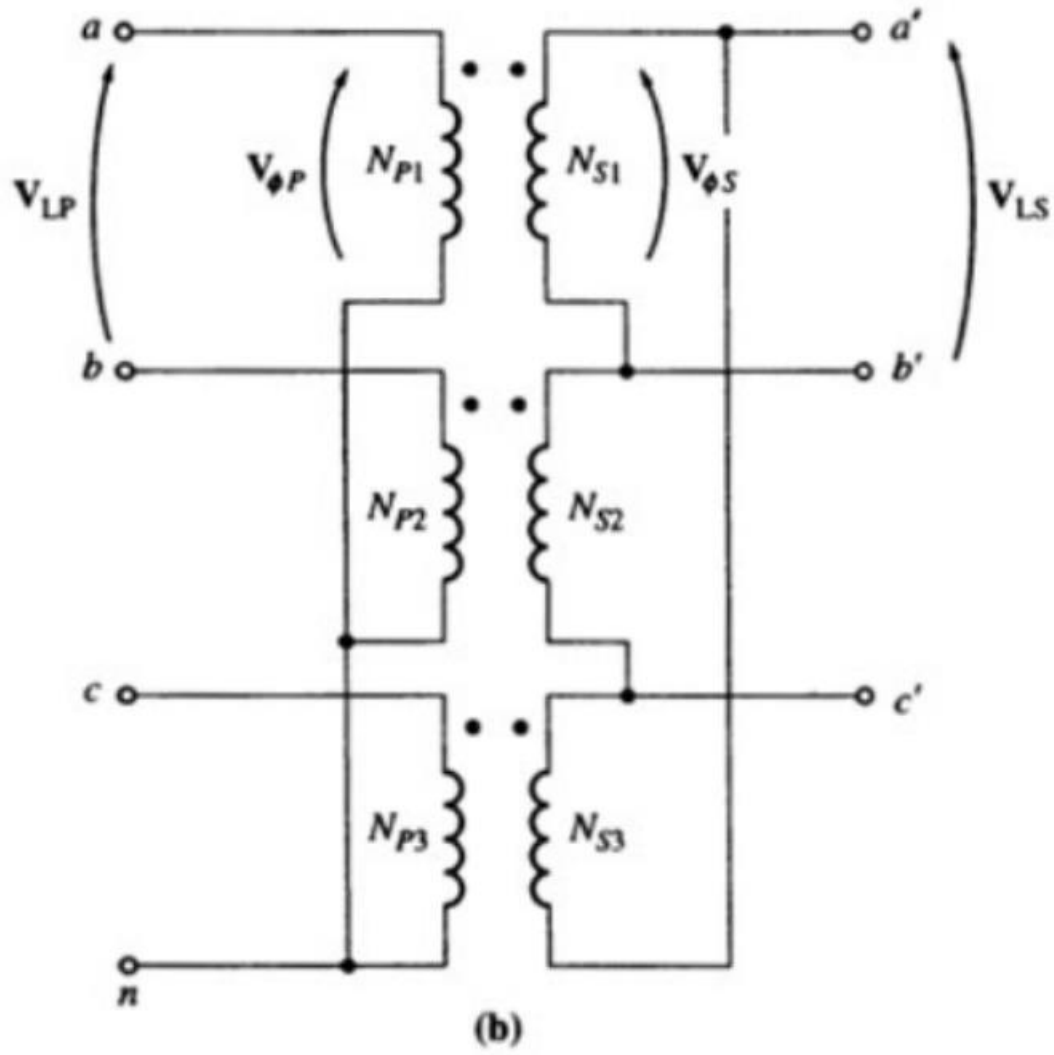
Wye-delta connection



$$\frac{V_{\phi P}}{V_{\phi S}} = a \qquad \frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{V_{\phi S}}$$

$$\boxed{\frac{V_{LP}}{V_{LS}} = \sqrt{3}a \qquad \text{Y}-\Delta}$$

Wye-delta connection

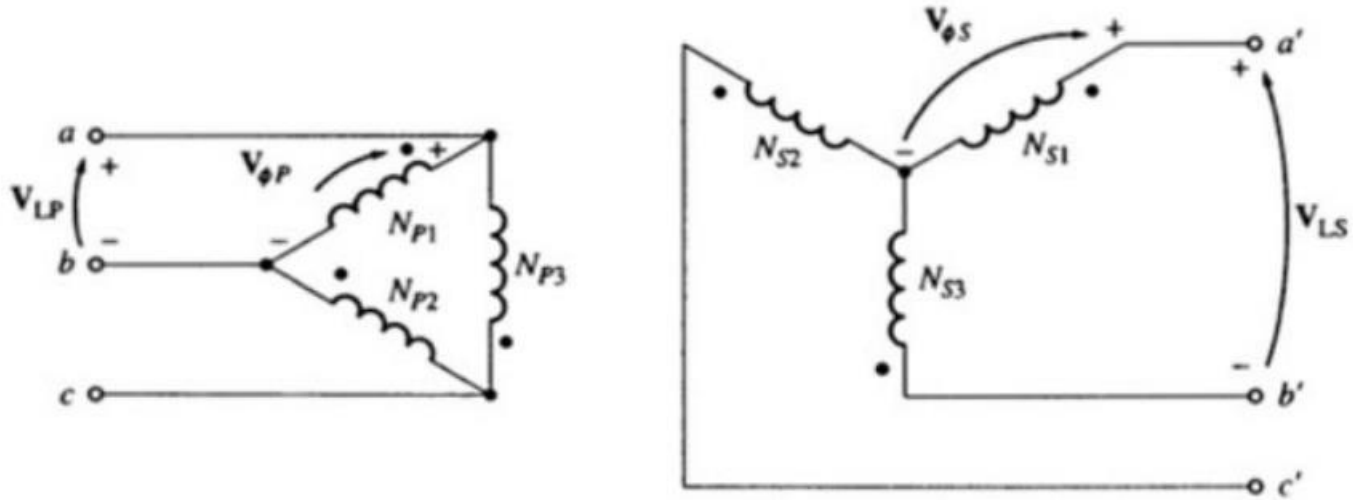


(b)

Wye-delta connection summarize

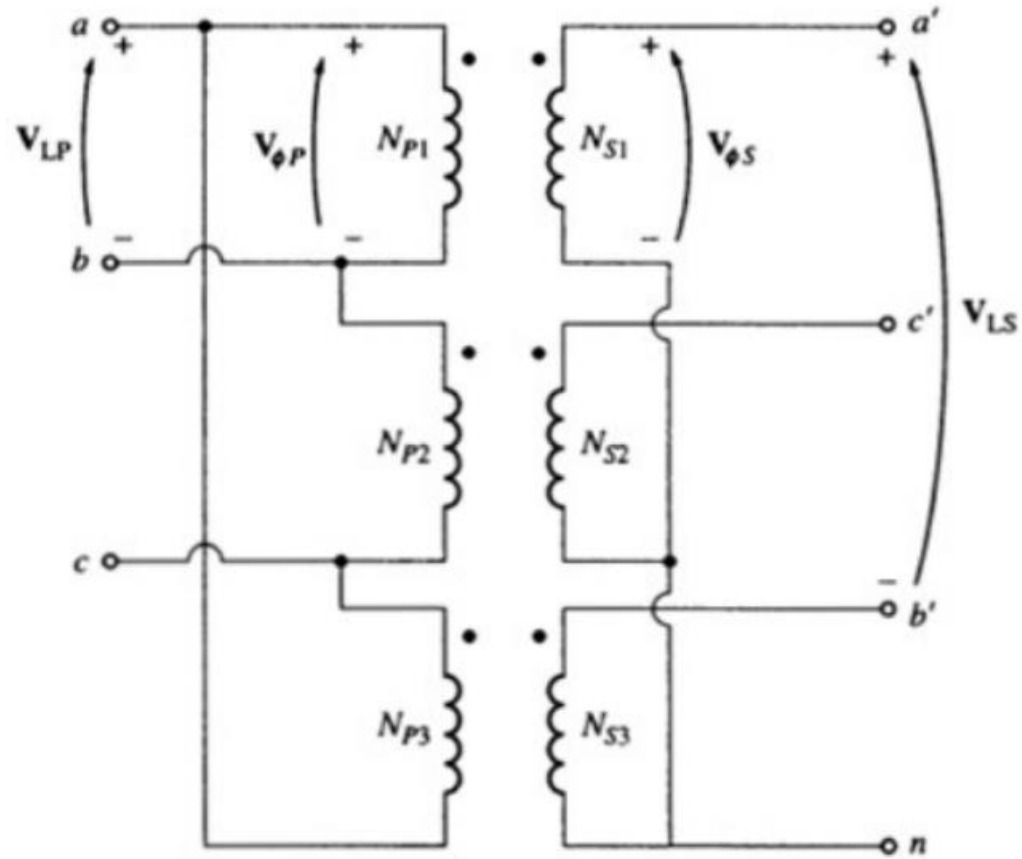
1. No 3rd harmonic component problem:
2. There are phase difference between each sides
3. In United state, the secondary voltage will lag the primary voltage 30 degrees

Delta-wye connection



$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{\sqrt{3}V_{\phi S}} = \frac{a}{\sqrt{3}}$$

Delta-wye connection



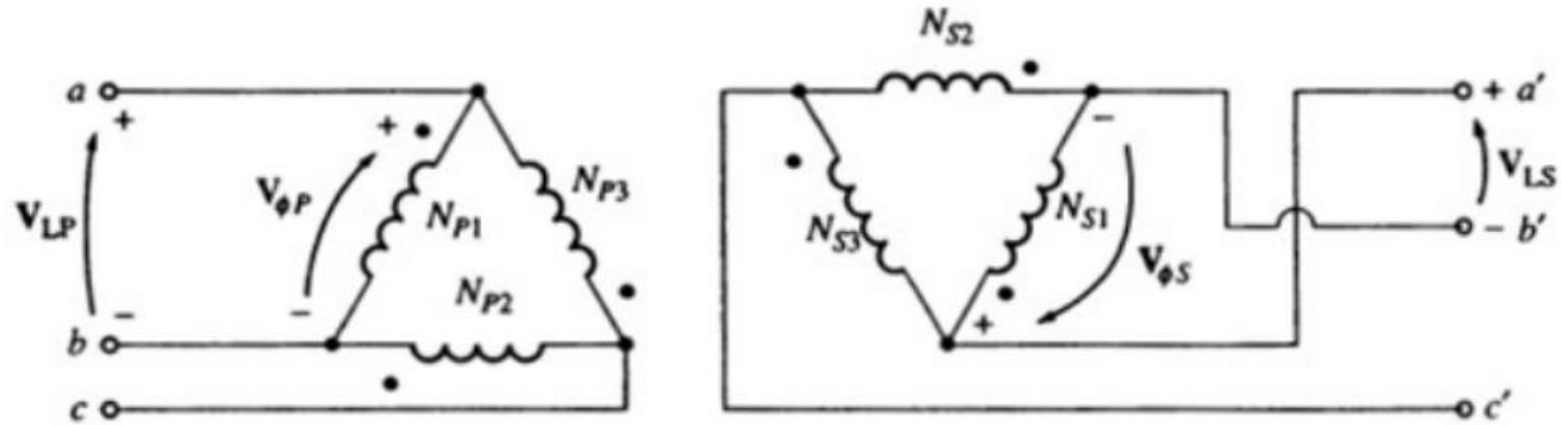
Delta-wye connection

- Common connection:
  - Used on three-wire (delta) to four wire (wye)
  - Used to isolate ground on wye side from source ground on delta side

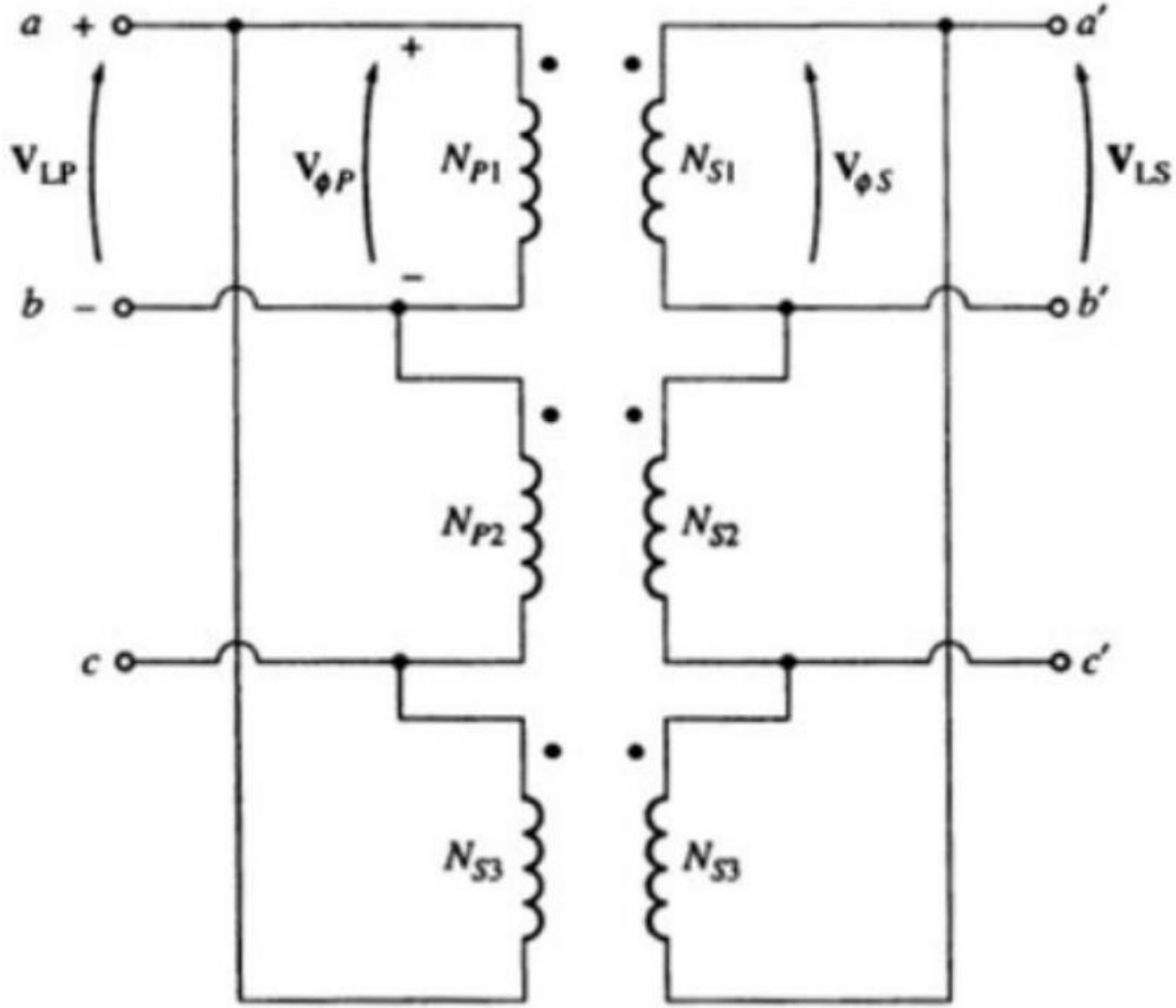
Delta-wye or wye-delta

- Common for wye-delta step-up transformer banks in generating plants
- Common for delta-wye step-down banks in substation

Delta-delta connection



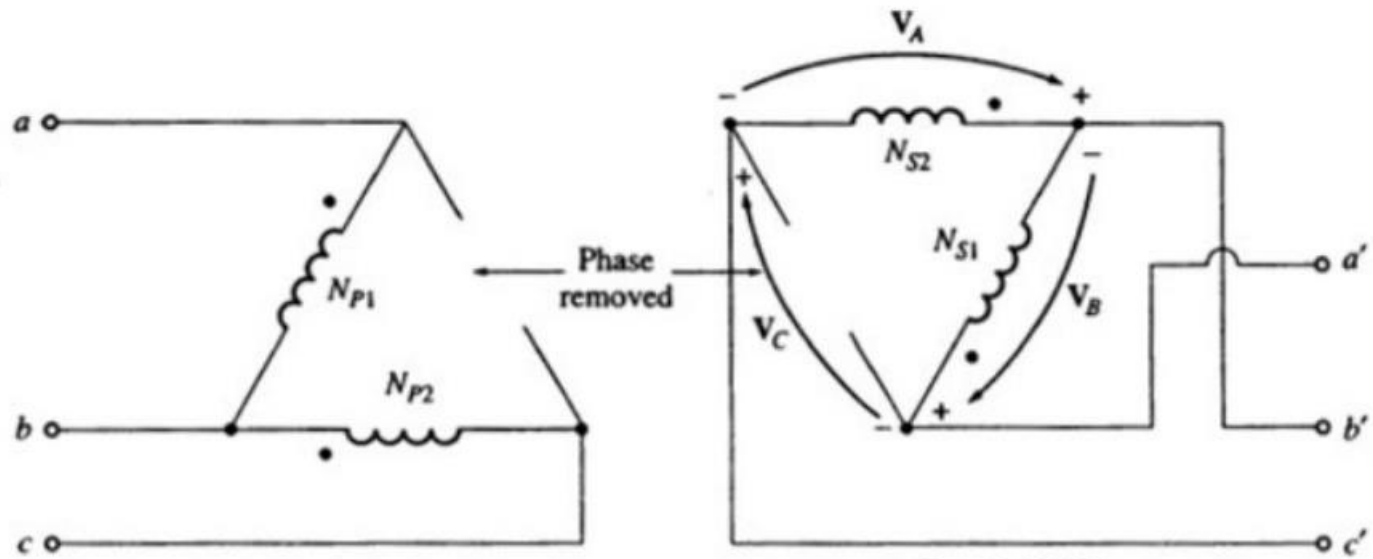
$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{V_{\phi S}} = a \qquad \Delta - \Delta$$


a
+
b
–
$\mathbf{V}_{LP}$
+
$\mathbf{V}_{\phi P}$
–
N_{P1}
N_{P2}
N_{P3}
c
N_{S1}
$\mathbf{V}_{\phi S}$
$\mathbf{V}_{LS}$
a′
b′
c′
N_{S2}
N_{S3}

Three-phase transformation using two transformers

1. The open-Δ (or V–V) connection
2. The open-Y–open-Δ connection
3. The Scott-T connection
4. The three-phase T connection

V-V connection

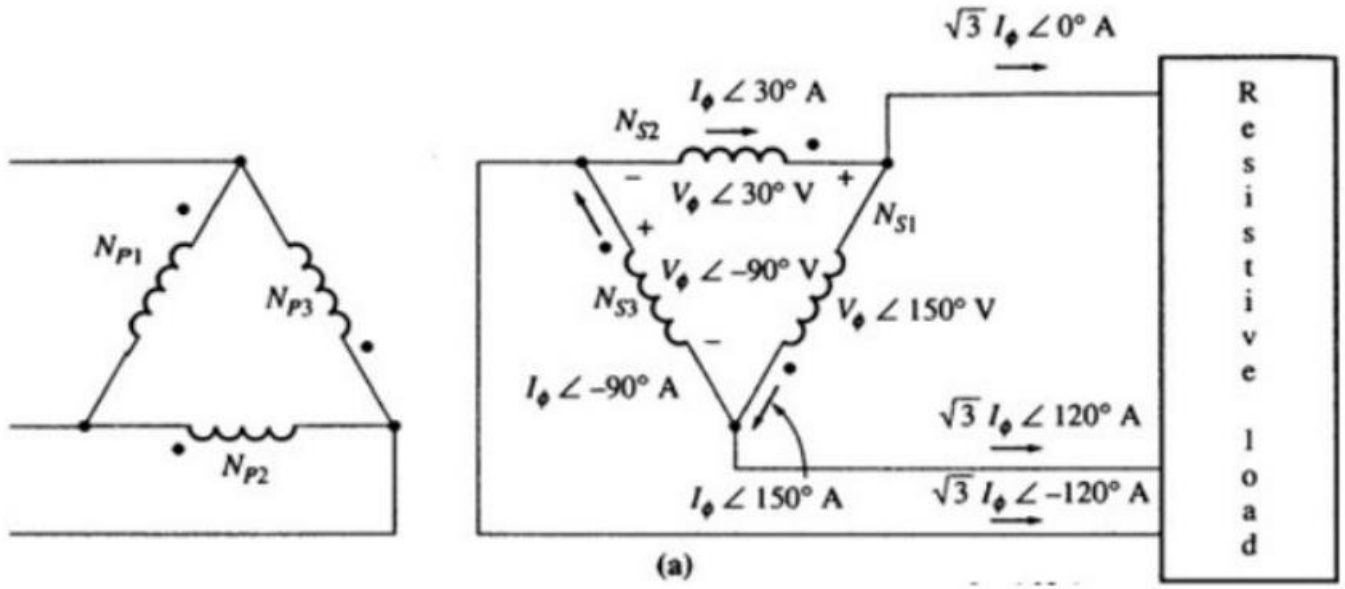


$\mathbf{V}_A = V \angle 0°$ V

$\mathbf{V}_B = V \angle 120°$ V

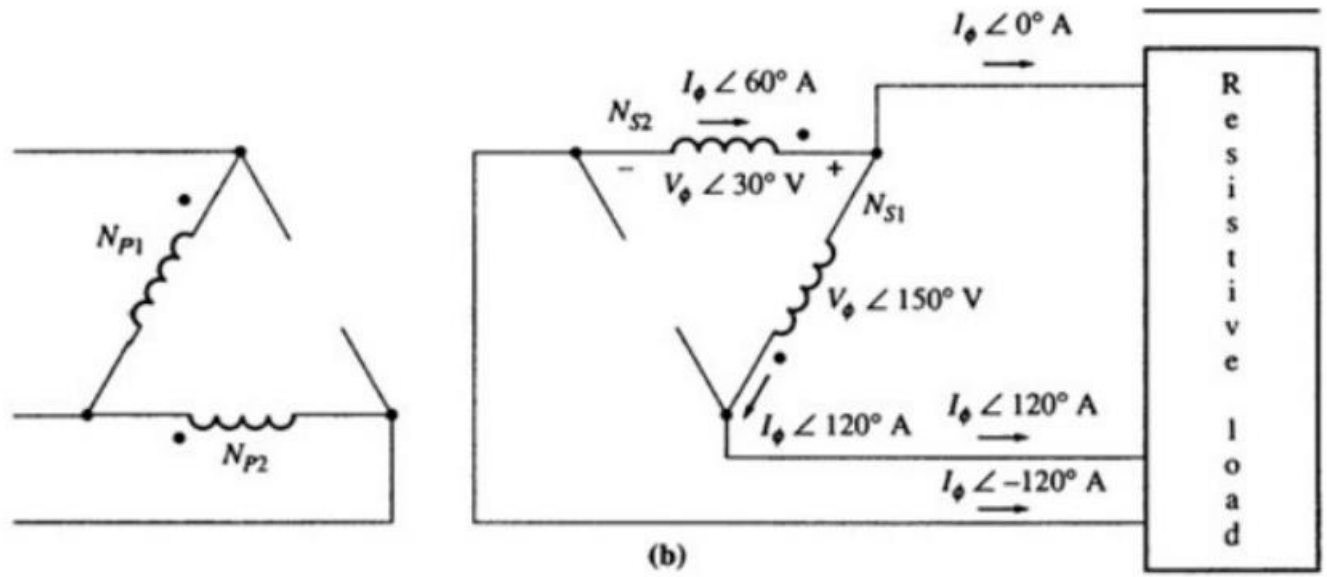
$$\begin{aligned}\mathbf{V}_C &= -\mathbf{V}_A - \mathbf{V}_B \\ &= -V\angle 0° - V\angle -120° \\ &= -V - (-0.5V - j0.866V) \\ &= -0.5V + j0.866V \\ &= V\angle 120° \quad \text{V}\end{aligned}$$

Power rating of Δ-Δ connection



$$P = 3V_\phi I_\phi \cos\theta$$
$$= 3V_\phi I_\phi$$

Power rating of V-V connection



$$P_1 = 3V_\phi I_\phi \cos(150° - 120°)$$
$$= 3V_\phi I_\phi \cos 30°$$
$$= \frac{\sqrt{3}}{2} V_\phi I_\phi$$

$$P_2 = 3V_\phi I_\phi \cos(30° - 60°)$$
$$= 3V_\phi I_\phi \cos(-30°)$$
$$= \frac{\sqrt{3}}{2} V_\phi I_\phi$$

$$P = \sqrt{3} V_\phi I_\phi$$

Comparison of power rating

1. The power rating comparison

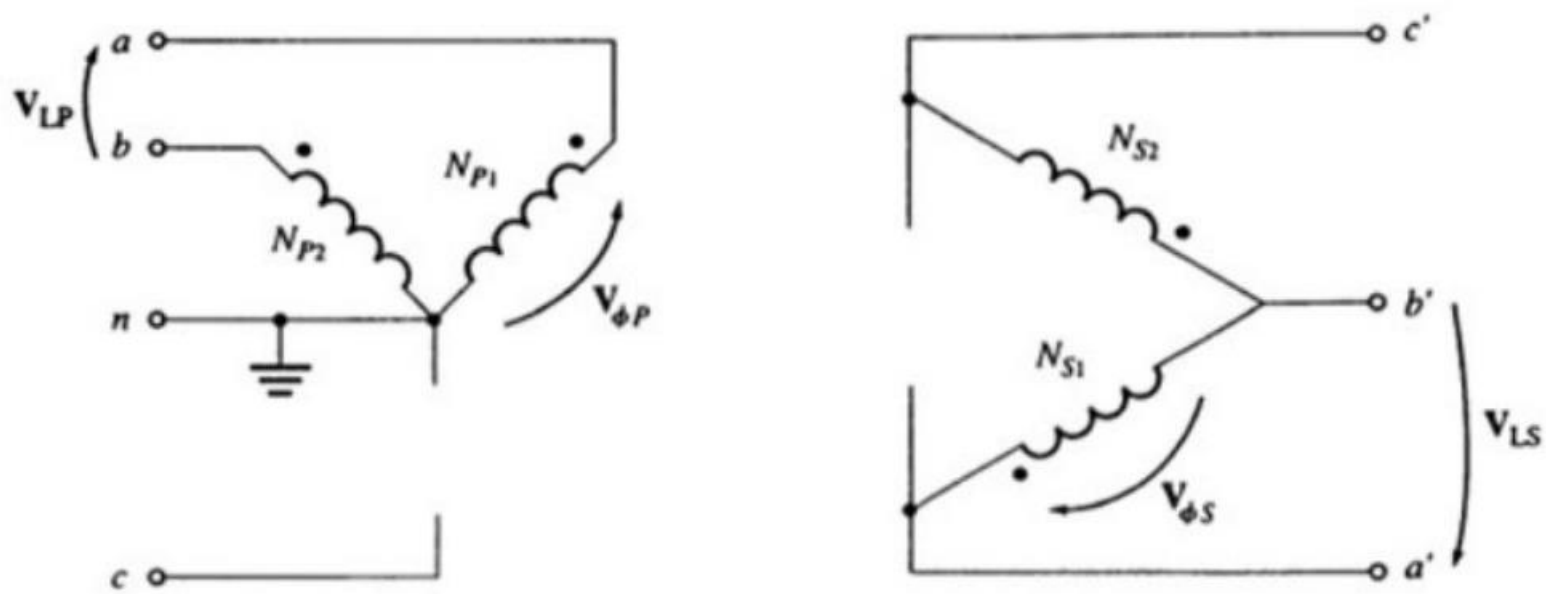
$$\frac{P_{\text{open }\Delta}}{P_{3\text{ phase}}} = \frac{\sqrt{3}V_\phi I_\phi}{3V_\phi I_\phi} = \frac{1}{\sqrt{3}} = 0.577$$

2. Where is the power rating ?
   - The existing reactive power will consume the power rating

$$\begin{aligned} Q_1 &= 3V_\phi I_\phi \sin(150° - 120°) \\ &= 3V_\phi I_\phi \sin 30° \\ &= \tfrac{1}{2} V_\phi I_\phi \end{aligned}$$

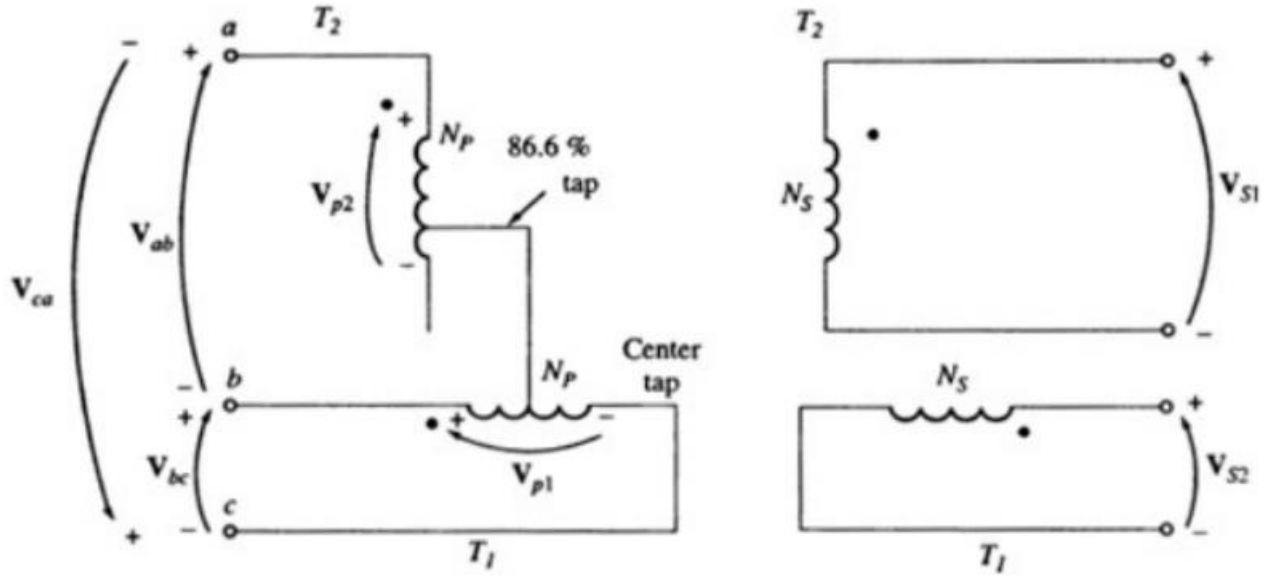
$$\begin{aligned} Q_2 &= 3V_\phi I_\phi \sin(30° - 60°) \\ &= 3V_\phi I_\phi \sin(-30°) \\ &= -\tfrac{1}{2} V_\phi I_\phi \end{aligned}$$

Open Y-open Δ connection



Scott-T connection – railroad applications

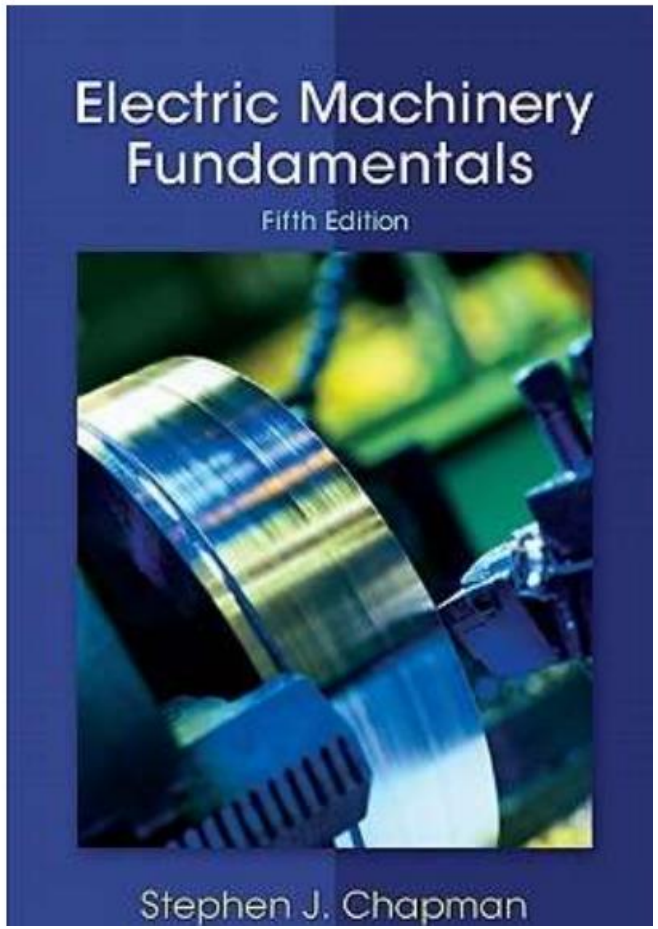
1. While in railroad applications, there always need two-phase power system for supporting northbound and southbound rails respectively.



(a)

Text book and supplementary materials of this course

- Stephen J. Chapman



7.4 PROBLEMS WITH COMMUTATION IN REAL MACHINES

Armature Reaction

L di/dt Voltages

Solutions to the Problems with Commutation

Three approaches have been developed to partially or completely correct the problems of armature reaction and *L di/dt* voltages:

1. Brush shifting
2. Commutating poles or interpoles
3. Compensating windings

Each of these techniques is explained below, together with its advantages and disadvantages.

7.5 THE INTERNAL GENERATED VOLTAGE AND INDUCED TORQUE EQUATIONS OF REAL DC MACHINES

How much voltage is produced by a real dc machine? The induced voltage in any given machine depends on three factors:

1. The flux ϕ in the machine
2. The speed ω_m of the machine's rotor
3. A constant depending on the construction of the machine

$$E_A = \frac{ZP}{2\pi a}\phi\omega_m$$

$$E_A = K\phi\omega_m$$

$$K = \frac{ZP}{2\pi a}$$

In modern industrial practice, it is common to express the speed of a machine in revolutions per minute instead of radians per second. The conversion from revolutions per minute to radians per second is

$$\omega_m = \frac{2\pi}{60} n_m \tag{7–40}$$

so the voltage equation with speed expressed in terms of revolutions per minute is

$$\boxed{E_A = K'\phi n_m} \tag{7–41}$$

where

$$\boxed{K' = \frac{ZP}{60a}} \tag{7–42}$$

How much torque is induced in the armature of a real dc machine? The torque in any dc machine depends on three factors:

1. The flux ϕ in the machine
2. The armature (or rotor) current I_A in the machine
3. A constant depending on the construction of the machine

$$\tau_{\text{ind}} = K \phi I_A$$

$$K = \frac{ZP}{2\pi a}$$

Example 7–3. A duplex lap-wound armature is used in a six-pole dc machine with six brush sets, each spanning two commutator segments. There are 72 coils on the armature, each containing 12 turns. The flux per pole in the machine is 0.039 Wb, and the machine spins at 400 r/min.

(a) How many current paths are there in this machine?
(b) What is its induced voltage E_A?

Solution

(a) The number of current paths in this machine is

$$a = mP = 2(6) = 12 \text{ current paths} \tag{7–26}$$

(b) The induced voltage in the machine is

$$E_A = K'\phi n_m \tag{7–41}$$

and

$$K' = \frac{ZP}{60a} \tag{7–42}$$

The number of conductors in this machine is

$$\begin{aligned} Z &= 2CN_C \qquad (7–22) \\ &= 2(72)(12) = 1728 \text{ conductors} \end{aligned}$$

Therefore, the constant K' is

$$K' = \frac{ZP}{60a} = \frac{(1728)(6)}{(60)(12)} = 14.4$$

and the voltage E_A is

$$\begin{aligned} E_A &= K'\phi n_m \\ &= (14.4)(0.039 \text{ Wb})(400 \text{ r/min}) \\ &= 224.6 \text{ V} \end{aligned}$$

Example 7–4. A 12-pole dc generator has a simplex wave-wound armature containing 144 coils of 10 turns each. The resistance of each turn is 0.011 Ω. Its flux per pole is 0.05 Wb, and it is turning at a speed of 200 r/min.

(a) How many current paths are there in this machine?
(b) What is the induced armature voltage of this machine?
(c) What is the effective armature resistance of this machine?
(d) If a 1-kΩ resistor is connected to the terminals of this generator, what is the resulting induced countertorque on the shaft of the machine? (Ignore the internal armature resistance of the machine.)

Solution

(a) There are $a = 2m = 2$ current paths in this winding.

(b) There are $Z = 2CN_c = 2(144)(10) = 2880$ conductors on this generator's rotor. Therefore,

$$K' = \frac{ZP}{60a} = \frac{(2880)(12)}{(60)(2)} = 288$$

Therefore, the induced voltage is

$$\begin{aligned} E_A &= K'\phi n_m \\ &= (288)(0.05 \text{ Wb})(200 \text{ r/min}) \\ &= 2880 \text{ V} \end{aligned}$$

(c) There are two parallel paths through the rotor of this machine, each one consisting of $Z/2 = 1440$ conductors, or 720 turns. Therefore, the resistance in each current path is

$$\text{Resistance/path} = (720 \text{ turns})(0.011\ \Omega\text{/turn}) = 7.92\ \Omega$$

Since there are two parallel paths, the effective armature resistance is

$$R_A = \frac{7.92\ \Omega}{2} = 3.96\ \Omega$$

(d) If a 1000-Ω load is connected to the terminals of the generator, and if R_A is ignored, then a current of $I = 2880 \text{ V}/1000\ \Omega = 2.88$ A flows. The constant K is given by

$$K = \frac{ZP}{2\pi a} = \frac{(2880)(12)}{(2\pi)(2)} = 2750.2$$

Therefore, the countertorque on the shaft of the generator is

$$\tau_{ind} = K\phi I_A = (2750.2)(0.05\ \text{Wb})(2.88\ \text{A})$$
$$= 396\ \text{N} \cdot \text{m}$$

7.6 THE CONSTRUCTION OF DC MACHINES

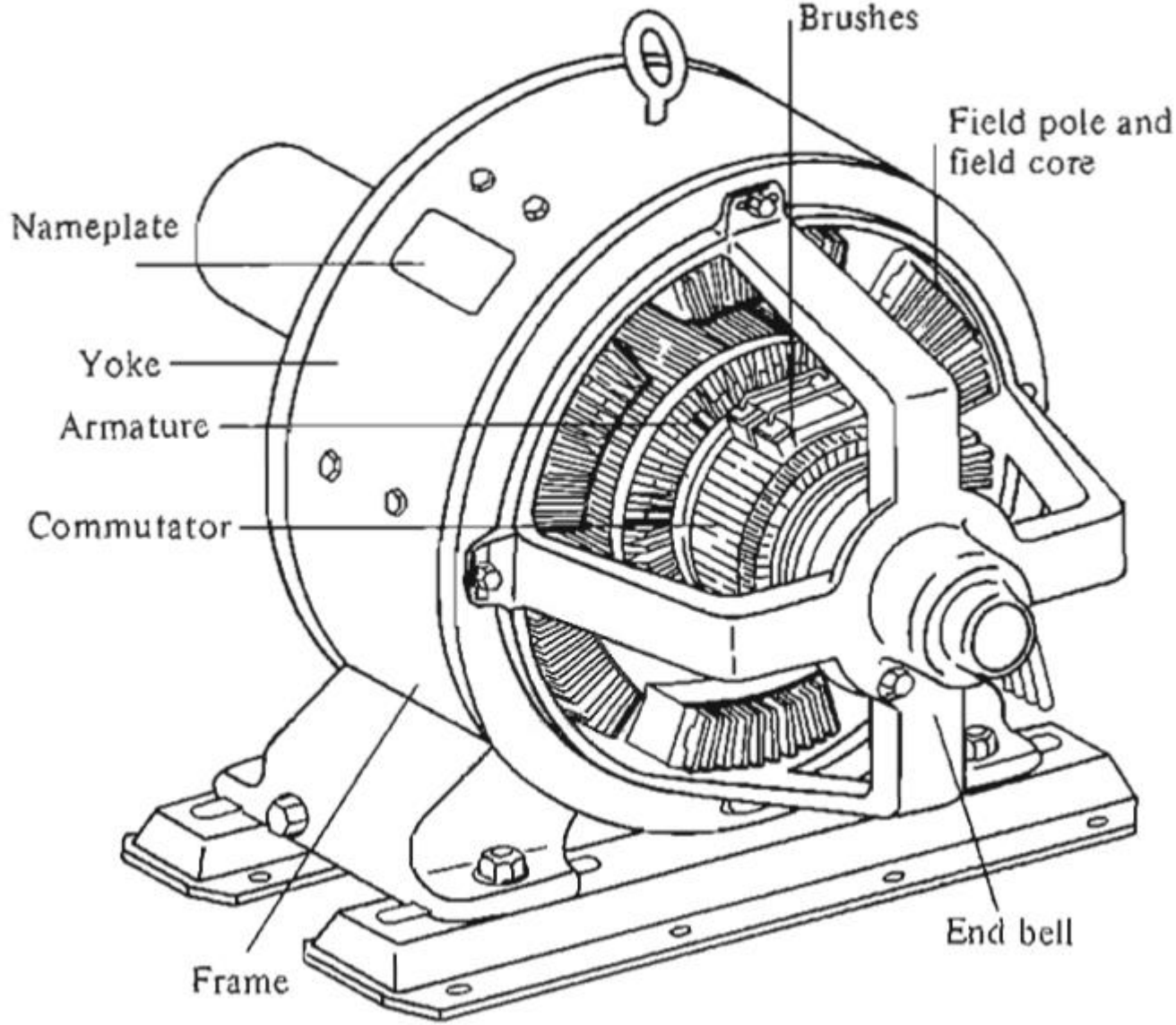


FIGURE 7–33
A simplified diagram of a dc machine.

7.7 POWER FLOW AND LOSSES IN DC MACHINES

DC generators take in mechanical power and produce electric power, while dc motors take in electric power and produce mechanical power. In either case, not all the power input to the machine appears in useful form at the other end—there is *always* some loss associated with the process.

The efficiency of a dc machine is defined by the equation

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% \tag{7–50}$$

The difference between the input power and the output power of a machine is the losses that occur inside it. Therefore,

$$\eta = \frac{P_{out} - P_{loss}}{P_{in}} \times 100\% \tag{7–51}$$

The Losses in DC Machines

The losses that occur in dc machines can be divided into five basic categories:

1. Electrical or copper losses (I^2R losses)
2. Brush losses
3. Core losses
4. Mechanical losses
5. Stray load losses

ELECTRICAL OR COPPER LOSSES. Copper losses are the losses that occur in the armature and field windings of the machine. The copper losses for the armature and field windings are given by

Armature loss:	$P_A = I_A^2 R_A$	(7–52)
Field loss:	$P_F = I_F^2 R_F$	(7–53)

where P_A = armature loss
P_F = field circuit loss

I_A = armature current

I_F = field current

R_A = armature resistance

R_F = field resistance

The resistance used in these calculations is usually the winding resistance at normal operating temperature.

BRUSH LOSSES. The brush drop loss is the power lost across the contact potential at the brushes of the machine. It is given by the equation

$$\boxed{P_{BD} = V_{BD} I_A} \tag{7–54}$$

where P_{BD} = brush drop loss
V_{BD} = brush voltage drop
I_A = armature current

The reason that the brush losses are calculated in this manner is that the voltage drop across a set of brushes is approximately constant over a large range of armature currents. Unless otherwise specified, the brush voltage drop is usually assumed to be about 2 V.

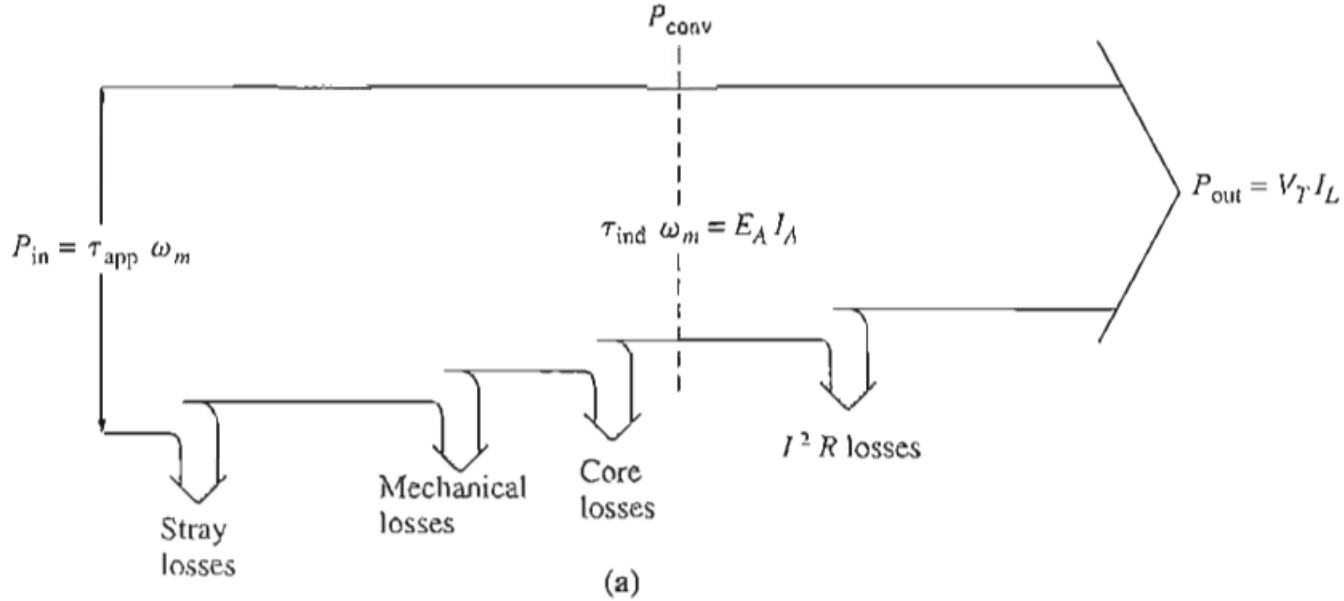
CORE LOSSES. The core losses are the hysteresis losses and eddy current losses occurring in the metal of the motor. These losses are described in Chapter 1. These losses vary as the square of the flux density (B^2) and, for the rotor, as the 1.5th power of the speed of rotation ($n^{1.5}$).

MECHANICAL LOSSES. The mechanical losses in a dc machine are the losses associated with mechanical effects. There are two basic types of mechanical losses: *friction* and *windage*. Friction losses are losses caused by the friction of the bearings in the machine, while windage losses are caused by the friction between the moving parts of the machine and the air inside the motor's casing. These losses vary as the cube of the speed of rotation of the machine.

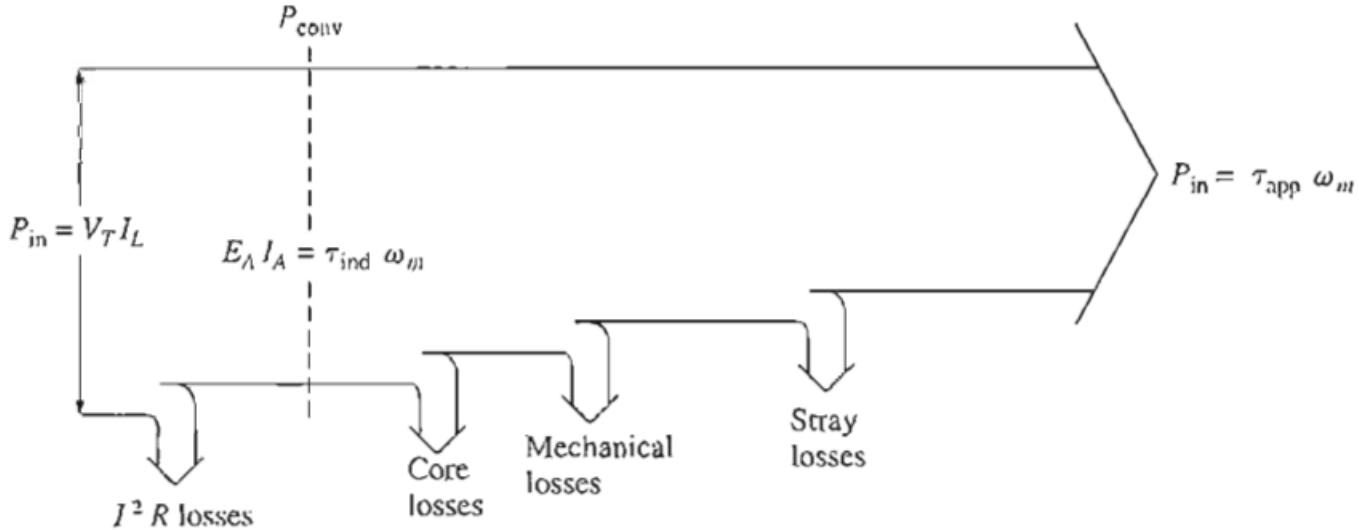
STRAY LOSSES (OR MISCELLANEOUS LOSSES). Stray losses are losses that cannot be placed in one of the previous categories. No matter how carefully losses are accounted for, some always escape inclusion in one of the above categories. All such losses are lumped into stray losses. For most machines, stray losses are taken by convention to be 1 percent of full load.

The Power-Flow Diagram

One of the most convenient techniques for accounting for power losses in a machine is the *power-flow diagram*. A power-flow diagram for a dc generator is shown in Figure 7–39a. In this figure, mechanical power is input into the machine,


P_{conv}
$P_{in} = \tau_{app}\ \omega_m$
$\tau_{ind}\ \omega_m = E_A\ I_A$
$P_{out} = V_T\ I_L$
Stray losses
Mechanical losses
Core losses
$I^2\ R$ losses

(a)


P_{conv}
$P_{in} = V_T\ I_L$
$E_A\ I_A = \tau_{ind}\ \omega_m$
$P_{in} = \tau_{app}\ \omega_m$
$I^2\ R$ losses
Core losses
Mechanical losses
Stray losses

and then the stray losses, mechanical losses, and core losses are subtracted. After they have been subtracted, the remaining power is ideally converted from mechanical to electrical form at the point labeled P_{conv}. The mechanical power that is converted is given by

$$P_{conv} = \tau_{ind}\omega_m \tag{7–55}$$

and the resulting electric power produced is given by

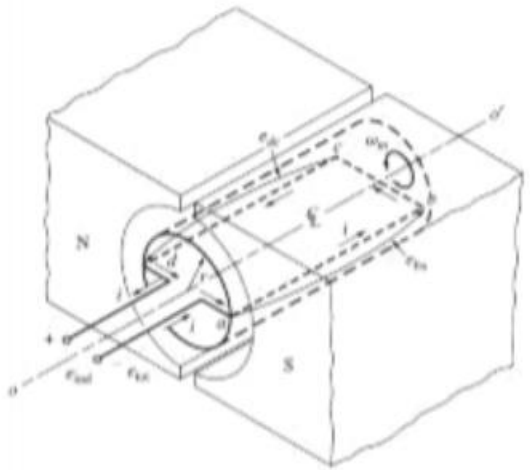
$$P_{conv} = E_A I_A \tag{7–56}$$

DC Machinery Fundamentals

This chapter will firstly explain the principles of dc machine operation using simple examples before considering some of the complications that occur in real dc machines.

8.1. A simple rotating loop between curved pole faces

The simplest rotating dc machine is shown below.



It consists of:

- A rotating single loop of wire in a slot carved in a cylindrical ferromagnetic core – **rotor**

- A pair of magnetic poles (north and south) – **stator**

The **magnetic field** is supplied by the stator, i.e. N and S poles.

The air gap between the rotor and stator is of **constant width** (**i.e.** $\Re_{gap}$ is the **same** everywhere under the pole faces).

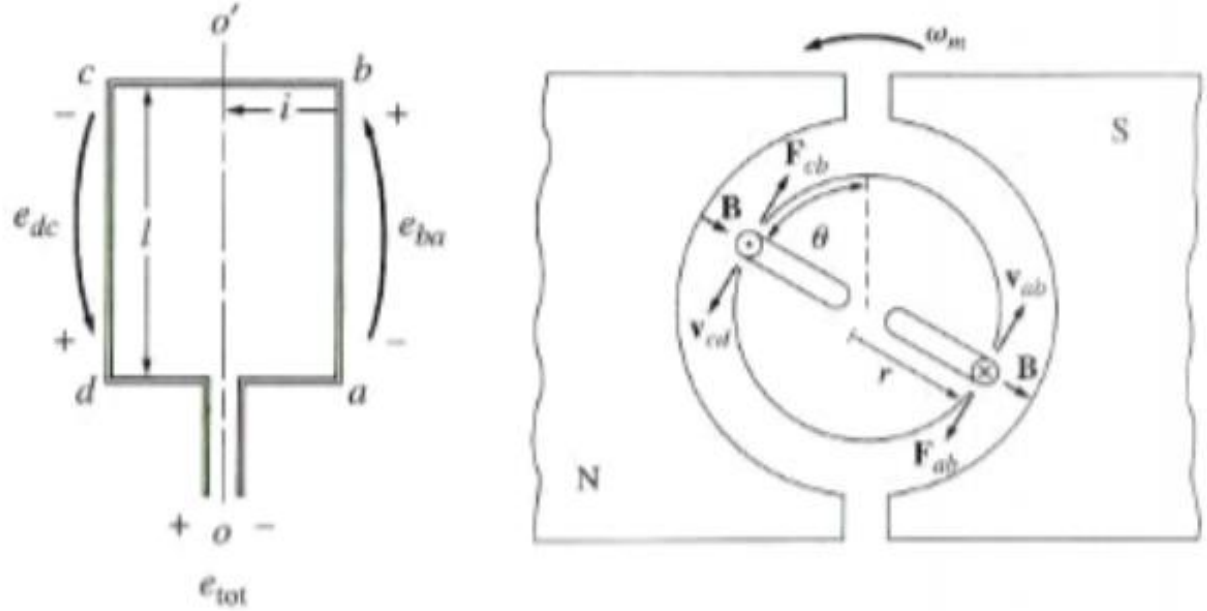
⇨ magnetic **flux density is** **constant** everywhere **under** the **pole surface**.

From Chapter 1, we know that $\Re_{gap} >> \Re_{iron}$, hence

⇨ magnetic flux takes **shortest possible path** through the air gap (i.e. **perpendicular** to the rotor surface).

The voltage induced in a rotating loop

If the rotor is rotated, a **voltage will be induced** in the wire loop. (Note: No voltage source applied to the rotor terminals.)



Top and front view of the simple dc machine

To determine the total voltage e_{tot} on the loop, each segment of the loop (as shown in the figure above) has to be examined separately and the resulting voltages summed up (see page 33).

The voltage on **each segment** is given by $e_{ind} = (\vec{v} \times \vec{B}) \bullet \vec{l}$

Hence, the **total induced voltage** in the loop:

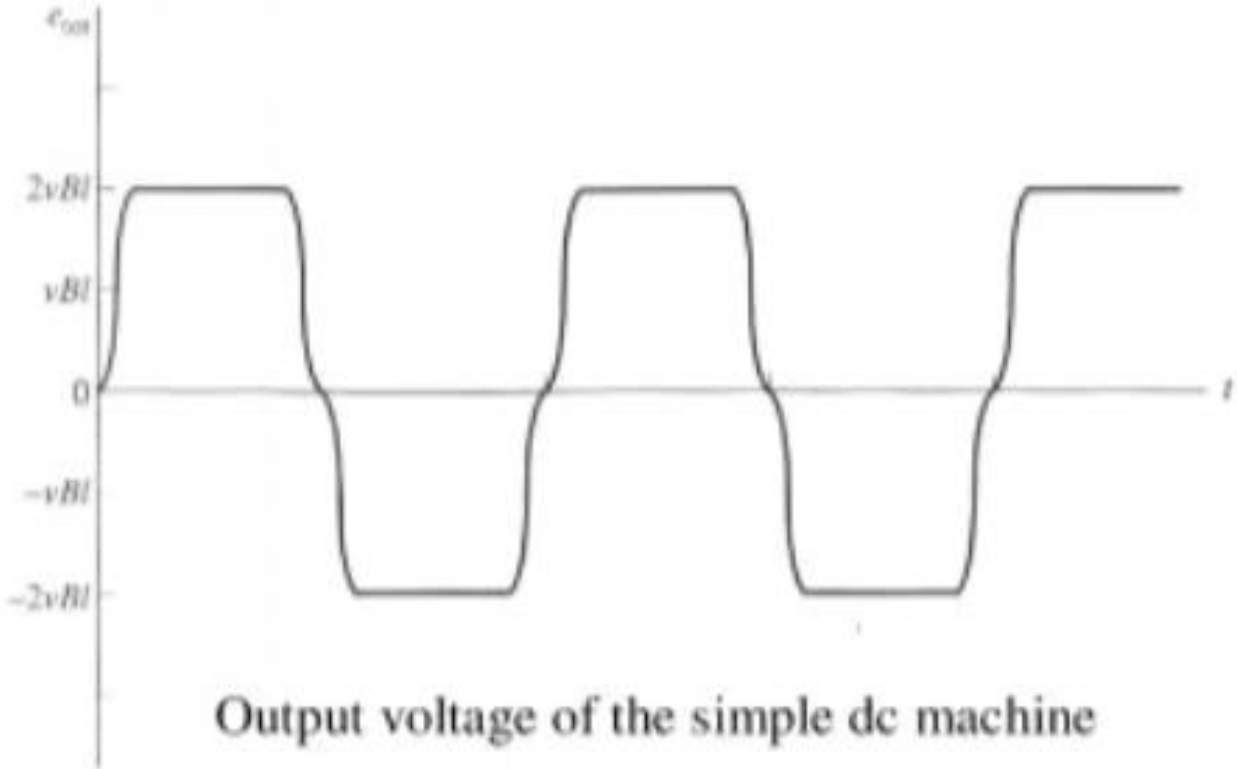
$$e_{ind} = e_{tot} = e_{ba} + e_{cb} + e_{dc} + e_{ad}$$

$$\therefore e_{ind} = \begin{cases} 2vBl & \text{under the pole faces} \\ 0 & \text{beyond the pole edges} \end{cases} \qquad (8.1)$$

When the **loop rotates through 180°**,

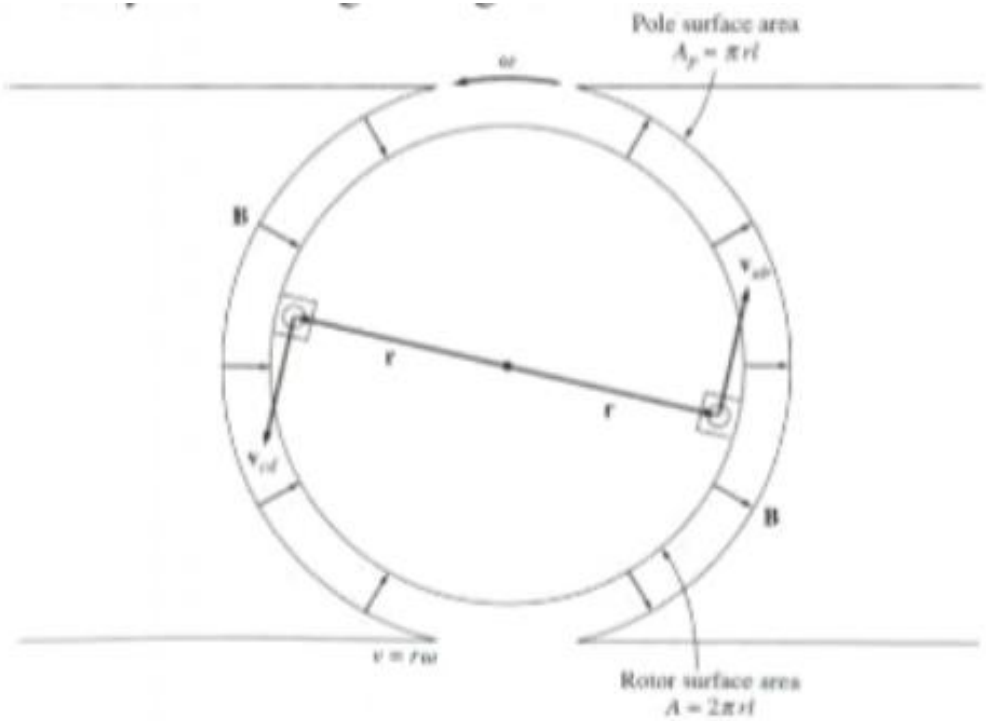
- segment *ab* is under the **north pole instead** of the south pole
- **direction of e_{ba} and e_{dc} _____ but magnitude remains constant**

The **resulting voltage** e_{tot} is shown in the figure below.



Output voltage of the simple dc machine

An **alternative expression** for e_{ind} that relates the behaviour of the single loop to the behaviour of larger, real dc machines can be obtained by examining the figure below:



Derivation of an alternative form of induced voltage equation.

The tangential velocity v of the loop edges can be expressed as $v = r\omega$. Therefore,

$$e_{ind} = \begin{cases} 2r\omega Bl & \text{under the pole face} \\ 0 & \text{beyond the pole edges} \end{cases}$$

For a 2-pole machine, if we assume that the gap between the poles is negligible (the gaps at the top and bottom of the diagram) then the surface area of the pole can be written as

$$A_p = \frac{2\pi rl}{2} = \pi rl$$

Therefore,

$$e_{ind} = \begin{cases} \frac{2}{\pi} A_p B\omega & \text{under the pole face} \\ 0 & \text{beyond the pole edges} \end{cases}$$

Since the air gap flux density B is constant everywhere under the pole faces, the total flux under each pole is:

$$\phi = A_p B$$

Thus, the **final form of the voltage equation** is:

$$e_{ind} = \begin{cases} \frac{2}{\pi}\phi\omega & \text{under the pole face} \\ 0 & \text{beyond the pole edges} \end{cases} \qquad (8.2)$$

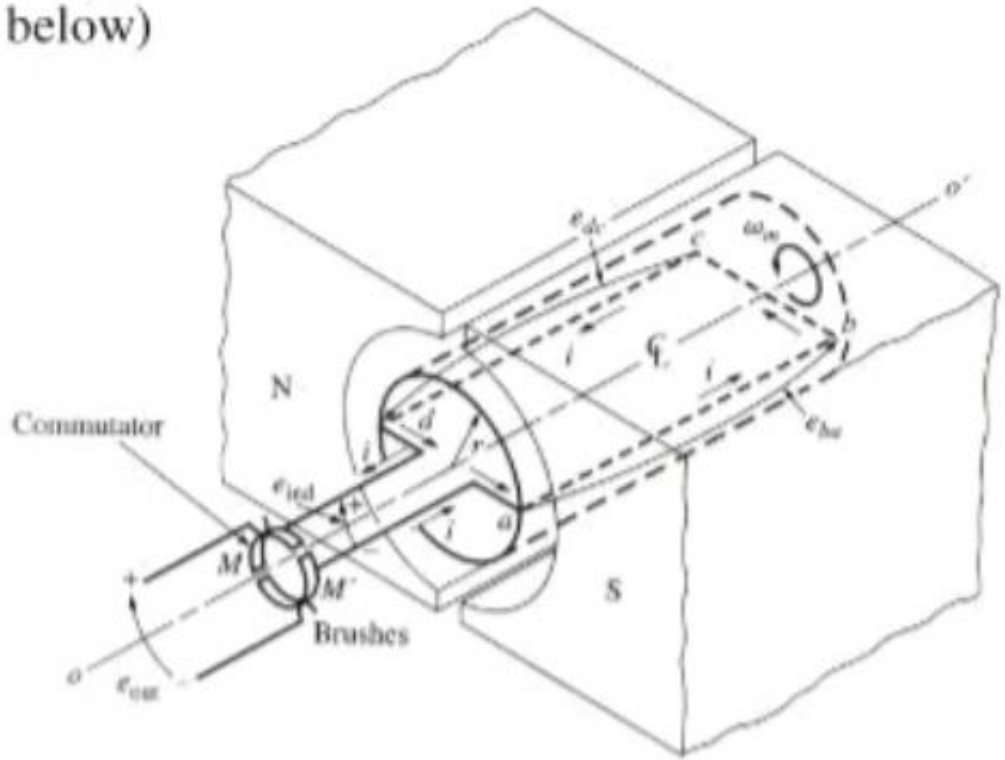
In general, the **voltage in any real machine will depend** on the same three factors:

- the **flux** in the machine
- the **speed** of rotation
- a constant representing the **machine construction**

Getting DC voltage out of the rotating loop

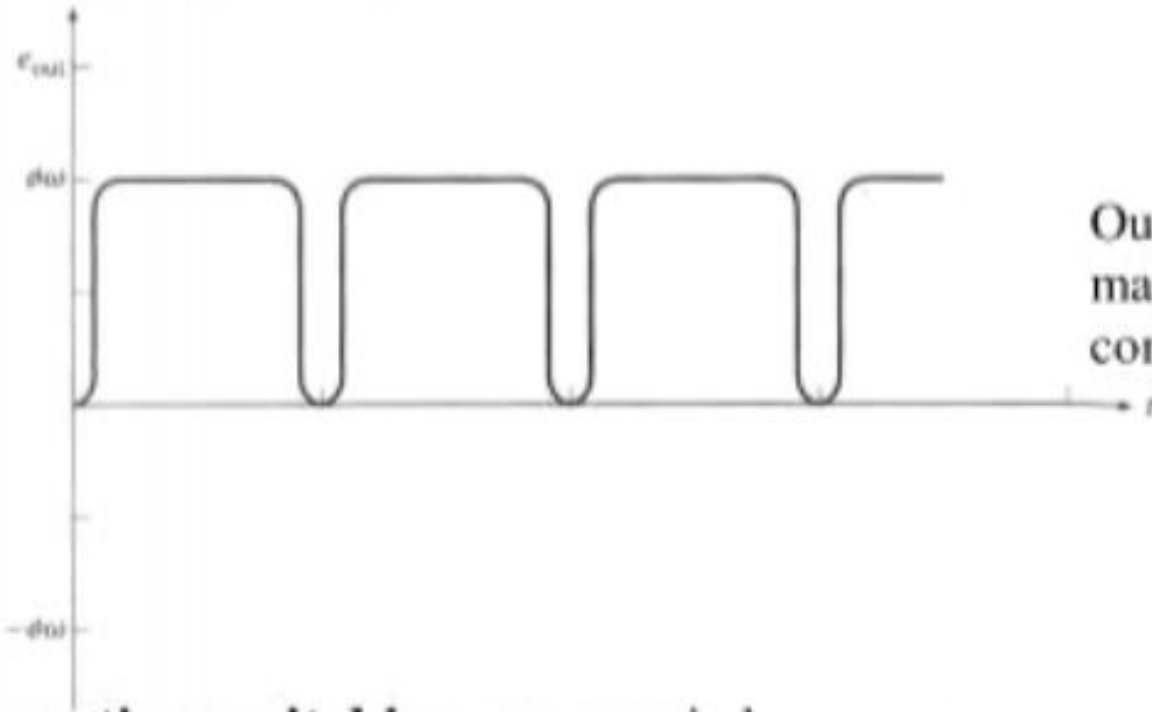
The voltage out of the loop is alternating between a constant positive value and a constant negative value, i.e. ac voltage.
How can this machine be modified to produce a dc voltage?

⇨ By using a mechanism called **commutator and brushes** (see below)



- two semicircular conducting segments (**commutator segments**) are added to the end of the loop
- two fixed contacts (**brushes**) are placed at an angle such that at the instant the voltage in the loop is zero, the contacts **short-circuit** the two segments

Every time voltage of the loop changes direction, the contacts also switch connections and the output of the contacts is always built up in the same way (as shown below).

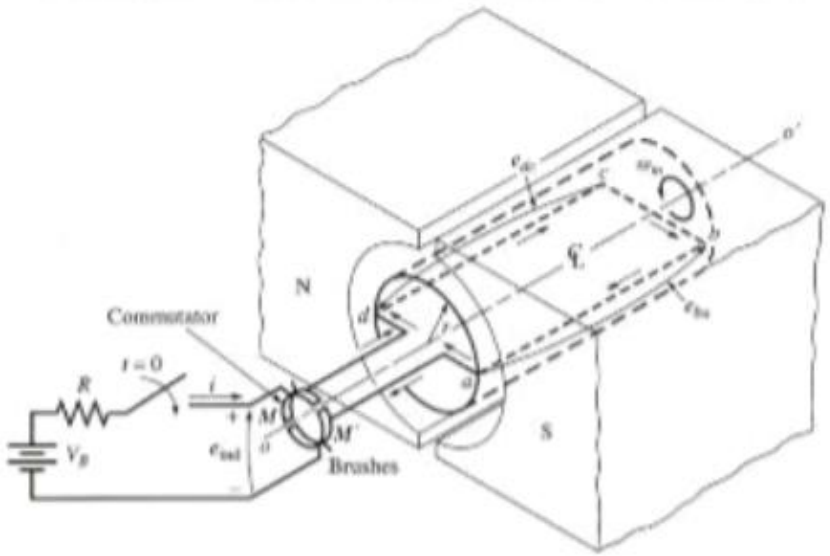


Output voltage of a dc machine with commutator and brushes.

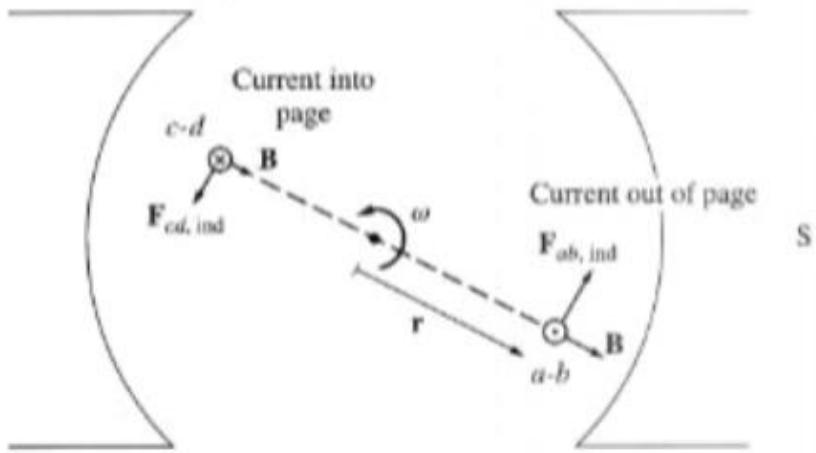
This **connection-switching process** is known as

The induced torque in the rotating loop

If the simple machine is connected to a battery, as shown below, **how much torque will be produced** in the stationary loop when the switch is closed and current is allowed to flow?



The simple dc machine connection for derivation of induced torque equation.



Front view of the machine (the iron core is omitted for clarity).

As before, the approach is to examine each segment of the loop and then sum the effects of all segments (see page 34).

The **force on each segment** is given by $\vec{F} = i(\vec{l} \times \vec{B})$.

And the **torque on the segment** is given by $T = \vec{r} \times \vec{F} = rF \sin\theta$, where θ is the angle between $\vec{r}$ and $\vec{F}$.

When the **loop is beyond the pole edges, $T = 0$** (since $\vec{B} = 0$).

Hence, the **total induced torque** in the loop is:

$$T_{ind} = T_{tot} = T_{ab} + T_{bc} + T_{cd} + T_{da}$$

$$\therefore T_{ind} = \begin{cases} 2rilB & \text{under the pole faces} \\ 0 & \text{beyond the pole edges} \end{cases} \qquad (8.3)$$

By employing the facts that $A_p \approx \pi r l$ and $\phi = A_p B$, the **torque expression** can be reduced to:

$$T_{ind} = \begin{cases} \frac{2}{\pi}\phi i & \text{under the pole face} \\ 0 & \text{beyond the pole edges} \end{cases} \qquad (8.4)$$

In general, the **torque in any real machine** will depend on the same three factors:

- the **flux** in the machine
- the **current** in the machine
- a constant representing the **machine construction**

Example 7–1. Figure 7–6 shows a simple rotating loop between curved pole faces connected to a battery and a resistor through a switch. The resistor shown models the total resistance of the battery and the wire in the machine. The physical dimensions and characteristics of this machine are

$$r = 0.5\ \text{m} \qquad l = 1.0\ \text{m}$$
$$R = 0.3\ \Omega \qquad B = 0.25\ \text{T}$$
$$V_B = 120\ \text{V}$$

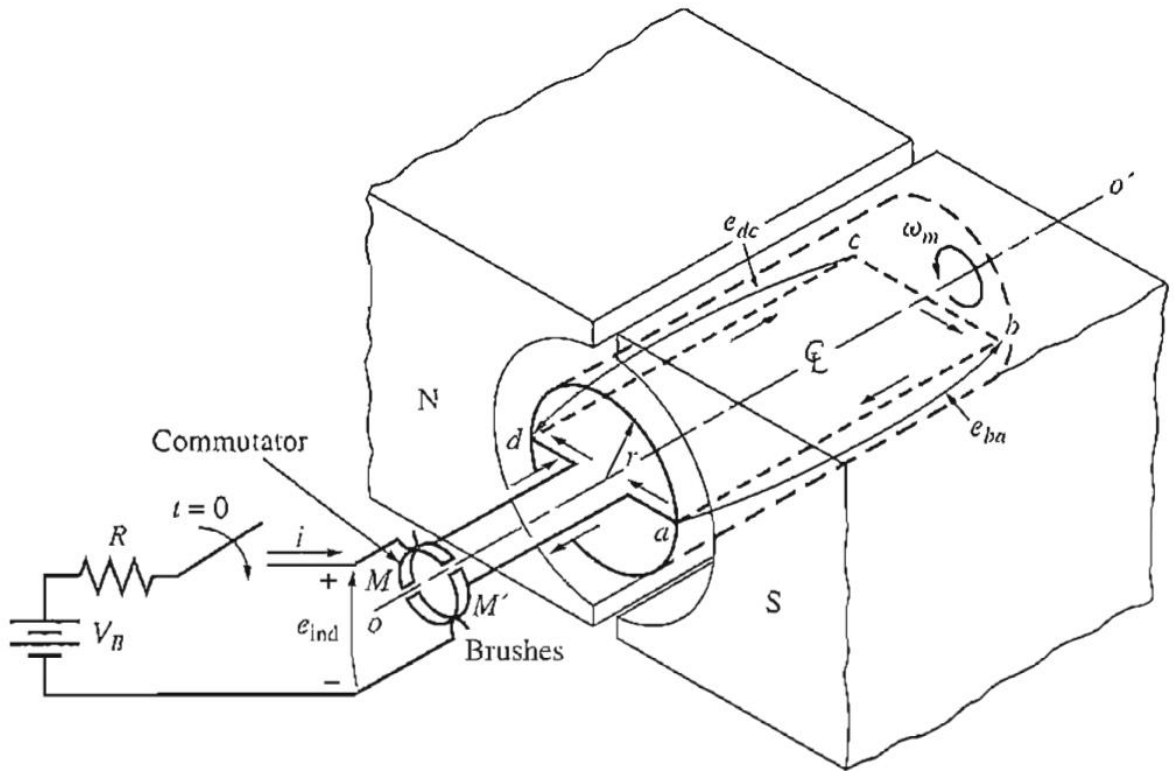
(a) What happens when the switch is closed?

(b) What is the machine's maximum starting current? What is its steady-state angular velocity at no load?

(c) Suppose a load is attached to the loop, and the resulting load torque is 10 N • m. What would the new steady-state speed be? How much power is supplied to the shaft of the machine? How much power is being supplied by the battery? Is this machine a motor or a generator?

(d) Suppose the machine is again unloaded, and a torque of 7.5 N • m is applied to the shaft in the direction of rotation. What is the new steady-state speed? Is this machine now a motor or a generator?

(e) Suppose the machine is running unloaded. What would the final steady-state speed of the rotor be if the flux density were reduced to 0.20 T?



Solution

(a) When the switch in Figure 7–6 is closed, a current will flow in the loop. Since the loop is initially stationary, $e_{ind} = 0$. Therefore, the current will be given by

$$i = \frac{V_B - e_{ind}}{R} = \frac{V_B}{R}$$

This current flows through the rotor loop, producing a torque

$$\tau_{ind} = \frac{2}{\pi} \phi i \qquad \text{CCW}$$

This induced torque produces an angular acceleration in a counterclockwise direction, so the rotor of the machine begins to turn. But as the rotor begins to turn, an induced voltage is produced in the motor, given by

$$e_{ind} = \frac{2}{\pi} \phi \omega_m$$

so the current i falls. As the current falls, $\tau_{ind} = (2/\pi)\phi i \downarrow$ decreases, and the machine winds up in steady state with $\tau_{ind} = 0$, and the battery voltage $V_B = e_{ind}$.

This is the same sort of starting behavior seen earlier in the linear dc machine.

(b) At *starting conditions,* the machine's current is

$$i = \frac{V_B}{R} = \frac{120 \text{ V}}{0.3 \ \Omega} = 400 \text{ A}$$

$$i = \frac{V_B}{R} = \frac{120\ \text{V}}{0.3\ \Omega} = 400\ \text{A}$$

At *no-load steady-state conditions,* the induced torque τ_{ind} must be zero. But $\tau_{\text{ind}} = 0$ implies that current i must equal zero, since $\tau_{\text{ind}} = (2/\pi)\phi i$, and the flux is nonzero. The fact that $i = 0$ A means that the battery voltage $V_B = e_{\text{ind}}$. Therefore, the speed of the rotor is

$$V_B = e_{\text{ind}} = \frac{2}{\pi}\phi\omega_m$$

$$\omega = \frac{V_B}{(2/\pi)\phi} = \frac{V_B}{2rlB}$$

$$= \frac{120\ \text{V}}{2(0.5\ \text{m})(1.0\ \text{m})(0.25\ \text{T})} = 480\ \text{rad/s}$$

(c) If a load torque of 10 N • m is applied to the shaft of the machine, it will begin to slow down. But as ω decreases, $e_{\text{ind}} = (2/\pi)\phi\,\omega\downarrow$ decreases and the rotor current increases $[i = (V_B - e_{\text{ind}}\downarrow)/R]$. As the rotor current increases, $|\tau_{\text{ind}}|$ increases too, until $|\tau_{\text{ind}}| = |\tau_{\text{load}}|$ at a lower speed ω.

At steady state, $|\tau_{\text{load}}| = |\tau_{\text{ind}}| = (2/\pi)\phi i$. Therefore,

$$i = \frac{\tau_{\text{ind}}}{(2/\pi)\phi} = \frac{\tau_{\text{ind}}}{2rlB}$$

$$= \frac{10\ \text{N} \cdot \text{m}}{(2)(0.5\ \text{m})(1.0\ \text{m})(0.25\ \text{T})} = 40\ \text{A}$$

By Kirchhoff's voltage law, $e_{\text{ind}} = V_B - iR$, so

$$e_{\text{ind}} = 120\ \text{V} - (40\ \text{A})(0.3\ \Omega) = 108\ \text{V}$$

Finally, the speed of the shaft is

$$\omega = \frac{e_{\text{ind}}}{(2/\pi)\phi} = \frac{e_{\text{ind}}}{2rlB}$$

$$= \frac{108\ \text{V}}{(2)(0.5\ \text{m})(1.0\ \text{m})(0.25\ \text{T})} = 432\ \text{rad/s}$$

The power supplied to the shaft is

$$P = \tau\omega_m$$

$$= (10\ \text{N} \cdot \text{m})(432\ \text{rad/s}) = 4320\ \text{W}$$

The power out of the battery is

$$P = V_B i = (120\ \text{V})(40\ \text{A}) = 4800\ \text{W}$$

This machine is operating as a *motor,* converting electric power to mechanical power.

(*d*) If a torque is applied in the direction of motion, the rotor accelerates. As the speed increases, the internal voltage e_{ind} increases and exceeds V_B, so the current flows out of the top of the bar and into the battery. This machine is now a *generator*. This current causes an induced torque opposite to the direction of motion. The induced torque opposes the external applied torque, and eventually $|\tau_{load}| = |\tau_{ind}|$ at a higher speed ω_m.

The current in the rotor will be

$$i = \frac{\tau_{ind}}{(2/\pi)\phi} = \frac{\tau_{ind}}{2rlB}$$

$$= \frac{7.5 \text{ N} \cdot \text{m}}{(2)(0.5 \text{ m})(1.0 \text{ m})(0.25 \text{ T})} = 30 \text{ A}$$

The induced voltage e_{ind} is

$$\begin{aligned} e_{ind} &= V_B + iR \\ &= 120 \text{ V} + (30 \text{ A})(0.3 \ \Omega) \\ &= 129 \text{ V} \end{aligned}$$

Finally, the speed of the shaft is

$$\omega = \frac{e_{ind}}{(2/\pi)\phi} = \frac{e_{ind}}{2rlB}$$

$$= \frac{129 \text{ V}}{(2)(0.5 \text{ m})(1.0 \text{ m})(0.25 \text{ T})} = 516 \text{ rad/s}$$

(e) Since the machine is initially unloaded at the original conditions, the speed $\omega_m = 480$ rad/s. If the flux decreases, there is a transient. However, after the transient is over, the machine must again have zero torque, since there is still no load on its shaft. If $\tau_{ind} = 0$, then the current in the rotor must be zero, and $V_B = e_{ind}$. The shaft speed is thus

$$\omega = \frac{e_{ind}}{(2/\pi)\phi} = \frac{e_{ind}}{2rlB}$$

$$= \frac{120 \text{ V}}{(2)(0.5 \text{ m})(1.0 \text{ m})(0.20 \text{ T})} = 600 \text{ rad/s}$$

Notice that when the flux in the machine is decreased, its speed increases. This is the same behavior seen in the linear machine and the same way that real dc motors behave.

The Rotor Coils

Regardless of the way in which the windings are connected to the commutator segments, most of the rotor windings themselves consist of diamond-shaped preformed coils which are inserted into the armature slots as a unit (see Figure 7–11). Each coil consists of a number of *turns* (loops) of wire, each turn taped and insulated from the other turns and from the rotor slot. Each side of a turn is called *a conductor*. The number of conductors on a machine's armature is given by

$$\boxed{Z = 2CN_C} \tag{7–22}$$

where Z = number of conductors on rotor

C = number of coils on rotor

N_C = number of turns per coil

Normally, a coil spans 180 electrical degrees. This means that when one side is under the center of a given magnetic pole, the other side is under the center of a pole of *opposite polarity*. The *physical* poles may not be located 180 mechanical degrees apart, but the magnetic field has completely reversed its polarity in traveling from under one pole to the next. The relationship between the electrical angle and mechanical angle in a given machine is given by

$$\theta_e = \frac{P}{2}\theta_m \tag{7–23}$$

where θ_e = electrical angle, in degrees
θ_m = mechanical angle, in degrees
P = number of magnetic poles on the machine

If a coil spans 180 electrical degrees, the voltages in the conductors on either side of the coil will be exactly the same in magnitude and opposite in direction at all times. Such a coil is called a *full-pitch coil.*

Sometimes a coil is built that spans less than 180 electrical degrees. Such a coil is called a *fractional-pitch coil*, and a rotor winding wound with fractional-pitch coils is called a *chorded winding*. The amount of chording in a winding is described by a *pitch factor p*, which is defined by the equation

$$p = \frac{\text{electrical angle of coil}}{180^\circ} \times 100\% \tag{7–24}$$

Sometimes a small amount of chording will be used in dc rotor windings to improve commutation.

Most rotor windings are *two-layer windings*, meaning that sides from two different coils are inserted into each slot. One side of each coil will be at the bottom of its slot, and the other side will be at the top of its slot. Such a construction requires the individual coils to be placed in the rotor slots by a very elaborate

Connections to the Commutator Segments

Once the windings are installed in the rotor slots, they must be connected to the commutator segments. There are a number of ways in which these connections can be made, and the different winding arrangements which result have different advantages and disadvantages.

The distance (in number of segments) between the commutator segments to which the two ends of a coil are connected is called the *commutator pitch* y_c. If the end of a coil (or a set number of coils, for wave construction) is connected to a commutator segment ahead of the one its beginning is connected to, the winding is called a *progressive winding* (see Figure 7–13a). If the end of a coil is connected to a commutator segment behind the one its beginning is connected to, the winding is called a *retrogressive winding* (see Figure 7–13b). If everything else is identical, the direction of rotation of a progressive-wound rotor will be opposite o the direction of rotation of a retrogressive-wound rotor.

Rotor (armature) windings are further classified according to the *plex* of their windings. A *simplex* rotor winding is a single, complete, closed winding wound on a rotor. A *duplex* rotor winding is a rotor with *two complete and independent sets* of rotor windings. If a rotor has a duplex winding, then each of the windings will be associated with every other commutator segment: One winding

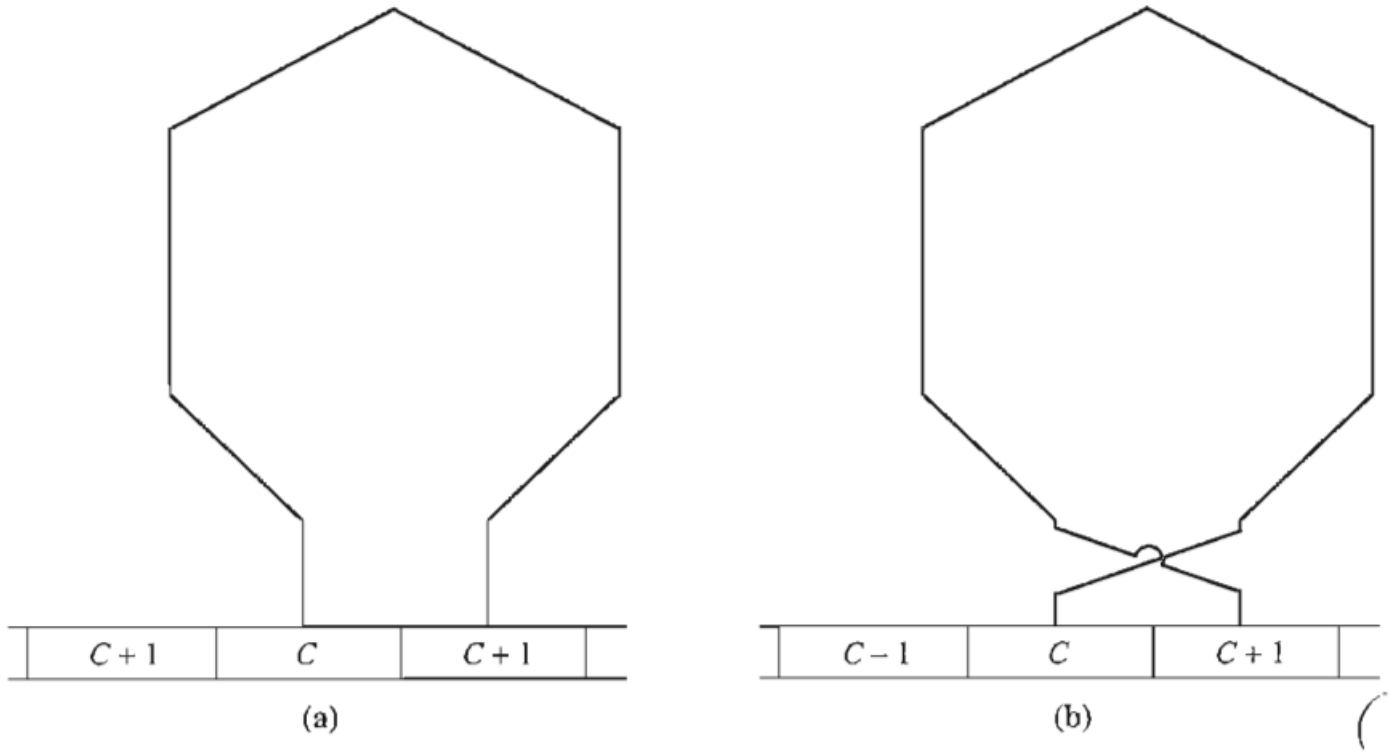
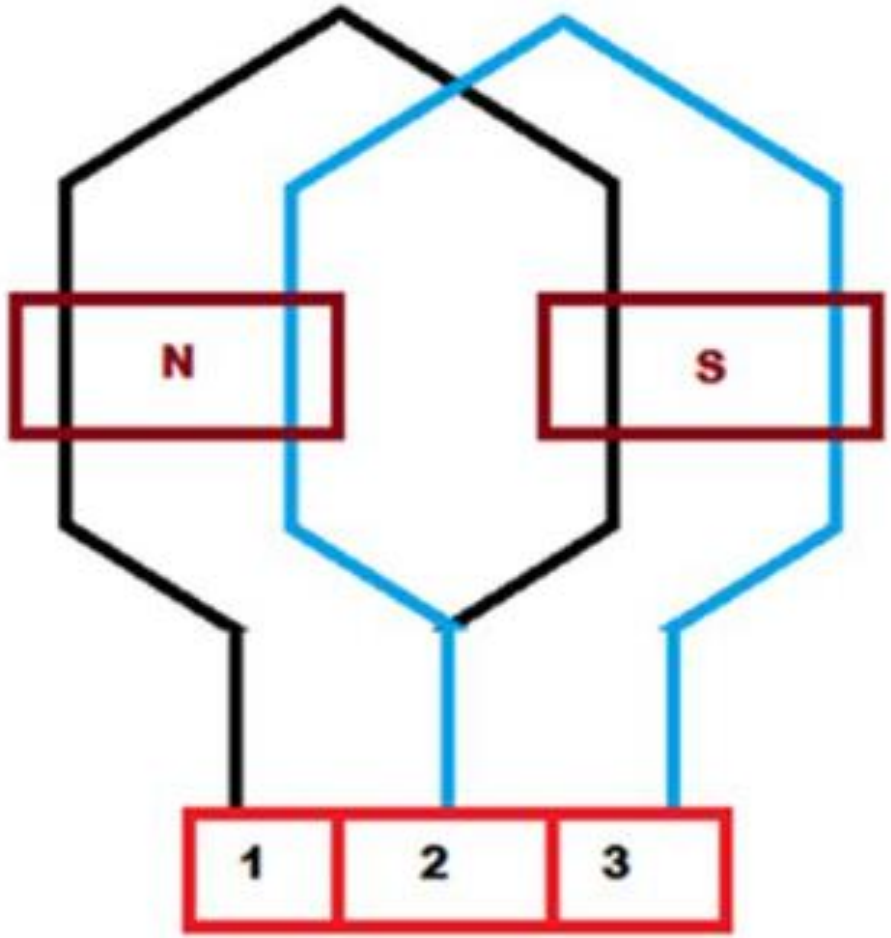


FIGURE 7–13
(a) A coil in a progressive rotor winding. (b) A coil in a retrogressive rotor winding.

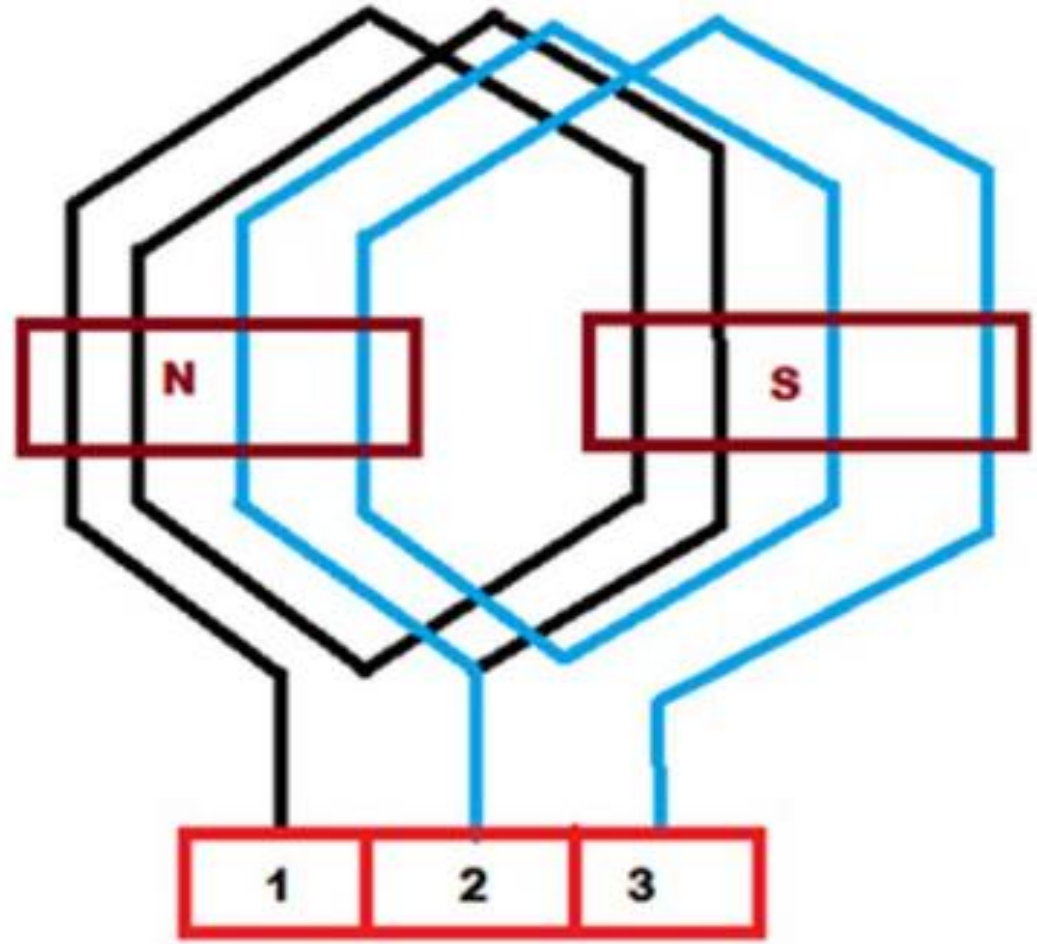
The Lap Winding

The simplest type of winding construction used in modern dc machines is the simplex *series* or *lap winding*. A simplex lap winding is a rotor winding consisting of coils containing one or more turns of wire with the two ends of each coil coming out at *adjacent commutator segments* (Figure 7–13). If the end of the coil is connected to the segment after the segment that the beginning of the coil is connected to, the winding is a progressive lap winding and $y_c = 1$; if the end of the coil is connected to the segment before the segment that the beginning of the coil is connected to, the winding is a retrogressive lap winding and $y_c = -1$. A simple two-pole machine with lap windings is shown in Figure 7–14.

An interesting feature of simplex lap windings is that *there are as many parallel current paths through the machine as there are poles on the machine.* If C is the number of coils and commutator segments present in the rotor and P is the



Simplex lap winding



Duplex lap winding

In general, for an m-plex lap winding, the commutator pitch y_c is

$$\boxed{y_c = \pm m} \qquad \text{lap winding} \tag{7–25}$$

and the number of current paths in a machine is

$$\boxed{a = mP} \qquad \text{lap winding} \tag{7–26}$$

where a = number of current paths in the rotor
m = plex of the windings (1, 2, 3, etc.)
P = number of poles on the machine

The Wave Winding

The general expression for commutator pitch in any simplex wave winding is

$$\boxed{y_c = \frac{2(C \pm 1)}{P}} \quad \text{simplex wave} \qquad (7\text{–}27)$$

$$\boxed{a = 2m} \quad \text{multiplex wave}$$

where C is the number of coils on the rotor and P is the number of poles on the machine. The plus sign is associated with progressive windings, and the minus sign is associated with retrogressive windings. A simplex wave winding is shown in Figure 7–21.

DC MOTORS AND GENERATORS

DC motors are often compared by their speed regulations. The *speed regulation* (SR) of a motor is defined by

$$\text{SR} = \frac{\omega_{\text{m,nl}} - \omega_{\text{m,fl}}}{\omega_{\text{m,fl}}} \times 100\% \tag{8–1}$$

$$\text{SR} = \frac{n_{\text{m,nl}} - n_{\text{m,fl}}}{n_{\text{m,fl}}} \times 100\% \tag{8–2}$$

It is a rough measure of the shape of a motor's torque–speed characteristic—a positive speed regulation means that a motor's speed drops with increasing load, and a negative speed regulation means a motor's speed increases with increasing load. The magnitude of the speed regulation tells approximately how steep the slope of the torque–speed curve is.

There are five major types of dc motors in general use:

1. The separately excited dc motor
2. The shunt dc motor
3. The permanent-magnet dc motor
4. The series dc motor
5. The compounded dc motor

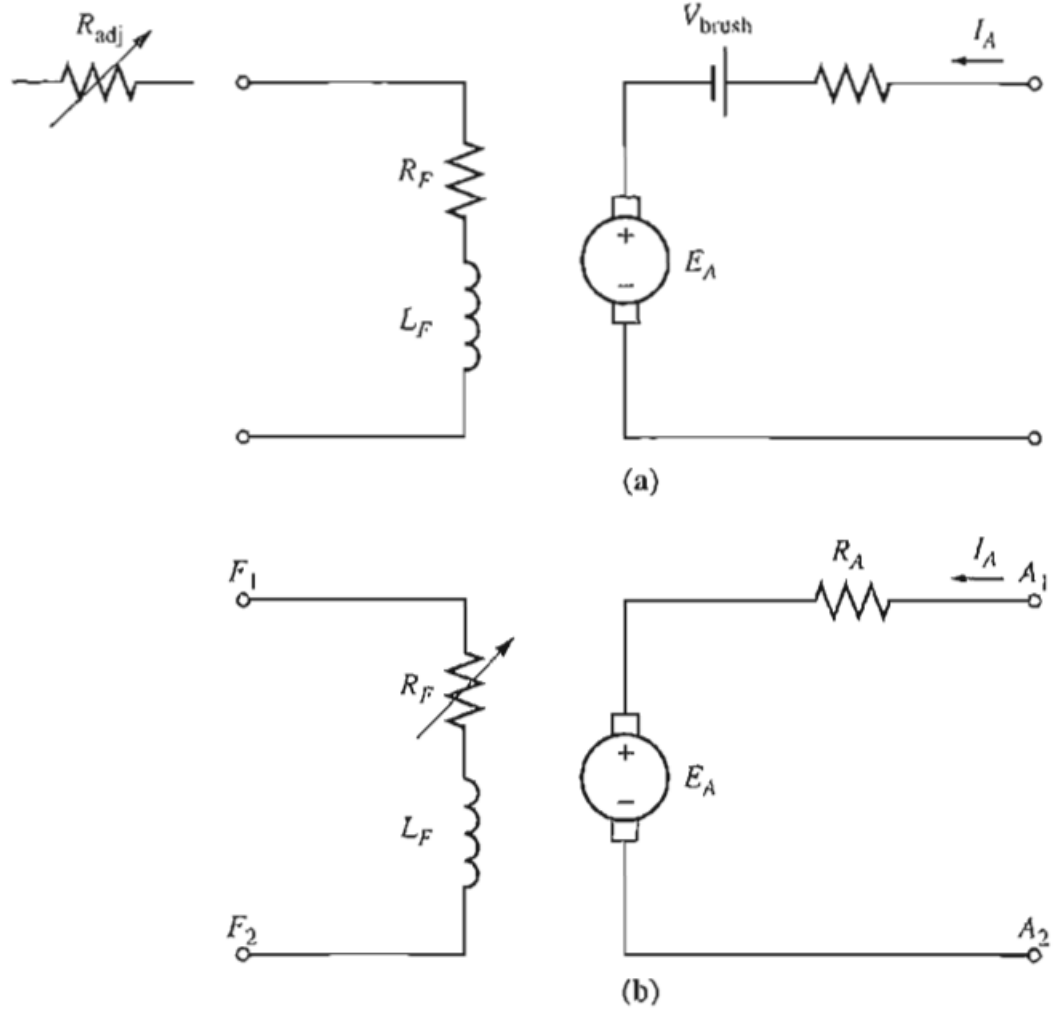


FIGURE 8–2
(a) The equivalent circuit of a dc motor. (b) A simplified equivalent circuit eliminating the brush voltage drop and combining R_{adj} with the field resistance.

The internal generated voltage in this machine is given by the equation

$$E_A = K\phi\omega_m \tag{7–38}$$

and the induced torque developed by the machine is given by

$$\tau_{ind} = K\phi I_A \tag{7–49}$$

These two equations, the Kirchhoff's voltage law equation of the armature circuit and the machine's magnetization curve, are all the tools necessary to analyze the behavior and performance of a dc motor.

The field current in a dc machine produces a field magnetomotive force given by $\mathcal{F} = N_F I_F$. This magnetomotive force produces a flux in the machine in accordance with its magnetization curve (Figure 8–3). Since the field current is directly proportional to the magnetomotive force and since E_A is directly proportional to the flux, it is customary to present the magnetization curve as a plot of E_A versus field current for a given speed ω_0 (Figure 8–4).

It is worth noting here that, to get the maximum possible power per pound of weight out of a machine, most motors and generators are designed to operate

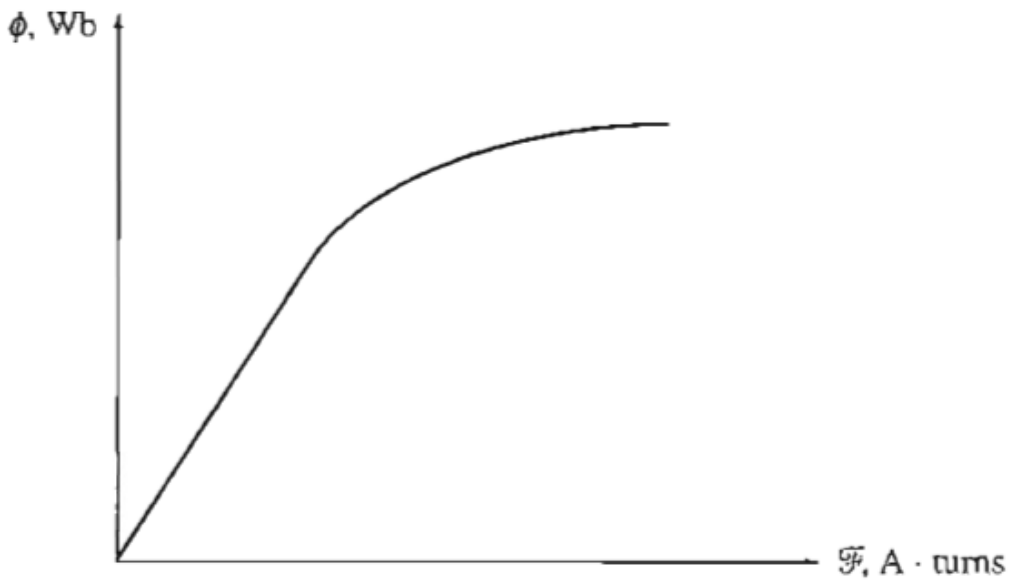


FIGURE 8–3
The magnetization curve of a ferromagnetic material (ϕ versus $\mathcal{F}$).

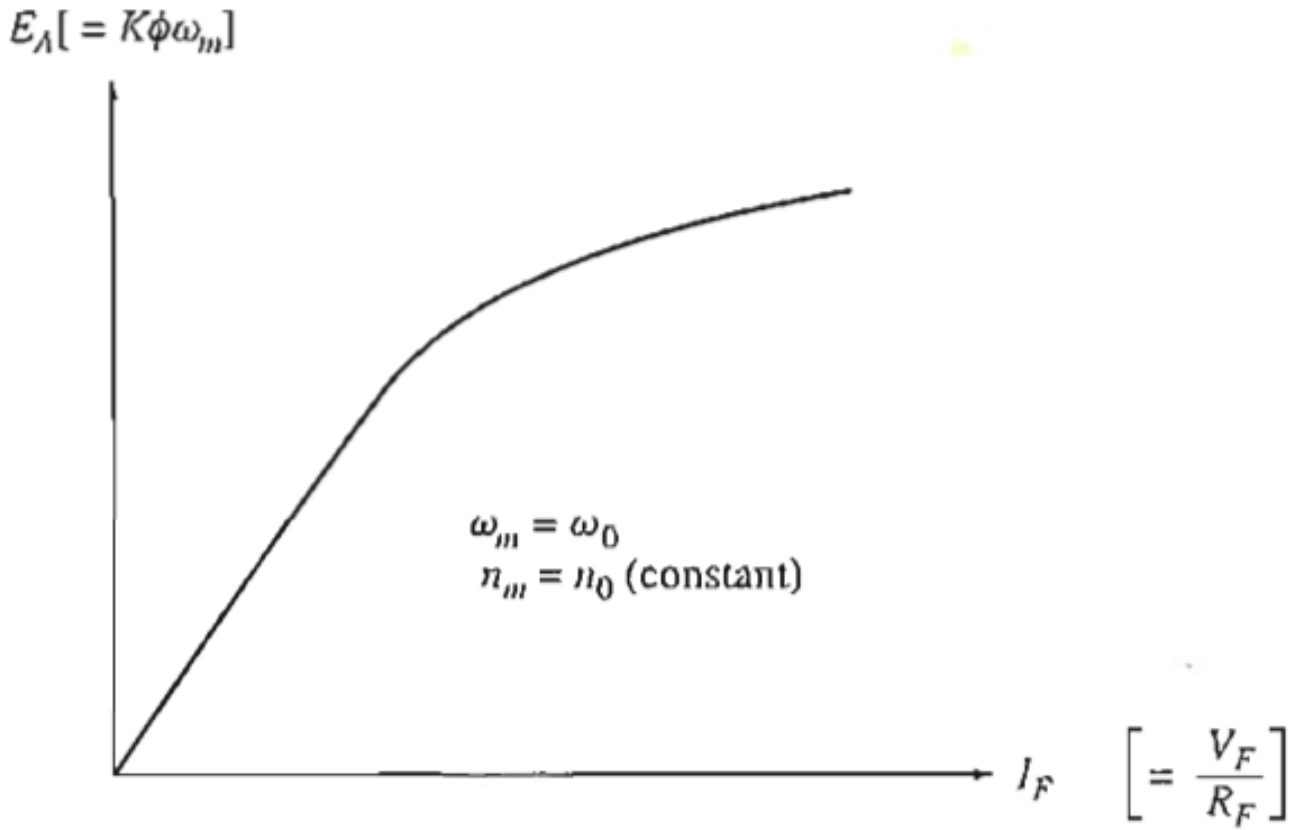
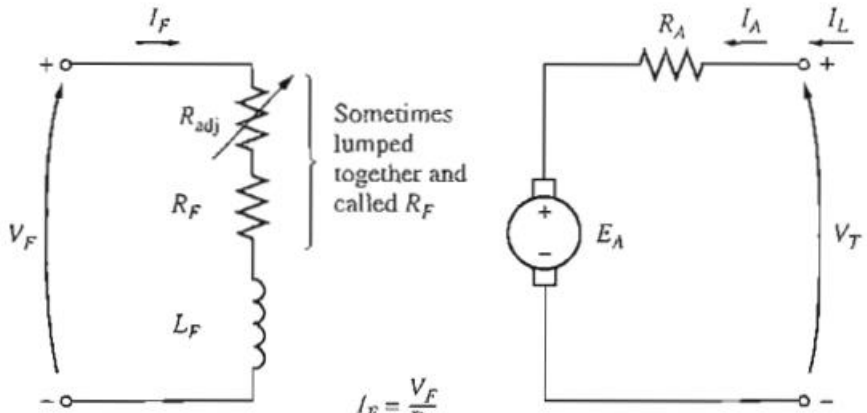


FIGURE 8–4
The magnetization curve of a dc machine expressed as a plot of E_A versus I_F, for a fixed speed ω_o.

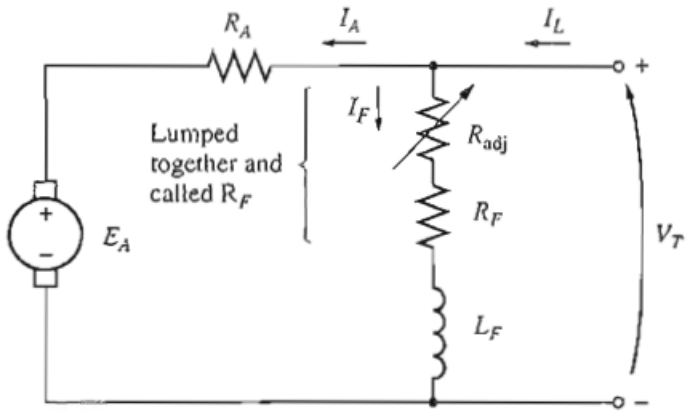
8.4 SEPARATELY EXCITED AND SHUNT DC MOTORS



$$I_F = \frac{V_F}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A$$



$$I_F = \frac{V_T}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A + I_F$$

$$\boxed{V_T = E_A + I_A R_A}$$

The Terminal Characteristic of a Shunt DC Motor

A terminal characteristic of a machine is a plot of the machine's output quantities versus each other. For a motor, the output quantities are shaft torque and speed, so the terminal characteristic of a motor is a plot of its output *torque versus speed.*

How does a shunt dc motor respond to a load? Suppose that the load on the shaft of a shunt motor is increased. Then the load torque τ_{load} will exceed the induced torque τ_{ind} in the machine, and the motor will start to slow down. When the motor slows down, its internal generated voltage drops ($E_A = K\phi\omega_m\downarrow$), so the armature current in the motor $I_A = (V_T - E_A\downarrow)/R_A$ increases. As the armature current rises, the induced torque in the motor increases ($\tau_{ind} = K\phi I_A\uparrow$), and finally the induced torque will equal the load torque at a lower mechanical speed of rotation ω_m.

The output characteristic of a shunt dc motor can be derived from the induced voltage and torque equations of the motor plus Kirchhoff's voltage law (KVL). The KVL equation for a shunt motor is

$$V_T = E_A + I_A R_A \tag{8–3}$$

The induced voltage $E_A = K\phi\omega_m$, so

$$V_T = K\phi\omega_m + I_A R_A \tag{8–4}$$

Since $\tau_{ind} = K\phi I_A$, current I_A can be expressed as

$$I_A = \frac{\tau_{ind}}{K\phi} \tag{8–5}$$

Combining Equations (8–4) and (8–5) produces

$$V_T = K\phi\omega_m + \frac{\tau_{ind}}{K\phi} R_A \tag{8–6}$$

Finally, solving for the motor's speed yields

$$\boxed{\omega_m = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2} \tau_{ind}} \tag{8–7}$$

Another effect *internal to the motor* that can also affect the shape of the torque–speed curve is armature reaction. If a motor has armature reaction, then as its load increases, the flux-weakening effects *reduce* its flux. As Equation (8–7) shows, the effect of a reduction in flux is to increase the motor's speed at any given load over the speed it would run at without armature reaction. The torque–speed characteristic of a shunt motor with armature reaction is shown in

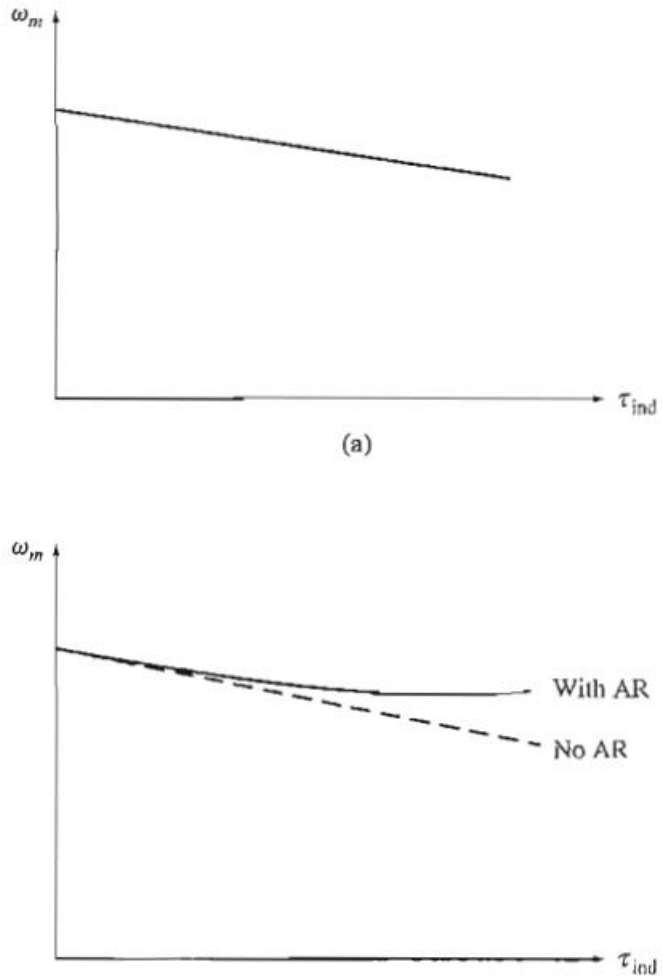


FIGURE 8–6
(a) Torque–speed characteristic of a shunt or separately excited dc motor with compensating windings to eliminate armature reaction. (b) Torque–speed characteristic of the motor with armature reaction present.

Figure 8–6b. If a motor has compensating windings, of course there will be no flux-weakening problems in the machine, and the flux in the machine will be constant.

If a shunt dc motor has compensating windings so that *its flux is constant regardless of load*, and the motor's speed and armature current are known at any one value of load, then it is possible to calculate its speed at any other value of load, as long as the armature current at that load is known or can be determined. Example 8–1 illustrates this calculation.

Example 8–1. A 50-hp, 250-V, 1200 r/min dc shunt motor with compensating windings has an armature resistance (including the brushes, compensating windings, and

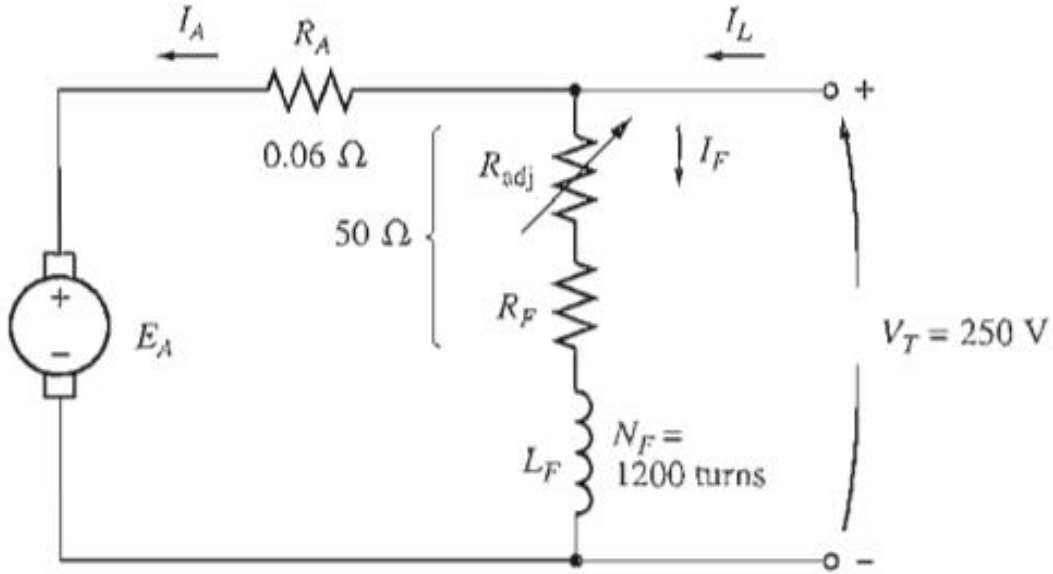


FIGURE 8–7
The shunt motor in Example 8–1.

interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_{adj} + R_F$ of 50 Ω, which produces a *no-load* speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding (see Figure 8–7).

(a) Find the speed of this motor when its input current is 100 A.
(b) Find the speed of this motor when its input current is 200 A.
(c) Find the speed of this motor when its input current is 300 A.
(d) Plot the torque–speed characteristic of this motor.

Solution

The internal generated voltage of a dc machine with its speed expressed in revolutions per minute is given by

$$E_A = K'\phi n_m \tag{7–41}$$

Since the field current in the machine is constant (because V_T and the field resistance are both constant), and since there are no armature reaction effects, *the flux in this motor is constant.* The relationship between the speeds and internal generated voltages of the motor at two different load conditions is thus

$$\frac{E_{A2}}{E_{A1}} = \frac{K'\phi n_{m2}}{K'\phi n_{m1}} \tag{8–8}$$

The constant K' cancels, since it is a constant for any given machine, and the flux ϕ cancels as described above. Therefore,

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} \tag{8–9}$$

At no load, the armature current is zero, so $E_{A1} = V_T = 250$ V, while the speed $n_{m1} = 1200$ r/min. If we can calculate the internal generated voltage at any other load, it will be possible to determine the motor speed at that load from Equation (8–9).

(a) If $I_L = 100$ A, then the armature current in the motor is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F}$$
$$= 100 \text{ A} - \frac{250 \text{ V}}{50 \ \Omega} = 95 \text{ A}$$

Therefore, E_A at this load will be

$$E_A = V_T - I_A R_A$$
$$= 250 \text{ V} - (95 \text{ A})(0.06 \ \Omega) = 244.3 \text{ V}$$

The resulting speed of the motor is

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{244.3 \text{ V}}{250 \text{ V}} 1200 \text{ r/min} = 1173 \text{ r/min}$$

(b) If $I_L = 200$ A, then the armature current in the motor is

$$I_A = 200 \text{ A} - \frac{250 \text{ V}}{50 \ \Omega} = 195 \text{ A}$$

Therefore, E_A at this load will be

$$E_A = V_T - I_A R_A$$
$$= 250 \text{ V} - (195 \text{ A})(0.06 \ \Omega) = 238.3 \text{ V}$$

The resulting speed of the motor is

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{238.3 \text{ V}}{250 \text{ V}} 1200 \text{ r/min} = 1144 \text{ r/min}$$

(c) If $I_L = 300$ A, then the armature current in the motor is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F}$$

$$= 300 \text{ A} - \frac{250 \text{ V}}{50 \ \Omega} = 295 \text{ A}$$

Therefore, E_A at this load will be

$$E_A = V_T - I_A R_A$$

$$= 250 \text{ V} - (295 \text{ A})(0.06 \ \Omega) = 232.3 \text{ V}$$

The resulting speed of the motor is

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{232.3 \text{ V}}{250 \text{ V}} 1200 \text{ r/min} = 1115 \text{ r/min}$$

(d) To plot the output characteristic of this motor, it is necessary to find the torque corresponding to each value of speed. At no load, the induced torque τ_{ind} is clearly zero. The induced torque for any other load can be found from the fact that power converted in a dc motor is

$$\boxed{P_{\text{conv}} = E_A I_A = \tau_{\text{ind}} \omega_m} \qquad (7\text{–}55, 7\text{–}56)$$

From this equation, the induced torque in a motor is

$$\tau_{\text{ind}} = \frac{E_A I_A}{\omega_m} \tag{8–10}$$

Therefore, the induced torque when $I_L = 100$ A is

$$\tau_{\text{ind}} = \frac{(244.3 \text{ V})(95 \text{ A})}{(1173 \text{ r/min})(1 \text{ min/60s})(2\pi \text{ rad/r})} = 190 \text{ N} \cdot \text{m}$$

The induced torque when $I_L = 200$ A is

$$\tau_{\text{ind}} = \frac{(238.3 \text{ V})(95 \text{ A})}{(1144 \text{ r/min})(1 \text{ min/60s})(2\pi \text{ rad/r})} = 388 \text{ N} \cdot \text{m}$$

The induced torque when $I_L = 300$ A is

$$\tau_{\text{ind}} = \frac{(232.3 \text{ V})(295 \text{ A})}{(1115 \text{ r/min})(1 \text{ min/60s})(2\pi \text{ rad/r})} = 587 \text{ N} \cdot \text{m}$$

The resulting torque–speed characteristic for this motor is plotted in Figure 8–8.

Nonlinear Analysis of a Shunt DC Motor

The flux ϕ and hence the internal generated voltage E_A of a dc machine is a *nonlinear* function of its magnetomotive force. Therefore, anything that changes the

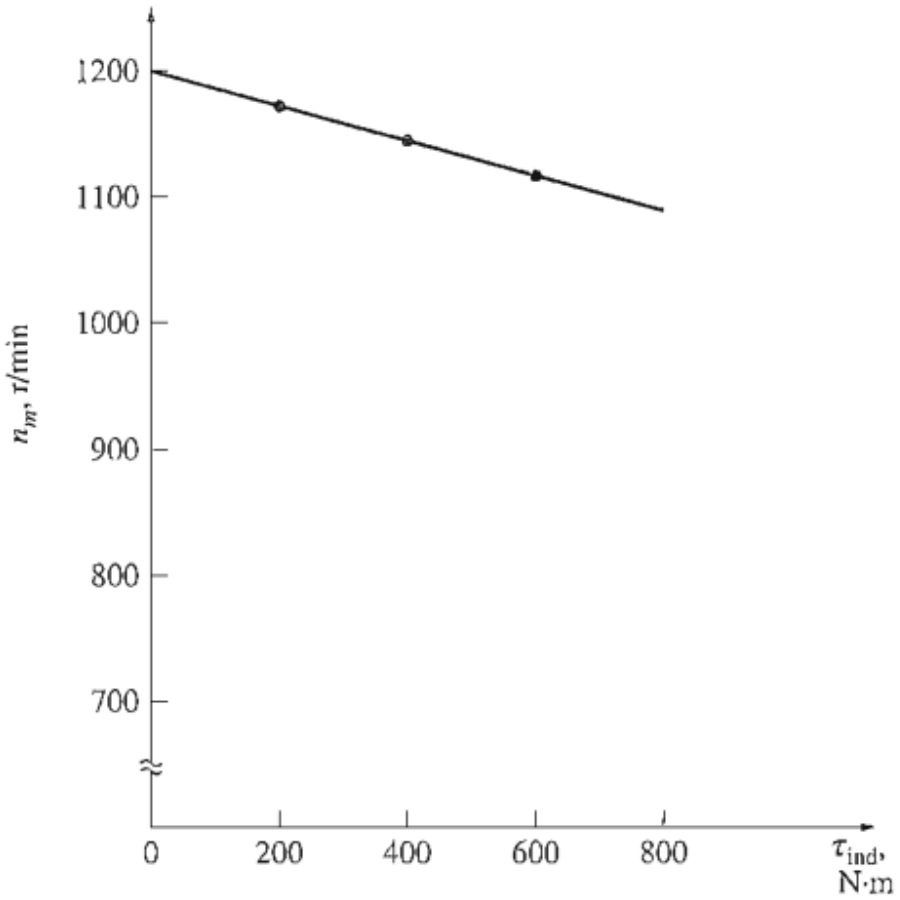


FIGURE 8–8
The torque–speed characteristic of the motor in Example 8–1.

$$\mathcal{F}_{net} = N_F I_F - \mathcal{F}_{AR} \tag{8–11}$$

$$I_F^* = I_F - \frac{\mathcal{F}_{AR}}{N_F} \tag{8–12}$$

The equation for the induced voltage in a dc machine when speed is expressed in revolutions per minute is

$$E_A = K'\phi n_m \tag{7–41}$$

For a given effective field current, the flux in a machine is fixed, so the internal generated voltage is related to speed by

$$\boxed{\frac{E_A}{E_{A0}} = \frac{n_m}{n_0}} \tag{8–13}$$

where E_{A0} and n_0 represent the reference values of voltage and speed, respectively. If the reference conditions are known from the magnetization curve and the actual E_A is known from Kirchhoff's voltage law, then it is possible to determine the actual speed n from Equation (8–13). The use of the magnetization curve and Equations (8–12) and (8–13) is illustrated in the following example, which analyzes a dc motor with armature reaction.

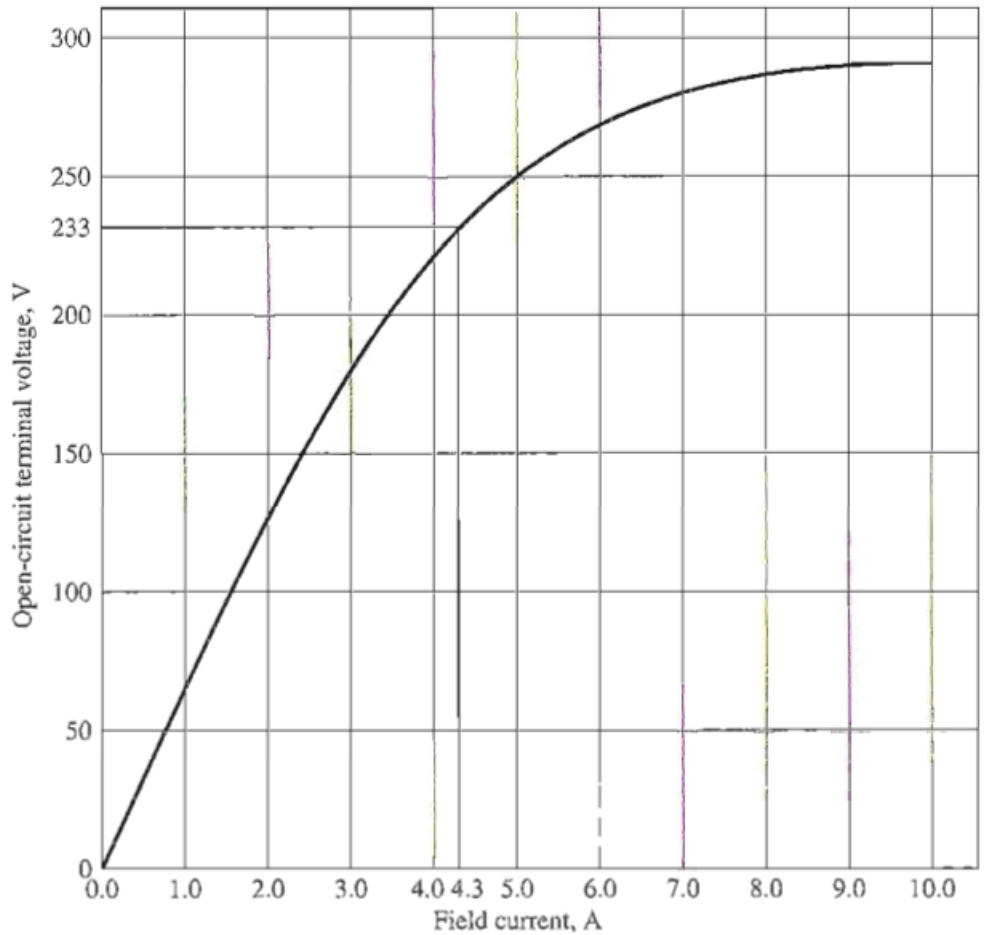


FIGURE 8–9
The magnetization curve of a typical 250-V dc motor, taken at a speed of 1200 r/min.

Example 8–2. A 50-hp, 250-V, 1200 r/min dc shunt motor *without* compensating windings has an armature resistance (including the brushes and interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_F + R_{adj}$ of 50 Ω, which produces a no-load speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding, and the armature reaction produces a demagnetizing magnetomotive force of 840 A • turns at a load current of 200 A. The magnetization curve of this machine is shown in Figure 8–9.

(a) Find the speed of this motor when its input current is 200 A.

(b) This motor is essentially identical to the one in Example 8–1 except for the absence of compensating windings. How does its speed compare to that of the previous motor at a load current of 200 A?

(c) Calculate and plot the torque–speed characteristic for this motor.

(a) If $I_L = 200$ A, then the armature current of the motor is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F}$$
$$= 200 \text{ A} - \frac{250 \text{ V}}{50 \ \Omega} = 195 \text{ A}$$

Therefore, the internal generated voltage of the machine is

$$E_A = V_T - I_A R_A$$
$$= 250 \text{ V} - (195 \text{ A})(0.06 \ \Omega) = 238.3 \text{ V}$$

At $I_L = 200$ A, the demagnetizing magnetomotive force due to armature reaction is 840 A • turns, so the effective shunt field current of the motor is

$$I_F^* = I_F - \frac{\mathcal{F}_{AR}}{N_F} \tag{8–12}$$
$$= 5.0 \text{ A} - \frac{840 \text{ A} \cdot \text{turns}}{1200 \text{ turns}} = 4.3 \text{ A}$$

From the magnetization curve, this effective field current would produce an internal generated voltage E_{A0} of 233 V at a speed n_0 of 1200 r/min.

We know that the internal generated voltage E_{A0} would be 233 V at a speed of 1200 r/min. Since the actual internal generated voltage E_A is 238.3 V, the actual operating speed of the motor must be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \tag{8–13}$$

$$n_m = \frac{E_A}{E_{A0}} n_0 = \frac{238.3 \text{ V}}{233 \text{ V}} (1200 \text{ r/min}) = 1227 \text{ r/min}$$

(*b*) At 200 A of load in Example 8–1, the motor's speed was $n_m = 1144$ r/min. In this example, the motor's speed is 1227 r/min. *Notice that the speed of the motor with armature reaction is higher than the speed of the motor with no armature reaction.* This relative increase in speed is due to the flux weakening in the machine with armature reaction.

Speed Control of Shunt DC Motors

How can the speed of a shunt dc motor be controlled? There are two common nethods and one less common method in use. The common methods have already been seen in the simple linear machine in Chapter 1 and the simple rotating loop in Chapter 7. The two common ways in which the speed of a shunt dc machine can be controlled are by

1. Adjusting the field resistance R_F (and thus the field flux)
2. Adjusting the terminal voltage applied to the armature.

The less common method of speed control is by

3. Inserting a resistor in series with the armature circuit.

CHANGING THE FIELD RESISTANCE. To understand what happens when the field resistor of a dc motor is changed, assume that the field resistor increases and observe the response. If the field resistance increases, then the field current decreases ($I_F = V_T / R_F \uparrow$), and as the field current decreases, the flux ϕ decreases with it. A decrease in flux causes an instantaneous decrease in the internal generated voltage $E_A (= K\phi\downarrow\omega_m)$, which causes a large increase in the machine's armature current, since

$$I_A\uparrow = \frac{V_T - E_A\downarrow}{R_A}$$

The induced torque in a motor is given by $\tau_{ind} = K\phi I_A$. Since the flux ϕ in this machine decreases while the current I_A increases, which way does the induced torque change? The easiest way to answer this question is to look at an example. Figure 8–11 shows a shunt dc motor with an internal resistance of 0.25 Ω. It is currently operating with a terminal voltage of 250 V and an internal generated voltage of 245 V. Therefore, the armature current flow is I_A = (250 V −

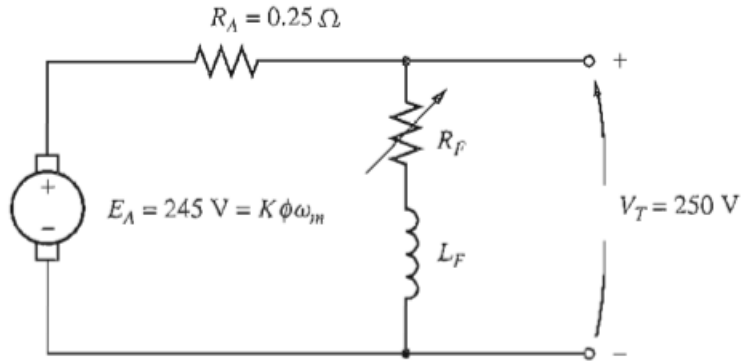


FIGURE 8–11
A 250-V shunt dc motor with typical values of E_A and R_A.

245 V)/0.25 Ω = 20 A. What happens in this motor *if there is a 1 percent decrease in flux?* If the flux decreases by 1 percent, then E_A must decrease by 1 percent too, because $E_A = K\phi\omega_m$. Therefore, E_A will drop to

$$E_{A2} = 0.99\ E_{A1} = 0.99(245\ \text{V}) = 242.55\ \text{V}$$

The armature current must then rise to

$$I_A = \frac{250\ \text{V} - 242.55\ \text{V}}{0.25\ \Omega} = 29.8\ \text{A}$$

Thus a 1 percent decrease in flux produced a 49 percent increase in armature current.

So to get back to the original discussion, the increase in current predominates over the decrease in flux, and the induced torque rises:

$$\tau_{\text{ind}} = K\overset{\downarrow}{\phi}\overset{\Uparrow}{I_A}$$

Since $\tau_{\text{ind}} > \tau_{\text{load}}$, the motor speeds up.

However, as the motor speeds up, the internal generated voltage E_A rises, causing I_A to fall. As I_A falls, the induced torque τ_{ind} falls too, and finally τ_{ind} again equals τ_{load} at a higher steady-state speed than originally.

To summarize the cause-and-effect behavior involved in this method of speed control:

1. Increasing R_F causes $I_F (= V_T / R_F \uparrow)$ to decrease.
2. Decreasing I_F decreases ϕ.
3. Decreasing ϕ lowers E_A $(= K\phi\downarrow\omega_m)$.
4. Decreasing E_A increases $I_A (= V_T - E_A \downarrow)/R_A$.
5. Increasing I_A increases $\tau_{ind} (= K\phi\downarrow I_A\Uparrow)$, with the change in I_A dominant over the change in flux).
6. Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$, and the speed ω_m increases.
7. Increasing ω_m increases $E_A = K\phi\omega_m\uparrow$ again.

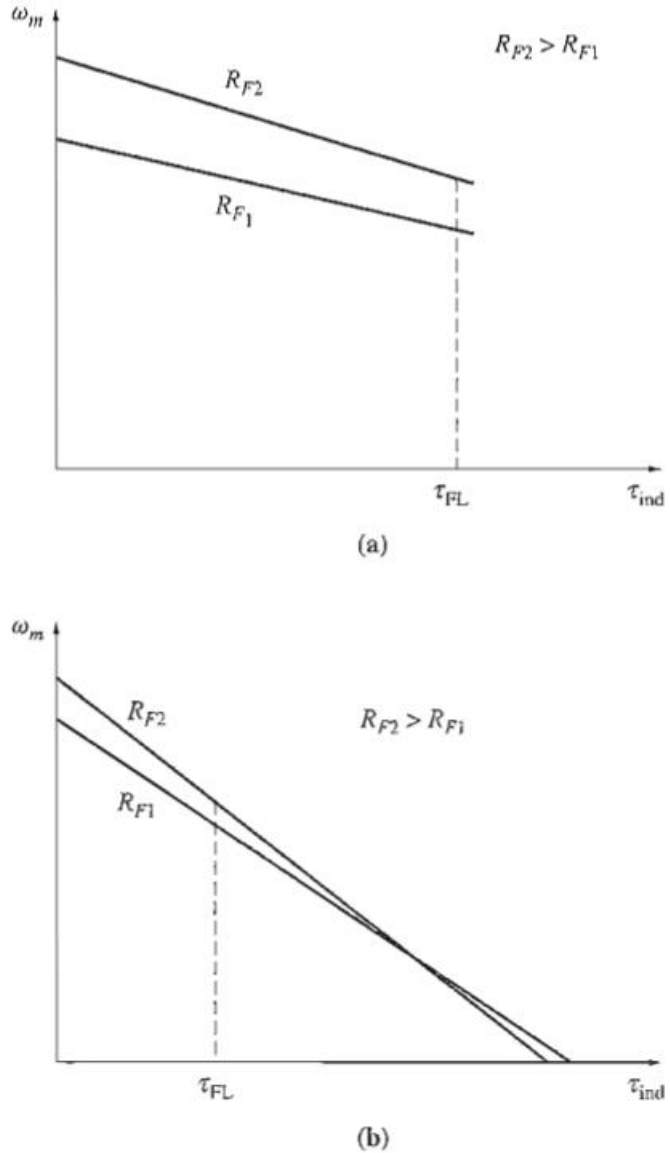


FIGURE 8–12
The effect of field resistance speed control on a shunt motor's torque–speed characteristic: (a) over the motor's normal operating range; (b) over the entire range from no-load to stall conditions.

The effect of increasing the field resistance on the output characteristic of a shunt motor is shown in Figure 8–12a. Notice that as the flux in the machine decreases, the no-load speed of the motor increases, while the slope of the torque–speed curve becomes steeper. Naturally, decreasing R_F would reverse the whole process, and the speed of the motor would drop.

8. Increasing E_A decreases I_A.
9. Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher speed ω_m.

CHANGING THE ARMATURE VOLTAGE. The second form of speed control involves changing the voltage applied to the armature of the motor *without changing the voltage applied to the field*. A connection similar to that in Figure 8–13 is necessary for this type of control. In effect, the motor must be *separately excited* to use armature voltage control.

If the voltage V_A is increased, then the armature current in the motor must rise [$I_A = (V_A \uparrow - E_A)/R_A$]. As I_A increases, the induced torque $\tau_{ind} = K\phi I_A \uparrow$ increases, making $\tau_{ind} > \tau_{load}$, and the speed ω_m of the motor increases.

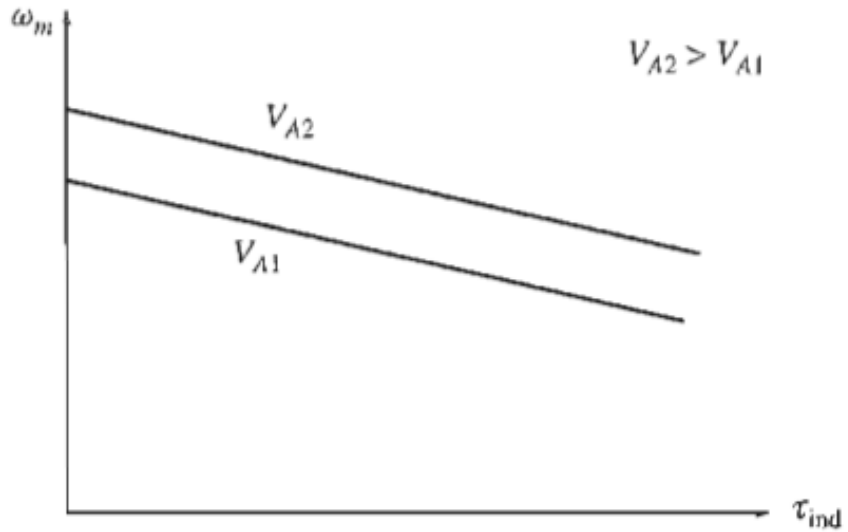


FIGURE 8–14
The effect of armature voltage speed control on a shunt motor's torque–speed characteristic.

But as the speed ω_m increases, the internal generated voltage $E_A(= K\phi\omega_m\uparrow)$ increases, causing the armature current to decrease. This decrease in I_A decreases the induced torque, causing τ_{ind} to equal τ_{load} at a higher rotational speed ω.

To summarize the cause-and-effect behavior in this method of speed control:

1. An increase in V_A increases I_A $[= (V_A \uparrow - E_A)/R_A]$.
2. Increasing I_A increases τ_{ind} $(= K\phi I_A\uparrow)$.
3. Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$ increasing ω_m.
4. Increasing ω_m increases $E_A(= K\phi\omega_m\uparrow)$.
5. Increasing E_A decreases I_A $[= (V_A \uparrow - E_A)/R_A]$.
6. Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher ω_m.

The effect of an increase in V_A on the torque–speed characteristic of a separately excited motor is shown in Figure 8–14. Notice that the no-load speed of the motor is shifted by this method of speed control, but the slope of the curve remains constant.

INSERTING A RESISTOR IN SERIES WITH THE ARMATURE CIRCUIT. If a resistor is inserted in series with the armature circuit, the effect is to drastically increase the slope of the motor's torque–speed characteristic, making it operate more slowly if loaded (Figure 8–15). This fact can easily be seen from Equation (8–7). The insertion of a resistor is a very wasteful method of speed control, since the losses in the inserted resistor are very large. For this reason, it is rarely used. It will be found only in applications in which the motor spends almost all its time operating at full speed or in applications too inexpensive to justify a better form of speed control.

The two most common methods of shunt motor speed control—field resistance variation and armature voltage variation—have different safe ranges of operation.

In field resistance control, the lower the field current in a shunt (or separately excited) dc motor, the faster it turns; and the higher the field current, the

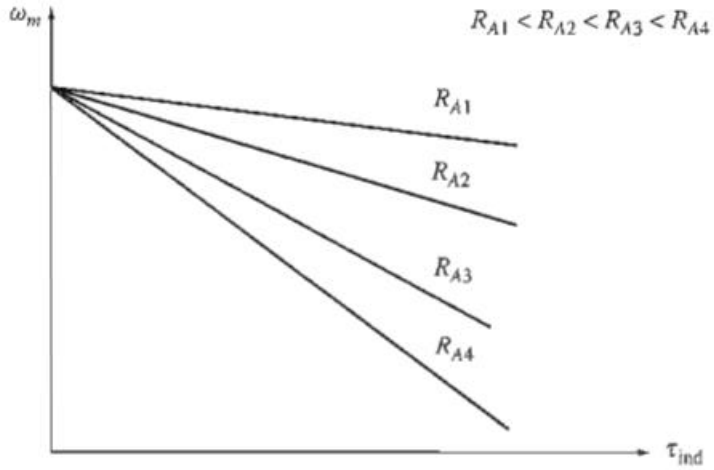


FIGURE 8–15
The effect of armature resistance speed control on a shunt motor's torque–speed characteristic.

slower it turns. Since an increase in field current causes a decrease in speed, there is always a minimum achievable speed by field circuit control. This minimum speed occurs when the motor's field circuit has the maximum permissible current flowing through it.

If a motor is operating at its rated terminal voltage, power, and field current, then it will be running at rated speed, also known as *base speed*. Field resistance control can control the speed of the motor for speeds above base speed but not for speeds below base speed. To achieve a speed slower than base speed by field circuit control would require excessive field current, possibly burning up the field windings.

In armature voltage control, the lower the armature voltage on a separately excited dc motor, the slower it turns; and the higher the armature voltage, the faster it turns. Since an increase in armature voltage causes an increase in speed, there is always a maximum achievable speed by armature voltage control. This maximum speed occurs when the motor's armature voltage reaches its maximum permissible level.

If the motor is operating at its rated voltage, field current, and power, it will be turning at base speed. Armature voltage control can control the speed of the motor for speeds below base speed but not for speeds above base speed. To achieve a speed faster than base speed by armature voltage control would require excessive armature voltage, possibly damaging the armature circuit.

These two techniques of speed control are obviously complementary. Armature voltage control works well for speeds below base speed, and field resistance or field current control works well for speeds above base speed. By combining the two speed-control techniques in the same motor, it is possible to get a ange of speed variations of up to 40 to 1 or more. Shunt and separately excited dc motors have excellent speed control characteristics.

There is a significant difference in the torque and power limits on the machine under these two types of speed control. The limiting factor in either case is the heating of the armature conductors, which places an upper limit on the magnitude of the armature current I_A.

For armature voltage control, *the flux in the motor is constant*, so the maximum torque in the motor is

$$\tau_{max} = K\phi I_{A,max} \tag{8–14}$$

This maximum torque is constant, regardless of the speed of the rotation of the motor. Since the power out of the motor is given by $P = \tau\omega$, the maximum power of the motor at any speed under armature voltage control is

$$P_{max} = \tau_{max}\omega_m \tag{8–15}$$

Thus *the maximum power out of the motor is directly proportional to its operating speed* under armature voltage control.

On the other hand, when field resistance control is used, the flux does change. In this form of control, a speed increase is caused by a decrease in the machine's flux. In order for the armature current limit not to be exceeded, the induced torque limit must decrease as the speed of the motor increases. Since the power out of the motor is given by $P = \tau\omega$, and the torque limit decreases as the speed of the motor increases, *the maximum power out of a dc motor under field current control is constant*, while *the maximum torque varies as the reciprocal of the motor's speed.*

These shunt dc motor power and torque limitations for safe operation as a function of speed are shown in Figure 8–16.

The following examples illustrate how to find the new speed of a dc motor if it is varied by field resistance or armature voltage control methods.

Example 8–3. Figure 8–17a shows a 100-hp, 250-V, 1200 r/min shunt dc motor with an armature resistance of 0.03 Ω and a field resistance of 41.67 Ω. The motor has compensating windings, so armature reaction can be ignored. Mechanical and core losses may be assumed to be negligible for the purposes of this problem. The motor is assumed to be driving a load with a line current of 126 A and an initial speed of 1103 r/min. To simplify the problem, assume that the amount of armature current drawn by the motor remains constant.

(*a*) If the machine's magnetization curve is shown in Figure 8–9, what is the motor's speed if the field resistance is raised to 50 Ω?

(*b*) Calculate and plot the speed of this motor as a function of the field resistance R_F assuming a constant-current load.

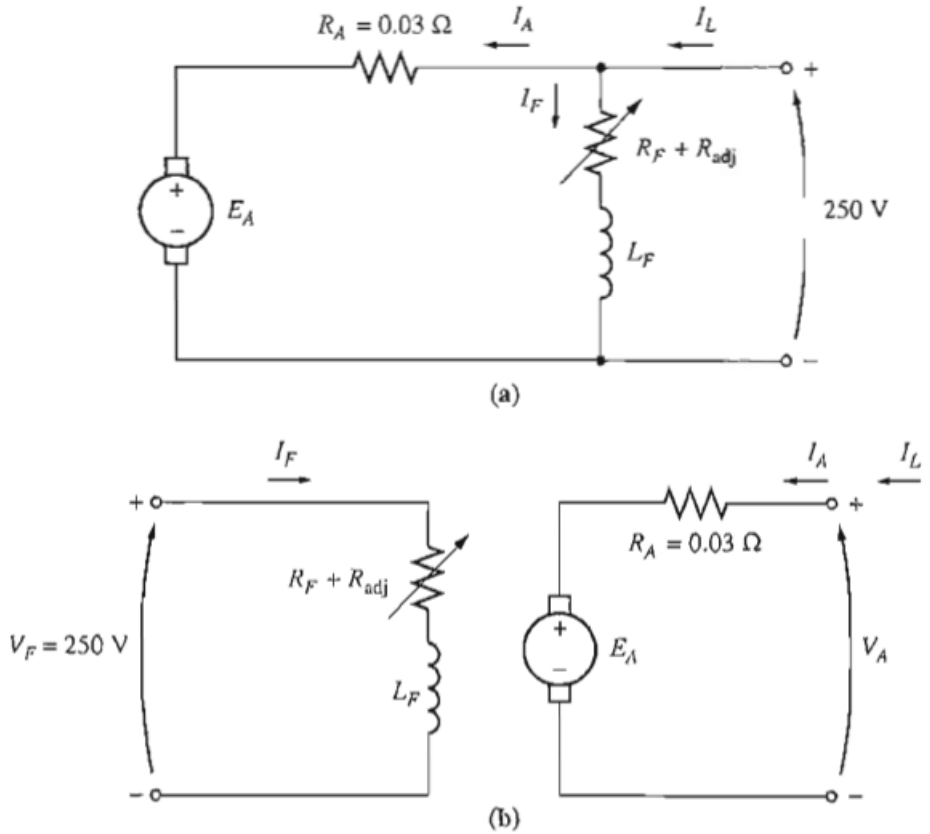


FIGURE 8–17
(a) The shunt motor in Example 8–3. (b) The separately excited dc motor in Example 8–4.

(a) The motor has an initial line current of 126 A, so the initial armature current is

$$I_{A1} = I_{L1} - I_{F1} = 126\ \text{A} - \frac{150\ \text{V}}{41.67\ \Omega} = 120\ \text{A}$$

Therefore, the internal generated voltage is

$$\begin{aligned} E_{A1} &= V_T - I_{A1}R_A = 250\ \text{V} - (120\ \text{A})(0.03\ \Omega) \\ &= 246.4\ \text{V} \end{aligned}$$

After the field resistance is increased to 50 Ω, the field current will become

$$I_{F2} = \frac{V_T}{R_F} = \frac{250\ \text{V}}{50\ \Omega} = 5\ \text{A}$$

The ratio of the internal generated voltage at one speed to the internal generated voltage at another speed is given by the ratio of Equation (7–41) at the two speeds:

$$\frac{E_{A2}}{E_{A1}} = \frac{K'\phi_2 n_{m2}}{K'\phi_1 n_{m1}} \tag{8–16}$$

Because the armature current is assumed constant, $E_{A1} = E_{A2}$, and this equation reduces to

$$1 = \frac{\phi_2 n_{m2}}{\phi_1 n_{m1}}$$

or

$$n_{m2} = \frac{\phi_1}{\phi_2} n_{m1} \tag{8–17}$$

A magnetization curve is a plot of E_A versus I_F for a given speed. Since the values of E_A on the curve are directly proportional to the flux, the ratio of the internal generated voltages read off the curve is equal to the ratio of the fluxes within the machine. At $I_F = 5$ A, $E_{A0} = 250$ V, while at $I_F = 6$ A, $E_{A0} = 268$ V. Therefore, the ratio of fluxes is given by

$$\frac{\phi_1}{\phi_2} = \frac{268 \text{ V}}{250 \text{ V}} = 1.076$$

and the new speed of the motor is

$$n_{m2} = \frac{\phi_1}{\phi_2} n_{m1} = (1.076)(1103 \text{ r/min}) = 1187 \text{ r/min}$$

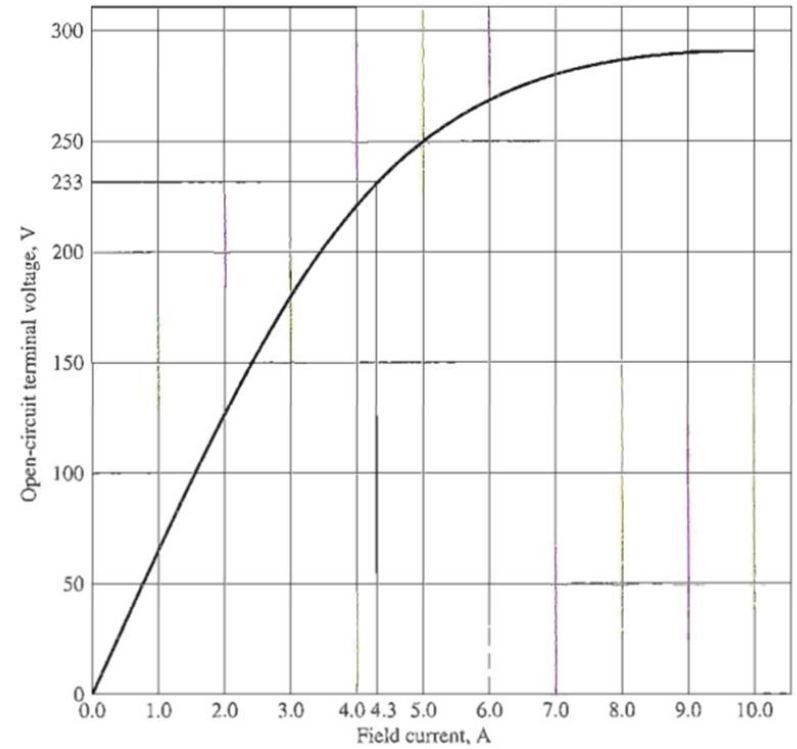


FIGURE 8–9
The magnetization curve of a typical 250-V dc motor, taken at a speed of 1200 r/min.

Example 8–4. The motor in Example 8–3 is now connected separately excited, as shown in Figure 8–17b. The motor is initially running with $V_A = 250$ V, $I_A = 120$ A, and $n = 1103$ r/min, while supplying a constant-torque load. What will the speed of this motor be if V_A is reduced to 200 V?

Solution

The motor has an initial line current of 120 A and an armature voltage V_A of 250 V, so the internal generated voltage E_A is

$$E_A = V_T - I_A R_A = 250\,\text{V} - (120\,\text{A})(0.03\,\Omega) = 246.4\,\text{V}$$

By applying Equation (8--16) and realizing that the flux ϕ is constant, the motor's speed can be expressed as

$$\frac{E_{A2}}{E_{A1}} = \frac{K'\phi_2 n_{m2}}{K'\phi_1 n_{m1}} \qquad (8\text{–}16)$$

$$= \frac{n_{m2}}{n_{m1}}$$

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1}$$

To find E_{A2} use Kirchhoff's voltage law:

$$E_{A2} = V_T - I_{A2} R_A$$

Since the torque is constant and the flux is constant, I_A is constant. This yields a voltage of

$$E_{A2} = 200\,\text{V} - (120\,\text{A})(0.03\,\Omega) = 196.4\,\text{V}$$

The final speed of the motor is thus

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{196.4\,\text{V}}{246.4\,\text{V}}\,1103\,\text{r/min} = 879\,\text{r/min}$$

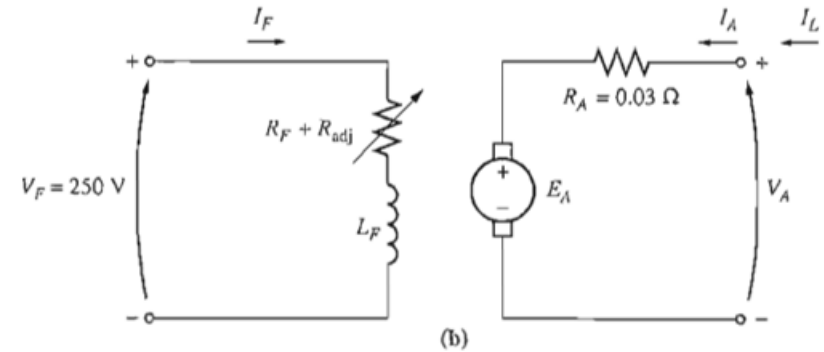


FIGURE 8–17
(a) The shunt motor in Example 8–3. (b) The separately excited dc motor in Example 8–4.

8.6 THE SERIES DC MOTOR

A series dc motor is a dc motor whose field windings consist of a relatively few turns connected in series with the armature circuit. The equivalent circuit of a series dc motor is shown in Figure 8–20. In a series motor, the armature current, field current, and line current are all the same. The Kirchhoff's voltage law equation for this motor is

$$V_T = E_A + I_A(R_A + R_S) \tag{8–18}$$

Induced Torque in a Series DC Motor

The terminal characteristic of a series dc motor is very different from that of the shunt motor previously studied. The basic behavior of a series dc motor is due to the fact that *the flux is directly proportional to the armature current*, at least until saturation is reached. As the load on the motor increases, its flux increases too. As seen earlier, an increase in flux in the motor causes a decrease in its speed. The result is that a series motor has a sharply drooping torque–speed characteristic.

The induced torque in this machine is given by Equation (7–49):

$$\tau_{\text{ind}} = K\phi I_A \tag{7–49}$$

The flux in this machine is directly proportional to its armature current (at least until the metal saturates). Therefore, the flux in the machine can be given by

$$\phi = cI_A \tag{8–19}$$

where c is a constant of proportionality. The induced torque in this machine is thus given by

$$\tau_{\text{ind}} = K\phi I_A = KcI_A^2 \tag{8–20}$$

In other words, the torque in the motor is proportional to the square of its armature current. As a result of this relationship, it is easy to see that a series motor gives more torque per ampere than any other dc motor. It is therefore used in applications requiring very high torques. Examples of such applications are the starter motors in cars, elevator motors, and tractor motors in locomotives.

The Terminal Characteristic of a Series DC Motor

To determine the terminal characteristic of a series dc motor, an analysis will be based on the assumption of a linear magnetization curve, and then the effects of saturation will be considered in a graphical analysis.

The assumption of a linear magnetization curve implies that the flux in the motor will be given by Equation (8–19):

$$\phi = cI_A \tag{8–19}$$

This equation will be used to derive the torque–speed characteristic curve for the series motor.

The derivation of a series motor's torque–speed characteristic starts with Kirchhoff's voltage law:

$$V_T = E_A + I_A(R_A + R_S) \tag{8–18}$$

From Equation (8–20), the armature current can be expressed as

$$I_A = \sqrt{\frac{\tau_{ind}}{Kc}}$$

Also, $E_A = K\phi\omega_m$. Substituting these expressions in Equation (8–18) yields

$$V_T = K\phi\omega_m + \sqrt{\frac{\tau_{ind}}{Kc}}(R_A + R_S) \tag{8–21}$$

If the flux can be eliminated from this expression, it will directly relate the torque of a motor to its speed. To eliminate the flux from the expression, notice that

$$I_A = \frac{\phi}{c}$$

and the induced torque equation can be rewritten as

$$\tau_{ind} = \frac{K}{c}\phi^2$$

Therefore, the flux in the motor can be rewritten as

$$\phi = \sqrt{\frac{c}{K}}\sqrt{\tau_{ind}} \tag{8–22}$$

Substituting Equation (8–22) into Equation (8–21) and solving for speed yields

$$V_T = K\sqrt{\frac{c}{K}}\sqrt{\tau_{ind}}\omega_m + \sqrt{\frac{\tau_{ind}}{Kc}}(R_A + R_S)$$

$$\sqrt{Kc}\sqrt{\tau_{ind}}\omega_m = V_T - \frac{R_A + R_S}{\sqrt{Kc}}\sqrt{\tau_{ind}}$$

$$\omega_m = \frac{V_T}{\sqrt{Kc}\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$

The resulting torque–speed relationship is

$$\omega_m = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc} \tag{8–23}$$

Notice that for an unsaturated series motor, the speed of the motor varies as the reciprocal of the square root of the torque. That is quite an unusual relationship! This ideal torque–speed characteristic is plotted in Figure 8–21.

One disadvantage of series motors can be seen immediately from this equation. When the torque on this motor goes to zero, its speed goes to infinity. In practice, the torque can never go entirely to zero because of the mechanical, core, and stray losses that must be overcome. However, if no other load is connected to the motor, it can turn fast enough to seriously damage itself. *Never* completely unload a series motor, and never connect one to a load by a belt or other mechanism that could break. If that were to happen and the motor were to become unloaded while running, the results could be serious.

The nonlinear analysis of a series dc motor with magnetic saturation effects, but ignoring armature reaction, is illustrated in Example 8–5.

Example 8–5. Figure 8–20 shows a 250-V series dc motor with compensating windings, and a total series resistance $R_A + R_S$ of 0.08 Ω. The series field consists of 25 turns per pole, with the magnetization curve shown in Figure 8–22.

(a) Find the speed and induced torque of this motor for when its armature current is 50 A.

(b) Calculate and plot the torque–speed characteristic for this motor.

Solution

(a) To analyze the behavior of a series motor with saturation, pick points along the operating curve and find the torque and speed for each point. Notice that the magnetization curve is given in units of magnetomotive force (ampere-turns) versus E_A for a speed of 1200 r/min, so calculated E_A values must be compared to the equivalent values at 1200 r/min to determine the actual motor speed.

For $I_A = 50$ A,

$$E_A = V_T - I_A(R_A + R_S) = 250\ \text{V} - (50\text{A})(0.08\ \Omega) = 246\ \text{V}$$

Since $I_A = I_F = 50$ A, the magnetomotive force is

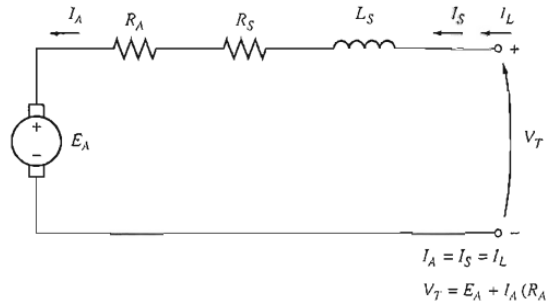
$$\mathcal{F} = NI = (25\ \text{turns})(50\ \text{A}) = 1250\ \text{A} \cdot \text{turns}$$

From the magnetization curve at $\mathcal{F} = 1250$ A • turns, $E_{A0} = 80$ V. To get the correct speed of the motor, remember that, from Equation (8–13),

$$n_m = \frac{E_A}{E_{A0}} n_0$$

$$= \frac{246\ \text{V}}{80\ \text{V}} 120\ \text{r/min} = 3690\ \text{r/min}$$

To find the induced torque supplied by the motor at that speed, recall that $P_{\text{conv}} = E_A I_A = \tau_{\text{ind}} \omega_m$. Therefore,



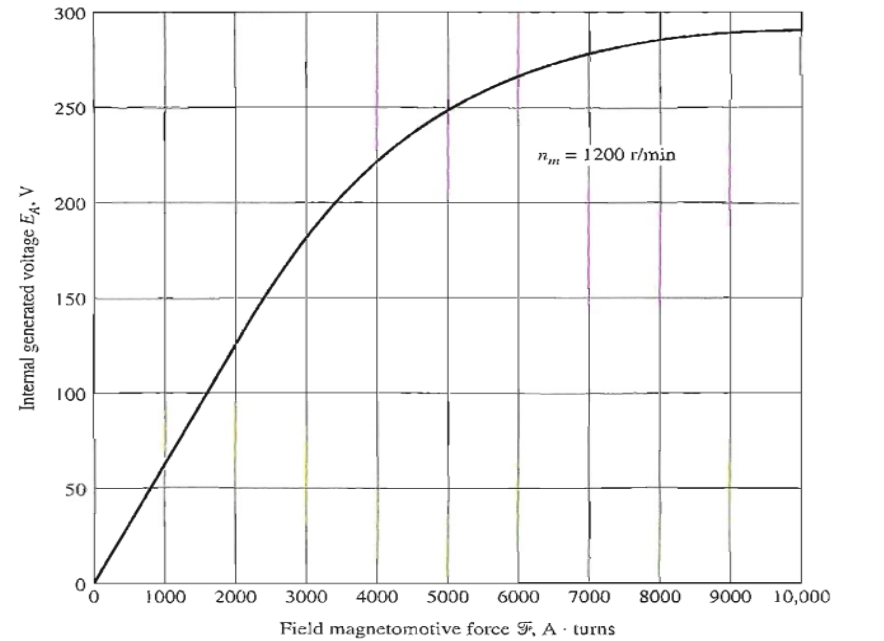


FIGURE 8–22
The magnetization curve of the motor in Example 8–5. This curve was taken at speed $n_m = 1200$ r/min.

$$\tau_{\text{ind}} = \frac{E_A I_A}{\omega_m}$$

$$= \frac{(246\ \text{V})(50\ \text{A})}{(3690\ \text{r/min})(1\ \text{min}/60\ \text{s})(2\pi\ \text{rad/r})} = 31.8\ \text{N} \cdot \text{m}$$

Speed Control of Series DC Motors

Unlike with the shunt dc motor, there is only one efficient way to change the speed of a series dc motor. That method is to change the terminal voltage of the motor. If the terminal voltage is increased, the first term in Equation (8–23) is increased, resulting in a *higher speed for any given torque*.

The speed of series dc motors can also be controlled by the insertion of a series resistor into the motor circuit, but this technique is very wasteful of power and is used only for intermittent periods during the start-up of some motors.

Until the last 40 years or so, there was no convenient way to change V_T, so the only method of speed control available was the wasteful series resistance method. That has all changed today with the introduction of solid-state control circuits.

$$\omega_m = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc} \qquad (8\text{–}23)$$

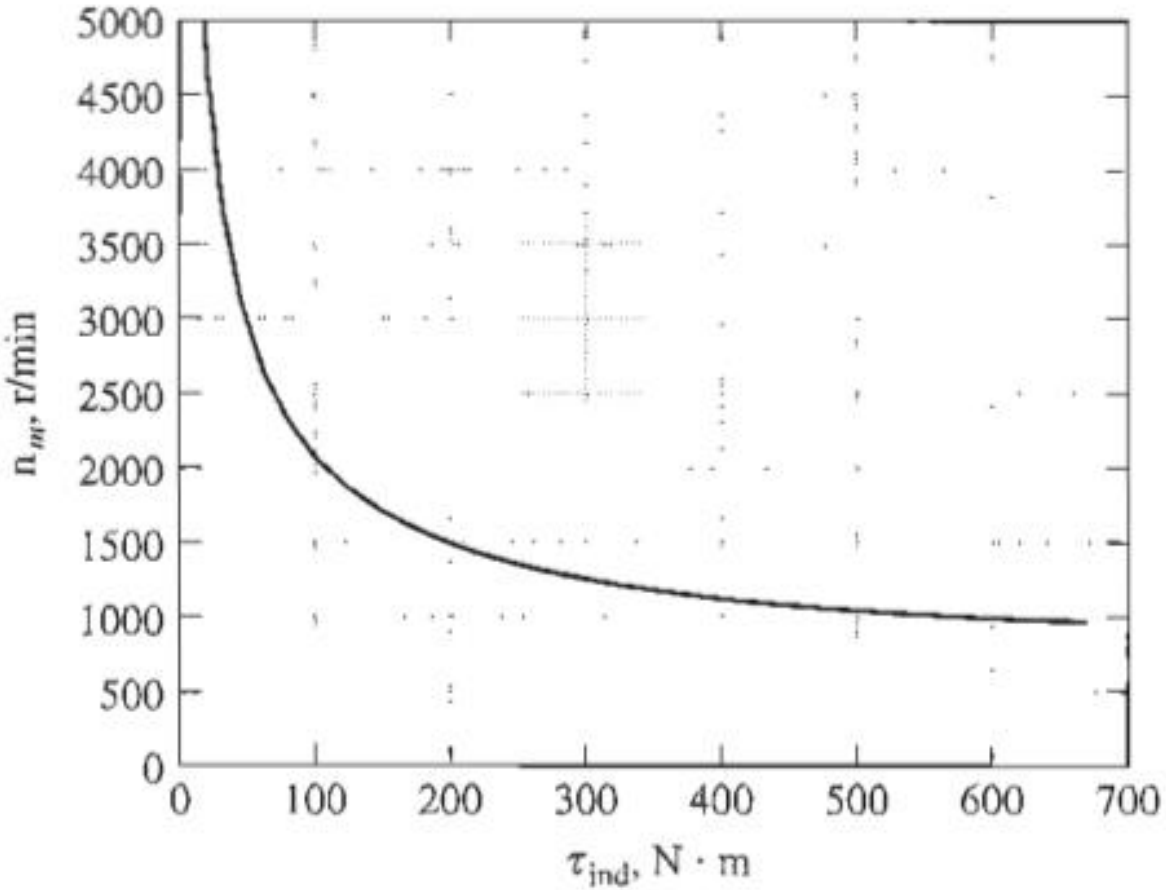


FIGURE 8–23
The torque–speed characteristic of the series dc motor in Example 8–5.

Parameter	DC Series Motor	Shunt Motor
Basic	In a dc series motor, the field winding is connected in series with the armature winding.	In a dc shunt motor, the field winding is connected in parallel with the armature winding.
Field current	In dc series motors, the field current is same as the armature current.	In dc shunt motors, the field current is not equal to armature current.
Voltage across armature and field windings	In dc series motors, the voltages across armature and field windings are different depending on their resistance.	In dc shunt motors, the voltage across the armature winding and field winding is same and is equal to the supply voltage.

Field winding design	A dc series motor has a field winding design with few turns of thick conductor wire.	The field winding of a dc shunt motor is designed with large number of turns of fine wire.
Resistance of field winding	The field winding of a dc series motor has very low resistance as it carries full armature current.	The field winding of a dc shunt motor has a very high resistance, so that it draws a very small current.
Staring torque	A dc series motor develops very high starting torque.	A dc shunt motor develops low and constant starting torque.

Torque-current relation	In dc series motors, the torque is directly proportional to the square of the armature current i.e.,	In dc shunt motors, the torque is directly proportional to the armature current i.e.,
Starting current	High starting current due to combined field and armature winding in series.	Lower starting current due to the parallel winding arrangement.
Speed	The dc series motors are variable speed dc motors.	The dc shunt motors are approximately constant speed motors.

Speed-load characteristics	DC series motors have low speed at high loads, whereas dangerously high speed at low loads.	There is not significant change in the speed of dc shunt motor with the change in load.
Speed regulation	Poor speed regulation	Its speed regulation is better as compared to the DC Series motor.
Dynamic response	DC series motors have a faster dynamic response due to high starting torque.	The dynamic response of dc shunt motors is lower as compared to dc series motors.
Rotor design	The design of rotor of dc series motors is optimized for high starting torque.	In dc shunt motors, the rotor design is optimized for constant speeds.

Energy efficiency	DC series motors have lower energy efficiency at high speeds and light loads.	DC shunt motors provide better energy efficiency at a wide range of speeds and loads.
Field current adjustment	In dc series motors, it is not possible to adjust the field current independently as it has same field and armature currents.	In dc shunt motors, the field current can be adjusted independently without affecting the armature current.
Need of load balancing mechanism	DC series motors require a load balancing mechanism to prevent over speeding.	There is no need of this kind of mechanism as they have almost constant speed.

Applications	DC series motors are used for applications where high starting torque is required like in electric locomotives, hoists & cranes, electric bikes and cars, etc.	DC shunt motors are used in applications where constant speed is required like in fans, blowers, elevators, centrifugal pumps, lath machines, etc.

8.7 THE COMPOUNDED DC MOTOR

A compounded dc motor is a motor with *both a shunt and a series field*. Such a motor is shown in Figure 8–24. The dots that appear on the two field coils have the same meaning as the dots on a transformer: *Current flowing into a dot produces a positive magnetomotive force*. If current flows into the dots on both field coils, the resulting magnetomotive forces combine to produce a larger total magnetomotive force. This situation is known as *cumulative compounding*. If current flows into the dot on one field coil and out of the dot on the other field coil, the resulting magnetomotive forces subtract. In Figure 8–24 the round dots correspond to cumulative compounding of the motor, and the squares correspond to differential compounding.

The Kirchhoff's voltage law equation for a compounded dc motor is

$$V_T = E_A + I_A(R_A + R_S) \tag{8–24}$$

The currents in the compounded motor are related by

$$I_A = I_L - I_F \tag{8–25}$$

$$I_F = \frac{V_T}{R_F} \tag{8–26}$$

The net magnetomotive force and the effective shunt field current in the compounded motor are given by

$$\boxed{\mathcal{F}_{\text{net}} = \mathcal{F}_F \pm \mathcal{F}_{\text{SE}} - \mathcal{F}_{\text{AR}}} \tag{8–27}$$

and

$$\boxed{I_F^* = I_F \pm \frac{N_{SE}}{N_F} I_A - \frac{\mathcal{F}_{\text{AR}}}{N_F}} \tag{8–28}$$

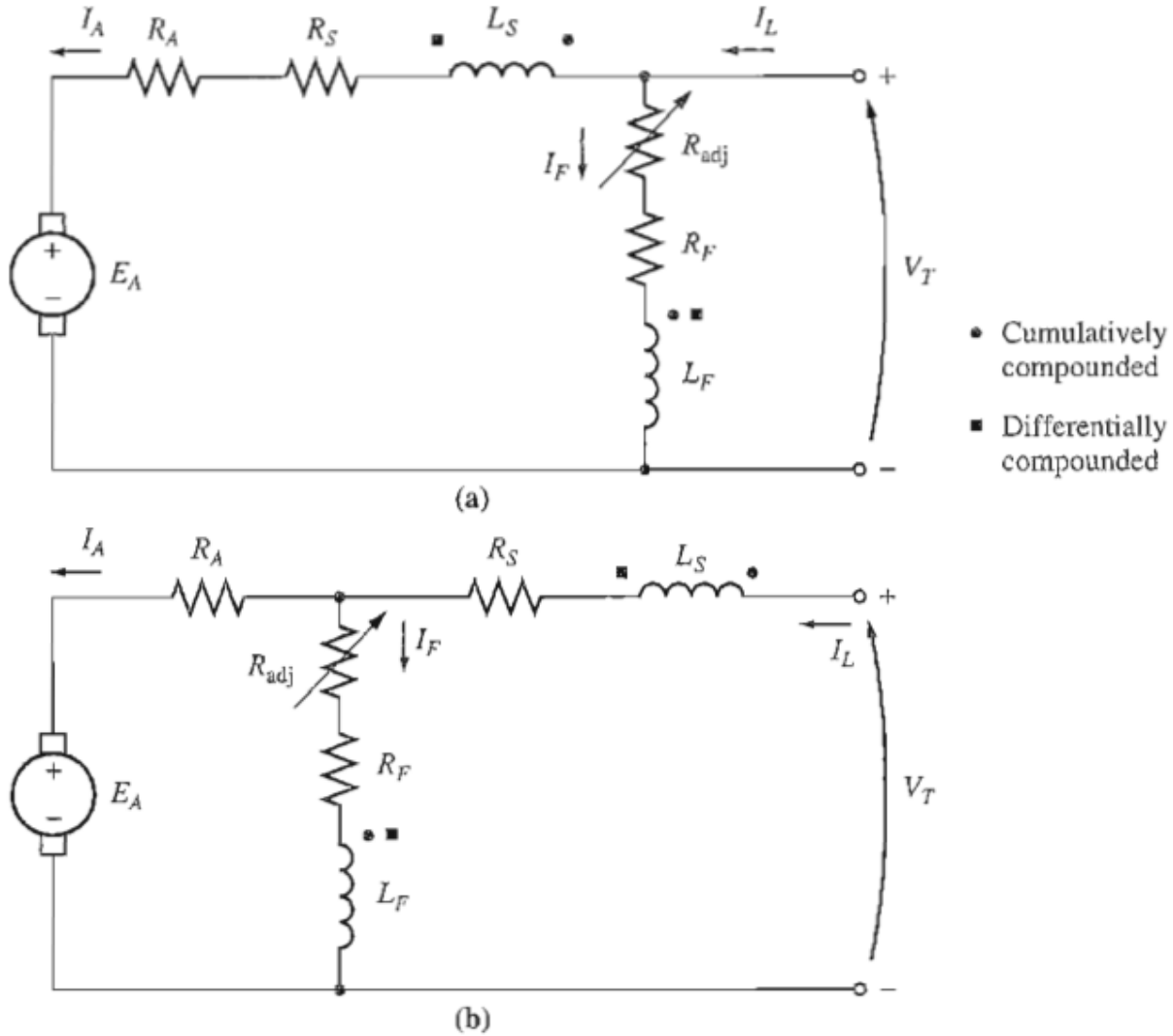


FIGURE 8–24
The equivalent circuit of compounded dc motors: (a) long-shunt connection; (b) short-shunt connection.

Example 8–8. A 50-hp, 250-V, 1200 r/min shunt dc motor has a rated armature current of 170 A and a rated field current of 5 A. When its rotor is blocked, an armature voltage of 10.2 V (exclusive of brushes) produces 170 A of current flow, and a field voltage of 250 V produces a field current flow of 5 A. The brush voltage drop is assumed to be 2 V. At no load with the terminal voltage equal to 240 V, the armature current is equal to 13.2 A, the field current is 4.8 A, and the motor's speed is 1150 r/min.

(a) How much power is output from this motor at rated conditions?
(b) What is the motor's efficiency?

Solution

The armature resistance of this machine is approximately

$$R_A = \frac{10.2\ \text{V}}{170\ \text{A}} = 0.06\ \Omega$$

and the field resistance is

$$R_F = \frac{250\ \text{V}}{5\ \text{A}} = 50\ \Omega$$

Therefore, at full load the armature I^2R losses are

$$P_A = (170\ \text{A})^2(0.06\ \Omega) = 1734\ \text{W}$$

and the field circuit I^2R losses are

$$P_F = (5\text{ A})^2(50\ \Omega) = 1250\text{ W}$$

The brush losses at full load are given by

$$P_{\text{brush}} = V_{\text{BD}} I_A = (2\text{ V})(170\text{ A}) = 340\text{ W}$$

The rotational losses at full load are essentially equivalent to the rotational losses at no load, since the no-load and full-load speeds of the motor do not differ too greatly. These losses may be ascertained by determining the input power to the armature circuit at no load and assuming that the armature copper and brush drop losses are negligible, meaning that the no-load armature input power is equal to the rotational losses:

$$P_{\text{tot}} = P_{\text{core}} + P_{\text{mech}} = (240\text{ V})(13.2\text{ A}) = 3168\text{ W}$$

(a) The input power of this motor at the rated load is given by

$$P_{\text{in}} = V_T I_L = (250\text{ V})(175\text{ A}) = 43{,}750\text{ W}$$

Its output power is given by

$$\begin{aligned} P_{\text{out}} &= P_{\text{in}} - P_{\text{brush}} - P_{\text{cu}} - P_{\text{core}} - P_{\text{mech}} - P_{\text{stray}} \\ &= 43{,}750\text{ W} - 340\text{ W} - 1734\text{ W} - 1250\text{ W} - 3168\text{ W} - (0.01)(43{,}750\text{ W}) \\ &= 36{,}820\text{ W} \end{aligned}$$

where the stray losses are taken to be 1 percent of the input power.

(b) The efficiency of this motor at full load is

$$\begin{aligned} \eta &= \frac{P_{\text{out}}}{P_{\text{out}}} \times 100\% \\ &= \frac{36{,}820\text{ W}}{43{,}750\text{ W}} \times 100\% = 84.2\% \end{aligned}$$

8.11 INTRODUCTION TO DC GENERATORS

DC generators are dc machines used as generators. As previously noted, there is no real difference between a generator and a motor except for the direction of power flow. There are five major types of dc generators, classified according to the manner in which their field flux is produced:

1. *Separately excited generator.* In a separately excited generator, the field flux is derived from a separate power source independent of the generator itself.
2. *Shunt generator.* In a shunt generator, the field flux is derived by connecting the field circuit directly across the terminals of the generator.
3. *Series generator.* In a series generator, the field flux is produced by connecting the field circuit in series with the armature of the generator.
4. *Cumulatively compounded generator.* In a cumulatively compounded generator, both a shunt and a series field are present, and their effects are additive.
5. *Differentially compounded generator.* In a differentially compounded generator, both a shunt and a series field are present, but their effects are subtractive.

These various types of dc generators differ in their terminal (voltage–current) characteristics, and therefore in the applications to which they are suited.

DC generators are compared by their voltages, power ratings, efficiencies, and voltage regulations. *Voltage regulation* (VR) is defined by the equation

$$\text{VR} = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\% \tag{8–39}$$

where V_{nl} is the no-load terminal voltage of the generator and V_{fl} is the full-load terminal voltage of the generator. It is a rough measure of the shape of the generator's voltage–current characteristic—a positive voltage regulation means a drooping characteristic, and a negative voltage regulation means a rising characteristic.

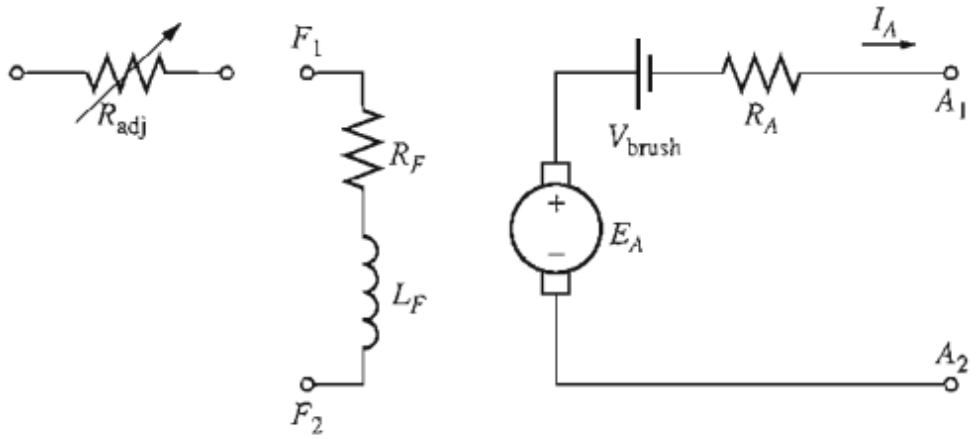


FIGURE 8–42
The equivalent circuit of a dc generator.

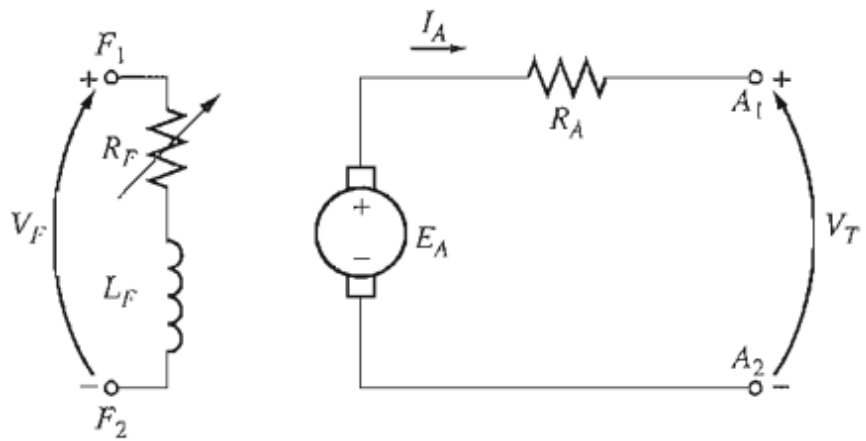


FIGURE 8–43
A simplified equivalent circuit of a dc generator, with R_F combining the resistances of the field coils and the variable control resistor.

8.12 THE SEPARATELY EXCITED GENERATOR

A separately excited dc generator is a generator whose field current is supplied by a separate external dc voltage source. The equivalent circuit of such a machine is shown in Figure 8–44. In this circuit, the voltage V_T represents the actual voltage measured at the terminals of the generator, and the current I_L represents the current flowing in the lines connected to the terminals. The internal generated voltage is E_A, and the armature current is I_A. It is clear that the armature current is equal to the line current in a separately excited generator:

$$I_A = I_L \tag{8–40}$$

The Terminal Characteristic of a Separately Excited DC Generator

The *terminal characteristic* of a device is a plot of the output quantities of the device versus each other. For a dc generator, the output quantities are its terminal voltage and line current. The terminal characteristic of a separately excited generator is

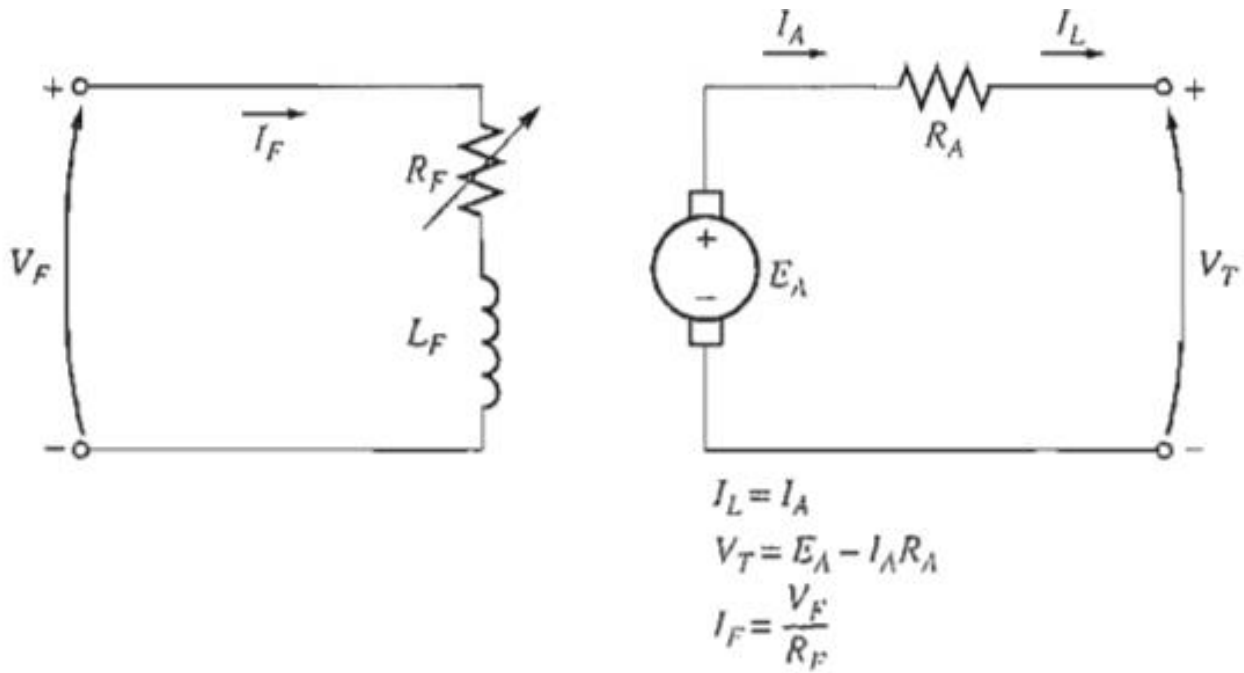


FIGURE 8–44
A separately excited dc generator.

thus a plot of V_T versus I_L for a constant speed ω. By Kirchhoff's voltage law, the terminal voltage is

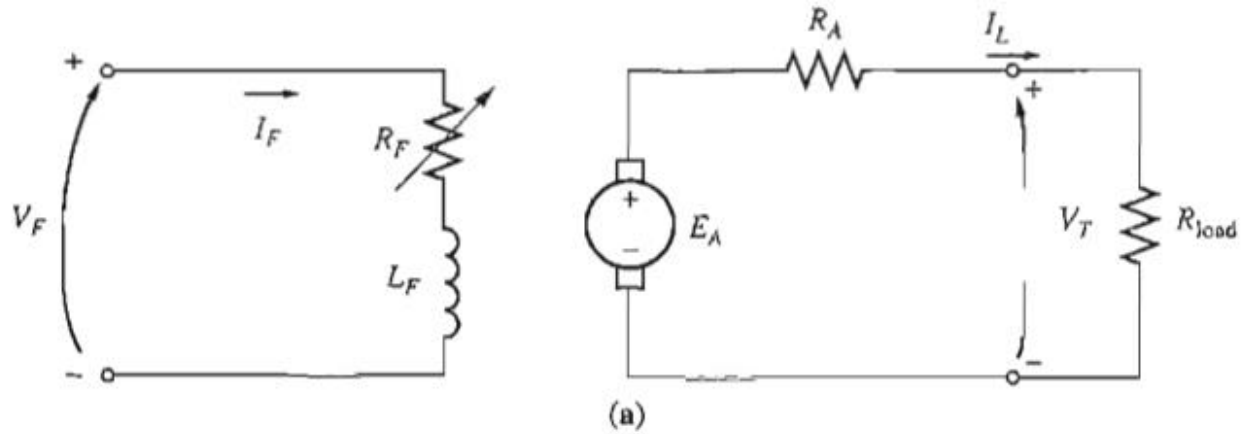
$$\boxed{V_T = E_A - I_A R_A} \tag{8–41}$$

Nonlinear Analysis of a Separately Excited DC Generator

Because the internal generated voltage of a generator is a nonlinear function of its magnetomotive force, it is not possible to calculate simply the value of E_A to be expected from a given field current. The magnetization curve of the generator must be used to accurately calculate its output voltage for a given input voltage.

In addition, if a machine has armature reaction, its flux will be reduced with each increase in load, causing E_A to decrease. The only way to accurately determine the output voltage in a machine with armature reaction is to use graphical analysis.

The total magnetomotive force in a separately excited generator is the field circuit magnetomotive force less the magnetomotive force due to armature reaction (AR):



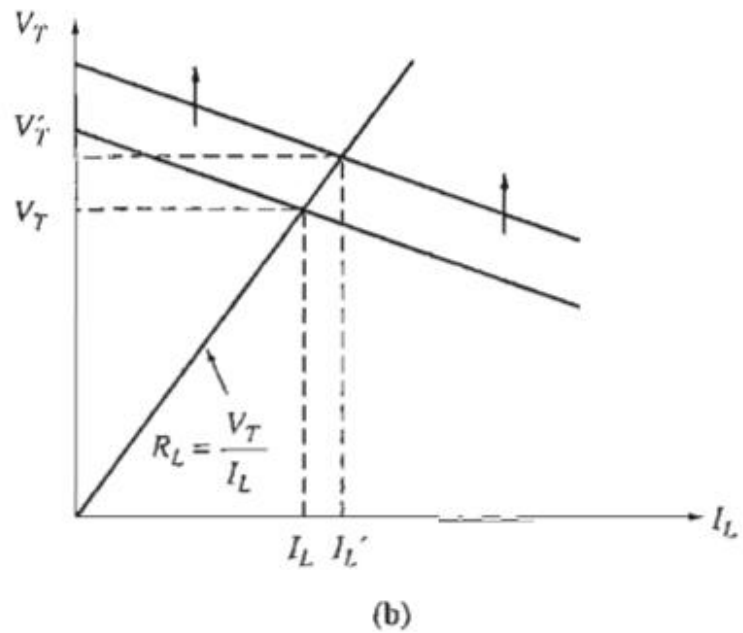


FIGURE 8–46
(a) A separately excited dc generator with a resistive load. (b) The effect of a decrease in field resistance on the output voltage of the generator.

$$\mathcal{F}_{net} = N_F I_F - \mathcal{F}_{AR} \tag{8–42}$$

As with dc motors, it is customary to define an *equivalent field current* that would produce the same output voltage as the combination of all the magnetomotive forces in the machine. The resulting voltage E_{A0} can then be determined by locating that equivalent field current on the magnetization curve. The equivalent field current of a separately excited dc generator is given by

$$I_F^* = I_F - \frac{\mathcal{F}_{AR}}{N_F} \tag{8–43}$$

Also, the difference between the speed of the magnetization curve and the real speed of the generator must be taken into account using Equation (8–13):

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \tag{8–13}$$

The following example illustrates the analysis of a separately excited dc generator.

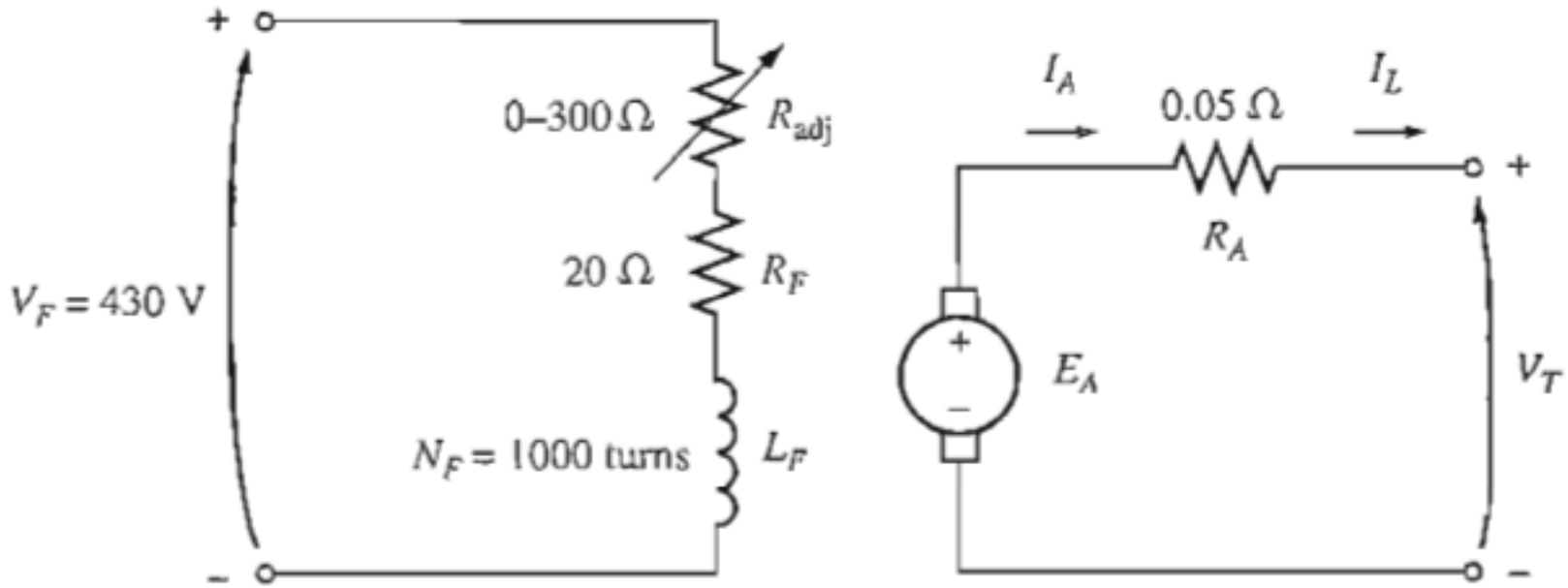


FIGURE 8–47
The separately excited dc generator in Example 8–9.

Example 8–9. A separately excited dc generator is rated at 172 kW, 430 V, 400 A, and 1800 r/min. It is shown in Figure 8–47, and its magnetization curve is shown in Figure 8–48. This machine has the following characteristics:

$$R_A = 0.05\ \Omega \qquad V_F = 430\ \text{V}$$
$$R_F = 20\ \Omega \qquad N_F = 1000\ \text{turns per pole}$$
$$R_{adj} = 0 \text{ to } 300\ \Omega$$

(a) If the variable resistor R_{adj} in this generator's field circuit is adjusted to 63 Ω and the generator's prime mover is driving it at 1600 r/min, what is this generator's no-load terminal voltage?

(b) What would its voltage be if a 360-A load were connected to its terminals? Assume that the generator has compensating windings.

(c) What would its voltage be if a 360-A load were connected to its terminals but the generator does not have compensating windings? Assume that its armature reaction at this load is 450 A • turns.

(d) What adjustment could be made to the generator to restore its terminal voltage to the value found in part *a*?

(e) How much field current would be needed to restore the terminal voltage to its no-load value? (Assume that the machine has compensating windings.) What is the required value for the resistor R_{adj} to accomplish this?

Solution

(a) If the generator's total field circuit resistance is

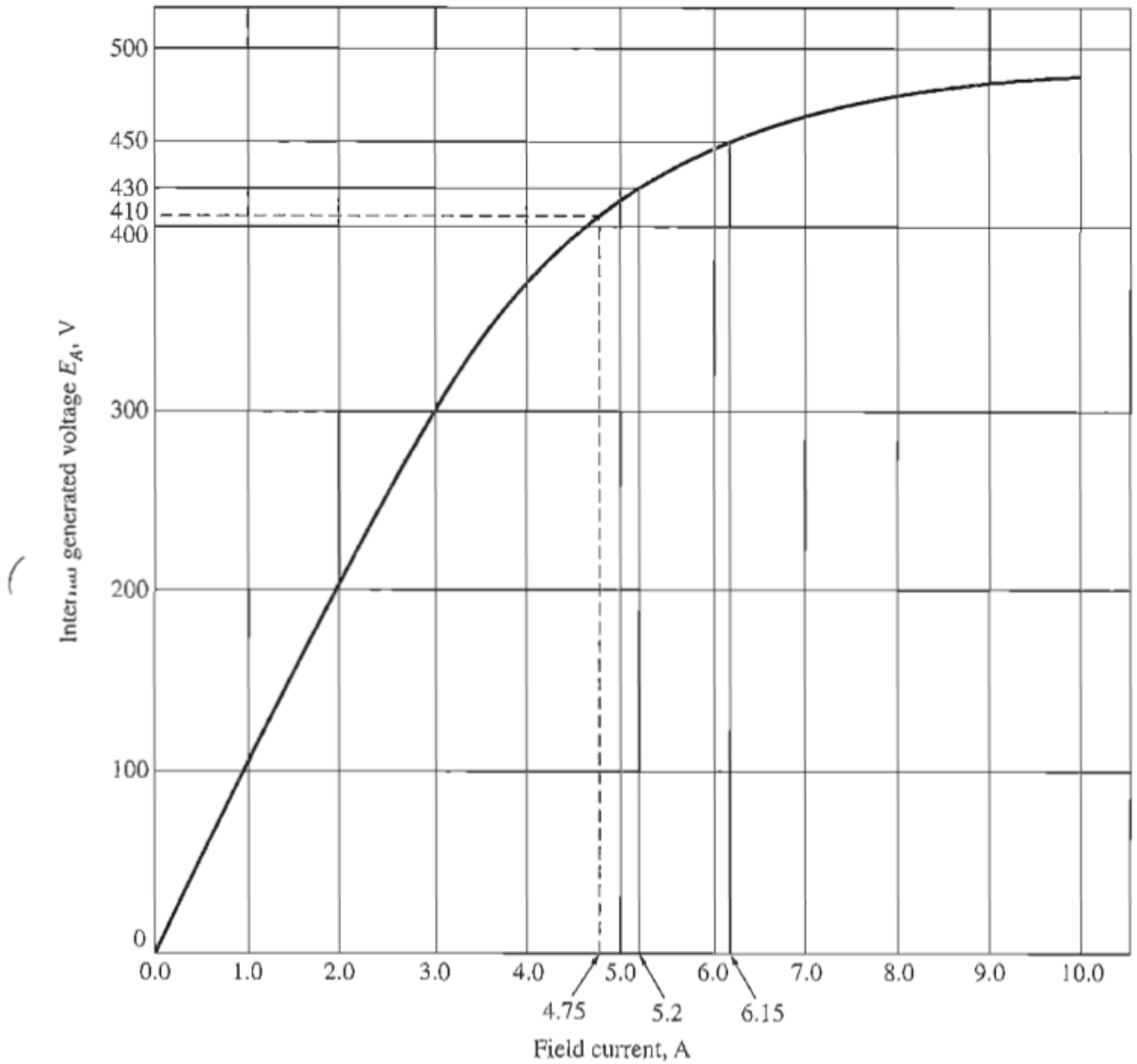
$$R_F + R_{adj} = 83\ \Omega$$

then the field current in the machine is

$$I_F = \frac{V_F}{R_F} = \frac{430\ \text{V}}{83\ \Omega} = 5.2\ \text{A}$$

From the machine's magnetization curve, this much current would produce a voltage $E_{A0} = 430$ V at a speed of 1800 r/min. Since this generator is actually turning at $n_m = 1600$ r/min, its internal generated voltage E_A will be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \qquad (8\text{–}13)$$



Note: When the field current is zero, E_A is about 3 V.

FIGURE 8–48
The magnetization curve for the generator in Example 8–9.

$$E_A = \frac{1600 \text{ r/min}}{1800 \text{ r/min}} 430 \text{ V} = 382 \text{ V}$$

Since $V_T = E_A$ at no-load conditions, the output voltage of the generator is $V_T = 382$ V.

(b) If a 360-A load were connected to this generator's terminals, the terminal voltage of the generator would be

$$V_T = E_A - I_A R_A = 382 \text{ V} - (360 \text{ A})(0.05 \ \Omega) = 364 \text{ V}$$

(c) If a 360-A load were connected to this generator's terminals and the generator had 450 A • turns of armature reaction, the effective field current would be

$$I_F^* = I_F - \frac{\mathcal{F}_{AR}}{N_F} = 5.2 \text{ A} - \frac{450 \text{ A} \cdot \text{turns}}{1000 \text{ turns}} = 4.75 \text{ A}$$

From the magnetization curve, $E_{A0} = 410$ V, so the internal generated voltage at 1600 r/min would be

$$\frac{E_A}{E_{A0}} = \frac{n}{n_0} \tag{8–13}$$

$$E_A = \frac{1600 \text{ r/min}}{1800 \text{ r/min}} 410 \text{ V} = 364 \text{ V}$$

Therefore, the terminal voltage of the generator would be

$$V_T = E_A - I_A R_A = 364 \text{ V} - (360 \text{ A})(0.05 \ \Omega) = 346 \text{ V}$$

It is lower than before due to the armature reaction.

(d) The voltage at the terminals of the generator has fallen, so to restore it to its original value, the voltage of the generator must be increased. This requires an increase in E_A, which implies that R_{adj} must be decreased to increase the field current of the generator.

(e) For the terminal voltage to go back up to 382 V, the required value of E_A is

$$E_A = V_T + I_A R_A = 382 \text{ V} + (360 \text{ A})(0.05 \ \Omega) = 400 \text{ V}$$

To get a voltage E_A of 400 V at $n_m = 1600$ r/min, the equivalent voltage at 1800 r/min would be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \tag{8–13}$$

$$E_{A0} = \frac{1800 \text{ r/min}}{1600 \text{ r/min}} 400 \text{ V} = 450 \text{ V}$$

1600 r/min

From the magnetization curve, this voltage would require a field current of I_F = 6.15 A. The field circuit resistance would have to be

$$R_F + R_{adj} = \frac{V_f}{I_F}$$

$$20\ \Omega + R_{adj} = \frac{430\ \text{V}}{6.15\ \text{A}} = 69.9\ \Omega$$

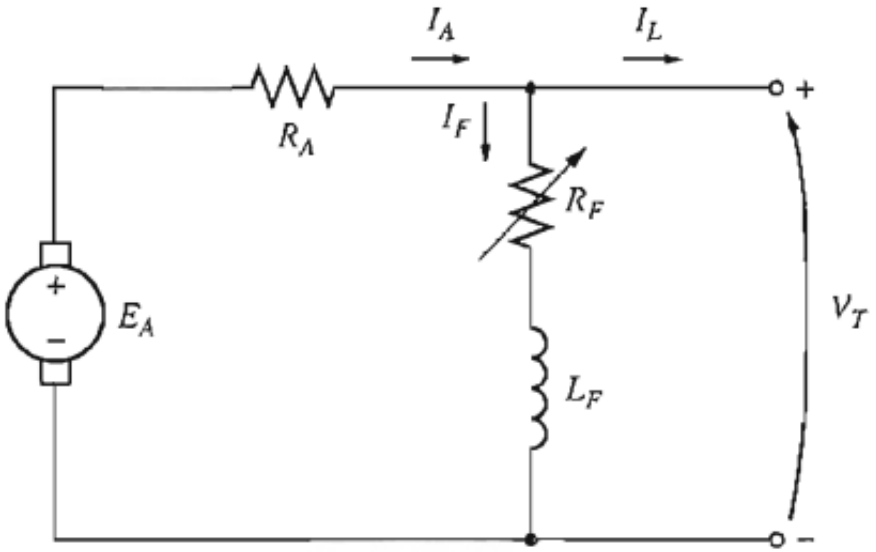
$$R_{adj} = 49.9\ \Omega \approx 50\ \Omega$$

Notice that, for the same field current and load current, the generator with armature reaction had a lower output voltage than the generator without armature reaction. The armature reaction in this generator is exaggerated to illustrate its effects—it is a good deal smaller in well-designed modern machines.

8.13 THE SHUNT DC GENERATOR

A shunt dc generator is a dc generator that supplies its own field current by having its field connected directly across the terminals of the machine. The equivalent circuit of a shunt dc generator is shown in Figure 8–49. In this circuit, the armature current of the machine supplies both the field circuit and the load attached to the machine:

$$\boxed{I_A = I_F + I_L} \qquad (8\text{–}44)$$



$$I_A = I_F + I_L$$

$$V_T = E_A - I_A R_A$$

$$I_F = \frac{V_T}{R_F}$$

FIGURE 8–49
The equivalent circuit of a shunt dc generator.

The Kirchhoff's voltage law equation for the armature circuit of this machine is

$$\boxed{V_T = E_A - I_A R_A} \tag{8–45}$$

This type of generator has a distinct advantage over the separately excited dc generator in that no external power supply is required for the field circuit. But that leaves an important question unanswered: If the generator supplies its own field current, how does it get the initial field flux to start when it is first turned on?

8.14 THE SERIES DC GENERATOR

A series dc generator is a generator whose field is connected in series with its armature. Since the armature has a *much* higher current than a shunt field, the series field in a generator of this sort will have only a very few turns of wire, and the wire used will be much thicker than the wire in a shunt field. Because magnetomotive force is given by the equation $\mathcal{F} = NI$, exactly the same magnetomotive force can be produced from a few turns with high current as can be produced from many turns with low current. Since the full-load current flows through it, a series field is designed to have the lowest possible resistance. The equivalent circuit of a series dc generator is shown in Figure 8–56. Here, the armature current, field

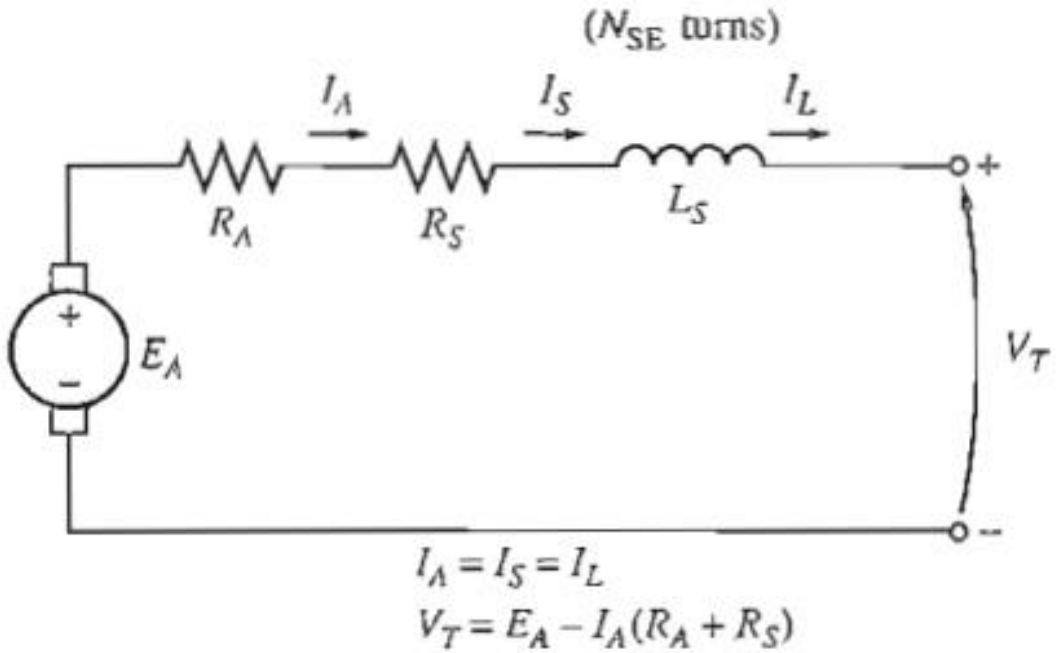
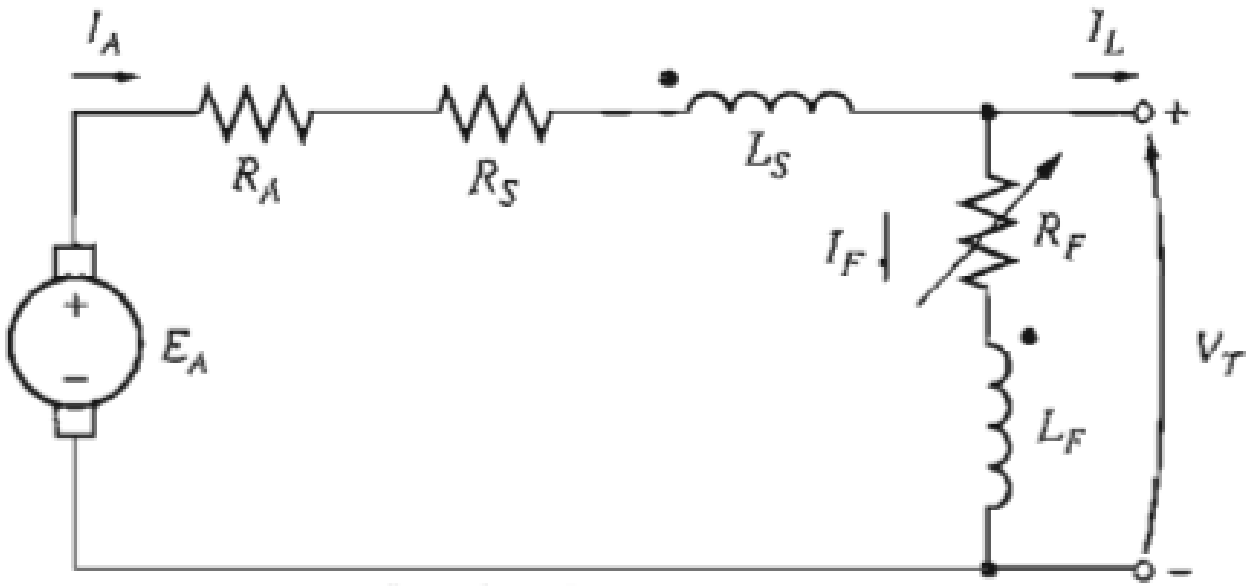


FIGURE 8–56
The equivalent circuit of a series dc generator.

current, and line current all have the same value. The Kirchhoff's voltage law equation for this machine is

$$V_T = E_A - I_A(R_A + R_S) \tag{8–47}$$



$$I_A = I_L + I_F$$

$$V_T = E_A - I_A(R_A + R_S)$$

$$I_F = \frac{V_T}{R_F}$$

$$\mathcal{F}_{net} = N_F I_F + N_{SE} I_A - \mathcal{F}_{AR}$$

FIGURE 8–59
The equivalent circuit of a cumulatively compounded dc generator with a long-shunt connection.